

MATHEMATICAL PAPERS.

LONDON: C. J. CLAY AND SONS,
CAMBRIDGE UNIVERSITY PRESS WAREHOUSE,
AVE MARIA LANE.
Glasgow: 263, ARGYLE STREET.

Leipzig: F. A. BROCKHAUS.
New York: THE MACMILLAN COMPANY.

THE COLLECTED
MATHEMATICAL PAPERS

OF

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VOL. XII.

CAMBRIDGE:
AT THE UNIVERSITY PRESS.
1897.

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AT THE UNIVERSITY PRESS.

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THE present volume contains 89 papers, numbered 799 to 887, published for the most part in the years 1883 to 1889.

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A. R. FORSYTH.

19 May, 1897.

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799.

ON CURVILINEAR COORDINATES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIX. (1883),
pp. 1—22.]

THE present memoir is based upon Mr Warren's "Exercises in Curvilinear and Normal Coordinates," *Camb. Phil. Trans.* t. XII. (1877), pp. 455—502, and has for a principal object the establishment of the six differential equations of the second order corresponding to his six equations for normal coordinates: but the notation is different; the results are more general, inasmuch as I use throughout general curvilinear coordinates instead of his normal coordinates; and as regards my six equations for general curvilinear coordinates, the terms containing differential coefficients of the first order are presented under a different form.

If the position of a point in space is determined by the rectangular coordinates x, y, z ; then p, q, r being each of them a given function of x, y, z , we have conversely x, y, z , each of them a given function of p, q, r , which are thus in effect coordinates serving to determine the position of the point, and are called curvilinear coordinates.

But it is not in the first instance necessary to regard x, y, z as rectangular coordinates, or even as Cartesian coordinates at all; we are simply concerned with the two sets of variables x, y, z and p, q, r , each variable of the one set being a given function of the variables of the other set; and, in particular, the x, y, z are regarded as being each of them a given function of the p, q, r .

Except as regards the symbols ξ, η, ζ presently mentioned, the suffixes 1, 2, 3 refer to the variables p, q, r respectively, and are used to denote differentiations in regard to these variables, viz.

$$x_1, x_2, x_3, x_{11}, x_{22}, x_{33}, \dots$$

are written to denote

$$\frac{dx}{dp}, \frac{dx}{dq}, \frac{dx}{dr}, \frac{d^2x}{dp^2}, \frac{d^2x}{dq^2}, \frac{d^2x}{dr^2}, \text{ \&c.},$$

and so in other cases; in particular,

$$\begin{aligned} x_1, x_2, x_3 \text{ denote } \frac{dx}{dp}, \frac{dx}{dq}, \frac{dx}{dr}, \\ y_1, y_2, y_3 \text{ ,, } \frac{dy}{dp}, \frac{dy}{dq}, \frac{dy}{dr}, \\ z_1, z_2, z_3 \text{ ,, } \frac{dz}{dp}, \frac{dz}{dq}, \frac{dz}{dr}. \end{aligned}$$

The minors formed with these differential coefficients are denoted by suffixed letters ξ, η, ζ , thus

$$\xi_1, \xi_2, \xi_3 \text{ denote } y_2z_3 - y_3z_2, y_3z_1 - y_1z_3, y_1z_2 - y_2z_1,$$

$$\eta_1, \eta_2, \eta_3 \text{ ,, } z_2x_3 - z_3x_2, z_3x_1 - z_1x_3, z_1x_2 - z_2x_1,$$

$$\zeta_1, \zeta_2, \zeta_3 \text{ ,, } x_2y_3 - x_3y_2, x_3y_1 - x_1y_3, x_1y_2 - x_2y_1,$$

so that, as regards these letters ξ, η, ζ , the suffixes do not denote differentiations.

The determinant

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \text{ is put } = L;$$

and the symbols $(a, b, c, f, g, h), (A, B, C, F, G, H)$ are defined as follows:

$$\begin{aligned} a &= x_1^2 + y_1^2 + z_1^2, & A &= \xi_1^2 + \eta_1^2 + \zeta_1^2, \\ b &= x_2^2 + y_2^2 + z_2^2, & B &= \xi_2^2 + \eta_2^2 + \zeta_2^2, \\ c &= x_3^2 + y_3^2 + z_3^2, & C &= \xi_3^2 + \eta_3^2 + \zeta_3^2, \\ f &= x_2x_3 + y_2y_3 + z_2z_3, & F &= \xi_2\xi_3 + \eta_2\eta_3 + \zeta_2\zeta_3, \\ g &= x_3x_1 + y_3y_1 + z_3z_1, & G &= \xi_3\xi_1 + \eta_3\eta_1 + \zeta_3\zeta_1, \\ h &= x_1x_2 + y_1y_2 + z_1z_2, & H &= \xi_1\xi_2 + \eta_1\eta_2 + \zeta_1\zeta_2. \end{aligned}$$

We have then, further,

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = L, \quad \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = L^2,$$

$$\begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix} = L^2, \quad \begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = \frac{1}{L^2};$$

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

$$L^2(a, b, c, f, g, h) = (BC - F^2, CA - G^2, AB - H^2, GH - AF, HF - BG, FG - CH),$$

which equations are at once proved, and are fundamental ones in the theory.

It is convenient to add that we have

$$\begin{vmatrix} \frac{dp}{dx} & \frac{dp}{dy} & \frac{dp}{dz} \\ \frac{dq}{dx} & \frac{dq}{dy} & \frac{dq}{dz} \\ \frac{dr}{dx} & \frac{dr}{dy} & \frac{dr}{dz} \end{vmatrix} = \frac{1}{L} \begin{vmatrix} \xi_1 & \eta_1 & \zeta_1 \\ \xi_2 & \eta_2 & \zeta_2 \\ \xi_3 & \eta_3 & \zeta_3 \end{vmatrix},$$

that is, $\frac{dp}{dx} = \frac{1}{L} \xi_1$, &c.; and, further,

$$dx^2 + dy^2 + dz^2 = (a, b, c, f, g, h \mid dp, dq, dr)^2,$$

$$dp^2 + dq^2 + dr^2 = \frac{1}{L^2} (A, B, C, F, G, H \mid dx, dy, dz)^2.$$

Differentiating the values of a, b, c, f, g, h , we have $\frac{1}{2}a_1 = x_1x_{11} + y_1y_{11} + z_1z_{11}$, which may be written in the abbreviated form $\frac{1}{2}a_1 = 1.11$; similarly

$$x_1x_{12} + x_2x_{11} + y_1y_{12} + y_2y_{11} + z_1z_{12} + z_2z_{11},$$

which in like manner may be written $f_1 = 2.13 + 3.12$, and so in other cases; observe that, in the duad part of any symbol, the order of the numbers is immaterial, $2.13 = 2.31$. The whole system of equations is

$\frac{1}{2}a_1 = 1.11,$	$\frac{1}{2}a_2 = 1.12,$	$\frac{1}{2}a_3 = 1.13,$
$\frac{1}{2}b_1 = 2.12,$	$\frac{1}{2}b_2 = 2.22,$	$\frac{1}{2}b_3 = 2.23,$
$\frac{1}{2}c_1 = 3.13,$	$\frac{1}{2}c_2 = 3.23,$	$\frac{1}{2}c_3 = 3.33,$
$f_1 = 2.13 + 3.12,$	$f_2 = 2.23 + 3.22,$	$f_3 = 2.33 + 3.23,$
$g_1 = 3.11 + 1.13,$	$g_2 = 3.12 + 1.23,$	$g_3 = 3.13 + 1.33,$
$h_1 = 1.12 + 2.11,$	$h_2 = 1.22 + 2.12,$	$h_3 = 1.23 + 2.13.$

These may also be written

$1.11 = \frac{1}{2}a_1,$	$1.22 = h_2 - \frac{1}{2}b_1,$	$1.33 = g_3 - \frac{1}{2}c_1,$
$2.11 = h_1 - \frac{1}{2}a_2,$	$2.22 = \frac{1}{2}b_2,$	$2.33 = f_3 - \frac{1}{2}c_2,$
$3.11 = g_1 - \frac{1}{2}a_3,$	$3.22 = f_2 - \frac{1}{2}b_3,$	$3.33 = \frac{1}{2}c_3,$
$1.23 = \frac{1}{2}(-f_1 + g_2 + h_3),$	$1.31 = \frac{1}{2}a_3,$	$1.12 = \frac{1}{2}a_2,$
$2.23 = \frac{1}{2}b_3,$	$2.31 = \frac{1}{2}(f_1 - g_2 + h_3),$	$2.12 = \frac{1}{2}b_1,$
$3.23 = \frac{1}{2}c_2,$	$3.31 = \frac{1}{2}c_1,$	$3.12 = \frac{1}{2}(f_1 + g_2 - h_3).$

It is to be observed that we can, from each system of three equations, express a set of second differential coefficients of the x, y, z in terms of the first differential

1—2

coefficients of the a, b, c, f, g, h ; thus the three equations containing 11, written at length, are

$$\begin{aligned} x_1x_{11} + y_1y_{11} + z_1z_{11} &= \frac{1}{2}a_1, \\ x_2 + y_2 + z_2 &= h_1 - \frac{1}{2}a_2, \\ x_3 + y_3 + z_3 &= g_1 - \frac{1}{2}a_3, \end{aligned}$$

three linear equations for the determination of x_{11}, y_{11}, z_{11} ; hence for x_{11} we have

$$x_{11} \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \frac{1}{2}a_1(y_2z_3 - y_3z_2) + (h_1 - \frac{1}{2}a_2)(y_3z_1 - y_1z_3) + (g_1 - \frac{1}{2}a_3)(y_1z_2 - y_2z_1),$$

or, what is the same thing,

$$L x_{11} = \frac{1}{2}a_1 \xi_1 + (h_1 - \frac{1}{2}a_2) \xi_2 + (g_1 - \frac{1}{2}a_3) \xi_3,$$

and so for y_{11}, z_{11} ; or if (as in the sequel) we desire the value of a linear function $\alpha x_{11} + \beta y_{11} + \gamma z_{11}$, calling this for a moment $\square$, we join to the foregoing a new equation

$$\alpha x_{11} + \beta y_{11} + \gamma z_{11} = \square,$$

and then, eliminating the three quantities, we have

$$\begin{vmatrix} \alpha & \beta & \gamma & \square \\ x_1 & y_1 & z_1 & \frac{1}{2}a_1 \\ x_2 & y_2 & z_2 & h_1 - \frac{1}{2}a_2 \\ x_3 & y_3 & z_3 & g_1 - \frac{1}{2}a_3 \end{vmatrix} = 0,$$

giving $L\square$ as a linear function of $\frac{1}{2}a_1, h_1 - \frac{1}{2}a_2, g_1 - \frac{1}{2}a_3$.

We can in like manner form the expressions for the second differential coefficients of the a, b, c, f, g, h : these will of course contain third differential coefficients of the x, y, z .

Writing down only what is wanted, we have

$$\begin{aligned} \frac{1}{2}a_{22} &= 12.12 + 1.122, \\ \frac{1}{2}b_{11} &= 12.12 + 2.112, \\ h_{12} &= 12.12 + 11.22 + 1.122 + 2.112, \end{aligned}$$

where of course 12.12 denotes $x_{12}^2 + y_{12}^2 + z_{12}^2$, 1.122 denotes $x_1.x_{122} + y_1.y_{122} + z_1.z_{122}$, and so in other cases: it follows that

$$\frac{1}{2}(a_{22} + b_{11} - 2h_{12}) = 12.12 - 11.22,$$

so that the third differential coefficients of x, y, z , which enter into the expression of $a_{22} + b_{11} - 2h_{12}$ destroy each other, and this combination contains really only second differential coefficients of x, y, z .

Similarly

$$\begin{aligned} f_{12} &= 31.23 + 33.12 + 3.123 + 2.133, \\ g_{31} &= 31.23 + 33.12 + 3.123 + 1.233, \\ c_{12} &= 2 \cdot 31.23 + 2 \cdot 3.123, \\ h_{33} &= 2 \cdot 31.23 + 1.233 + 2.133, \end{aligned}$$

and thence

$$f_{12} + g_{31} - c_{12} - h_{33} = -2(31.23 - 33.12),$$

so that here again we have a combination containing only second differential coefficients of x, y, z .

There are thus, in all, the six combinations

$$\begin{aligned} b_{33} + c_{22} - 2f_{23} &\dots\dots\dots (\mathfrak{A}), \\ c_{11} + a_{33} - 2g_{31} &\dots\dots\dots (\mathfrak{B}), \\ a_{22} + b_{11} - 2h_{12} &\dots\dots\dots (\mathfrak{C}), \\ g_{23} + h_{32} - a_{23} - f_{11} &\dots\dots\dots (\mathfrak{f}), \\ h_{31} + f_{13} - b_{31} - g_{22} &\dots\dots\dots (\mathfrak{g}), \\ f_{12} + g_{21} - c_{12} - h_{33} &\dots\dots\dots (\mathfrak{h}), \end{aligned}$$

each really containing only second differential coefficients of x, y, z ; and we thus understand how each of these combinations may be expressible in terms of the first differential coefficients of (a, b, c, f, g, h) . We have, in fact, for thus expressing these combinations the six equations called $(\mathfrak{A}), (\mathfrak{B}), (\mathfrak{C}), (\mathfrak{f}), (\mathfrak{g}), (\mathfrak{h})$ about to be obtained, and which are the generalisations of Warren's six equations for normal coordinates.

I consider the several determinants of the form

$$\begin{vmatrix} x_{\alpha\beta} & y_{\alpha\beta} & z_{\alpha\beta} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix},$$

in all 18, since the suffixes for the top row may be 11, 22, 33, 23, 31, 12, and those for the second and third rows 2, 3; 3, 1; or 1, 2. Each determinant is a linear function of second differential coefficients of x, y, z (thus the determinant written down is $\xi_1 x_{11} + \eta_1 y_{11} + \zeta_1 z_{11}$, and as such it can, by what precedes, be expressed by means of the first differential coefficients of the a, b, c, f, g, h . Thus, if the determinant above written down be called D , writing ξ_3, η_3, ζ_3 , for α, β, γ , we have

$$\begin{vmatrix} \xi_3 & \eta_3 & \zeta_3 & \square \\ x_1 & y_1 & z_1 & \frac{1}{2}a_1 \\ x_2 & y_2 & z_2 & h_1 - \frac{1}{2}a_2 \\ x_3 & y_3 & z_3 & g_1 - \frac{1}{2}a_3 \end{vmatrix} = 0,$$

that is,

$$L\square = - \begin{vmatrix} & \xi_3 & \eta_3 & \zeta_3 \\ \frac{1}{2}a_1 & x_1 & y_1 & z_1 \\ h_1 - \frac{1}{2}a_2 & x_2 & y_2 & z_2 \\ g_1 - \frac{1}{2}a_3 & x_3 & y_3 & z_3 \end{vmatrix},$$

where the right-hand side is

$$\begin{aligned}
&= \frac{1}{2}a_1 \cdot \xi_3 + \eta_3 \cdot \zeta\zeta \\
&= (h_1 - \frac{1}{2}a_2)(-\frac{1}{2}\xi_2 + \eta_2 \cdot \zeta\zeta) \\
&\quad + (g_1 - \frac{1}{2}a_3)(\xi_1^2 + \eta_1^2 + \zeta_1^2),
\end{aligned}$$

which is

$$= G \cdot \frac{1}{2}a_1 + F(h_1 - \frac{1}{2}a_2) + C(g_1 - \frac{1}{2}a_3),$$

or, as this might be written,

$$= (G, F, C \langle \frac{1}{2}a_1, h_1 - \frac{1}{2}a_2, g_1 - \frac{1}{2}a_3 \rangle).$$

Retaining the former form, the result is

$$L\Box = G \cdot \frac{1}{2}a_1 + F(h_1 - \frac{1}{2}a_2) + C(g_1 - \frac{1}{2}a_3),$$

and it is now very easy to write down the complete system of the 18 equations; viz. if the determinant above written down be called 11.1.2, and so in other cases, then we have

	2.3	3.1	1.2
	A	H	G
	H	B	F
	G	F	C
$L.11$	$\frac{1}{2}a_1$	$h_1 - \frac{1}{2}a_2$	$g_1 - \frac{1}{2}a_3$
$.22$	$h_2 - \frac{1}{2}b_1$	$\frac{1}{2}b_2$	$f_2 - \frac{1}{2}b_3$
$.33$	$g_3 - \frac{1}{2}c_1$	$f_3 - \frac{1}{2}c_2$	$\frac{1}{2}c_3$
$.23$	$-\frac{1}{2}(-f_1 + g_2 + h_3)$	$\frac{1}{2}c_1$	$\frac{1}{2}b_3$
$.31$	$\frac{1}{2}c_2$	$\frac{1}{2}(f_1 - g_2 + h_3)$	$\frac{1}{2}a_3$
$.12$	$\frac{1}{2}b_3$	$\frac{1}{2}a_2$	$\frac{1}{2}(f_1 + g_2 - h_3)$

read for instance

$$L.11.1.2 = G \cdot \frac{1}{2}a_1 + F(h_1 - \frac{1}{2}a_2) + C(g_1 - \frac{1}{2}a_3),$$

the equation obtained above.

There are eighteen functions as shown by the following diagram:

	2.3	3.1	1.2
$(22)(33) - (23)^2$		2.3	
$(33)(11) - (31)^2$		3.1	
$(11)(22) - (12)^2$		1.2	
$(31)(12) - (11)(23)$		2.3	
$(12)(23) - (22)(31)$		3.1	
$(23)(31) - (33)(12)$		1.2,	

viz. in any line of the diagram the bracketed duads may belong to any one at pleasure, but all to the same, of the three pairs 2.3, 3.1, 1.2; thus the first line might be

$$(22.3.1)(33.3.1) - (23.3.1)^2,$$

or, instead of the 3.1, we might have 2.3 or 1.2. But of the 18 functions I distinguish 6, viz. those in which for the six lines respectively the pairs are 2.3, 3.1, 1.2, 2.3, 3.1, 1.2, as shown in the diagram. Each of these six functions can be obtained under two different forms, and by equating these we have the equations ($\mathfrak{A}$), ($\mathfrak{B}$), ($\mathfrak{C}$), ($\mathfrak{f}$), ($\mathfrak{g}$), ($\mathfrak{h}$), before referred to; thus ($\mathfrak{C}$) is obtained by equating two different forms of the function $(11.1.2)(22.1.2) - (12.1.2)^2$, and ($\mathfrak{f}$) is obtained by equating two different forms of the function $(23.1.2)(31.1.2) - (33.1.2)(12.1.2)$.

The determinants 11.1.2, &c., may be denoted by accented letters a, b, c, f, g, h, as follows:

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}, \dots = a', b', c', f', g', h',$$

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_3 & y_3 & z_3 \\ x_1 & y_1 & z_1 \end{vmatrix}, \dots = a'', b'', c'', f'', g'', h'',$$

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix}, \dots = a''', b''', c''', f''', g''', h''';$$

viz.

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = a', \quad \&c.$$

In this notation, ($\mathfrak{C}$) is obtained by equating two values of $a''b'' - (h'')^2$, and ($\mathfrak{f}$) by equating two values of $f'f' - g''h''$.

The forms which I call the second are those given by the immediate substitution of the foregoing values of (11.1.2), &c.; to obtain the first forms, I proceed to calculate those of ($\mathfrak{C}$) and ($\mathfrak{f}$).

Forming by the ordinary rule the product of the determinants (11.1.2) and (22.1.2), which are

$$\begin{vmatrix} x_{11} & y_{11} & z_{11} \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \quad \text{and} \quad \begin{vmatrix} x_{22} & y_{22} & z_{22} \\ x_1 & y_1 & z_1 \\ x_3 & y_3 & z_3 \end{vmatrix},$$

this is

$$\begin{vmatrix} 11.22 & 1.11 & 2.11 \\ 1.22 & 1.1 & 1.2 \\ 2.22 & 1.2 & 2.2 \end{vmatrix},$$

where 11.22 denotes $x_{11}x_{22} + y_{11}y_{22} + z_{11}z_{22}$, and the like for the other symbols. In like manner, the square of the determinant (12.1.2), that is, of

$$\begin{vmatrix} x_{12} & y_{12} & z_{12} \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{vmatrix},$$

is

$$\begin{vmatrix} 12.12 & 1.12 & 2.12 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 2.1 & 2.2 \end{vmatrix},$$

or, observing that in the two resulting determinants the terms 11.22 and 12.12 are multiplied by the same factor, the expression for the difference gives

$$\begin{aligned} & (11.1.2)(22.1.2) - (12.1.2)^2 \\ &= (11.22 - 12.12) \begin{vmatrix} 1.1 & 1.2 \\ 1.2 & 2.2 \end{vmatrix} + \begin{vmatrix} 1.22 & 1.11 & 2.11 \\ 2.22 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix}, \end{aligned}$$

containing $11.22 - 12.12$, which, by what precedes, is

$$= -\frac{1}{2}(a_{22} + b_{11} - 2h_{12});$$

the other terms are also known, viz. the whole value is

$$\begin{aligned} &= -\frac{1}{2}(a_{22} + b_{11} - 2h_{12})(ab - h^2) \\ &+ a \begin{vmatrix} 0 & \frac{1}{2}b_1 & \frac{1}{2}(b_2 - h_1) \\ \frac{1}{2}a_2 - h_1 & a & h \\ \frac{1}{2}b_2 - h_1 & h & b \end{vmatrix} - h \begin{vmatrix} 0 & \frac{1}{2}a_1 & \frac{1}{2}a_2 \\ \frac{1}{2}b_1 & a & h \\ \frac{1}{2}b_2 - h_1 & h & b \end{vmatrix}, \end{aligned}$$

which is

$$\begin{aligned} &= -\frac{1}{2}(a_{22} + b_{11} - 2h_{12})(ab - h^2) \\ &+ a\left(\frac{1}{2}b_1^2 - \frac{1}{2}b_2h_1 + \frac{1}{2}a_2b_2\right) \\ &+ b\left(-\frac{1}{2}a_2^2 - \frac{1}{2}h_1^2 + \frac{1}{2}a_1h_2\right) \\ &+ h\left[\frac{1}{2}(a_2b_2 - a_1b_2) + h_1h_2 - a_2h_2 - b_1h_1\right] \\ &= (ab - h^2)k, \quad \text{where } k \text{ is the measure of curvature.} \end{aligned}$$

In the same way, we have

$$\begin{aligned} & (23.1.2)(31.1.2) - (33.1.2)(12.1.2) \\ &= \begin{vmatrix} 23.31 & 1.31 & 2.31 \\ 1.23 & 1.1 & 1.2 \\ 2.23 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} 12.33 & 1.33 & 2.33 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix}, \end{aligned}$$

which is

$$= (23.31 - 12.33) \begin{vmatrix} 1.1 & 1.2 \\ 1.2 & 2.2 \end{vmatrix} + \begin{vmatrix} 0 & 1.13 & 2.13 \\ 1.23 & 1.1 & 1.2 \\ 2.23 & 1.2 & 2.2 \end{vmatrix} - \begin{vmatrix} 0 & 1.33 & 2.33 \\ 1.12 & 1.1 & 1.2 \\ 2.12 & 1.2 & 2.2 \end{vmatrix},$$

containing $23.31 - 12.33$, which by what precedes is

$$= -\frac{1}{2}(g_{23} + f_{31} - c_{12} - h_{33});$$

the other terms are also known, and the value is

$$\begin{aligned} &= -\frac{1}{2}(g_{23} + f_{31} - c_{12} - h_{33})(ab - h^2) \\ &\quad + a \begin{vmatrix} 0 & \frac{1}{2}a_3 & \frac{1}{2}(f_1 - g_2 + h_3) \\ \frac{1}{2}(-f_1 + g_2 + h_3) & a & h \\ \frac{1}{2}b_3 & h & b \end{vmatrix} \\ &\quad - h \begin{vmatrix} 0 & \frac{1}{2}c_1 & \frac{1}{2}c_2 \\ \frac{1}{2}a_2 & a & h \\ \frac{1}{2}b_3 & h & b \end{vmatrix}, \end{aligned}$$

or finally this is

$$\begin{aligned} &= -\frac{1}{2}(g_{23} + f_{31} - c_{12} - h_{33})(ab - h^2) \\ &\quad + a\left(\frac{1}{2}b_3f_1 - \frac{1}{2}b_3g_2 - \frac{1}{2}b_3h_3 + \frac{1}{2}a_3b_2 - b_3h_3\right) \\ &\quad + b\left(\frac{1}{2}a_3g_2 + \dots\right) \\ &\quad + h(\dots). \end{aligned}$$

The remaining four of the six functions can of course be obtained from the two just found by a cyclical interchange of the letters and suffix-numbers 1, 2, 3, and it is not worth while to write down the values.

The two values of $(11.1.2)(22.1.2) - (12.1.2)^2$ are

$$\begin{aligned} &= -\frac{1}{2}(a_{22} + b_{11} - 2h_{12})(ab - h^2) + a\left(\frac{1}{2}b_1^2 - \frac{1}{2}b_2h_1 + \frac{1}{2}a_2b_2\right) \\ &\quad + b\left(-\frac{1}{2}a_2^2 - \frac{1}{2}h_1^2 + \frac{1}{2}a_1h_2\right) \\ &\quad + h\left[\frac{1}{2}(a_2b_2 - a_1b_2) + h_1h_2 - a_2h_2 - b_1h_1\right], \end{aligned}$$

and

$$\{G \cdot \frac{1}{2}a_1 + F(h_1 - \frac{1}{2}a_2) + C(g_1 - \frac{1}{2}a_3)\} \cdot \{G(h_2 - \frac{1}{2}b_1) + F \cdot \frac{1}{2}b_2 + C(f_2 - \frac{1}{2}b_3)\},$$

where, in the first value, for $a, b, h, ab - h^2$ we must write $L^{-2}(BC - F^2)$, $L^{-2}(CA - G^2)$, $L^{-2}(FG - CH)$, and $L^{-2}C^2$; making this change, multiplying by 4 and equating, we obtain

$$\begin{aligned} &-2L^2C(a_{22} + b_{11} - 2h_{12}) \\ &\quad + (BC - F^2)(b_1^2 - 2b_2h_1 + a_2b_2) \\ &\quad + (CA - G^2)(a_2^2 - 2a_1h_2 + a_1a_1?) \\ &\quad + (FG - CH)(a_2b_2 - a_1b_2 + 4h_1h_2 - 2b_1h_1 - 2a_2h_2) \\ &\quad - \{G \cdot a_1 + F(2h_1 - a_2) + C(2g_1 - a_3)\}\{G(2h_2 - b_1) + Fb_2 + C(2f_2 - b_3)\} \\ &\quad + \{Gb_1 + Fb_2 + C(f_1 + g_2 - h_3)\}^2 = 0. \end{aligned}$$

Developing the fourth and fifth lines, it appears that in this expression the coefficients of F^2 , G^2 and FG each of them vanish; the whole equation is thus divisible by C , and omitting this factor throughout, the equation becomes

$$\begin{aligned}
0 = & -2L(a_{22} + b_{11} - 2h_{12}) \\
& + A(a_2^2 - 2a_2h_1 + a_1b_1) \\
& + B(b_1^2 - 2b_1h_2 + a_2b_2) \\
& + C\{-a_3b_3 + 2a_2f_3 + 2b_1g_3 - 4f_3g_3 + (f_1 + g_2 - h_3)^2\} \\
& + F\{(a_3b_2 - a_2b_3) + 2(b_1g_3 - b_2g_1) + 2(b_1h_3 - b_3h_1) + 2(a_2f_2 + b_1f_2 - 2h_1f_2)\} \\
& + G\{a_3b_2 - a_2b_3 + 2(a_2f_3 - a_3f_2) + 2(a_2h_3 - a_3h_2) + 2(a_2g_2 + b_1g_1 - 2g_1h_2)\} \\
& + H\{2(a_2h_2 + b_1h_1 - 2h_1h_2)\};
\end{aligned}$$

this is substantially the required equation ($\mathfrak{C}$), but the form of it may be greatly simplified.

Forming the identity,

$$\begin{aligned}
0 = & 2L[(a_2 - h_1)L_1 + (b_1 - h_2)L_2] \\
= & -A(a_1h_2 - a_2h_2 - a_1h_1 + a_2^2) \\
& - B(b_1^2 - b_1h_2 - b_2h_1 + a_2b_2) \\
& - C(b_1c_2 + a_2c_1 - c_1h_2 - c_2h_1) \\
& - 2F(b_1f_2 + a_2f_1 - f_1h_2 - f_2h_1) \\
& - 2G(b_1g_2 + a_2g_1 - g_1h_2 - g_2h_1) \\
& - 2H(a_2h_2 + b_1h_1 - 2h_1h_2),
\end{aligned}$$

we add hereto the last preceding equation; the coefficients of A , B , F , G , H thus assume new and simple forms, but the coefficient of C requires a further transformation.

Assume

$$\begin{aligned}
\Omega = & (b_1c_1 - c_2h_1) + (c_2a_2 - c_1g_2) + (a_3b_3 - h_3^2) \\
& + (g_2h_3 + g_3h_2 - a_2f_3 - a_3f_2) + (h_3f_1 + h_1f_3 - b_1g_3 - b_3g_1) + (f_1g_2 + f_2g_1 - c_2h_1 - c_1h_2);
\end{aligned}$$

then, if we add to the equation C multiplied by this value and subtract $C\Omega$, the coefficient of C takes its proper form, and the equation is

$$\begin{aligned}
0 = & -2L(a_{22} + b_{11} - 2h_{12}) \\
& + 2L[(b_1 - h_2)L_2 + (a_2 - h_1)L_1] - C\Omega \\
& + A\{ (a_1h_2 - a_2h_1) \} \\
& + B\{ (b_1h_2 - b_2h_1) \} \\
& + C\{-(a_2f_2 - a_1f_1) + (b_1g_1 - b_2g_2) - (g_2h_1 - g_1h_2) - (h_1f_2 - h_2f_1) + 3(f_1g_2 - f_2g_1)\} \\
& + F\{-(a_2b_2 - a_2b_3) + 2(b_1g_3 - b_3g_1) + 2(b_1h_3 - b_3h_1) - 2(h_1f_3 - h_3f_1)\} \\
& + G\{-(a_2b_3 - a_3b_2) - 2(a_2f_3 - a_3f_2) - 2(a_2h_3 - a_3h_2) - 2(g_1h_3 - g_3h_1)\} \\
& + H\{ \dots \} \\
= & 0;
\end{aligned}$$

where, as a verification, observe that the letters (a, b), (f, g) may be interchanged if at the same time we interchange the suffixes 1 and 2.

The two values of (23.1.2)(31.1.2) – (33.1.2)(12.1.2) are

$$\begin{aligned} & -\frac{1}{2}(g_{23} + f_{31} - c_{12} - h_{33})(ab - h^2) \\ & + a\left(\frac{1}{2}b_1f_3 - \frac{1}{2}b_3f_1 - \frac{1}{2}b_1c_2 + \frac{1}{2}b_2g_3 - b_3h_3\right) \\ & + b\left(\frac{1}{2}a_3g_2 + \dots - a_3h_3\right) + h\left[\dots - \frac{1}{2}(f_1 + g_2)^2\right], \end{aligned}$$

and

$$\{G \cdot \frac{1}{2}a_3 + F(f_2 - \frac{1}{2}c_1) + C \cdot \frac{1}{2}c_3\} \cdot \{G \cdot \frac{1}{2}a_2 + F \cdot \frac{1}{2}b_3 + C \cdot \frac{1}{2}(f_1 + g_2 - h_3)\},$$

and here for a, b, h, $ab - h^2$ substituting their values, multiplying by 4, and equating, we have

$$\begin{aligned} & -2L^2C(g_{23} + f_{31} - c_{12} - h_{33}) \\ & + (BC - F^2)(2b_1f_3 - b_3f_1 - b_1c_2 + b_2g_3 - b_3h_3) \\ & + (CA - G^2)(2a_2g_3 - a_3g_2 - a_2c_1 + a_1f_3 - a_3h_3) \\ & + (FG - CH)(a_3b_3 + b_1c_1 + a_2c_2 - 2b_1g_3 - 2a_2f_3 + h_3^2 - f_2g_1 + 2f_1g_2 - g_1^2a_2^2) \\ & - \{G(-f_1 + g_2 + h_3) + Fb_3 + Cc_2\}\{Ga_3 + F(f_2 - g_1 + h_2) + Cc_1\} \\ & + \{G(2g_3 - c_2) + F(2f_3 - c_2) + Cc_3\}\{Ga_2 + Fb_2 + C(f_1 + g_2 - h_3)\} = 0. \end{aligned}$$

Here again the terms in F^2 , FG , G^2 all vanish, hence the whole equation divides by C ; throwing this factor out, we have

$$\begin{aligned} 0 = & -2L(g_{23} + f_{31} - c_{12} - h_{33}) \\ & + A(2a_2g_3 - a_3g_2 - a_2c_1 + a_1f_3 - a_3h_3) \\ & + B(2b_1f_3 - b_3f_1 - b_1c_2 + b_2g_3 - b_3h_3) \\ & + C(-c_1c_2 + c_3f_1 + c_3g_2 - c_3h_3) \\ & + F\{-(b_3c_1 - b_1c_3) - 2c_2f_1 + 2f_1f_3 + 2g_2f_3 - 2f_3h_3\} \\ & + G(a_2c_3 - a_3c_2 - 2c_1g_3 + 2f_2g_3 + 2g_2g_3 - 2g_3h_3) \\ & + H(-a_3b_3 - b_1c_1 - a_2c_2 + 2b_1g_3 + 2a_2f_3 - h_3^2 + f_2g_1 - 2f_1g_2 + g_1^2a_2^2) = 0, \end{aligned}$$

which is substantially the required equation (f); but the form has to be altered. First, multiply by 2, then forming the expression

$$\begin{aligned} & 2L[(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3] \\ & = (f_3 - c_2)(Aa_1 + Bb_1 + Cc_1 + 2Ff_1 + \dots) \\ & + (g_3 - c_1)(Aa_2 + Bb_2 + Cc_2 + 2Ff_2 + \dots) \\ & + (f_1 + g_2 - 2h_3)(Aa_3 + Bb_3 + Cc_3 + 2Ff_3 + 2Gg_3 + \dots), \end{aligned}$$

2—2

this is

$$\begin{aligned}
&= A(-a_1c_2 - a_2c_1 + a_1f_1 + a_2f_1 + a_2g_2 + a_3g_3 - 2a_3h_3) \\
&\quad + B(-b_1c_2 - b_2c_1 + b_1f_1 + b_1f_3 + b_2g_3 + b_3g_2 - 2b_3h_3) \\
&\quad + C(-2c_1c_2 + c_1f_1 + c_3f_1 + c_2g_2 + c_3g_2 - 2c_3h_3) \\
&\quad + 2F(-c_1f_1 - c_2f_2 + 2f_1f_3 + f_2g_3 + f_3g_2 - 2f_3h_3) \\
&\quad + 2G(-c_1g_2 - c_2g_1 + f_1g_2 + f_2g_1 + 2g_2g_3 - 2g_3h_3) \\
&\quad + 2H(-c_1h_1 - c_2h_2 + g_1h_2 + g_2h_1 + h_1f_2 + h_2f_1 - 2h_3^2);
\end{aligned}$$

then adding this expression

$$(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3$$

and subtracting its foregoing value, we have an equation with new coefficients of A, B, C, F, G, H , all of which, except that of H , are in the proper form, and for the coefficient of H , recurring to the foregoing value of Ω , we must add to the equation $2H$ multiplied by the value of Ω and subtract $2H\Omega$. The final result is

$$\begin{aligned}
0 = & -4L(g_{23} + f_{31} - c_{12} - h_{33}) \\
& + 2L[(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3] - 2H\Omega \\
& + A\{ 3(a_2g_3 - a_3g_2) - (c_1a_2 - c_2a_1) + (a_1f_3 - a_3f_1) \} \\
& + B\{ -3(b_3f_1 - b_1f_3) - (b_1c_2 - b_2c_1) - (b_3g_2 - b_2g_3) \} \\
& + C\{ (c_2f_1 - c_1f_2) - (c_1g_2 - c_2g_1) \} \\
& + 2F\{ -(b_3c_1 - b_1c_3) + (c_1f_2 - c_2f_1) - (f_2g_3 - f_3g_2) \} \\
& + 2G\{ -(c_3a_2 - c_2a_3) - (c_1g_2 - c_2g_1) - (f_1g_3 - f_3g_1) \} \\
& + 2H\{ -(b_3g_1 - b_1g_3) + (a_2f_3 - a_3f_2) \} = 0,
\end{aligned}$$

which, it will be observed, remains unaltered by the interchange of a and b , f and g , A and B , F and G , the suffix numbers 1 and 2 being at the same time interchanged.

We have thus the required six equations, in which

$$\mathfrak{K} = abc - af^2 - bg^2 - ch^2 + 2fgh,$$

$$\begin{aligned}
\Omega = & b_1c_1 - c_2h_1 + c_2a_2 - c_1g_2 + a_3b_3 - h_3^2 \\
& + (g_2h_3 + g_3h_2 - a_2f_3 - a_3f_2) + (h_3f_1 + h_1f_3 - b_1g_3 - b_3g_1) + (f_1g_2 + f_2g_1 - c_2h_1 - c_1h_2),
\end{aligned}$$

and where I write also, for shortness, $ab|2$ for $a_1b_2 - a_2b_1$, &c.; viz. the equations are

$$\begin{aligned}
0 = & -2L(b_{33} + c_{22} - 2f_{23}) \\
& + 2L[(c_3 - f_2)L_2 + (b_2 - f_3)L_3] - A\Omega \\
& + A\{ -bg|31 + ch|2 + 3 \cdot gh|23 - hf|31 - fg|12 \} \\
& + B\{ -bf|23 \} \\
& + C\{ +cf|23 \} \\
& + F\{ -bc|23 \} \\
& + G\{ -bc|31 + 2 \cdot cf|12 + 2 \cdot ch|23 - 2 \cdot fg|23 \} \\
& + H\{ -2 \cdot bg|23 - 2 \cdot hf|23 \} \quad (\mathfrak{A}).
\end{aligned}$$

$$\begin{aligned}
0 = & -2L(c_{11} + a_{33} - 2g_{31}) \\
& + 2L[(c_3 - g_1)L_1 + (a_1 - g_3)L_3] - B\Omega \\
& + A\{ ag|31 \} \\
& + B\{ af|23 - ch|12 - gh|23 + 3 \cdot hf|31 - fg|31 \} \\
& + C\{ -cg|31 \} \\
& + F\{ -ca|23 - 2 \cdot cg|12 - 2 \cdot ch|31 - 2 \cdot fg|31 \} \\
& + G\{ -ca|31 - 2 \cdot gh|31 \} \\
& + H\{ \dots \} \quad (\mathfrak{B}).
\end{aligned}$$

$$\begin{aligned}
0 = & -2L(a_{22} + b_{11} - 2h_{12}) \\
& + 2L[(b_1 - h_2)L_2 + (a_2 - h_1)L_1] - C\Omega \\
& + A\{ -ah|12 \} \\
& + B\{ +bh|12 \} \\
& + C\{ -af|23 + bg|31 - gh|23 - hf|31 + 3 \cdot fg|12 \} \\
& + F\{ -ab|23 + 2 \cdot bg|12 + 2 \cdot bh|31 - 2 \cdot hf|12 \} \\
& + G\{ -ab|31 - 2 \cdot af|12 + 2 \cdot ah|23 - 2 \cdot gh|12 \} \\
& + H\{ -ab|12 \} \quad (\mathfrak{C}).
\end{aligned}$$

$$\begin{aligned}
0 = & -4L(g_{12} + h_{31} - a_{23} - f_{11}) \\
& + 2L[(g_2 + h_3 - 2f_1)L_1 + (g_3 - a_1)L_2 + (h_1 - a_1)L_3] - 2F\Omega \\
& + A\{ -ag|12 - ah|31 \} \\
& + B\{ -ab|23 + bg|12 + 3 \cdot bh|31 \} \\
& + C\{ -ca|23 - 3 \cdot cg|12 - ch|31 \} \\
& + 2F\{ \dots + bg|31 - ch|12 \} \\
& + 2G\{ -ca|12 + ag|23 - gh|31 \} \\
& + 2H\{ -ab|31 - ah|23 - gh|12 \} \quad (\mathfrak{f}).
\end{aligned}$$

$$\begin{aligned}
0 = & -4L(h_{12} + f_{31} - b_{23} - g_{22}) \\
& + 2L[(f_2 - b_3)L_1 + (h_1 + f_3 - 2g_2)L_2 + (h_2 - b_1)L_3] - 2G\Omega \\
& + A\{ -ab|31 - 3 \cdot ah|23 - af|12 \} \\
& + B\{ bh|23 - bf|12 \} \\
& + C\{ -bc|31 + ch|23 + 3 \cdot cf|12 \} \\
& + 2F\{ -bc|12 - bf|31 - hf|23 \} \\
& + 2G\{ \dots + ch|12 - af|23 \} \\
& + 2H\{ -ab|23 + bh|31 - hf|12 \} \quad (\mathfrak{g}).
\end{aligned}$$

$$\begin{aligned}
0 = & -4L(f_{12} + g_{31} - c_{23} - h_{33}) \\
& + 2L[(f_3 - c_2)L_1 + (g_3 - c_1)L_2 + (f_1 + g_2 - 2h_3)L_3] - 2H\Omega \\
& + A\{ -ca|12 + af|31 + 3 \cdot ag|23 \} \\
& + B\{ -bc|12 - 3 \cdot bf|31 - bg|23 \} \\
& + C\{ cf|31 - cg|23 \} \\
& + F\{ -bc|31 + cf|12 - fg|23 \} \\
& + G\{ -ca|23 - cg|12 - fg|31 \} \\
& + H\{ \dots + af|23 - bg|31 \} \quad (h).
\end{aligned}$$

It would be possible in these equations to introduce the symbols $AB|12$, &c., in place of $ab|12$, &c., and then writing $A = B = C = 1$, all these symbols other than those where the letters are GH , HF or FG would vanish, and we should obtain Mr Warren's six equations for normal coordinates. But in the general case it would seem that there is not any advantage in the introduction of the new symbols $AB|12$, &c., and I retain by preference the equations in the form in which I have given them.

To the foregoing may be joined a symmetrical equation obtained (as by Mr Warren) by multiplying the several equations by a, b, c, f, g, h respectively, and adding; the result is in the first instance obtained in the form

$$\mathfrak{X} - \mathfrak{Y} + \mathfrak{Z} = 0,$$

where

$$\begin{aligned}
\mathfrak{X} = & a(b_{33} + c_{22} - 2f_{23}) \\
& + b(c_{11} + a_{33} - 2g_{31}) \\
& + c(a_{22} + b_{11} - 2h_{12}) \\
& + 2f(g_{12} + h_{31} - a_{23} - f_{11}) \\
& + 2g(h_{12} + f_{31} - b_{23} - g_{22}) \\
& + 2h(f_{12} + g_{31} - c_{23} - h_{33}).
\end{aligned}$$

For $\mathfrak{Y}$, collecting the terms which contain L_1, L_2, L_3 respectively, and attending to the values of (A, B, C, F, G, H) ,

$$= (bc - f^2, \quad ca - g^2, \quad ab - h^2, \quad gh - af, \quad hf - bg, \quad fg - ch),$$

this is easily reduced to the form

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The term in Ω is

$$= -(Aa + \dots) = -3\mathfrak{K}\Omega, \text{ as above.}$$

For the calculation of D , collecting the terms, we have $D =$

A, B, C, F, G, H	A, B, C, F, G, H	A, B, C, F, G, H
bc - a	bc - $2g_2h_3$	bc - $2g_3h_2$
ca - b	ca - $2h_3f_1$	ca - $2h_1f_3$
ab - c	ab - $2f_1g_2$	ab - $2f_2g_1$
gh - f_1	gh - $2g_2f_1 ?$	gh - $2f_2g_2 ?$
hf - g_2	hf + $ag_3 - bg_1 ?$	hf - $2h_1f_2$
fg - h_3	fg + $ab - ch ?$	fg - $2h_2f_1$

[The full table on book p. 15 displays the eighteen coefficient slots used in the expression of D ; the entries combine the symbols $A, \dots, H$ with the six binary products bc, ca, ab, gh, hf, fg and with the second differential coefficients of a, b, c, f, g, h .]

Some however of the coefficients require reduction; for instance, that of $hf|2$ is $-(a, b, f|C, F, -G)$, that is $-(ac + bF + fG)$, but we again rename this term in the form

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and where I write also, for shortness, $abc|2$ for $a_1b_2c_2 - a_2b_1c_1$, &c, the foregoing expressions for the entries of the table reduce, in fact, to combinations $abc|3, abh|31, bch|12$, etc., used in the next page.

or, arranging the terms in a different order, this may be written

$$\Omega =$$

$$\begin{aligned} & A\{ a(3 \cdot gh|23 - hf|31 - fg|12) + b \cdot ag|31 - c \cdot ah|12 - c \cdot ah|12 \\ & \quad + f(ag|12 - ah|31) + g(2 \cdot gh|12 - ah|23) + h(2 \cdot gh|31 + ag|23) \} \\ & + B\{ -a \cdot bf|23 + b(-gh|23 + 3 \cdot hf|31 - fg|12) + c \cdot bh|12 \\ & \quad + f(2 \cdot hf|12 + bh|31) + g(bh|23 - bf|12) + h(2 \cdot hf|23 - bf|31) \} \\ & + C\{ a \cdot cf|23 - b \cdot cg|31 + c(-gh|23 - hf|31 + 3 \cdot fg|12) \\ & \quad + f(2 \cdot fg|31 - cg|12) + g(2 \cdot fg|23 + cf|12) + h(cf|31 - cg|23) \} \\ & + F\{ -a \cdot bc|23 - b \cdot ch|31 + c \cdot bg|12 \\ & \quad + f(bg|31 - ch|12) + g(bg|23 - bc|12) - h(ch|23 + bc|31) \} \\ & + G\{ +a \cdot ch|23 - b \cdot ca|31 - c \cdot af|12 \\ & \quad - f(af|31 + ca|12) + g(ch|12 - af|23) + h(ch|31 - ca|23) \} \\ & + H\{ -a \cdot bg|23 + b \cdot af|31 - c \cdot ab|12 \\ & \quad + f(af|12 - ab|31) - g(bg|12 + ab|23) + h(af|23 - bg|31) \}. \end{aligned}$$

Attending to the values of A, B, C, F, G, H , we have

$$\begin{aligned} A_1 + B_2 + C_3 + 2F_4 + 2G_5 + 2H_6 = & bc_{11} + cb_{11} - 2ff_{11} + 2(b_1c_1 - f_1^2) \\ & + ca_{22} + ac_{22} - 2gg_{22} + 2(c_2a_2 - g_2^2) \\ & + ab_{33} + ba_{33} - 2hh_{33} + 2(a_3b_3 - h_3^2) \\ & + 2\{gh_{23} + hg_{23} - af_{23} - fa_{23} + g_2h_3 + g_3h_2 - a_2f_3 - a_3f_2\} \\ & + 2\{hf_{31} + fh_{31} - bg_{31} - gb_{31} + h_3f_1 + h_1f_3 - b_3g_1 - b_1g_3\} \\ & + 2\{fg_{12} + gf_{12} - ch_{12} - hc_{12} + f_1g_2 + f_2g_1 - c_1h_2 - c_2h_1\}, \end{aligned}$$

which is, in fact, $= 2\mathfrak{X} + 2\Omega$.

Hence the foregoing equation ($\mathfrak{X}$) may also be written

$$-L(Aa_{11} + Bb_{22} + Cc_{33} + 2Ff_{23} + 2Gg_{31} + 2Hh_{12}) + 2L \cdot \Omega + D = 0,$$

where $\mathfrak{X}, \Omega, D$ have their before-mentioned values.

In the particular case where $f = 0, g = 0, h = 0$, we have

$$A, B, C, F, G, H = bc, ca, ab, 0, 0, 0; \quad L^2 = abc;$$

the equation ($\mathfrak{X}$) becomes

$$-2abc(b_{33} + c_{22}) + [c(a_1bc + b_1ca + c_1ab) + b(a_1bc + b_1ca + c_1ab)] - bc(b_1c_2 + c_2a_2 + a_3b_3) = 0,$$

that is,

$$-2abc(b_{33} + c_{22}) - bc b_2c_1 + ca(b_1c_1 + b_3^2) + ab(b_2c_3 + c_1^2) = 0,$$

and the equation (f) becomes

$$4abc \cdot a_{23} - [a_3(a_2bc + b_2ca + c_2ab) + a_2(a_3bc + b_3ca + c_3ab)] \\ - ca(a_2b_3 - a_3b_2) - ab(c_2a_3 - c_3a_2) = 0,$$

that is,

$$4abc \cdot a_{23} - 2bc \cdot a_2a_3 - 2ca \cdot a_2b_3 - 2ab \cdot a_3b_2 = 0.$$

Dividing by $2abc$ and $4abc$ respectively, and completing the system, the six equations are

$$b_{33} + c_{22} - \frac{b_2^2}{2b} - \frac{c_3^2}{2c} - \frac{bc_1}{2a} + \frac{bc_2}{2b} + \frac{bc_3}{2c} = 0 \quad (\mathfrak{A})$$

$$c_{11} + a_{33} - \frac{c_3^2}{2c} - \frac{a_1^2}{2a} - \frac{ca_1}{2a} + \frac{ca_2}{2b} + \frac{ca_3}{2c} = 0 \quad (\mathfrak{B})$$

$$a_{22} + b_{11} - \frac{a_1^2}{2a} - \frac{b_2^2}{2b} - \frac{ab_1}{2a} + \frac{ab_2}{2b} + \frac{ab_3}{2c} = 0 \quad (\mathfrak{C})$$

$$a_{23} - \frac{a_2a_3}{2a} - \frac{a_2b_3}{2b} - \frac{a_3c_2}{2c} = 0 \quad (\mathfrak{f})$$

$$b_{31} - \frac{b_3b_1}{2b} - \frac{b_3a_1}{2a} - \frac{b_1c_3}{2c} = 0 \quad (\mathfrak{g})$$

$$c_{12} - \frac{c_1c_2}{2c} - \frac{c_1b_2}{2b} - \frac{c_2a_1}{2a} = 0. \quad (\mathfrak{h})$$

These are in fact Lamé's equations, *Leçons sur les coordonnées curvilignes*, etc., Paris (1859), pp. 76, 78, viz. the first of the equations (8), p. 76, is

$$\frac{d^2H}{dp_2 dp_3} = \frac{1}{H_2} \frac{dH_2}{dp_3} \frac{dH}{dp_2} + \frac{1}{H_3} \frac{dH_3}{dp_2} \frac{dH}{dp_3},$$

which, in the notation of the present paper, is

$$(\sqrt{a})_{23} = \frac{1}{\sqrt{b}}(\sqrt{b})_3(\sqrt{a})_2 + \frac{1}{\sqrt{c}}(\sqrt{c})_2(\sqrt{a})_3.$$

Here

$$(\sqrt{a})_2 = \frac{1}{2} \frac{a_2}{\sqrt{a}},$$

$$(\sqrt{a})_{23} = \frac{1}{2} \left(\frac{a_{23}}{\sqrt{a}} - \frac{1}{2} \frac{a_2a_3}{a\sqrt{a}} \right);$$

and the equation therefore is

$$\frac{a_{23}}{\sqrt{a}} - \frac{1}{2} \frac{a_2a_3}{a\sqrt{a}} = \frac{1}{2} \frac{a_2b_3}{\sqrt{b}\sqrt{a}} + \frac{1}{2} \frac{a_3c_2}{\sqrt{c}\sqrt{a}},$$

which, multiplying by $\sqrt{a}$, gives the foregoing equation

$$a_{23} - \frac{a_2a_3}{2a} - \frac{a_2b_3}{2b} - \frac{a_3c_2}{2c} = 0.$$

The first of the equations (9) p. 78 is

$$\frac{d}{dp_1} \left(\frac{1}{H_1} \frac{dH}{dp_1} \right) + \frac{d}{dp_2} \left(\frac{1}{H_2} \frac{dH}{dp_2} \right) = \frac{1}{H_2} \frac{dH_1}{dp_1} \frac{dH_1}{dp_2},$$

which, in the notation of the present paper, is

$$\left(\frac{(\sqrt{c})_1}{\sqrt{a}} \right)_1 + \left(\frac{(\sqrt{c})_2}{\sqrt{b}} \right)_2 = \frac{1}{\sqrt{a}\sqrt{b}} (\sqrt{a})_2 (\sqrt{b})_1.$$

This gives first

$$\frac{(\sqrt{c})_{11}}{\sqrt{a}} - \frac{1}{2} \frac{(\sqrt{c})_1 a_1}{a\sqrt{a}} + \frac{(\sqrt{c})_{22}}{\sqrt{b}} - \frac{1}{2} \frac{(\sqrt{c})_2 b_2}{b\sqrt{b}} = \frac{1}{\sqrt{a}\sqrt{b}} (\sqrt{a})_2 (\sqrt{b})_1,$$

and then

$$\frac{1}{2} \frac{c_{11}}{\sqrt{c}\sqrt{a}} - \frac{1}{4} \frac{c_1^2}{c\sqrt{c}\sqrt{a}} - \frac{1}{4} \frac{c_1 a_1}{a\sqrt{c}\sqrt{a}} + \frac{1}{2} \frac{c_{22}}{\sqrt{c}\sqrt{b}} - \frac{1}{4} \frac{c_2^2}{c\sqrt{c}\sqrt{b}} - \frac{1}{4} \frac{c_2 b_2}{b\sqrt{c}\sqrt{b}} = \frac{1}{4} \frac{a_2 b_1}{\sqrt{a}\sqrt{b}},$$

which, on multiplying by $\sqrt{a}\sqrt{b}$ and reducing, gives the foregoing equation

$$a_{22} + b_{11} - \frac{a_1^2}{2a} - \frac{b_2^2}{2b} - \frac{ab_1}{2a} + \frac{ab_2}{2b} - \frac{a_2 b_1 + a_1 b_2}{2c} = 0.$$

NOTE ON THE STANDARD SOLUTIONS OF A SYSTEM OF
LINEAR EQUATIONS.[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 38–40.]

To fix the ideas, the equations are assumed to be without constant terms. Supposing the system to be insufficient for the determination of the ratios of the unknown quantities, then regarding the unknown quantities as having a definite order of arrangement, there are certain solutions which may be regarded as standard solutions. Take the unknown quantities to be A, B, C, D, E, F, G , &c.; then assuming $A = 0$, or else A, B , each $= 0$, or else A, B, C , each $= 0$, as many equations as possible, the system as thus modified will have a definite solution; for instance, the assumed equations A, B, C, D, E , each $= 0$, may give a definite solution in which F is not $= 0$, and it may then for convenience be put $= 1$. We have thus a solution beginning with $F = 1$. This being so, there will be a solution or solutions with $F = 0$; we cannot then have A, B, C, D, E , each $= 0$, but we again assume $A = 0$, or else A, B , each $= 0$, as many equations as possible; suppose A, B, C , each $= 0$ give a definite solution, with D not $= 0$; and then taking it for convenience to be $= 1$, we have a solution beginning $D = 1$, and for which $F = 0$. Going on in this manner we obtain, it may be, a solution beginning $B = 1$, and for which $D = 0, F = 0$; and so on, the process stopping, if not sooner, with a solution beginning $A = 1$, and with the initial letters of the preceding solutions, each $= 0$. We have in this manner the system of standard solutions, of a form such as

A	B	C	D	E	F	G	
0	0	0	0	0	1	*	
0	0	0	1	*	0	*	3—2
0	1	*	0	*	0	*	
1	0	*	0	*	0	*	

where the $*$ denotes a value which is not in general $= 0$, but which may in any particular case happen to be so.

For instance, let it be required for the binary quartic $(a, b, c, d, e)x, y)^4$, to find the aszygetic seminvariants of the degree 4 and weight 6. Assuming for the seminvariant the value in the left-hand column of the diagram, the unknown coefficients being A, B, C, D, E, F, G , this must be reduced to zero by the operation

$$\Delta = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e;$$

and we thus obtain as many equations as there are terms of the degree 4 and weight 5, as appearing by the second column

		A	B	C	D	E	F	G
a^2e^2	$ad^2b + 2.$	.	.	.	.	.	.	.
A	$= a^2ce$	1						
B	$= ab^2e$		1					
C	$= abcd$			1				
D	$= ac^3$				1			
E	$= b^3d$					1		
F	$= b^2c^2$						1	

[The table on book p. 20 records, for each of the seven monomials in the diagram, the corresponding column giving the result of Δ acting upon it; the resulting standard solutions $A = 1, B = -1, \dots, F = 1$ enumerate the irreducible composites P and Q used on p. 21.]

As is known, there is no irreducible solution, but only the composite forms

$$\begin{aligned} \text{I} &= a(ace - ad^2 - b^2e + 2bcd - c^3), \\ \text{II} &= (ac - b^2)(ae - 4bd + 3c^2), \end{aligned}$$

the developed values of which are given above: II (as a form beginning $A = 1$ and with $B = 0$) can be nothing else than, and is in fact $= Q$: and so I (as a form beginning with $A = 1, B = -1$) can be nothing else than, and is in fact $= Q - P$; that is, we have

$$\begin{array}{ll} P = -\text{I} + \text{II}, & \text{or } \text{I} = -P + Q, \\ Q = \text{II}, & \text{II} = Q; \end{array}$$

and so, in general, we have a standard set of values for the asyzygetic seminvariants of a given degree and weight; or, what is the same thing, for the covariants of a given deg-order.

ON SEMINVARIANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 131–138.]

The present paper is a somewhat fragmentary one, but it contains some results which seem to me to be worth putting on record.

I consider here not any binary quantic in particular, but the whole series $(a, b, c \rangle x, y)^2$, $(a, b, c, d \rangle x, y)^3$, &c.; or in a somewhat different point of view, I consider the indefinite series of coefficients $(a, b, c, d, e, \dots)$; here, instead of covariants and invariants, we have only seminvariants; viz. a seminvariant is a function reduced to zero by the operator

$$\Delta = a\partial_b + 2b\partial_c + 3c\partial_d + 4d\partial_e + 5e\partial_f + \dots,$$

for instance, seminvariants are

$$\begin{aligned} a, \quad ac - b^2, \quad a^2d - 3abc + 2b^3, \quad a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2, \\ ae - 4bd + 3c^2, \quad ace - ad^2 - b^2e + 2bcd - c^3, \quad \&c. \end{aligned}$$

A seminvariant is of a certain degree θ in the coefficients, and of a certain weight w (viz. the coefficients $a, b, c, d, \dots$ are reckoned as being of the weights 0, 1, 2, 3, $\dots$ respectively); it is, moreover, of a certain rank ρ ; viz. according as the highest letter therein is $a, c, d, e, \dots$ (it is never b), the rank is taken to be 0, 2, 3, 4, $\dots$, and we have $w = \frac{1}{2}\rho\theta$. The seminvariant... may be regarded as belonging to a quantic $(a, \dots \rangle x, y)^n$, the order of which is n , equal to or greater than ρ ; viz. in regard to such quantic the seminvariant, say A , is the leading coefficient of a covariant

$$(A, B, \dots, K \rangle x, y)^{n\theta-2w},$$

where the weights of the successive coefficients are $w, w+1, \dots$ up to $n\theta-w$; hence the number of terms less unity, that is, μ , is $= n\theta - 2w$; the least value of μ is thus $= \rho\theta - 2w$, which is either zero, or positive; in the former case, $w = \frac{1}{2}\rho\theta$, the seminvariant is an invariant of the quantic $(a, \dots \rangle x, y)^\rho$, the order of which is equal to the rank of the seminvariant; but if $w < \frac{1}{2}\rho\theta$, then it is the leading coefficient of a covariant $(A, B, \dots, E \rangle x, y)^{\rho\theta-2w}$ of the same quantic $(a, \dots \rangle x, y)^\rho$; and in every case, taking $n > \rho$, the seminvariant is the leading coefficient A of a covariant $(A, B, \dots, K \rangle x, y)^{n\theta-2w}$ of a quantic $(a, \dots \rangle x, y)^n$.

Take A as belonging to the quantic $(a, \dots)x, y)^n$; corresponding to such quantic, we have an operator Δ of the same rank n , viz.

$$\begin{aligned}\Delta &= 2b\partial_a + c\partial_b && \text{for } n = 2, \\ &= 3b\partial_a + 2c\partial_b + d\partial_c && n = 3, \\ &= 4b\partial_a + 3c\partial_b + 2d\partial_c + e\partial_d && n = 4,\end{aligned}$$

Operating with Δ on A , we have a series of terms

$$A, \Delta A, \Delta^2 A, \dots, \Delta^{n\theta-2w} A,$$

but the next term $\Delta^{n\theta-2w+1} A$, and of course every succeeding term, is $= 0$, and this being so, the coefficients of the covariant $(A, B, \dots, K)x, y)^{n\theta-2w}$ are

$$\left(1, \frac{1}{1}\Delta, \frac{1}{1 \cdot 2}\Delta^2, \dots\right)A,$$

or what is the same thing, each coefficient is obtained from the next preceding one by the formulae

.....

The coefficients A and K , B and J , ... are derived one from the other by reversal of the order of the coefficients of $(a, b, \dots)x, y)^n$, with or without a change of sign, and thus it is only necessary to calculate up to the middle coefficient, or pair of coefficients; and we obtain, moreover, a verification.

Calculating in this manner the covariant

$$(A, B, \dots, K)x, y)^{n\theta-2w}$$

which belongs to the quantic $(a, \dots)x, y)^n$, if we herein change $a, b, c, \dots$ into $ax + by, bx + cy, cx + dy, \dots$ we obtain the covariant belonging to the quantic $(a, \dots)x, y)^{n+1}$; and in this covariant making the like change, or what is the same thing, in the first-mentioned covariant changing $a, b, c, \dots$ into $(a, b, c)x, y)^2, (b, c, d)x, y)^2, (c, d, e)x, y)^2, \dots$ we have the covariant belonging to $(a, \dots)x, y)^{n+2}$; and in like manner we obtain the covariant belonging to the quantic $(a, \dots)x, y)^n$ of any given order n .

In particular, if $w = \frac{1}{2}\rho\theta$, that is, if the given seminvariant be an invariant of $(a, \dots)x, y)^\rho$, then we obtain the series of covariants directly from A , by therein changing $a, b, c, \dots$ into $ax + by, bx + cy, cx + dy, \dots$ and in the result making the like change; or what is the same thing, in A changing $a, b, c, \dots$ into $(a, b, c)x, y)^2, (b, c, d)x, y)^2, (c, d, e)x, y)^2, \dots$ and so on until we obtain the covariant for the quantic $(a, \dots)x, y)^n$ of the given order n .

A seminvariant which cannot be expressed as a rational and integral function of lower seminvariants is said to be irreducible. The theory is distinct from that of the irreducible covariants of a quantic of a given order; for instance, as regards the cubic $(a, b, c, d)x, y)^3$, we have the irreducible covariant (invariant)

$$a^2d^2 + 4ac^3 + 4b^3d - 6abcd - 3b^2c^2,$$

but this is not an irreducible seminvariant; it is

$$= (ac - b^2)(ae - 4bd + 3c^2) - a(ace - ad^2 - b^2e + 2bcd),$$

or, what is the same thing, there is not for the quartic $(a, b, c, d, e)x, y)^4$ or for the higher quantics, any irreducible covariant having this for the leading coefficient.

We may consider the question to determine the number of aszygetic seminvariants of a given degree and weight. For instance, taking the weights up to 12, so that the series of letters extends as far as m , then for the degrees 1, 2, 3 we have as follows:

$w =$	0	1	2	3	4	5	6	7	8	9	10	11	12
Deg. 1	a	b	c	d	e	f	g	h	i	j	k	l	m
Num.	1	1	1	1	1	1	1	1	1	1	1	1	1
Diff.	0	0	0	0	0	0	0	0	0	0	0	0	0
Deg. 2	a^2	ab b^2	ac bc c^2	ad bd cd d^2	ae be ce de e^2	af bf cf df ef f^2	ag bg cg dg eg fg g^2	ah bh ch dh eh fh gh h^2	ai bi ci di ei fi gi hi i^2	aj bj cj dj ej fj gj ij ij	ak bk ck dk ek	al bl cl	am
Num.	1	1	2	2	3	3	4	4	5	5	6	6	7
Diff.	1	0	1	0	1	0	1	0	1	0	1	0	1
Deg. 3	a^3	a^2b ab^2	a^2c abc b^3	a^2d abd ac^2	a^2e abe acd b^2d bc^2	a^2f abf ace b^2e bcd c^3	a^2g abg acf ad^2 ae^2 bce bd^2 c^2d	a^2h abh acg ade b^2h bch bdf cde d^2e	a^2i abi ach adf aef b^2i bci cd^2 ce^2 d^2g	a^2j abj aci adg aeg b^2j bdi cei def	a^2k abk acj adh aeh	a^2l abl ack adi	a^2m
Num.	1	1	2	3	4	5	7	8	10	12	14	16	19
Diff.	1	0	1	1	1	1	2	1	2	2	2	2	3

For the degree 1, the line of differences shows that the only seminvariant is ($W = 0$), the seminvariant a .

For the degree 2, the line of differences 0, 1, 1, 0, $\dots$, shows that the number of seminvariants is $= 1$ for each even degree, $= 0$ for each odd degree; thus for the weight there is a seminvariant $= a^2$ which of course is not, irreducible; while for each of the other even weights we have a single irreducible seminvariant; as is well known, the forms are

$W =$	2	4	6	8	10
	$ac - b^2$	$ae - 4bd + 3c^2$	$ag - 6bf + 15ce - 10d^2$	$ai - 8bh + 28cg - 56df + 35e^2$	$ak - \dots \quad am - 12b^2$

For degree 3, the line of differences shows that the

W	=	0	1	2	3	4	5	6	7	8	9	10	11	12
Nos. irr. =		0	1	1	1	1	1	2	1	2	2	2	2	3

but inasmuch as for each even weight there is a quadric seminvariant, which multiplied by a given a cubic seminvariant, to obtain the number of irreducible cubic seminvariants we subtract

	1	0	1	0	1	0	1	0	1	0	1	0	1
	0	0	0	1	0	1	1	1	1	2	1	2	2

or the numbers of irreducible cubic seminvariants are as in the line last written down.

C. XII. 4

There is a convenience however in giving, for each even weight, as well the rejected reducible covariant; and the entire series of results is found to be

$W =$	0	2	4	6	8	10	12	14	16	18	20	22	24
$a^2 + 1$	$ac + 1$	$a^2d + 1$	$ad^2 + 1$	$ae^2 + 1$	$af^2 + 1$	$ag^2 + 1$	$ah^2 + 1$	...	...	...	...	...	...
$\vdots$													

[Book p. 26 (PDF p. 46) gives an extensive multi-column tabulation, indexed by even weights from $W = 0$ up to $W = 12$ and beyond, of the aszygetic and irreducible cubic seminvariants with the integer multipliers needed for their reductions. The verifications carried out by Cayley involve degrees 1, 2, 3 cubic seminvariants whose entries are sums of monomials in $a, b, c, \dots$ with small integer coefficients, in the format shown above.]

The canonical form given for the quintic in my Tenth Memoir on Quantics [693] belongs to a series, viz. writing now the small roman letters (instead of the italic letters) for the series of coefficients, and using the italic letters $a, c, d, e, f, \dots$ to denote seminvariants, they are as follows:

$$\begin{aligned}
0 \quad a &= a, \\
2 \quad c &= ac - b^2, \\
3 \quad d &= a^2d - 3abc + 2b^3 \quad (= f \text{ in the tenth memoir}), \\
4 \quad e &= ae - 4bd + 3c^2 \quad (= b \text{ in the tenth memoir}), \\
5 \quad f &= a^2f - 5abe + 2acd + 8b^2d - 6bc^2, \\
6 \quad g &= ag - 6bf + 15ce - 10d^2, \\
7 \quad h &= a^2h - 7abg + 9acf - 5ade + 12b^2f - 30bce + 20bd^2, \\
8 \quad i &= ai - 8bh + 28cg - 56df + 35e^2, \\
&\&c.
\end{aligned}$$

Writing also (instead of d in the tenth memoir)

$$\bar{e} = ace - ad^2 - b^2e + 2bcd - c^3,$$

so that the equation $a^2d^2 + a^2bc - 4c^3 \cdot f = 0$ of the tenth memoir is in the present notation $a^2\bar{e} - a^2ec + 4c^3 - d^2 = 0$, then the series of canonical forms is

$$\begin{aligned}
&\text{Quadric } (1, 0, c \rangle x, y)^2, \\
&\text{Cubic } (1, 0, c, d \rangle x, y)^3, \\
&\text{Quartic } (1, 0, c, d, a^2e - 3c^2 \rangle x, y)^4, \\
&\text{Quintic } (1, 0, c, d, a^2e - 3c^2, a^2f - 2cd \rangle x, y)^5, \\
&\&c.
\end{aligned}$$

the series of coefficients being

$$\begin{array}{ccccccc}
1, & 0, & c, & d, & a^2e + 1, & a^2f + 1, & a^2g + 1, & a^2h + 1, & a^2i + 1, \\
& & & & c^2 - 3, & cd - 2, & a^2ce - 15, & & \\
& & & & c^3 + 45, & a^2de + 5, & a^4e^2 - 35, & & \\
& & & & d^2 + 10, & c^2d + 3, & a^2c^2e + 630, & & \\
& & & & & & d^2f + 56, & & \\
& & & & & & c^4 - 1575, & & \\
& & & & & & cd^2 - 392 & &
\end{array}$$

these values being, in fact, the expressions in terms of the seminvariants $a, c, d, \&c.$ of

$$1, \quad 0, \quad ac, \quad a^2d, \quad a^3e, \quad a^4f, \quad a^5g, \quad a^6h, \quad a^7i.$$

I annex verifications of the foregoing values. The original page tabulates, for the canonical-form coefficients $a^2e - 3c^2$, $a^2f - 2cd$, $a^2g - 15a^2ce + 45c^3 + 10d^2$, $a^2h - 28a^2cg + 35a^2df - 630ab^2f + 112c^2e + \dots$, the integer multipliers attached to each monomial a^2e , abd , ac^2 , b^2c , a^2f , abe , $abcd$, ac^3 , a^2g , abf , acf , ade , b^3 , b^2d , b^2e , bce , bcf , bd^2 , c^2d , c^2e , c^3 , cd^2 , d^2 , $\dots$, displayed as a sequence of small tables side by side. The first such block (for $a^2e - 3c^2$) is

	a^2e	$-3c^2$		a^2f	$-2cd$
a^2e	+1		a^2f	+1	
abd	-4		a^2cd		-3
ac^2	+3	-3	abe	-5	
b^2c		+8	$abcd$	-2	
			ac^3		+10

	a^2g	$-15a^2ce$	$+45c^3$	$+10d^2$
a^2g	+1			
abf	-6			
acf		-15		
ade		-10		
abe	+5			
b^3			+45	
b^2d			-100	+10
b^2e				
bce			+45	
bd^2			-30	+5
c^2e			-15	
c^3			+45	
cd^2			-30	+5
d^2				+6

The remaining block, for $a^2h - 28a^2cg + 35a^2df - 630ab^2f + 112c^2e + \dots$, follows the same layout; the entries continue on book p. 29.

	a^2i	$-36a^2bh$	$-56a^2dg$	$+35a^2e^2$	$+630a^2cf$	$-1575c^4$
ai	+1					
abh	-8					
ach						
acg	+28	-36				
adg	-56		+56			
adf	-56					
ae^2	+35			+35		
b^2g		+290				
b^2h						
bcg		-460	+1200			
bdf		+840	-840			
be^2			-315	+112		
c^2g		+1890			+630	-1575
cd^2			+1080			
ce^2				+672	-315	
d^2f					+6300	-6275
de^2						+1004

It would be interesting to obtain the general law for the expressions of the canonical coefficients in terms of the seminvariants.

802.

NOTE ON CAPTAIN MACMAHON'S PAPER, "ON THE DIFFERENTIAL EQUATION $X^{-\frac{1}{3}}dx + Y^{-\frac{1}{3}}dy + Z^{-\frac{1}{3}}dz = 0$."

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 182–184.]

In general, if $f = (x, y, 1)^3, = 0$ be the equation of a cubic curve, and if

$$d\omega = \frac{dx}{\frac{df}{dy}} = -\frac{dy}{\frac{df}{dx}},$$

then if 1, 2, 3 are the intersections of the curve by an arbitrary right line, the coordinates of these points being $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ respectively, we have, by Abel's theorem,

$$d\omega_1 + d\omega_2 + d\omega_3 = 0,$$

viz. this is the differential relation corresponding to the integral relation which expresses that the three points are the intersections of the cubic curve by a right line, or say to the integral equation

$$\nabla, = \begin{vmatrix} y_1 & y_2 & y_3 \\ x_1 & x_2 & x_3 \\ 1 & 1 & 1 \end{vmatrix} = 0,$$

in which equation y_1, y_2, y_3 are regarded as functions of x_1, x_2, x_3 respectively, given by means of the equations $f_1 = 0, f_2 = 0, f_3 = 0$, which express that the points are on the cubic curve. See my "Memoir on the Abelian and Theta Functions," [819].

In particular, if the equation of the curve is

$$f, = \frac{1}{3}y^3 - (A + 3Bx + 3Cx^2 + Dx^3), = \frac{1}{3}(y^3 - X), = 0,$$

then

$$d\omega = \frac{dx}{y^2}, \quad = \frac{dx}{X^{\frac{2}{3}}};$$

and corresponding to the differential relation

$$X_1^{-\frac{1}{3}} dx_1 + X_2^{-\frac{1}{3}} dx_2 + X_3^{-\frac{1}{3}} dx_3 = 0,$$

we have the integral relation

$$\nabla, = \begin{vmatrix} X_1^{\frac{1}{3}}, & X_2^{\frac{1}{3}}, & X_3^{\frac{1}{3}} \\ x_1, & x_2, & x_3 \\ 1, & 1, & 1 \end{vmatrix} = 0,$$

viz. this last equation, as containing no arbitrary constant, is a particular integral of the differential equation.

If instead of x_1, x_2, x_3 we write x, y, z , then we have

$$\nabla = \begin{vmatrix} X^{\frac{1}{3}}, & Y^{\frac{1}{3}}, & Z^{\frac{1}{3}} \\ x, & y, & z \\ 1, & 1, & 1 \end{vmatrix}, = 0,$$

as a particular integral of the differential equation

$$X^{-\frac{1}{3}} dx + Y^{-\frac{1}{3}} dy + Z^{-\frac{1}{3}} dz = 0.$$

To rationalize the integral equation, write

$$\alpha, \beta, \gamma = y - z, z - x, x - y \quad (\text{so that } \alpha + \beta + \gamma = 0),$$

the equation is

$$\alpha X^{\frac{1}{3}} + \beta Y^{\frac{1}{3}} + \gamma Z^{\frac{1}{3}} = 0;$$

and we thence have

$$\alpha^3 X + \beta^3 Y + \gamma^3 Z = 3\alpha\beta\gamma X^{\frac{1}{3}} Y^{\frac{1}{3}} Z^{\frac{1}{3}}.$$

The left-hand side is

$$\begin{aligned} & \alpha^3 (A + 3Bx + 3Cx^2 + Dx^3) \\ & + \beta^3 (A + 3By + 3Cy^2 + Dy^3) \\ & + \gamma^3 (A + 3Bz + 3Cz^2 + Dz^3); \end{aligned}$$

or assuming

$$\alpha', \beta', \gamma' = x(y - z), y(z - x), z(x - y) \quad (\text{so that } \alpha' + \beta' + \gamma' = 0),$$

this is

$$= A(\alpha^3 + \beta^3 + \gamma^3) + 3B(\alpha^2\alpha' + \beta^2\beta' + \gamma^2\gamma') + 3C(\alpha\alpha'^2 + \beta\beta'^2 + \gamma\gamma'^2) + D(\alpha'^3 + \beta'^3 + \gamma'^3).$$

But taking λ arbitrary, and

$$a, b, c = \alpha + \lambda\alpha', \beta + \lambda\beta', \gamma + \lambda\gamma',$$

then

$$a + b + c = 0,$$

whence

$$a^3 + b^3 + c^3 = 3abc;$$

or substituting for a, b, c their values, and comparing the coefficients of the several powers of λ ,

$$\begin{aligned}\alpha^3 + \beta^3 + \gamma^3 &= 3\alpha\beta\gamma, \\ \alpha^2\alpha' + \beta^2\beta' + \gamma^2\gamma' &= \alpha^2\beta'\gamma' + \beta^2\gamma'\alpha' + \gamma^2\alpha'\beta' = \alpha\beta\gamma(x + y + z), \\ \alpha\alpha'^2 + \beta\beta'^2 + \gamma\gamma'^2 &= \alpha\alpha'^2\beta\gamma + \beta\beta'^2\gamma\alpha + \gamma\gamma'^2\alpha\beta = \alpha\beta\gamma(yz + zx + xy), \\ \alpha'^3 + \beta'^3 + \gamma'^3 &= 3\alpha'\beta'\gamma' = 3\alpha\beta\gamma \cdot xyz.\end{aligned}$$

Hence we have

$$\alpha^3 X + \beta^3 Y + \gamma^3 Z = 3\alpha\beta\gamma [A + B(x + y + z) + C(yz + zx + xy) + Dxyz],$$

or the integral equation is

$$[A + B(x + y + z) + C(yz + zx + xy) + Dxyz]^3 = XYZ,$$

that is,

$$\sqrt[3]{\{A + B(x + y + z) + C(yz + zx + xy) + Dxyz\}^3} = \sqrt[3]{XYZ},$$

the elegant result given by Capt. MacMahon at the beginning of his paper.

The author in a letter to me, dated Jan. 13, 1883, remarks that the particular integral of the equation in question

$$X^{-\frac{1}{3}} dx + Y^{-\frac{1}{3}} dy + Z^{-\frac{1}{3}} dz = 0,$$

is expressible as a determinant in a rational form as follows. Writing it $XYZ = P^3$, where

$$P = A + B(x + y + z) + C(yz + zx + xy) + Dxyz,$$

then the form is

$$\nabla, = \begin{vmatrix} 1, & \left(\frac{1}{3}P\frac{dX}{dx} - X\frac{dP}{dx}\right), & X \\ 1, & \left(\frac{1}{3}P\frac{dY}{dy} - Y\frac{dP}{dy}\right), & Y \\ 1, & \left(\frac{1}{3}P\frac{dZ}{dz} - Z\frac{dP}{dz}\right), & Z \end{vmatrix} = 0,$$

for, as shown by Captain MacMahon in his paper, each of the three terms such as

$$Z\left(\frac{1}{3}P\frac{dY}{dy} - Y\frac{dP}{dy}\right) - Y\left(\frac{1}{3}P\frac{dZ}{dz} - Z\frac{dP}{dz}\right),$$

which compose the determinant, is divisible by $XYZ - P^3$.

It may be added that we have identically

$$\frac{1}{3}P\frac{dX}{dx} - X\frac{dP}{dx} = (AC - B^2)(2x - y - z) + (AD - BC)(x^2 - yz) + (BD - C^2)(x^2y + x^2z - 2xyz),$$

and of course like values for the other two expressions in the determinant.

803.

ON MR ANGLIN'S FORMULA FOR THE SUCCESSIVE POWERS OF THE ROOT OF AN ALGEBRAICAL EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xix. (1883), pp. 223, 224.]

Suppose $x^m - px^{m-1} + qx^{m-2} - \dots = 0$, then the successive powers x^m, x^{m+1}, x^{m+2} , &c. of x can be expressed in the form $Px^{m-1} - Qx^{m-2} + Rx^{m-3} - \dots$. Mr Anglin has obtained for this purpose a very elegant formula, with a demonstration which (it occurred to me) might be presented under a somewhat simplified form; and he has permitted me to draw up the present Note.

Take, for greater convenience, the equation to be

$$x^4 - px^3 + qx^2 - rx + s = 0,$$

and let $h_1, h_2, h_3, \dots$ be the sums of the homogeneous products of the roots, of the orders 1, 2, 3, &c. respectively; then, writing also $h_0 = 1$, we have

$$\begin{aligned} h_1 &= h_0p, \\ h_2 &= h_1p - h_0q, \\ h_3 &= h_2p - h_1q + h_0r, \\ h_4 &= h_3p - h_2q + h_1r - h_0s, \\ h_5 &= h_4p - h_3q + h_2r - h_1s, \\ &\dots \end{aligned}$$

And this being so, starting from the equation

$$x^4 = px^3 - qx^2 + rx - s,$$

that is,

$$= h_1x^3 - h_0qx^2 + h_0rx - h_0s,$$

we obtain successively

$$\begin{aligned}
x^5 &= h_1(px^3 - qx^2 + rx - s) \\
&\quad - h_0qx^3 + h_0rx^2 - h_0sx \\
&= h_2x^3 - (h_1q - h_0r)x^2 + (h_1r - h_0s)x - h_1s, \\
x^6 &= h_2(px^3 - qx^2 + rx - s) \\
&\quad - (h_1q - h_0r)x^3 + (h_1r - h_0s)x^2 - h_1sx \\
&= h_3x^3 - (h_2q - h_1r + h_0s)x^2 + (h_2r - h_1s)x - h_2s, \\
x^7 &= h_3(px^3 - qx^2 + rx - s) \\
&\quad - (h_2q - h_1r + h_0s)x^3 + (h_2r - h_1s)x^2 - h_2sx \\
&= h_4x^3 - (h_3q - h_2r + h_1s)x^2 + (h_3r - h_2s)x - h_3s,
\end{aligned}$$

and so on, the characteristic feature being that by the introduction of the symbols h , the coefficient of x^3 presents itself at each step as a monomial, and the coefficients of the lower powers require no reduction. It is obvious that the process is a perfectly general one, and that for the equation

$$x^m - p_1x^{m-1} + p_2x^{m-2} - \dots + (-)^mp_m = 0,$$

the formula is

$$\begin{aligned}
x^{m+s} &= h_{s+1}x^{m-1} \\
&\quad - (h_sp_2 - h_{s-1}p_3 + \dots)x^{m-2} \\
&\quad + (h_sp_3 - h_{s-1}p_4 + \dots)x^{m-3} \\
&\quad \vdots \\
&\quad + (-)^{m-1}(h_sp_{m-1} - h_{s-1}p_m + \dots)x \\
&\quad + (-)^{m-1}h_sp_m,
\end{aligned}$$

where, as regards each power of x , the series forming the coefficient thereof is continued as far as possible, that is, up to the term which contains p_m or h_0 as the case may be.

804.

ON THE ELLIPTIC-FUNCTION SOLUTION OF THE EQUATION

$$x^3 + y^3 - 1 = 0.$$

[From the *Proceedings of the Cambridge Philosophical Society*, vol. iv. (1883), pp. 106–109.]

I HAD occasion to find elliptic-function expressions for the coordinates (x, y) of a point on the cubic curve $x^3 + y^3 = 1$. These are derivable from the formulæ given, Legendre, *Fonctions Elliptiques*, t. I. pp. 185, 186, for the reduction to elliptic integrals of the integral $R = \int \frac{dz}{(1 - z^3)^{\frac{1}{2}}}$. Legendre, writing

$$z = \frac{\sqrt{4y^3 - 1} - \sqrt{3}}{\sqrt{4y^3 - 1} + \sqrt{3}},$$

and then

$$m'^2 = 2 \quad \text{and} \quad m'^2 y = 1 + x',$$

finds first

$$R = m\sqrt{3} \int \frac{dx'}{\sqrt{x'^3 + 3x'^2 + 3}},$$

and then writing $r = \sqrt[6]{3}$, $x = \tan \frac{1}{2}\phi$, and $c^2 = \frac{1}{4}(2 - r^2)$, finds

$$R = \frac{1}{2}mr \int \frac{d\phi}{\sqrt{1 - c^2 \sin^2 \phi}};$$

we have therefore only to write $\sin \phi = \text{sn } u$, to modulus

$$c, = \frac{1}{2}\sqrt{2 - \sqrt[3]{3}},$$

and we thence obtain an expression for z in terms of the elliptic functions $\text{sn } u$, $\text{cn } u$, $\text{dn } u$.

Writing x instead of z , and k for c , then

$$m = \sqrt[3]{2}, \quad r = \sqrt[6]{3}; \quad k = \frac{1}{2}\sqrt{2 - r^2}, \quad k' = \frac{1}{2}\sqrt{2 + r^2}.$$

5-2

Working out the substitutions, the resulting formulæ are

$$x = \frac{2r \operatorname{sn} u \operatorname{dn} u - (1 + \operatorname{cn} u)^2}{2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2},$$

$$y = \frac{m(1 + \operatorname{cn} u) [1 + r^2 + (1 - r^2) \operatorname{cn} u]}{2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2},$$

where the modulus is k as above; and these values give

$$x^3 + y^3 = 1, \quad \frac{dx}{(1 - x^3)^{\frac{2}{3}}}, = \frac{-dy}{(1 - y^3)^{\frac{2}{3}}} = \frac{1}{2}mr du.$$

The verification is interesting enough; starting from the expression for x , and for shortness representing it by

$$x = \frac{A - B}{A + B},$$

we have

$$1 - x^3 = \frac{2B(3A^2 + B^2)}{(A + B)^3}, \quad = \frac{m^3(1 + \operatorname{cn} u)^3(3A^2 + B^2)}{[2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2]^3}.$$

We find

$$3A^2 + B^2 = 12r^2 \operatorname{cn}^2 u \operatorname{dn}^2 u + (1 + \operatorname{cn} u)^4,$$

$$= (1 + \operatorname{cn} u) [12r^2(1 - \operatorname{cn} u)(k'^2 + k^2 \operatorname{cn}^2 u) + (1 + \operatorname{cn} u)^3],$$

where the term in $\{ \}$ is a perfect cube

$$= [1 + \operatorname{cn} u + r^2(1 - \operatorname{cn} u)]^3.$$

The last-mentioned expression is, in fact,

$$(1 + \operatorname{cn} u)^3 + r^2(1 - \operatorname{cn} u) [3(1 + \operatorname{cn} u)^2 + 3r^2(1 - \operatorname{cn} u)(1 + \operatorname{cn} u) + r^4(1 - \operatorname{cn} u)^2],$$

where the second term is

$$= 12r^2(1 - \operatorname{cn} u) [\frac{1}{4}(1 + \operatorname{cn} u)^2 + \frac{1}{4}r^2(1 - \operatorname{cn}^2 u)],$$

that is, it is

$$= 12r^2(1 - \operatorname{cn} u) (k'^2 + k^2 \operatorname{cn}^2 u).$$

We have consequently

$$1 - x^3 = \frac{m^3(1 + \operatorname{cn} u)^3 [1 + r^2 + (1 - r^2) \operatorname{cn} u]^3}{[2r \operatorname{sn} u \operatorname{dn} u + (1 + \operatorname{cn} u)^2]^3},$$

or extracting the cube root $y, = \sqrt[3]{1 - x^3}$, has its foregoing value: and the differential expressions are then verified.

Suppose $y = 1$, we have

$$(m - 1)(1 + \operatorname{cn} u)^2 + mr^2(1 - \operatorname{cn}^2 u) = 2r \operatorname{sn} u \operatorname{dn} u,$$

that is,

$$(m-1)^2(1+\operatorname{cn} u)^4 + 2m(m-1)r^2(1+\operatorname{cn} u)^2(1-\operatorname{cn} u) \\ + 3m^2(1+\operatorname{cn} u)(1-\operatorname{cn} u)^2 = r^2(1-\operatorname{cn} u) \{4 - 4k^2(1-\operatorname{cn}^2 u)\},$$

or observing that the right-hand side is

$$= r^2(1-\operatorname{cn} u) \{(1+\operatorname{cn} u)^2 + (1-\operatorname{cn} u)^2 + r^2(1+\operatorname{cn} u)(1-\operatorname{cn} u)\},$$

and multiplying by $\frac{1}{1+\operatorname{cn} u}$, the equation becomes

$$0 = \frac{1}{2}(m-1)^2 r^2 (1+\operatorname{cn} u)^3 + (2m^2 - 2m + 1)(1+\operatorname{cn} u)^2(1-\operatorname{cn} u) \\ + (m^2 - 1)r^2(1+\operatorname{cn} u)(1-\operatorname{cn} u)^2 - (1-\operatorname{cn} u)^3;$$

viz. this is

$$0 = \frac{1}{2}r^2(m^2 - 1)(1+\operatorname{cn} u) - (1-\operatorname{cn} u)^3,$$

as is immediately verified: hence writing $\frac{1}{2}r^2 = \frac{1}{m^2}$, we have for the value in question, $y = 1$,

$$(m^2 - 1)(1+\operatorname{cn} u) - r^2(1-\operatorname{cn} u) = 0,$$

or say

$$m^2(1+\operatorname{cn} u) = (1+\operatorname{cn} u) + r^2(1-\operatorname{cn} u),$$

that is,

$$\operatorname{cn} u = \frac{r^2 + 1 - m^2}{r^2 - 1 + m^2},$$

which is one of the values of $\operatorname{cn} u$ derived from the equation $x = 0$; but this equation $x = 0$ gives, not the foregoing equation, but

$$m^2(1+\operatorname{cn} u)^3 = [(1+\operatorname{cn} u) + r^2(1-\operatorname{cn} u)]^3,$$

viz. the three values of $\operatorname{cn} u$ are the foregoing value and the two values obtained therefrom by changing m into ωm and $\omega^2 m$ respectively, ω being an imaginary cube root of unity. In fact, the curve $x^3 + y^3 - 1 = 0$ has at the point $x = 0, y = 1$ an inflexion, the tangent being $y = 1$, so that this line meets the curve in the point counting three times; but the line $x = 0$ meets the curve in the point, and besides in two imaginary points.

805.

NOTE ON ABEL'S THEOREM.

[From the *Proceedings of the Cambridge Philosophical Society*, vol. IV. (1883), pp. 119–122.]

CONSIDERING Abel's theorem in so far as it relates to the first kind of integrals, and as a differential instead of an integral theorem, the theorem may be stated as follows:

We have a fixed curve $f(x, y, 1) = 0$ of the order m ; this implies a relation $f'(x) dx + f'(y) dy = 0$, between the differentials dx, dy of the coordinates of a point on the curve; and we may therefore write

$$d\omega = \frac{dx}{f'(y)} = -\frac{dy}{f'(x)},$$

and, instead of dx or dy , use $d\omega$ to denote the displacement of a point (x, y) on the curve.

Taking for greater simplicity the fixed curve to be a curve without nodes or cusps, and therefore of the deficiency $\frac{1}{2}(m-1)(m-2)$, we consider its intersections by a variable curve $\phi(x, y, 1) = 0$ of the order n . And then, if $(x, y, 1)^{m-3}$ denote an arbitrary rational and integral function of (x, y) of the order $m-3$, the theorem is that we have between the displacements $d\omega_1, d\omega_2, \dots, d\omega_{mn}$ of the mn points of intersection, the relation

$$\Sigma(x, y, 1)^{m-3} d\omega = 0,$$

where the left-hand side is the sum of the values of $(x, y, 1)^{m-3} d\omega$ belonging to the mn points of intersection respectively.

For the proof, observe that, varying in any manner the curve ϕ , we obtain

$$\frac{d\phi}{dx} dx + \frac{d\phi}{dy} dy + \delta\phi = 0,$$

where $\delta\phi$ is that part which depends on the variation of the coefficients, of the whole variation of ϕ ; viz. if $\phi = ax^n + bx^{n-1}y + \dots$, then $\delta\phi = x^n da + x^{n-1}y db + \dots$; $\delta\phi$ is thus, in regard to the coordinates (x, y) , a rational and integral function of the order n .

Writing in this equation

$$dx, dy = \frac{df}{dy} d\omega, -\frac{df}{dx} d\omega,$$

the equation becomes

$$\left(\frac{d\phi}{dx} \frac{df}{dy} - \frac{d\phi}{dy} \frac{df}{dx} \right) d\omega + \delta\phi = 0,$$

or say

$$-J(f, \phi) d\omega + \delta\phi = 0,$$

that is,

$$d\omega = \frac{\delta\phi}{J(f, \phi)};$$

and then multiplying each side by the arbitrary function $(x, y, 1)^{m-3}$, we have

$$\Sigma(x, y, 1)^{m-3} d\omega = \Sigma \frac{(x, y, 1)^{m-3}}{J(f, \phi)} \delta\phi,$$

where $\delta\phi$ being of the order n in the variables, the numerator is a rational and integral function of (x, y) of the order $m + n - 3$: hence by a theorem contained in Jacobi's paper "Theoremata nova algebraica circa systema duarum æquationum inter duas variables," *Crelle*, t. xiv. (1835), pp. 281–288, [*Ges. Werke*, t. iii., pp. 285–294], the sum on the right-hand side is $= 0$: hence the required result $\Sigma(x, y, 1)^{m-3} d\omega = 0$.

Observing that $(x, y, 1)^{m-3}$ is an arbitrary function, the equation just obtained breaks up into the equations

$$\Sigma d\omega = 0, \quad \Sigma x d\omega = 0, \quad \Sigma y d\omega = 0, \quad \dots, \quad \Sigma x^{m-3} d\omega = 0, \quad \dots, \quad \Sigma y^{m-3} d\omega = 0,$$

viz. the number of equations is

$$1 + 2 + \dots + (m-1), = \frac{1}{2}(m-1)(m-2),$$

which is $= p$, the deficiency of the curve.

Suppose the fixed curve $f(x, y, 1) = 0$ is a cubic, $m = 3$, and we have the single relation $\Sigma d\omega = 0$, where the summation refers to the $3n$ points of intersection of the cubic and of the variable curve of the order n , $\phi(x, y, 1) = 0$.

In particular, if this curve be a line, $n = 1$, and the equation is $d\omega_1 + d\omega_2 + d\omega_3 = 0$; here the two points $(x_1, y_1), (x_2, y_2)$ taken at pleasure on the cubic, determine the line, and they consequently determine uniquely the third point of intersection (x_3, y_3) ; there should thus be a single equation giving the displacement $d\omega_3$ in terms of the displacements $d\omega_1, d\omega_2$; viz. this is the equation just found

$$d\omega_1 + d\omega_2 + d\omega_3 = 0.$$

So if the variable curve be a conic, $n = 2$; and we have between the displacements of the six points the relation

$$d\omega_1 + d\omega_2 + \dots + d\omega_6 = 0 :$$

here five of the points determine the conic, and they therefore determine uniquely the sixth point; and there should be between the displacements a single relation as just found.

If the variable curve be a cubic, $n = 3$, and we have between the displacements of the nine points the relation

$$d\omega_1 + d\omega_2 + \dots + d\omega_9 = 0 :$$

here eight of the points do *not* determine the cubic ϕ , but they nevertheless determine the ninth point, viz. (reproducing the reasoning which establishes this well-known and fundamental theorem as to cubic curves) if $\phi_0 = 0$ be a particular cubic through the 8 points, then the general cubic is $\phi_0 + kf = 0$, and the intersections with $f = 0$ are given by the equations $\phi_0 = 0, f = 0$; whence the ninth point is independent of k , and is determined uniquely by the 8 points. There should thus be a single relation between the displacements, viz. this is the relation just found.

And so if the variable curve be a quartic, or curve of any higher order, it appears in like manner that there should be a single relation between the displacements; this relation being in fact the foregoing relation $\Sigma d\omega = 0$.

But take the fixed curve to be a quartic, $m = 4$: then we have between the displacements $d\omega$ the relation

$$\Sigma(x, y, 1) d\omega = 0,$$

that is, the three equations

$$\Sigma x d\omega = 0, \quad \Sigma y d\omega = 0, \quad \Sigma d\omega = 0.$$

If the variable curve is a conic, $n = 2$, then there are 8 points of intersection; 5 of these taken at pleasure determine the conic, and they consequently determine the remaining 3 points of intersection: hence there should be 3 equations. And so if the variable curve be a curve of any higher order, then by considerations similar to those made use of in the case where the first curve is a cubic it appears that the number of equations between the displacements $d\omega$ should always be $= 3$.

But if the variable curve be a line, $n = 1$, then the number of the points of intersection is $= 4$. 2 of these taken at pleasure determine the line, and they consequently determine the remaining 2 points of intersection; and the number of equations between the displacements $d\omega$ should thus be $= 2$. But by what precedes, we have the 3 equations

$$d\omega_1 + d\omega_2 + d\omega_3 + d\omega_4 = 0,$$

$$x_1 d\omega_1 + x_2 d\omega_2 + x_3 d\omega_3 + x_4 d\omega_4 = 0,$$

$$y_1 d\omega_1 + y_2 d\omega_2 + y_3 d\omega_3 + y_4 d\omega_4 = 0;$$

here the 4 points of intersection are on a line $y = ax + b$; we have therefore $y_1 = ax_1 + b, \dots, y_4 = ax_4 + b$; the equations between the $d\omega$'s give

$$(y_1 - ax_1 - b) d\omega_1 + \dots + (y_4 - ax_4 - b) d\omega_4 = 0,$$

that is, is a single relation $0 = 0$; or the 3 equations thus reduce themselves to 2 independent equations.

Again, if the fixed curve be a quintic, $m = 5$, there are here between the displacements the 6 equations

$$\Sigma x^2 d\omega = 0, \quad \Sigma xy d\omega = 0, \quad \Sigma y^2 d\omega = 0,$$

$$\Sigma x d\omega = 0, \quad \Sigma y d\omega = 0, \quad \Sigma d\omega = 0;$$

the two cases in which the number of independent equations is less than 6 are (i) when the variable curve is a line, and (ii) when the variable curve is a conic. For the line $n = 1$, and the number should be $= 3$. We have the above 6 equations; but the equation of the line is $ax + by + c = 0$, that is, we have $ax_1 + by_1 + c = 0$, &c.; we deduce the 3 identical equations

$$\Sigma x(ax + by + c) = 0, \quad \Sigma y(ax + by + c) = 0, \quad \Sigma (ax + by + c) = 0,$$

and the number of independent equations is thus $6 - 3, = 3$ as it should be.

So when the variable curve is a conic, $n = 2$; the number of independent equations should be $= 5$. The points of intersection lie on a conic $(a, b, c, f, g, h\langle x, y, 1 \rangle)^2 = 0$; we have therefore the several equations $(a, b, c, f, g, h\langle x_1, y_1, 1 \rangle)^2 = 0$, &c.: we have therefore the single identical equation

$$\Sigma (a, b, c, f, g, h\langle x, y, 1 \rangle)^2 d\omega = 0,$$

and the number of independent equations is $6 - 1, = 5$ as it should be.

Obviously the like considerations apply to the case where the fixed curve is a curve of any given order whatever.

806.

DETERMINATION OF THE ORDER OF A SURFACE.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 29–32.]

[On p. lxx of the Prolegomena to C. Taylor's *Introduction to the Geometry of Conics* (1881) occurs the following passage:

“*Proof and extension of Newton's Descriptio Organica.*

“Let two angles AOB and $A\omega B$ of given magnitudes turn about O and ω respectively, and let the intersection A trace a curve of the n th order. For a given position of the arm OB there are n positions of A and therefore n of B . When OB is in the position $O\omega$ the n B 's coincide with ω , which is therefore an n -fold point on the locus of B , as is also the point O ; and since any line through O (or ω) meets the locus of B in n other points, the locus is of the order $2n$. Its order is the same when $A\omega B$ is a zero-angle or straight line.

“Let a given trihedral angle $O(ABC)$ —or a plane OBC and a line OA rigidly attached to it—turn about O , and let a variable plane through a fixed point ω meet OA in A and the plane OBC in BC ; then if the line BC describes a ruled surface of the order n the point A describes a surface of the order $4n$.”

And in a foot-note it is stated that the author is indebted to Professor Cayley for the determination of the order of this surface. The following paper contains Professor Cayley's determination, which was communicated by him to Mr Taylor.]*

LEMMA. Take (a, b, c, f, g, h) the six coordinates of a line; these are connected by the equation

$$af + bg + ch = 0.$$

[* l.c., p. 29.]

In order that the line may belong to a ruled surface, we must have between the coordinates three more equations, say these are

$$\begin{aligned} F(a, b, c, f, g, h) &= 0, \\ G(\quad \quad \quad) &= 0, \\ H(\quad \quad \quad) &= 0, \end{aligned}$$

of the orders p, q, r respectively; then the scroll is of the order $n = 2pqr$.

For expressing that the line meets an arbitrary line (a', b', c', f', g', h') , we have the linear relation

$$f'a + g'b + h'c + a'f + b'g + c'h = 0;$$

the five relations determine the ratios $a : b : c : f : g : h$, and the number of systems of values is = product of orders $= 2 \cdot p \cdot q \cdot r \cdot 1, = 2pqr$, viz. this is the number of lines meeting the arbitrary line; or, what is the same thing, it is = order of ruled surface.

Consider now a trihedral angle $OABC$ rotating about a fixed point O which may be taken for the origin, and consider a fixed point ω . Let the lines OB, OC each meet a line L , and the plane ωL intersect OA in a point P ; then it is to be shown that, if L is a line of a ruled surface of the order n , the locus of P is a surface of the order $4n$.

Observe that, for a given position of the line L , the position of the lines OB, OC is not determinate, but that the angle BOC has any position at pleasure in the plane OL , or say it rotates round the line ON which is the normal at O to the plane OL ; the line OA therefore also rotates about ON , being always inclined to it at a determinate angle θ , or the locus of OA is a right cone axis ON and semi-aperture $= \theta$.

The points P are the intersections of the several lines of the cone with the fixed plane ωL , or say that (for the given position of L) the locus of P is the conic C which is the intersection of the cone in question by the plane ωL . And then varying the position of L , the required surface is the locus of the corresponding conics C .

Take now

$$Ax + By + Cz + D = 0, \quad A'x + B'y + C'z + D' = 0,$$

for the equations of a particular line L ; then writing

$$AD' - A'D, \quad BD' - B'D, \quad CD' - C'D, \quad BC' - B'C, \quad CA' - C'A, \quad AB' - A'B = (a, b, c, f, g, h)$$

respectively, these are the six coordinates of the line L .

The equation of the plane OL is

$$(AD' - A'D)x + (BD' - B'D)y + (CD' - C'D)z = 0,$$

that is,

$$ax + by + cz = 0; 6 - 2$$

and the equations of the normal ON are therefore

$$\frac{x}{a} = \frac{y}{b} = \frac{z}{c};$$

we have therefore for the cone

$$\frac{ax + by + cz}{\sqrt{(a^2 + b^2 + c^2)} \sqrt{(x^2 + y^2 + z^2)}} = \cos \theta;$$

or if, for convenience, $\cos^2 \theta = k$, then the equation of the cone is

$$(ax + by + cz)^2 = k(a^2 + b^2 + c^2)(x^2 + y^2 + z^2);$$

say this is

$$[a^2 - k(a^2 + b^2 + c^2), b^2 - k(a^2 + b^2 + c^2), c^2 - k(a^2 + b^2 + c^2), bc, ca, ab\langle x, y, z \rangle]^2 = 0. \quad \dots (1)$$

Taking then (x_0, y_0, z_0) for the coordinates of the point ω , the equation of the plane ωL is

$$\frac{Ax + By + Cz + D}{Ax_0 + By_0 + Cz_0 + D} = \frac{A'x + B'y + C'z + D'}{A'x_0 + B'y_0 + C'z_0 + D'},$$

viz. it is

$$(BC' - B'C)(yz_0 - y_0z) + \dots + (AD' - A'D)(x - x_0) + \dots = 0,$$

that is,

$$f(yz_0 - y_0z) + g(zx_0 - z_0x) + h(xy_0 - x_0y) + a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

or say

$$x(hy_0 - gz_0 + a) + y(-hx_0 + fz_0 + b) + z(gx_0 - fy_0 + c) + (-ax_0 - by_0 - cz_0) = 0, \quad \dots (2)$$

viz. (1) and (2) are the equations of the conic C , and the coordinates (a, b, c, f, g, h) satisfy of course the equation

$$af + bg + ch = 0. \quad \dots\dots (3)$$

Considering now the line L as belonging to a ruled surface, the coordinates $(a, \dots)$ satisfy as before three equations

$$F(a, b, c, f, g, h) = 0, \quad \dots\dots (4)$$

$$G(\quad \quad) = 0, \quad \dots\dots (5)$$

$$H(\quad \quad) = 0, \quad \dots\dots (6)$$

of the orders p, q, r respectively, and we can from the six equations eliminate a, b, c, f, g, h . The resulting equation $\nabla = 0$ contains the coefficients of (1) in the order $1 \cdot 2 \cdot p \cdot q \cdot r = 2pqr$ (which is the product of the orders of the other 5 equations), and the coefficients of (2) in the order $2 \cdot 2 \cdot p \cdot q \cdot r$ (which is the product of the orders of the other 5 equations). But the coefficients of (1) being quadric functions of (x, y, z) , and those of (2) being linear functions of (x, y, z) , the aggregate order in (x, y, z) is

$$2 \cdot 2pqr + 1 \cdot 4pqr, = 8pqr;$$

or, since the order of the ruled surface is $n, = 2pqr$, the order of the locus is $4n$; which is the above-stated theorem.

807.

A PROOF OF WILSON'S THEOREM.

[From the *Messenger of Mathematics*, vol. xii. (1883), p. 41.]

LET n be a prime number; and imagine n points, the vertices of a regular polygon; any polygon which can be formed with these n points as vertices is either regular or else it is one of a set of n equal and similar polygons. For instance, $n = 5$, the polygon as shown in the figure is one of a set of 5 equal and similar

[Figure: 5-pointed star polygon labelled 1 2 3 4 5 with sides 13254]

polygons; in fact, if the points taken in their cyclical order, but beginning at pleasure with any one of the 5 points are called 1, 2, 3, 4, 5, then we have 5 such polygons 13254; and so in general. The whole number of polygons is $\frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \dots (n-1)$; and the number of the regular polygons is $\frac{1}{2}(n-1)$; hence the number of the remaining polygons is $= \frac{1}{2}(n-1)[1 \cdot 2 \dots (n-2) - 1]$; and this number must therefore be divisible by n ; that is, $1 \cdot 2 \dots (n-1) - n + 1$ is divisible by n ; or, what is the same thing, $1 \cdot 2 \dots (n-1) + 1$ is divisible by n , which is the theorem in question.

808.

NOTE ON A FORM OF THE MODULAR EQUATION IN THE TRANSFORMATION OF THE THIRD ORDER.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 173, 174.]

In my *Treatise on Elliptic Functions*, pp. 214–216, writing only $\frac{1}{J}$, $\frac{1}{J'}$ instead of Ω , Ω' , and α , β instead of α' , β' , I have shown as follows: viz. if k , λ denote as usual the original modulus, and the transformed modulus, and if

$$J = \frac{(k^4 + 14k^2 + 1)^3}{108k^4(1 - k^2)^4}, \quad J' = \frac{(\lambda^4 + 14\lambda^2 + 1)^3}{108\lambda^4(1 - \lambda^2)^4},$$

then the relation between J and J' can be found by the elimination of α , β from the equations

$$\alpha + \beta = 1,$$

$$J = \frac{(1 + 8\alpha)^3}{64\alpha(1 - \alpha)^3}, \quad J' = \frac{(1 + 8\beta)^3}{64\beta(1 - \beta)^3}.$$

By a very slight change we obtain the result given by Prof. Klein in his paper, “Ueber die Transformation der elliptischen Functionen, &c.,” *Math. Ann.* t. xiv. (1879), pp. 111–172; viz., see p. 143, the relation is to be obtained by the elimination of τ , τ' from the equation $\tau\tau' = 1$, and the equations

$$J : J - 1 : 1 = (\tau - 1)(9\tau - 1)^3 : (27\tau^2 - 18\tau - 1)^2 : -64\tau;$$

$$J' : J' - 1 : 1 = (\tau' - 1)(9\tau' - 1)^3 : (27\tau'^2 - 18\tau' - 1)^2 : -64\tau';$$

these last equations being equivalent to two equations only in virtue of the identity

$$(\tau - 1)(9\tau - 1)^3 + 64\tau = (27\tau^2 - 18\tau - 1)^2,$$

and the like identity in τ' .

In fact, writing $\alpha = \frac{\tau}{\tau - 1}$, $\beta = \frac{\tau'}{\tau' - 1}$, the equation $\alpha + \beta = 1$ becomes $\tau\tau' = 1$; and then for α , β substituting their values, we have

$$J = \frac{(9\tau - 1)^3(\tau - 1)}{-64\tau}, \quad J' = \frac{(9\tau' - 1)^3(\tau' - 1)}{-64\tau'},$$

which are the formulæ in question.

809.

SCHRÖTER'S CONSTRUCTION OF THE REGULAR PENTAGON.

[From the *Messenger of Mathematics*, vol. xii. (1883), p. 177.]

THE following construction of the regular pentagon, analogous to the more complicated one for the polygon of 17 sides, is given in the paper, H. Schröter, Zur v. Staudtschen "Construction des regulären Siebenzehnecks," *Crelle*, t. lxxv. (1873), pp. 13–24.

Take in a circle AB , CD diameters at right angles to each other, and at C , D draw in the directions A to B and B to A respectively, the lines Cc , $= 4$ radii, and Dd , $= 1$ radius; draw cd meeting the circle in the points E and F ; draw CE and CF cutting AB in the points e and f respectively, and through these at right angles to AB draw the chords 34 and 25 respectively, then we have $A2345$ a regular pentagon.

810.

NOTE ON A SYSTEM OF EQUATIONS.

[From the *Messenger of Mathematics*, vol. xii. (1883), pp. 191, 192.]

THE equations are

$$x^2 = ax + by, \quad xy = cx + dy, \quad y^2 = ex + fy,$$

where

$$\frac{b}{d} = \frac{a-d}{c-f} = \frac{c}{e};$$

or, what is the same thing, if a, b, c, d are given, then

$$e = \frac{cd}{b}, \quad f = c - \frac{d(a-d)}{b};$$

and this being so, the equations are equivalent to two independent equations; viz. starting from the first and the second equations, we have

$$dx^2 - bxy = (ad - bc)x,$$

that is,

$$dx - by = (ad - bc);$$

and thence

$$dxy - by^2 = (ad - bc)y,$$

or

$$d(ex + dy) - by^2 = (ad - bc)y;$$

which, attending to the values of e and f , is the third equation

$$y^2 = ex + fy.$$

We have

$$\frac{x}{y} = \frac{ax + by}{cx + dy}, \quad \frac{y}{x} = \frac{ex + fy}{cx + dy};$$

that is,

$$cx^2 - (a - d)xy - by^2 = 0,$$

$$ex^2 - (c - f)xy - dy^2 = 0,$$

and eliminating the (x, y) from these equations we have an equation $\Omega = 0$ as the condition that the original three equations may have a single common root; the before-mentioned equations $\frac{b}{d} = \frac{a-d}{c-f} = \frac{c}{e}$ are the conditions in order that the three equations may have two common roots, that is, that there may be two systems of (x, y) satisfying the three equations.

We have, moreover, $y(x - d) = ex$, $x(y - c) = dy$, and substituting these values, say $y = \frac{ex}{x - d}$ and $x = \frac{dy}{y - c}$, in the first and third equations respectively, they become

$$x - a = \frac{bc}{x - d}, \quad y - f = \frac{de}{y - c},$$

that is,

$$x^2 - (a + d)x + ad - bc = 0,$$

$$y^2 - (c + f)y + cf - de = 0,$$

which are quadric equations for x and y respectively; it is easy to express the second equation (like the first) in terms of (a, b, c, d) , and the first equation (like the second) in terms of (c, d, e, f) , but the forms are less simple.

Suppose $(a, b, c, d) = (-1, -1, 1, 0)$, then we have $(e, f) = (0, 1)$, the two equations in $x : y$ become $x^2 + xy + y^2 = 0$, and $0 = 0$ respectively; those in x and y become $x^2 + x + 1 = 0$, $y^2 - 2y + 1 = 0$ respectively; this is right, for the three equations are

$$x^2 = -x - y, \quad xy = x, \quad y^2 = y;$$

viz. from the third equation we have $y = 1$, a value satisfying the second equation, and then the first equation becomes $x^2 + x + 1 = 0$, or, if we please, $x^2 + xy + y^2 = 0$; the values in fact being $x = \omega$, an imaginary cube root of unity, and $y = 1$.

In the general case, the values (x, y) may be regarded as units in a complex numerical theory, viz. if (a, b, c, d, e, f) are integers, and p, q, p', q', P, Q are also integers, then the product of the two complex integers $px + qy$ and $p'x + q'y$ will be a complex integer $Px + Qy$.

811.

ON THE LINEAR TRANSFORMATION OF THE THETA FUNCTIONS.

[From the *Messenger of Mathematics*, vol. xiii. (1884), pp. 54–60.]

THE functions referred to are the single Theta Functions; these may be defined as doubly infinite products, as was in fact done in my “Mémoire sur les fonctions doublement périodiques,” *Liouv. t. x.* (1845), pp. 385–420, [25]; and it is interesting to consider from this point of view the theory of their linear transformation: this I propose to do in the present paper, adopting throughout the notation of Smith’s* “Memoir on the Theta and Omega Functions.”

The periods K, iK' are, in general, imaginary quantities

$$K = A + Bi,$$

$$iK' = C + Di,$$

where $AD - BC$ is positive; writing then $\omega = \frac{iK'}{K}$, and $q = e^{i\pi\omega}$, also for shortness

$$(q1) = 2q^{\frac{1}{8}} \prod_1^{\infty} (1 - q^{2n})^3,$$

where $q^{\frac{1}{8}}$ denotes $e^{\frac{i\pi\omega}{8}}$, the expression of the odd theta-function $\vartheta_1(x, \omega)$ as a doubly infinite product is

$$\vartheta_1(x, \omega) = (q1) x \prod' \left(1 + \frac{x}{m\pi + n\pi\omega} \right), \quad \left(\frac{\mu}{\nu} = \infty \right),$$

where (m, n) have any positive or negative integer values (the combination $m = 0, n = 0$ excluded) from $m = -\mu$ to μ , and $n = -\nu$ to ν , μ and ν being each ultimately infinite but so that μ is infinite in comparison with ν ; this condition in regard to the limits is indicated by $\mu/\nu = \infty$; and similarly $\nu/\mu = \infty$ would indicate that ν was infinite in comparison with μ .

[* Smith’s *Collected Mathematical Papers*, vol. ii., pp. 415–621.]

The condition as to the limits might be that (m, n) have any positive or negative values (excluding as before) such that the modulus of $m + n\omega$ does not exceed a positive value T , which is ultimately taken to be infinite; this condition may be indicated by $\text{mod} = \infty$.

The values of the double product corresponding to the different conditions as to the limits are not equal, but they differ only by an exponential factor, the exponent being a multiple of x^2 ; we thus have

$$x \text{ III} \left(1 + \frac{x}{mK + niK'} \right) \left(\frac{\mu}{\nu} = \infty \right) = \exp(\nabla x^2) x \text{ III} \left(1 + \frac{x}{mK + niK'} \right) (\text{mod} = \infty),$$

where ∇ is a determinate value, depending on K and K' ; and similarly

$$x \text{ III} \left(1 + \frac{x}{m\Lambda + ni\Lambda'} \right) \left(\frac{\mu}{\nu} = \infty \right) = \exp(\square x^2) x \text{ III} \left(1 + \frac{x}{m\Lambda + ni\Lambda'} \right) (\text{mod} = \infty),$$

where $\square$ is a determinate value depending in like manner on Λ, Λ' .

We have, then, as above

$$\begin{aligned} \vartheta_1(x, \omega) &= (q1) x \text{ III} \left(1 + \frac{x}{m\pi + n\pi\omega} \right) \left(\frac{\mu}{\nu} = \infty \right) \\ &= (q1) \frac{\pi}{K} \cdot \frac{Kx}{\pi} \text{ III} \left(1 + \frac{\frac{Kx}{\pi}}{mK + niK'} \right) \\ &= (q1) \frac{\pi}{K} \exp\left(\nabla \frac{K^2 x^2}{\pi^2}\right) \frac{Kx}{\pi} \text{ III} \left(1 + \frac{\frac{Kx}{\pi}}{mK + niK'} \right) (\text{mod} = \infty), \end{aligned}$$

viz. we have thus defined $\vartheta_1(x, \omega)$ as a doubly infinite product with the limiting condition $(\text{mod} = \infty)$; if for x we write $\frac{\pi x}{h}$, h arbitrary, we have

$$\vartheta_1\left(\frac{\pi x}{h}, \omega\right) = (q1) \frac{\pi}{K} \exp\left(\nabla \frac{K^2 x^2}{h^2}\right) \frac{Kx}{h} \text{ III} \left(1 + \frac{\frac{Kx}{h}}{mK + niK'} \right) (\text{mod} = \infty),$$

and similarly, if $\Omega = \frac{i\Lambda'}{\Lambda}$, $Q = e^{i\pi\Omega}$, then

$$\begin{aligned} \vartheta_1\left\{(a + b\Omega) \frac{\pi x}{h}, \Omega\right\} &= (Q1) \frac{\pi}{\Lambda} \exp\left\{(a + b\Omega)^2 \square \Lambda^2 \frac{x^2}{h^2}\right\} \\ &\times (a + b\Omega) \frac{\Lambda x}{h} \text{ III} \left(1 + \frac{(a + b\Omega) \frac{\Lambda x}{h}}{m\Lambda + ni\Lambda'} \right) (\text{mod} = \infty). \end{aligned}$$

In the case of a linear transformation, we have

$$\omega = \begin{vmatrix} a, & b \\ c, & d \end{vmatrix} \Theta \Omega, \quad \text{that is,} \quad \omega = \frac{c + d\Omega}{a + b\Omega}, \quad 7-2$$

where a, b, c, d are positive or negative integers such that $ad - bc = +1$; it is to be shown that the two infinite products are in this case identical; this being so, we have

$$\frac{\vartheta_1\left\{(a+b\Omega)\frac{\pi x}{h}, \Omega\right\}}{\vartheta_1\left\{\frac{\pi x}{h}, \omega\right\}} = \frac{(Q1)}{(q1)} \exp\{(a+b\Omega)^2 \square \Lambda^2 - \nabla K^2\} \frac{x^2}{h^2},$$

viz. the two functions differ only by a constant factor and by an exponential factor, the exponent being a multiple of x^2 : after all reductions, this factor is found to be

$$= \exp\left(-i\pi b(a+b\Omega) \frac{x^2}{h^2}\right).$$

We have

$$\omega = \frac{c+d\Omega}{a+b\Omega},$$

or since

$$\omega = \frac{iK'}{K}, \quad \Omega = \frac{i\Lambda'}{\Lambda},$$

this is

$$\frac{iK'}{K} = \frac{c\Lambda + di\Lambda'}{a\Lambda + bi\Lambda'},$$

or say

$$\begin{aligned} \frac{1}{M}K &= a\Lambda + bi\Lambda', \\ \frac{1}{M}iK' &= c\Lambda + di\Lambda', \end{aligned}$$

either of which equations may be taken as a definition of the multiplier M . We have

$$\frac{K}{M\Lambda} = a + bi\Omega,$$

$$\begin{aligned} \frac{1}{M}(mK + niK') &= (ma + cn)\Lambda + (bm + dn)i\Lambda' \\ &= m'\Lambda + n'i\Lambda', \end{aligned}$$

if

$$\begin{aligned} m' &= am + cn, \\ n' &= bm + dn. \end{aligned}$$

Here to any integer values of (m, n) there correspond integer values of m', n' ; and conversely, in virtue of the equation $ad - bc = 1$, to any integer values of m', n' there correspond integer values of m, n . The two products are

$$\begin{aligned} &\text{III} \left(1 + \frac{\frac{Kx}{h}}{m'\Lambda + n'i\Lambda'} \right), \quad (\text{mod} = \infty), \\ &\text{III} \left(1 + \frac{(a+b\Omega)\frac{\Lambda x}{h}}{m\Lambda + ni\Lambda'} \right), \quad (\text{mod} = \infty). \end{aligned}$$

But, as above, we have $\frac{K}{M} = (a + b\Omega)\Lambda$: and then, observing that in the first of the two products we may for m', n' write m, n , it at once appears that the two products are identical.

The exponential factor, writing therein $(a + b\Omega)\Lambda = \frac{K}{M}$, becomes

$$\exp\left[\left(\frac{\square}{M^2} - \nabla\right) \frac{K^2 x^2}{h^2}\right].$$

The values of $\nabla, \square$ are at once obtained by means of a formula* given in my Memoir, viz. we have

$$\nabla = -\frac{1}{2}(B + \beta),$$

where

$$B = \frac{\pi(\omega\nu + \omega'\nu')}{\Pi \Upsilon \bmod (\omega\nu - \omega'\nu)},$$

$$\beta = \frac{\pi i}{\Pi \Upsilon \bmod (\omega\nu - \omega'\nu)} \frac{(\omega\nu - \omega'\nu')}{\omega\nu - \omega'\nu}.$$

Comparing with the present notation

$$\Omega = \omega + \omega'i, = A + Bi = K,$$

$$\Upsilon = \nu + \nu'i, = C + Di = K'i,$$

so that Ω, Υ denote $K, K'i$, and $\omega, \omega', \nu, \nu'$ denote A, B, C, D respectively: $\omega\nu - \omega'\nu'$ is thus $= AD - BC$, which has been assumed to be positive; hence also $\bmod (\omega\nu - \omega'\nu) = AD - BC$, and the formula becomes

$$\nabla = -\frac{1}{2}\pi\left\{\frac{AC + BD}{i(AD - BC)} + 1\right\} \frac{1}{KK'}.$$

Now writing

$$\Lambda = A_1 + B_1i,$$

$$i\Lambda' = C_1 + D_1i,$$

then we have

$$\frac{1}{M}(A + Bi) = a\Lambda + bi\Lambda' = a(A_1 + B_1i) + b(C_1 + D_1i),$$

$$\frac{1}{M}(C + Di) = c\Lambda + di\Lambda' = c(A_1 + B_1i) + d(C_1 + D_1i);$$

consequently, if

$$M = \rho(\cos\theta + i\sin\theta),$$

[* *Collected Mathematical Papers*, t. i., p. 164. The denominator factor $i^2\Upsilon$ has been omitted (p. 165) by mistake.]

we have

$$\begin{aligned}\frac{1}{\rho}(A \cos \theta + B \sin \theta) &= aA_1 + bC_1, \\ \frac{1}{\rho}(-A \sin \theta + B \cos \theta) &= aB_1 + bD_1, \\ \frac{1}{\rho}(C \cos \theta + D \sin \theta) &= cA_1 + dC_1, \\ \frac{1}{\rho}(-C \sin \theta + D \cos \theta) &= cB_1 + dD_1,\end{aligned}$$

and thence

$$\frac{1}{\rho^2}(AD - BC) = (ad - bc)(A_1D_1 - B_1C_1), = (A_1D_1 - B_1C_1).$$

Hence $A_1D_1 - B_1C_1$ is positive, and we have

$$\square = -\frac{1}{2}\pi \left\{ \frac{A_1C_1 + B_1D_1}{i(A_1D_1 - B_1C_1)} + 1 \right\} \frac{1}{\Lambda\Lambda'}.$$

Take K_1 the conjugate of K , Λ_1 the conjugate of Λ , then

$$\begin{aligned}K_1 &= A - Bi, \quad \Lambda_1 = A_1 - B_1i, \\ iK' &= C + Di, \quad i\Lambda' = C_1 + D_1i.\end{aligned}$$

We have

$$iK_1K' = AC + BD + i(AD - BC),$$

and therefore

$$\frac{K_1K'}{AD - BC} = \frac{AC + BD}{i(AD - BC)} + 1, \quad \nabla = \frac{-\frac{1}{2}\pi}{AD - BC} \cdot \frac{K_1}{K},$$

and similarly

$$\square = \frac{-\frac{1}{2}\pi}{A_1D_1 - B_1C_1} \cdot \frac{\Lambda_1}{\Lambda}.$$

The exponential is

$$\left(\frac{\square}{M^2} - \nabla \right) \frac{K^2 x^2}{h^2},$$

and we have

$$\frac{\square}{M^2} - \nabla = \frac{-\frac{1}{2}\pi}{M^2(A_1D_1 - B_1C_1)} \frac{\Lambda_1}{\Lambda} + \frac{\frac{1}{2}\pi}{AD - BC} \frac{K_1}{K},$$

which is

$$\begin{aligned}&= \frac{-\frac{1}{2}\pi}{M^2(A_1D_1 - B_1C_1)} \frac{\Lambda_1}{\Lambda} + \frac{\frac{1}{2}\pi}{\rho^2(A_1D_1 - B_1C_1)} \frac{K_1}{K} \\ &= \frac{-\frac{1}{2}\pi}{A_1D_1 - B_1C_1} \left(\frac{1}{M^2} \frac{\Lambda_1}{\Lambda} - \frac{1}{\rho^2} \frac{K_1}{K} \right).\end{aligned}$$

But $\rho/M = \cos \theta - i \sin \theta$, or calling this for a moment P , then $1/M^2 = P^2/\rho^2$, and the formula may be written

$$\begin{aligned} \frac{\square}{M^2} - \nabla &= \frac{-\frac{1}{2}\pi P}{\rho^2(A_1 D_1 - B_1 C_1)} \left(P \frac{\Lambda_1}{\Lambda} - P^{-1} \frac{K_1}{K} \right) \\ &= \frac{-\frac{1}{2}\pi P}{\rho^2(A_1 D_1 - B_1 C_1)} \frac{1}{K\Lambda} \{ (\cos \theta - i \sin \theta) \Lambda_1 K - (\cos \theta + i \sin \theta) \Lambda K_1 \}. \end{aligned}$$

The term in $\{ \}$ is

$$\begin{aligned} &(\cos \theta - i \sin \theta) (A + Bi) (A_1 - B_1 i) - (\cos \theta + i \sin \theta) (A - Bi) (A_1 + B_1 i) \\ &= 2 \cos \theta [-(AB_1 - A_1 B) i] - 2i \sin \theta (AA_1 + BB_1), \\ &= -2i \{ (AB_1 - A_1 B) \cos \theta + (AA_1 + BB_1) \sin \theta \}, \\ &= -2i \{ B_1 (A \cos \theta + B \sin \theta) - A_1 (-A \sin \theta + B \cos \theta) \}, \\ &= -2i \rho \{ B_1 (aA_1 + bC_1) - A_1 (aB_1 + bD_1) \}, \\ &= +2ib\rho (A_1 D_1 - B_1 C_1). \end{aligned}$$

Hence

$$\begin{aligned} \frac{\square}{M^2} - \nabla &= \frac{-\frac{1}{2}\pi P}{\rho^2(A_1 D_1 - B_1 C_1)} 2ib\rho (A_1 D_1 - B_1 C_1) \frac{1}{K\Lambda} \\ &= -\frac{i\pi b P}{\rho} \cdot \frac{1}{K\Lambda} = -\frac{i\pi b}{M} \cdot \frac{1}{K\Lambda}, \end{aligned}$$

and the exponential thus is

$$= \exp \left(-\frac{i\pi b}{M} \cdot \frac{1}{K\Lambda} \cdot \frac{K^2 x^2}{h^2} \right) = \exp \left(-i\pi b \frac{K}{M\Lambda} \cdot \frac{x^2}{h^2} \right);$$

or, since $\frac{K}{M\Lambda} = (a + b\Omega)$, this is

$$= \exp \left(-i\pi b (a + b\Omega) \frac{x^2}{h^2} \right);$$

and we have thus the required formula

$$\frac{\vartheta_1 \left\{ (a + b\Omega) \frac{\pi x}{h}, \Omega \right\}}{\vartheta_1 \left\{ \frac{\pi x}{h}, \omega \right\}} = \frac{(Q1)}{(q1)} (a + b\Omega) \exp \left(-i\pi b (a + b\Omega) \frac{x^2}{h^2} \right).$$

812.

ON ARCHIMEDES' THEOREM FOR THE SURFACE OF A CYLINDER.

[From the *Messenger of Mathematics*, vol. xiii. (1884), pp. 107, 108.]

The measure of the surface of a cylinder was first obtained by Archimedes in his Treatise on the Sphere and Cylinder (Book i., Prop. xiv.), *Œuvres d'Archimède*, par F. Peyrard, 4° Paris, 1807, pp. 26–31; viz. Archimedes showed that the surface of the cylinder was equal to the area of a circle, having its radius a mean proportional between the height and the diameter of the circular base [$S = 2\pi ah$, $= \pi \{ \sqrt{(2a \cdot h)} \}^2$].

The following is in effect his demonstration:

He considers regular polygons (with the same number of sides) inscribed in and circumscribed about a circle; and, as regards the cylinder, the prisms standing on these polygons.

Say for the circular base of the cylinder we have

$$\begin{array}{ll} S^{\times} & \text{surface of circumscribed prism,} \\ S & \text{cylinder,} \\ S_{\times} & \text{inscribed prism;} \end{array}$$

and for the circle, having its radius a mean proportional between the height and the diameter of the circular base,

$$\begin{array}{ll} B^{\times} & \text{area of circumscribed polygon,} \\ B & \text{circle,} \\ B_{\times} & \text{inscribed polygon,} \end{array}$$

where the four polygons referred to by $S^{\times}$, $S_{\times}$, $B^{\times}$, $B_{\times}$ have all of them the same number of sides.

It is in the preceding propositions (by means of an axiom as to curve lines) shown that

$$S^{\times} > S > S_{\times}, \quad B^{\times} > B > B_{\times};$$

and it is further shown that

$$S^{\times} = B^{\times}, \quad S_{\times} = B_{\times}.$$

It is moreover shown that, by taking the number of sides sufficiently large, the ratio $B^{\times} : B_{\times}$, or say the fraction $B^{\times}/B_{\times}$ (which is greater than 1) may be made less than any given quantity $1 + \varepsilon$.

It is then to be shown that $S = B$.

If not, then

either

$$B < S.$$

This being so, it is possible to make

$$B_{\times}/B^{\times} < S/B,$$

that is,

$$S^{\times}/B^{\times} < S/B,$$

or

$$S^{\times}/S < B^{\times}/B,$$

which is absurd, since

$$S^{\times}/S > 1; \quad B^{\times}/B < 1;$$

or else

$$B > S.$$

This being so, it is possible to make

$$B_{\times}/B^{\times} < B/S,$$

that is,

$$B^{\times}/S^{\times} < B/S,$$

or

$$B^{\times}/B < S^{\times}/S,$$

which is absurd, since

$$B^{\times}/B > 1; \quad S^{\times}/S < 1;$$

and consequently $S = B$, the theorem in question.

I take the opportunity of referring to two theorems by Archimedes, Lemmas, Prop. v. and vi., Peyrard, pp. 429–435, which relate to the contacts of circles. We have in each of them the figure which he calls the Arbelon, viz. if A, C, B are points in this order on the same straight line, then the figure consists of the three semicircles on the diameters AC, CB , and AB respectively, and the Arbelon is the space included between the three semi-circumferences.

In Prop. v., we have the common tangent at C to the two semicircles AC, CB ; this divides the Arbelon into two mixtilinear triangles (each bounded by the common tangent, one of the smaller semicircles, and a portion of the larger semi-circle), and inscribing each of these a circle, the theorem is that the two inscribed circles are of equal magnitude.

In Prop. vi., the theorem is that the radii of the smaller semicircles being as $3 : 2$, then the radius of the circle inscribed in the Arbelon is to the diameter of the larger semicircle as 6 to 19. But it is noticed that the demonstration would apply to any other value of the ratio; and, in fact, if the radii of the two smaller circles are as $a : b$, then the radius of the inscribed circle is to the diameter of the larger semicircle as ab to $a^2 + ab + b^2$, which is the general form of the theorem.

813.

[NOTE ON MR GRIFFITHS' PAPER "ON A DEDUCTION FROM
THE ELLIPTIC-INTEGRAL FORMULA $y = \sin(A + B + C + \dots)$ ".]

[From the *Proceedings of the London Mathematical Society*, vol. xv. (1884), p. 81.]

Consider, for instance,

the cubic transformation

$$y = \frac{x [1 + 2\alpha' - (1 + \alpha')^2 x^2]}{1 - \alpha^2 x^2},$$

where $\alpha^2 + \alpha'^2 = 1$.

This implies

$$\sqrt{1 - y^2} = \frac{\sqrt{1 - x^2} [1 - (1 + \alpha')^2 x^2]}{1 - \alpha^2 x^2},$$

viz., $\sqrt{1 - y^2} =$ a rational multiple of $\sqrt{1 - x^2}$.

Also the quadric transformation

$$z = \frac{1 - (1 + \beta'^2) x^2}{1 - \beta^2 x^2};$$

where $\beta^2 + \beta'^2 = 1$.

This implies

$$\sqrt{1 - z^2} = \frac{\sqrt{1 - x^2} \cdot 2\beta' x}{1 - \beta^2 x^2},$$

viz., $\sqrt{1 - z^2} =$ a rational multiple of $\sqrt{1 - x^2}$.

Hence, assuming

$$u = yz - \sqrt{1 - y^2} \sqrt{1 - z^2},$$

which is a rational function

$$= \frac{x(a_0 - a_2 x^2 + a_4 x^4)}{1 - \alpha^2 x^2 \cdot 1 - \beta^2 y^2},$$

we have

$$\sqrt{1 - u^2} = y\sqrt{1 - z^2} + z\sqrt{1 - y^2},$$

which is $= \sqrt{1 - x^2}$ multiplied by a like rational function.

That is, in defining the a_0, a_2, a_4 functions of the two arbitrary coefficients α, β , as above, we have in effect so determined them that $\sqrt{1-u^2}$ shall be $= \sqrt{1-x^2}$ multiplied by a rational function of x .

We can then further determine a_0, a_2, a_4 in such wise that the change of x into $\frac{1}{kx}$ shall change u into $\frac{1}{\lambda u}$; and, this being so, making the change in $\sqrt{1-x^2}$, we obtain $\sqrt{1-\lambda^2 u^2}$ in the form, $\sqrt{1-k^2 x^2}$ multiplied by a rational function of x ; viz. u is a function of x such that

$$\frac{du}{\sqrt{1-u^2} \sqrt{1-\lambda^2 u^2}} = \frac{M dx}{\sqrt{1-x^2} \sqrt{1-k^2 x^2}}.$$

The theory is thus in effect Jacobi's—with the novelty of combining two lower transformations in such wise that the assumed expression for u as a rational function of x shall give

$$\sqrt{1-u^2} = \sqrt{1-x^2} \text{ multiplied by a rational function of } x.$$

It is not necessary that the equations

$$y = \text{rational function of } x \quad \text{and} \quad z = \text{rational function of } x$$

should be elliptic-function transformations. All that is required is that they should be such as to give $\sqrt{1-y^2}$ and $\sqrt{1-z^2}$ each $= \sqrt{1-x^2}$ multiplied by a rational function of x .

814.

ON DOUBLE ALGEBRA.

[From the *Proceedings of the London Mathematical Society*, vol. xv. (1884), pp. 185–197. Read April 3, 1884.]

1. I consider the Double Algebra formed with the extraordinary symbols, or “extraordinaries” x , y , which are such that

$$x^2 = ax + by,$$

$$xy = cx + dy,$$

$$yx = ex + fy,$$

$$y^2 = gx + hy,$$

or, as these equations may also be written,

$$\begin{array}{c|cc} & x & y \\ \hline x & (a, b) & (c, d) \\ y & (e, f) & (g, h) \end{array},$$

where a, b, c, d, e, f, g, h are ordinary symbols, or say coefficients; all coefficients being commutative and associative *inter se* and with the extraordinaries x, y .

The system depends in the first instance on the eight parameters a, b, c, d, e, f, g, h ; but we may, instead of the extraordinaries x, y , consider the new extraordinaries connected therewith by the linear relations $\xi = \alpha x + \beta y$, $\eta = \gamma x + \delta y$, where the coefficients $\alpha, \beta, \gamma, \delta$ may be determined so as to establish between the eight parameters any four relations at pleasure (or, what is the same thing, $\alpha, \beta, \gamma, \delta$ are what I call “apocatastic” constants): and the number of parameters is thus properly $8 - 4, = 4$.

2. The extraordinaries here considered are not in general associative; differing herein from the imaginaries of Peirce’s Memoir, “Linear Associative Algebra” (1870),

reprinted in the *American Mathematical Journal*, t. iv. (1881), pp. 97–227, which as appears by the title, refers only to associative imaginaries. I recall some definitions and results. The symbol x is said to be *idempotent* if $x^2 = x$, *nilpotent* if $x^2 = 0$; and the systems of associative symbols are expressed as much as may be by means of such idempotent and nilpotent symbols: thus the linear systems are $(a_1) x^2 = x$, $(b_1) x^2 = 0$. A double linear system composed of independent symbols, that is, symbols x, y each belonging to its own linear system, and moreover such that $xy = yx = 0$, is said to be “mixed”; thus the mixed double systems are

$$\begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & 0 & y \end{array}, \quad \begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & 0 & 0 \end{array}, \quad \begin{array}{c|cc} & x & y \\ \hline x & 0 & 0 \\ y & 0 & 0 \end{array}.$$

But Peirce excludes those from consideration, attending only to the pure systems, which he finds to be

$$(a_2) \begin{array}{c|cc} & x & y \\ \hline x & x & y \\ y & y & 0 \end{array}, \quad (b_2) \begin{array}{c|cc} & x & y \\ \hline x & x & y \\ y & 0 & 0 \end{array}, \quad (c_2) \begin{array}{c|cc} & x & y \\ \hline x & 0 & x \\ y & 0 & 0 \end{array}.$$

To these, however, should be added the system

$$(d_2) \begin{array}{c|cc} & x & y \\ \hline x & x & 0 \\ y & y & 0 \end{array};$$

see *post*, No. 19.

3. In the general theory, where the symbols are not in the first instance taken to be associative, we may of course establish between the coefficients such relations as will make the symbols associative; and the question presents itself to show how in this case the system reduces itself to one of Peirce’s systems. This I considered in my note “On Associative Imaginaries,” *Johns Hopkins University Circular*, No. 15 (1882), p. 211 [822]; I there obtained, as the general form of the commutative and associative system,

$$\begin{aligned} x^2 &= ax + by, \\ xy = yx &= cx + dy, \\ y^2 &= \frac{cd}{b} x + \frac{d^2 + bc - cd}{b} y, \end{aligned}$$

the relation of which to Peirce's system was, as I there remarked, pointed out to me by Mr C. S. Peirce: this will be considered in the sequel, Nos. 13 to 19.

4. Starting now with the general equations

$$\begin{aligned}x^2 &= ax + by, \\xy &= cx + dy, \\yx &= ex + fy, \\y^2 &= gx + hy,\end{aligned}$$

we may attempt to find an extraordinary; ξ , $= \alpha x + \beta y$ (α, β coefficients), such that $\xi^2 = K\xi$ (K a coefficient). In general, K is not $= 0$, and when it is not $= 0$, it may without loss of generality be taken to be $= 1$; we have then $\xi^2 = \xi$, the *idempotent* symbol. But K may be $= 0$; and then $\xi^2 = 0$, ξ a *nilpotent* symbol. To include the two cases, I retain K , it being understood that, when K is not $= 0$, it may be taken to be $= 1$. We have

$$\begin{aligned}\xi^2 &= \alpha^2 (ax + by) + \alpha\beta (\overline{c+e}x + \overline{d+f}y) + \beta^2 (gx + hy) \\&= [\alpha^2 a + (c+e)\alpha\beta + g\beta^2]x + [\alpha^2 b + (d+f)\alpha\beta + h\beta^2]y.\end{aligned}$$

Hence, when this is $= K\xi$, we have

$$\begin{aligned}\frac{\alpha}{\beta} \cdot K(\alpha x + \beta y), \\ \frac{\alpha}{\beta} = \frac{\alpha^2 a + (c+e)\alpha\beta + g\beta^2}{\alpha^2 b + (d+f)\alpha\beta + h\beta^2},\end{aligned}$$

a cubic equation for the determination of the ratio $\alpha : \beta$; and, for any particular value of the ratio, we can in general determine the absolute magnitudes, so that

$$K, = \frac{1}{\alpha} [\alpha^2 a + (c+e)\alpha\beta + g\beta^2], = \frac{1}{\beta} [b\alpha^2 + (d+f)\alpha\beta + h\beta^2],$$

shall be $= 1$. If, however, for the given value of the ratio we have only

$$\alpha^2 a + (c+e)\alpha\beta + g\beta^2 = 0, \quad b\alpha^2 + (d+f)\alpha\beta + h\beta^2 = 0,$$

one of these equations, of course, implying the other, then the value of K is $= 0$.

5. It follows that there are in general three idempotent symbols ξ, η, ζ , that is, extraordinaries such that $\xi^2 = \xi, \eta^2 = \eta, \zeta^2 = \zeta$. The cubic equation may, however, have two equal roots, or three equal roots identically; in this last case, any extraordinary $\alpha x + \beta y$ is an idempotent symbol, and we have to consider in detail further on we may, instead of an idempotent symbol or symbols, have a nilpotent symbol or symbols. It might be convenient to use the term Pierce's symbol which is in general idempotent, but which may be also nilpotent. Writing $\frac{\alpha}{\beta} = \frac{-y}{x}$, we obtain a cubic equation $\Omega = (x, y)^3 = 0$, where obviously the linear factors of Ω are the just-mentioned functions ξ, η, ζ ; that is, we have ξ, η, ζ as the linear factors of the cubic function

$$\Omega = gx^3 + (h - c - e)x^2y + (a - d - f)xy^2 + by^3;$$

each such factor, except in the case where it is nilpotent, being determined so that it shall be idempotent. The cubic function of course vanishes identically if

$$g = 0, \quad h - c - e = 0, \quad a - d - f = 0, \quad b = 0.$$

6. Two extraordinaries $\xi, = \alpha x + \beta y; \eta, = \gamma x + \delta y$ ($\alpha, \beta, \gamma, \delta$ coefficients); may be such that $\xi\eta = 0$; this, of course, does not imply $\eta\xi = 0$; we have, in fact,

$$\begin{aligned} \xi\eta &= (\alpha x + \beta y)(\gamma x + \delta y) \\ &= \alpha\gamma x^2 + \alpha\delta xy + \beta\gamma yx + \beta\delta y^2 \\ &= \alpha\gamma(ax + by) + \alpha\delta(cx + dy) + \beta\gamma(ex + fy) + \beta\delta(gx + hy) \\ &= (\alpha\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta)x + (b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta)y; \end{aligned}$$

and the required condition is satisfied if

$$a\alpha\gamma + c\alpha\delta + e\beta\gamma + g\beta\delta = 0,$$

$$b\alpha\gamma + d\alpha\delta + f\beta\gamma + h\beta\delta = 0.$$

Writing these equations first under the form

$$\gamma(a\alpha + e\beta) + \delta(c\alpha + g\beta) = 0,$$

$$\gamma(b\alpha + f\beta) + \delta(d\alpha + h\beta) = 0,$$

and then under the form

$$\alpha(a\gamma + c\delta) + \beta(e\gamma + g\delta) = 0,$$

$$\alpha(b\gamma + d\delta) + \beta(f\gamma + h\delta) = 0,$$

we have

$$(a\alpha + e\beta)(d\alpha + h\beta) - (b\alpha + f\beta)(c\alpha + g\beta) = 0,$$

a quadric equation for the determination of $\alpha : \beta$; and similarly

$$(a\gamma + c\delta)(f\gamma + h\delta) - (b\gamma + d\delta)(e\gamma + g\delta) = 0,$$

a quadric equation for the determination of $\gamma : \delta$; that is, there are two values of the left-hand factor ξ , and two values of the right-hand factor η . But, of course, these correspond each to each, viz. either factor being given, the other factor is determined uniquely.

Writing successively $\frac{\alpha}{\beta} = \frac{-y}{x}$, and $\frac{\gamma}{\delta} = \frac{-y'}{x'}$, we have the quadric functions $(x, y)^2$,

$$\Phi = (ah - fg)x^2 + (-ah - cd + bg + cf)xy + (cd - be)y^2,$$

$$\Phi_1 = (ah - dg)x'^2 + (-ah - cf + bg + de)x'y' + (cf - be)y'^2,$$

where the linear factors of Φ are the two values ξ_1, ξ_2 of the left-hand factor ξ , and the linear factors of Φ_1 are the two values η_1, η_2 of the right-hand factor η .

7. In the commutative case, $c = e$, and $d = f$, we have

$$\Phi = \Phi_1 = (ah - dg)x^2 + (-ah + bg)xy + (cd - be)y^2;$$

here $\xi\eta = 0$, $\eta\xi = 0$, and the values may be taken to be $(\xi_1, \eta_1) = (\xi, \eta)$, $(\xi_2, \eta_2) = (\eta, \xi)$, so that $\xi_1\xi_2 = \xi\eta$, as above. The value of Ω is

$$\Omega = gx^3 + (h - 2c)x^2y + (a - 2d)xy^2 + by^3.$$

8. In the commutative and associative case, taking ξ, η, ζ to be the three idempotent symbols, $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$; we have

$$\xi(\eta - \xi\eta) = \xi\eta - \xi^2\eta = \xi\eta - \xi\eta = 0;$$

and in like manner we have the six equations

$$\xi(\eta - \xi\eta) = 0, \xi(\zeta - \xi\zeta) = 0; \eta(\xi - \eta\xi) = 0, \eta(\zeta - \eta\zeta) = 0; \zeta(\xi - \zeta\xi) = 0, \zeta(\eta - \zeta\eta) = 0;$$

viz. regarding the right-hand factor as being in each case expressed as a linear function of x, y , we have apparently six distinct factors $\eta - \xi\eta$, $\zeta - \xi\zeta$, &c. each of these products would be in any one case identically = 0, viz. the case of the second factor being = 0. But we know there are only two right-hand factors η_1, η_2 , that is, $\eta - \xi\eta$ must be in some way or other identically a linear function of ξ ; the same as $\zeta - \xi\zeta$, but here suppose (a, b) coefficients, neither of them = 0; we thence have

$$\xi(a\xi + b\eta) = a\xi^2 + b\xi\eta, = a\xi + b\eta, = a\xi + b\eta,$$

that is, $(a^2 - a)\xi + (b^2 - b)\eta = 0$; whence $a = 1$, $b = 1$, and therefore $\xi\eta = \xi + \eta$; hence also $\xi\zeta$ and $\eta\zeta = \xi + \zeta$ and $\eta + \zeta$; $\xi\eta\eta\zeta$. The six products consequently are $\xi\eta, \eta\zeta, \eta\xi, \zeta\xi, \zeta\eta$, each = 0 or identically = 0.

9. In verification of the theorem that for the commutative and associative system the cubic function Ω contains the quadric function Φ as a factor, we may write, as above,

$$y = \frac{cd}{b}, \quad h = \frac{d^2 + bc - cd}{b},$$

values which give

$$\begin{aligned} b\Phi &= (bc^2 - acd)x^2 + (-ad^2 - abc + ad + bcd)xy + (bcd - b^2e)y^2 \\ &= -(ad - bc)[cx^2 + (d - a)xy - by^2], \\ b\Omega &= cdx^3 + (d^2 - bc)x^2y + (ab - 2bd)xy^2 + b^2y^3, \\ &= (dx - by)[cx^2 + (d - a)xy - by^2], \end{aligned}$$

which gives the theorem in question. And observe further that

$$\begin{aligned} (dx - by)\xi &= d^2(ax + by)c - 2bd(cx + dy) + b^2\left(\frac{cd}{b}x + \frac{d^2 + bc - cd}{b}y\right), \\ &= -(ad - bc)(dx - by), \end{aligned}$$

that is, disregarding coefficients, the two idempotent symbols ξ, η are the linear factors of $cx^2 + (d - a)xy - by^2$, and the third idempotent symbol ζ is $dx - by$.

10. Introducing coefficients in order to make the symbols ξ , η , ζ idempotent, and writing accordingly

$$\xi = \frac{1}{K} [cx + \frac{1}{2} (d - a + \sqrt{\nabla}) y], \quad \nabla = (d - a)^2 + 4bc,$$

so that

$$\eta = \frac{1}{L} [cx + \frac{1}{2} (d - a - \sqrt{\nabla}) y], \quad \xi\eta = \frac{c}{KL} [cx^2 + (d - a)xy - by^2],$$

$$\zeta = \frac{1}{P} (dx - by),$$

we have to verify that it is possible to determine K , L , P so that $\xi^2 = \xi$, $\eta^2 = \eta$, $\zeta^2 = \zeta$, $\xi\eta = \xi + \eta$. The last equation gives

$$\frac{d}{P} = \frac{c}{K} + \frac{c}{L},$$

$$-\frac{2b}{P} = \frac{d - a + \sqrt{\nabla}}{K} + \frac{d - a - \sqrt{\nabla}}{L},$$

and we thence have

$$\frac{d(d - a) + 2bc - d\sqrt{\nabla}}{P} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{d(d - a) + 2bc + d\sqrt{\nabla}}{P} = \frac{2c\sqrt{\nabla}}{L},$$

and we can from the equation $\xi^2 = \xi$ find P ; viz. comparing the coefficients of x , we have

$$\frac{d}{P} = \frac{1}{P^2} \left(ad^2 - 2bdc + b\frac{cd}{b} \right), = \frac{d}{P^2} (ad - bc), \text{ that is, } P = (ad - bc),$$

or the values of K and L are

$$\frac{[d(d - a) + 2bc - d\sqrt{\nabla}]}{ad - bc} = -\frac{2c\sqrt{\nabla}}{K},$$

$$\frac{[d(d - a) + 2bc + d\sqrt{\nabla}]}{ad - bc} = +\frac{2c\sqrt{\nabla}}{L},$$

which should agree with the values of K and L found from the equations $\xi^2 = \xi$, $\eta^2 = \eta$, respectively. Comparing the coefficients of x , the first of these equations gives

$$\begin{aligned} \frac{c}{K} &= \frac{1}{K^2} \left\{ c^2a + c(d - a + \sqrt{\nabla})c + \frac{1}{4}(d - a + \sqrt{\nabla})^2 \frac{cd}{b} \right\} \\ &= \frac{c}{4bK^2} \left\{ 4abc + 4bc(d - a) + d[(d - a)^2 + \nabla] + 2[2bc + d(d - a)]\sqrt{\nabla} \right\} \\ &= \frac{c}{2bK^2} \left\{ d\nabla + [d(d - a) + 2bc]\sqrt{\nabla} \right\}, \end{aligned}$$

that is,

$$K = \frac{1}{2b} \sqrt{\nabla} [d(d - a) + 2bc + d\sqrt{\nabla}],$$

and the equation for K becomes

$$\frac{d(d-a) + 2bc - d\sqrt{\nabla}}{ad - bc} = \frac{-4bc}{d(d-a) + 2bc + d\sqrt{\nabla}};$$

that is,

$$[d(d-a) + 2bc]^2 - d^2[(d-a)^2 + 4bc] = -4bc(ad - bc),$$

which is right; and similarly the equation for L leads to this same equation.

11. We may now establish, on the principles appearing in No. 5, the different forms of the system. Using ideas and aid as abbreviations for idempotent and nilpotent respectively, there are in all 11 cases.

(i) 3 idems. Taking two of these to be x and y , the system is

$$x^2 = x, \quad xy = ex + dy, \quad yx = ex + fy, \quad y^2 = y.$$

Hence $\Omega = (1 - c - e)x^2y + (1 - d - f)xy^2 = 0$, so that the third factor is

$$(1 - c - e)x + (1 - d - f)y.$$

This must not reduce itself to x or y , for, if so, there would be a twofold idem.; viz. as negative conditions we must have $c + e \neq 1$, $d + f \neq 1$.

And we have

$$[(1 - c - e)x + (1 - d - f)y]^2 = [1 - (c + e)(d + f)][(1 - c - e)x + (1 - d - f)y],$$

which must be an idem.; viz. we have the further negative condition $(c + e)(d + f) \neq 1$.

(ii) 2 idems and 1 nil. This arises from (i) by assuming therein

$$(c + e)(d + f) = 1, \text{ say } d + f = \frac{1}{c + e};$$

for then, writing $z = -(c + e)x + y$, we have

$$\begin{aligned} z^2 &= (c + e)^2 x^2 - (c + e)[(c + e)x + (d + f)y] + y, \\ &= [1 - (c + e)(d + f)]y, = 0; \end{aligned}$$

viz. z is a nil. And, if in the equations instead of the idem y we introduce the nil z , then the equations assume the form

$$x^2 = x, \quad xz = [(c + e)d - c]x + dz, \quad zx = [(c + e)f - c]x + fz, \quad z^2 = 0;$$

with the idem $y = (c + e)x + z$, we have the negative conditions $c + e \neq 0$, implying $d + f \neq 0$.

But the equations are obtained in a more simple form by taking a for the idem and y for the nil; viz. we then have $x^2 = x$, $xy = ax + dy$, $yx = ex + fy$, $y^2 = 0$: we must then have z , $= -(c + e)x + (1 - d - f)y$, for an idem; this gives $z^2 = -(c + e)(d + f)z$, and we have the negative conditions $c + e \neq 0$, $d + f \neq 0$ or 1.

(iii) 1 idem and 2 nils. This may be deduced from (ii) by writing therein $d + f = 0$; for then $z, = -(c + e)x + y$, is a nil. The equations are $x^2 = x, xy = ax - dy, y^2 = 0$: and if, instead of the idem y , we introduce therein z by the equation $z = -(c + e)x + y$, the equations become $x^2 = x, yz = [-(c + e)d + a]x + dz, zx = [-(c + e)f + c]x + cz, z^2 = 0$, with the negative condition $c + e \neq 0$.

But it is more simple to take x, y as the nils: the equations then are $x^2 = 0, xy = cx + dy, yx = ex + fy, y^2 = 0$. We must have $z, = ex + cy$, for an idem; this gives $z^2 = (c + e + f + d)xy$, becomes $= z$ and $= y$; so that x or y is a twofold nil. Or, what comes to the same thing, we have

$$\Omega = -(c + e)x^2y - (d + f)xy^2,$$

and Ω has as twofold factor if $c + e = 0$ or $d + f = 0$.

(iv) A twofold idem and a onefold idem. Taking x for the twofold idem and y for the onefold idem, Ω must reduce itself to $(1 - c - e)x^2y$, we must have $d + f = 1$, or say $f = 1 - d$. The equations are $x^2 = x, xy = cx + dy, yx = ex + (1 - d)y, y^2 = y$; and we have the negative conditions $c + e \neq 0$, for otherwise Ω would vanish identically.

(v) A twofold idem and a onefold nil. Taking x for the twofold idem and y for the onefold nil, the equations are $x^2 = x, xy = cx + dy, yx = ex + fy, y^2 = 0$; and we have the negative conditions $c + e \neq 0, d + f \neq 0$.

(vi) A twofold nil and a onefold idem. Taking those to be x and y , then $d + f = 0$, and the equations are $x^2 = 0, xy = cx + dy, yx = ex - dy, y^2 = y$; we have the negative condition $c + e \neq 0$.

(vii) A twofold nil and a onefold nil. Taking those to be x and y , we have $d + f = 0$ and the equations are $x^2 = 0, xy = cx + dy, yx = ex - dy, y^2 = 0$; with the negative condition $c + e \neq 0$.

(viii) A threefold idem. Taking this to be x , then Ω must reduce itself to gx^3 , viz. we must have $h = c + e, 1 = d + f$; and the equations are $x^2 = x, xy = cx + dy, yx = ex + (c + e)y, y^2 = gx + (c + e)y$; we have the negative condition $g \neq 0$, for otherwise Ω would vanish identically.

(ix) A threefold nil. Taking this to be x , then we must have $h = c + e, 0 = d + f$; the equations are $x^2 = 0, xy = cx + dy, yx = ex - dy, y^2 = gx + (c + e)y$; and there is again the negative condition $g \neq 0$.

(x) $\Omega = 0$ identically; infinity of idems, 1 nil. Ω will vanish identically if $g = 0, h = c + e, a = d + f, b = 0$. If there is 1 idem, there will be no infinity of idems, and 1 nil. For, assume an idem $x, x^2 = x$; and, if possible, let there be no other idem; then there will be a nil $y, y^2 = 0$. We have $c + e = 0, d + f = 1$; and the equations

are $x^2 = x$, $xy = cx + dy$, $yx = -cx + (1 - d)y$, $y^2 = 0$; whence $xy + yx = y$. Taking α, β arbitrary coefficients, we have

$$(\alpha x + \beta y)^2 = \alpha^2 x + \alpha\beta y, = \alpha(\alpha x + \beta y);$$

hence for $\alpha x + \beta y$ to be an idem, except in the case $\alpha = 0$, when it is the original nil y .

If besides the idem x we have an idem y , then the conditions are $c + e = 1$, $d + f = 1$; the equations are

$$x^2 = x, \quad xy = cx + dy, \quad yx = (1 - c)x + (1 - d)y, \quad y^2 = y;$$

whence $xy + yx = x + y$. Considering the combination $\alpha x + \beta y$, we have

$$(\alpha x + \beta y)^2 = \alpha^2 x + \alpha\beta(x + y) + \beta^2 y, = (\alpha + \beta)(\alpha x + \beta y).$$

This is an idem, except in the case $\alpha + \beta = 0$, when it is a nil; or, say we have the single nil $x - y$. We have thus again an infinity of idems, 1 nil.

(xi) $\Omega = 0$ identically; an infinity of nils. Taking the two nils x and y , the conditions are $c + e = 0$, $d + f = 0$, the equations are $x^2 = 0$, $xy = cx + dy$, $yx = -cx - dy$, $y^2 = 0$; whence $xy + yx = 0$, or there are an infinity of nils.

viz. $\alpha x + \beta y$ is a nil; or there are an infinity of nils.

12. The different cases may be grouped together as follows:—

A. 2 idems, (i), (ii), (iv), (x).

$$\text{Equations } x^2 = x, \quad xy = cx + dy, \quad yx = ex + fy, \quad y^2 = y.$$

B. 1 idem and 1 nil, (ii), (iii), (v), (vi), (x).

$$\text{Equations } x^2 = x, \quad xy = cx + dy, \quad yx = ex + fy, \quad y^2 = 0.$$

C. 2 nils, (iii), (vii), (xi).

$$\text{Equations } x^2 = 0, \quad xy = cx + dy, \quad yx = ex + fy, \quad y^2 = 0.$$

D. Threefold idem, (viii).

$$\text{Equations } x^2 = x, \quad xy = cx + dy, \quad yx = ex - (1 - d)y, \quad y^2 = gx + (c + e)y.$$

E. Threefold nil, (ix).

$$\text{Equations } x^2 = 0, \quad xy = cx + dy, \quad yx = ex - dy, \quad y^2 = gx + (c + e)y.$$

The several cases of A, B, C respectively are distinguished by negative conditions which need not be here repeated.

13. I consider, as in my Note before referred to, the conditions in order that the system may be associative. We have the 8 products, x^3 , x^2y , xyx , xy^2 , yx^2 , yxy , y^2x , y^3 , giving rise to equations $x.x^2 = x^2.x$, $x.xy = xy.x$, $\dots$, $y.y^2 = y^2.y$, which, on putting therein for x^2 , xy , yx , y^2 their values, must be satisfied identically. We thus obtain in

the first instance 16 relations, but some of these are repeated, and we have actually only 12 relations; viz. the relations are

$$\begin{aligned}
& \text{(twice)} & b(c - e) &= 0, \\
& & b(f - d) &= 0, \\
& & g(c - e) &= 0, \\
& \text{(twice)} & g(f - d) &= 0, \\
& \text{(twice)} & bg - cd &= 0, \\
& \text{(twice)} & bg - ef &= 0, \\
& & c(e - h) + g(d - a) &= 0, \\
& & d(d - a) + b(c - b) &= 0, \\
& & e(e - h) + g(f - a) &= 0, \\
& & b(e - h) + f(f - a) &= 0, \\
& & a(e - c) - cf + de &= 0, \\
& & h(f - d) - cf + de &= 0.
\end{aligned}$$

14. From the first four equations it appears that either $b = 0$ and $g = 0$, or else $c = e$ and $d = f$. I attend first to the latter case, viz. we here have the commutative system

$$x^2 = ax + by, \quad xy = yx = cx + dy, \quad y^2 = gx + hy.$$

In order that this may be associative, we must still have the relations

$$\begin{aligned}
& bg - cd = 0, \\
& c(c - h) + g(d - a) = 0, \\
& d(d - a) + b(c - b) = 0,
\end{aligned}$$

or, as they may also be written,

$$\begin{vmatrix} b, & -c, & d - a \\ -d, & g, & c - b \end{vmatrix} = 0.$$

These are satisfied by $g = \frac{cd}{b}$, $h = \frac{d^2 + bc - cd}{b}$, and we have thus the commutative and associative system of the Note.

Every system is of the form A, B, C, D, or E; and it can be shown that the commutative and associative system is not of the form D. For, if D were commutative, we should have $c = e$, $d = \frac{1}{2}$, viz. the equations will be $x^2 = x$, $xy = yx = cx + \frac{1}{2}y$, $y^2 = gx + 2cy$, that is,

$$a, b, c, d, y, h = 1, 0, c, \frac{1}{2}, 2c;$$

and the last of the three relations, viz. $d(d - a) + b(c - b) = 0$, would thus be $\frac{1}{2}(\frac{1}{2} - 1) = 0$, which is not satisfied. Hence the commutative and associative system can only be of one of the forms A, B, C, and E.

15. First, of the form be A, B, or C; there will be the two idem-or-nil symbols x and y , that is, we may assume $b = 0$, $g = 0$, and the associative conditions then become $cd = 0$, $c(c - h) = 0$, $d(d - a) = 0$, viz. for the forms A, B, C,

$$\text{A. } x^2 = x, \quad y^2 = y,$$

$$\text{B. } x^2 = x, \quad y^2 = 0,$$

$$\text{C. } x^2 = 0, \quad y^2 = 0.$$

these are $cd = 0$, $c(c - 1) = 0$, $d(d - 1) = 0$; $c = 0$ or 1 , $d = 0$ or 1 ,

$$,, \quad cd = 0, \quad c^2 = 0, \quad d(d - 1) = 0; \quad c = 0, \quad d = 0 \text{ or } 1,$$

$$,, \quad cd = 0, \quad c^2 = 0, \quad d^2 = 0; \quad c = 0, \quad d = 0.$$

But for the form A, if $c = 0$, $d = 1$, then is $xy = yx = y$, then, writing $z = x - y$, we have $z^2 = x$, $yz = zy = 0$, then, writing $x' = -y$, $z = z$, we have $x'^2 = -x'$, then $x' = -x$, that is, in each of these is reduced to the first case $c = 0$, $d = 0$, that is, $x^2 = x$, $xy = 0$, $y^2 = y$.

For the form B, if $c = 0$, $d = 1$, the system is $x^2 = x$, $xy = yx = y$, $y^2 = 0$; and this cannot be reduced to the first case $x^2 = x$, $xy = yx = 0$, $y^2 = 0$.

For the form C, the system is $x^2 = 0$, $xy = yx = 0$, $y^2 = 0$. In each of these is only the commutative, but not associative, conditions $b = 0$, $g = 0$, as above.

For the form E, here we have $a = 0$, $b = 0$, $c = e$, $d = f$, and the system may be commutative, $h = 2c$, viz. the equations must be $x^2 = 0$, $xy = yx = cx$, $y^2 = gx + 2cy$. The associative conditions then give $c = 0$; the equations are $x^2 = 0$, $xy = yx = 0$, $y^2 = gx$. Writing $\frac{x}{g}$ instead of x , and for convenience interchangingly x and y , the equations are

$$x^2 = y, \quad xy = yx = 0, \quad y^2 = 0.$$

16. The commutative associative system is thus seen to be reducible as follows:—

A system is $x^2 = x$, $xy = yx = 0$, $y^2 = y$, first mixed system, see No. 2.

B. ,, $x^2 = x$, $xy = yx = y$, $y^2 = 0$, Peirce's system (a_1).

or else

B. ,, $x^2 = x$, $xy = yx = 0$, $y^2 = 0$, second mixed system.

C. ,, $x^2 = 0$, $xy = yx = 0$, $y^2 = 0$, third mixed system.

E. ,, $x^2 = y$, $xy = yx = 0$, $y^2 = 0$, Peirce's system (c_1).

I said, at the end of my Note before referred to, that it had been pointed out to me “that my system [the commutative associative system], in the general case $ad - bc \neq 0$, is expressible as a mixture of two algebras of the form (a_1), see *American Journal of Mathematics*, vol. iv., p. 120; whereas, if $ad - bc = 0$, it is reducible to the form (a), see p. 122 (l.c.).” The accurate conclusion is as above, that the commutative associative system is either a mixed system of one of the three forms, or else a system (a_1), or (c_1).

17. Considering next the non-commutative associative system, we have here, vide No. 14, $b = 0$, $g = 0$; and the relations which remain to be satisfied then are

$$cd = 0, \quad ef = 0, \quad c(c - h) = 0, \quad e(e - h) = 0, \quad d(d - a) = 0, \quad f(f - a) = 0,$$

$$a(e - c) - cf + de = 0, \quad h(f - d) - cf + de = 0.$$

The first equation gives $cd = 0$, viz. $c = 0$ or $d = 0$; but we may attend exclusively to the case $c = 0$, for the case $d = 0$ may be deduced from this by the interchange of x, y . We have then $ef = 0$; and it will be convenient to separate the cases

- I. $c = 0, e \neq 0, x^2 = ax, xy = yx = 0, y^2 = hy$;
- II. $c = 0, e \neq 0, f = 0, ;$
- III. $c = 0, e = 0, f \neq 0, .$

- I. (a) $d = 0; x^2 = ax, xy = yx = 0, y^2 = hy$: commutative, and so included in what precedes.
- I. (b) $d = a, h = 0; x^2 = ax, xy = ay, yx = 0, y^2 = 0$, which is Peirce's system (b_1) .
- I. (c) $d = a, h = a; x^2 = ax, xy = ay, yx = 0, y^2 = ay$: commutative, and so included in what precedes.
- II. (d) $d = 0, f = a; x^2 = ax, xy = 0, yx = ay, y^2 = 0$, writing as we may do $a = 1$, this is $x^2 = x, xy = 0$,
- II. (e) $d = a, e = h; x^2 = ax, xy = ay, yx = hx, y^2 = hy$, writing as we may do $a = 1, h = 1$, this is $x^2 = x, xy = y, yx = 0, y^2 = 0$.

Thus x, z form the system $x^2 = x, xz = z, zx = 0, z^2 = 0$ (or, what is the same thing, y, z form a system $y^2 = y, yz = 0, zy = z, z^2 = 0$); each of these is Peirce's system (b_1) .

The conclusion is that every non-commutative associative system is either Peirce's system (b_1) , or else the omitted system (d_1) . Hence, disregarding the mixed systems, every associative system is either $(a_1), (b_1), (c_1)$, or (d_1) .

19. It may be proper to show that the systems $(b_1), x^2 = x, xy = y, yx = 0, y^2 = 0$, and $(d_1), x^2 = x, xy = 0, yx = y, y^2 = 0$, or say

	a	b	c	d	e	f	g	h
(b_1)	1	0	0	1	0	0	0	0
(d_1)	1	0	0	0	0	1	0	0

are really distinct from each other. Observe that they each belong to the case (x), $\Omega = 0$, an infinity of idoms and 1 nil; viz. in each of them writing $z = x + \beta y$, β an arbitrary coefficient, we have $z^2 = x^2 + \beta(xy + yx) + \beta^2 y^2, = x + \beta y, = z$, we have z an idom, and y is the only nil. And, this being so, in the second system, $xy = 0, yx = y$; viz. the system is $x^2 = x, xy = 0, yx = y, y^2 = 0$, retaining, when we write therein z for x , its original form. And similarly, in the second system, $xy = 0, yx = y$; viz. the system is $x^2 = x, xy = y, yx = 0, y^2 = 0$, retaining, when we write therein z for x , its original form. The two are thus distinct systems, in no wise transformable the one into the other.

815.

THE BINOMIAL EQUATION $x^p - 1 = 0$; QUINQUESECTION. SECOND PART.

[From the *Proceedings of the London Mathematical Society*, vol. xvi. (1885), pp. 61–63.]

In the paper, “The binomial equation $x^p - 1 = 0$; quinquesection,” *Proc. Lond. Math. Soc.*, t. xii. (1881), pp. 15, 16, [744], I considered for an exponent $p = 5n + 1$, the five periods X, Y, Z, W, T connected by the equations

$$\begin{array}{rcccl} & X, & Y, & Z, & W, & T \\ X^2 & = & a, & b, & c, & d, & e \\ XY & = & f, & g, & h, & i, & j \\ XZ & = & k, & l, & m, & n, & o, \end{array}$$

and the equations deduced from these by cyclical permutations of the periods and of the coefficients of each set; but I did not obtain completely even the linear relations connecting the coefficients. I since found, by induction from the examples given in the Table 1, that the coefficients could be expressed linearly in terms of the linearly independent integer numbers a, β, f, k as follows: viz. introducing for convenience the new number θ , such that

$$a + \beta + \theta = \frac{1}{5}(p - 1),$$

then the expressions in question are

$$\begin{array}{rcccccc} a, b, c, d, e & = & -1 - 2\theta + a + \beta, & -\theta - a + \beta + f, & -\theta - a + \beta + k, & -a - 2\beta - k, & -2a - \beta - f, \\ f, g, h, i, j & = & f, & \theta - a - f, & a, & \beta, & a, \\ k, l, m, n, o & = & k, & a, & \theta - \beta - k, & \beta, & a, \end{array}$$

and I found further that, substituting these values of the coefficients in the 20 quadric relations referred to in the former paper, the 20 relations reduced themselves to two equations only, viz. these were

$$\begin{aligned} \theta(-2a + \beta + k) + 3a^2 - \beta^2 + \alpha(f - k - 1) - \beta f + f^2 - 2fk &= 0, \\ -\theta^2 + \theta(3\beta + 2k + f) + a^2 - a\beta - 3\beta - \alpha k + \beta(1 - f - k) - k^2 - 2fk &= 0. \end{aligned}$$

The final result thus is that the coefficients are expressed as functions of the five numbers a, β, f, k, θ , connected by the linear equation $\alpha + \beta + \theta = \frac{1}{5}(p - 1)$, and the two quadric equations. I remark that formulae equivalent to these were obtained and proved by Mr F. S. Carey in his Trinity Fellowship Dissertation, 1884; viz. writing $n = \frac{1}{5}(p - 1)$, his formulae were

$$\begin{aligned} a, b, c, d, e &= n - a, \beta - n, \gamma - n, \delta - n, \varepsilon - n, \\ f, g, h, i, j &= f, \theta - a - f, \alpha, \beta, \rho, \\ k, l, m, n, o &= \gamma, \alpha, \delta, \beta, \sigma, \alpha, \end{aligned}$$

with the three linear relations

$$\begin{aligned} \alpha + \beta + \gamma + \delta + \varepsilon &= n - 1, \\ \beta + \varepsilon + 2\rho + \sigma &= n, \\ \gamma + \delta + \rho + 2\sigma &= n, \end{aligned}$$

and the two quadric relations

$$\begin{aligned} \delta^2 + \gamma^2 + 2\sigma a + (\rho - \sigma)(\delta + \gamma) - 2\rho(\rho + \sigma) &= (\delta - \gamma)(\beta - \varepsilon), \\ \beta^2 + \varepsilon^2 + 2\rho a + (\sigma - \rho)(\beta + \varepsilon) - 2\sigma(\rho + \sigma) &= (\gamma - \delta)(\beta - \varepsilon), \end{aligned}$$

the coefficients being thus expressed in terms of the seven numbers $a, \beta, \gamma, \delta, \varepsilon, \rho, \sigma$ connected by five equations. The equivalence of the two sets of formulae may be shown without difficulty.

To the Table 2 of the Quintic Equations, given in the paper, may be added the following result from Legendre's *Théorie des Nombres*, Ed. 3, t. ii., p. 213,

$$\frac{p}{641} \left| \begin{array}{cccccc} q^6 & q^5 & q^4 & q^3 & q^2 & q & 1 \\ 1 & +1 & -256 & -564 & +3258 & -3120 & \end{array} \right| = 0,$$

calculated by him for the isolated case $p = 641$.

$$\frac{x}{1-\beta y} + \frac{y}{1-\gamma z} + \frac{z}{1-\alpha\beta} = 0 \quad \text{and} \quad \frac{\xi}{a(\gamma-\beta)} + \frac{\eta}{\beta(\gamma-a)} + \frac{\zeta}{\gamma(\beta-a)} = 0$$

respectively, we have 6+6 equations of like forms; but these two systems each of 6 equations are equivalent to each other, so that instead of $6+16+3+3=28$, we have $6+16+6(=6)=28$ equations for the 28 bitangents¹. I make another slight change of notation by introducing the single letters f, g, h to denote the reciprocals $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$; and I consider the whole question as follows. The theory is based on the equations

$$\begin{aligned} ax + by + cz + f\xi + g\eta + h\zeta &= 0, \\ a_1x + b_1y + c_1z + f_1\xi + g_1\eta + h_1\zeta &= 0, \\ a_2x + b_2y + c_2z + f_2\xi + g_2\eta + h_2\zeta &= 0, \\ a_3x + b_3y + c_3z + f_3\xi + g_3\eta + h_3\zeta &= 0, \end{aligned}$$

where $af = bg = ch = a_1f_1 \&c., = a_3h_3, = 1$: and the coefficients $a, b, c, a_1, b_1, c_1, a_2, b_2, c_2$ are arbitrary. The equations $x = 0, y = 0, z = 0$ represent any three given lines: and

816.

ON THE BITANGENTS OF A PLANE QUARTIC.

[From *Crelle's Journal der Mathem.*, t. xciv. (1883), pp. 93–115; *Camb. Phil. Soc. Proc.*, t. iv. (1883), p. 321]

Riemann in the paper “Zur Theorie der Abelschen Functionen für den Fall $p = 3$,” *Werke*, pp. 456–479, has given a remarkably elegant solution of the problem of the bitangents of a plane quartic. But his formulae may be improved by a slight change; viz. we may in his first equation $x + y + z + f\xi + g\eta + h\zeta = 0$ introduce coefficients so as to bring this into the same form as the other three equations. It thus appears that, instead of his 3+3 equations of the forms

$$\frac{x}{1-\beta y} + \frac{y}{1-\gamma z} + \frac{z}{1-\alpha\beta} = 0 \quad \text{and} \quad \frac{\xi}{a(\gamma-\beta)} + \frac{\eta}{\beta(\gamma-a)} + \frac{\zeta}{\gamma(\beta-a)} = 0$$

¹It will appear further on that the equation of each of the last-mentioned 6 bitangents can be expressed in 6 different forms.

¹It will appear further on that the equation of each of the last-mentioned 6 bitangents can be expressed in 6 different forms.

considering ξ, η, ζ to be determined as linear functions of x, y, z by means of the first three equations, then (the coefficients $a, b, c, a_1, b_1, c_1, a_2, b_2, c_2$ being determined accordingly) the equations $\xi = 0, \eta = 0, \zeta = 0$ will also represent any three given lines. But observe that, if the equation of the first of these lines is $\alpha x + my + nz = 0$, ξ is not = an arbitrary multiple $\theta(\alpha + my + nz)$ of the linear function $\alpha + my + nz$, but is = $G(\alpha + my + nz)$, G being a completely determinate value: and the like as regards η and ζ respectively.

The coefficients a_3, b_3, c_3 of the fourth equation are such that this fourth equation is a mere consequence of the other three: viz. we must have

$$\begin{aligned} a_3 &= \lambda a + \lambda_1 a_1 + \lambda_2 a_2, \\ b_3 &= \lambda b + \lambda_1 b_1 + \lambda_2 b_2, \\ c_3 &= \lambda c + \lambda_1 c_1 + \lambda_2 c_2, \\ f_3 &= \lambda f + \lambda_1 f_1 + \lambda_2 f_2, \\ g_3 &= \lambda g + \lambda_1 g_1 + \lambda_2 g_2, \\ h_3 &= \lambda h + \lambda_1 h_1 + \lambda_2 h_2, \end{aligned}$$

or, what is the same thing, we must have

$$\left\| \begin{array}{cccccc} a, & b, & c, & f, & g, & h \\ a_1, & b_1, & c_1, & f_1, & g_1, & h_1 \\ a_2, & b_2, & c_2, & f_2, & g_2, & h_2 \\ a_3, & b_3, & c_3, & f_3, & g_3, & h_3 \end{array} \right\| = 0,$$

viz. each of the determinants formed with four columns out of this matrix is equal to 0.

Using the equations in $\lambda, \lambda_1, \lambda_2$, and writing for shortness

$$a_3 f_3 = a_0 f_0, \quad a_3 g_3 + a_0 f_1 = a f_1 + a_1 f_0 = F, \quad F_1, \quad F_2,$$

$$b_3 g_3 = b_0 g_0, \quad b_3 g_3 + b_0 g_1 = b g_1 + b_1 g_0 = G, \quad G_1, \quad G_2,$$

$$c_3 h_3 = c_0 h_0, \quad c_3 h_3 + c_0 h_1 = c h_1 + c_1 h_0 = H, \quad H_1, \quad H_2;$$

then, forming the products $a_3 f_3, b_3 g_3, c_3 h_3$, we find

$$\begin{aligned} 1 - \lambda^2 - \lambda_1^2 - \lambda_2^2 &= P\lambda\lambda_1 + F_1\lambda\lambda_2 + F_2\lambda_1\lambda_2, \\ , , &= G\lambda\lambda_1 + G_1\lambda\lambda_2 + G_2\lambda_1\lambda_2, \\ , , &= H\lambda\lambda_1 + H_1\lambda\lambda_2 + H_2\lambda_1\lambda_2, \end{aligned}$$

or, as these equations may be written,

$$\lambda\lambda_1 : \lambda\lambda_2 : \lambda_1\lambda_2 = \left| \begin{array}{ccc} 1, & F, & F_2 \\ 1, & G, & G_2 \\ 1, & H, & H_2 \end{array} \right| : \left| \begin{array}{ccc} 1, & F, & F_1 \\ 1, & G, & G_1 \\ 1, & H, & H_1 \end{array} \right| : \left| \begin{array}{ccc} 1, & F_1, & F_2 \\ 1, & G_1, & G_2 \\ 1, & H_1, & H_2 \end{array} \right|.$$

say for shortness

$$= \Pi : \Lambda : \Lambda_1 : \Lambda_2.$$

We have thus

$$\begin{aligned}\frac{\lambda^2}{1 - \lambda^2 - \lambda_1^2 - \lambda_2^2} &= \frac{\Lambda_1 \Lambda_2}{\Pi \Lambda}, \\ \frac{\lambda_1^2}{1 - \lambda^2 - \lambda_1^2 - \lambda_2^2} &= \frac{\Lambda \Lambda_2}{\Pi \Lambda_1}, \\ \frac{\lambda_2^2}{1 - \lambda^2 - \lambda_1^2 - \lambda_2^2} &= \frac{\Lambda \Lambda_1}{\Pi \Lambda_2},\end{aligned}$$

and thence

$$\frac{1}{1 - \lambda^2 - \lambda_1^2 - \lambda_2^2} = \frac{1}{\Pi} \left(\frac{\Lambda_1 \Lambda_2}{\Lambda} + \frac{\Lambda \Lambda_2}{\Lambda_1} + \frac{\Lambda \Lambda_1}{\Lambda_2} \right),$$

equations which give $\lambda, \lambda_1, \lambda_2$, and consequently $a_3, b_3, c_3, f_3, g_3, h_3$ in terms of known quantities, the several expressions depending on the single radical

$$\sqrt{\frac{1}{\Pi} \left(\frac{\Lambda_1 \Lambda_2}{\Lambda} + \frac{\Lambda \Lambda_2}{\Lambda_1} + \frac{\Lambda \Lambda_1}{\Lambda_2} \right)}.$$

Observe that we have rationally

$$\lambda : \lambda_1 : \lambda_2 = \frac{1}{\Lambda} : \frac{1}{\Lambda_1} : \frac{1}{\Lambda_2}.$$

I consider now the quartic curve

$$\sqrt{x\xi} + \sqrt{y\eta} + \sqrt{z\zeta} = 0,$$

and I write down the equations of the 28 bitangents, each with its triple-theta characteristic as given by Riemann, and also with its dual symbol, derived from Hesse's method. I assume that these characteristics and dual symbols are properly attached to the several lines: and I insert also the current nos. 1 to 28. The equations are:

Current No.	Dual Symbol	Characteristic	
1	18	$\begin{smallmatrix} 111 \\ 111 \end{smallmatrix}$	$x = 0,$
2	28	$\begin{smallmatrix} 001 \\ 011 \end{smallmatrix}$	$y = 0,$
3	38	$\begin{smallmatrix} 011 \\ 001 \end{smallmatrix}$	$z = 0,$
4	23	$\begin{smallmatrix} 010 \\ 010 \end{smallmatrix}$	$\xi = 0,$
5	15	$\begin{smallmatrix} 100 \\ 110 \end{smallmatrix}$	$\eta = 0,$
6	12	$\begin{smallmatrix} 110 \\ 100 \end{smallmatrix}$	$\zeta = 0,$

Current No.	Dual Symbol	Characteristic	
7	49	$\begin{smallmatrix} 101 \\ 100 \end{smallmatrix}$	$ax + by + cz = 0, f\xi + g\eta + h\zeta = 0,$
8	14	$\begin{smallmatrix} 010 \\ 011 \end{smallmatrix}$	$f_1\xi + b_1y + c_1z = 0, a_1x + g_1\eta + h_1\zeta = 0,$
9	24	$\begin{smallmatrix} 100 \\ 011 \end{smallmatrix}$	$a_2x + g_2\eta + c_2z = 0, f_2\xi + b_2y + h_2\zeta = 0,$
10	34	$\begin{smallmatrix} 100 \\ 101 \end{smallmatrix}$	$a_3x + b_3y + h_3\zeta = 0, f_3\xi + g_3\eta + c_3z = 0,$
11	58	$\begin{smallmatrix} 100 \\ 010 \end{smallmatrix}$	$a_1x + f_1\xi + c_1z = 0,$
12	16	$\begin{smallmatrix} 001 \\ 010 \end{smallmatrix}$	$f_1\xi + b_1y + h_1\zeta = 0,$
13	25	$\begin{smallmatrix} 101 \\ 010 \end{smallmatrix}$	$a_2x + g_2\eta + h_2\zeta = 0,$
14	35	$\begin{smallmatrix} 111 \\ 110 \end{smallmatrix}$	$a_3x + g_3\eta + h_3\zeta = 0,$
15	68	$\begin{smallmatrix} 110 \\ 010 \end{smallmatrix}$	$a_1x + b_1y + h_1\zeta = 0,$
16	16	$\begin{smallmatrix} 001 \\ 101 \end{smallmatrix}$	$f_1\xi + g_1\eta + h_1\zeta = 0,$
17	26	$\begin{smallmatrix} 111 \\ 101 \end{smallmatrix}$	$a_2x + b_2y + h_2\zeta = 0,$
18	36	$\begin{smallmatrix} 101 \\ 011 \end{smallmatrix}$	$a_3x + b_3y + h_3\zeta = 0,$
19	78	$\begin{smallmatrix} 010 \\ 110 \end{smallmatrix}$	$a_1x + b_1y + c_1z = 0,$
20	17	$\begin{smallmatrix} 101 \\ 110 \end{smallmatrix}$	$f_1\xi + b_1y + c_1z = 0,$
21	27	$\begin{smallmatrix} 011 \\ 110 \end{smallmatrix}$	$a_2x + f_2\xi + c_2z = 0,$
22	37	$\begin{smallmatrix} 000 \\ 111 \end{smallmatrix}$	$a_3x + b_3y + h_3\zeta = 0,$
23	67	$\begin{smallmatrix} 100 \\ 100 \end{smallmatrix}$	$\frac{x}{bc-b_1c_1} + \frac{y}{ca-c_1a_1} + \frac{z}{ab-a_1b_1} = 0, \frac{1}{c_2h_2-c_3h_3} + \frac{1}{b_2f_2-b_3f_3} + \frac{1}{f_2g_2-f_3g_3} = 0,$ $\frac{x}{bc-b_2c_2} + \frac{y}{ca-c_2a_2} + \frac{z}{ab-a_2b_2} = 0, \frac{1}{c_1h_1-c_3h_3} + \frac{1}{a_1h_1-a_3h_3} + \frac{1}{f_1g_1-f_3g_3} = 0,$
24	57	$\begin{smallmatrix} 110 \\ 001 \end{smallmatrix}$	$\frac{x}{bc-b_3c_3} + \frac{y}{ca-c_3a_3} + \frac{z}{ab-a_3b_3} = 0,$
25	56	$\begin{smallmatrix} 010 \\ 001 \end{smallmatrix}$	$\frac{x}{bc-b_1c_1} + \frac{y}{ca-c_1a_1} + \frac{z}{ab-a_1b_1} = 0,$
26	45	$\begin{smallmatrix} 000 \\ 011 \end{smallmatrix}$	$\frac{x}{b_2f_2-b_3f_3} = \frac{y}{c_2g_2-c_3g_3} = \frac{z}{a_2h_2-a_3h_3},$
27	46	$\begin{smallmatrix} 001 \\ 110 \end{smallmatrix}$	$\frac{x}{b_1f_1-b_3f_3} = \frac{y}{c_1g_1-c_3g_3} = \frac{z}{a_1h_1-a_3h_3},$
28	47	$\begin{smallmatrix} 111 \\ 010 \end{smallmatrix}$	$\frac{x}{b_1f_1-b_2f_2} = \frac{y}{c_1g_1-c_2g_2} = \frac{z}{a_1h_1-a_2h_2},$

As regards each of the bitangents 7 to 22, the equivalence of the two forms of equation is at once seen by means of the fundamental equations $ax + by + cz + f\xi + g\eta + h\zeta = 0$: as regards the remaining bitangents 23 to 28, the equation of each of these is expressible in eight different forms, which are written down in full for the bitangent 23; the equivalence of the eight forms of equation must of course be proved.

I proceed to show that each of the 28 lines is, in fact, a bitangent to the quartic curve

$$\sqrt{x\xi} + \sqrt{y\eta} + \sqrt{z\zeta} = 0,$$

or, what is the same thing, that writing this equation in the form

$$\Omega = x^2\xi^2 + y^2\eta^2 + z^2\zeta^2 - 2yz\eta\zeta - 2zx\zeta\xi - 2xy\xi\eta = 0,$$

then that by means of any one of these equations Ω becomes a perfect square.

This is obviously the case for each of the equations 1, 2, 3, 4, 5, 6.

For the equations 7, 8, 9, 10, observe that the equation of the quartic may be written:

$$\sqrt{axf\xi} + \sqrt{byg\eta} + \sqrt{czh\zeta} = 0,$$

or, what is the same thing,

$$\Omega = ax^2\xi^2 + f^2\xi^2 + \dots = 0.$$

Hence writing $x, y, z, \xi, \eta, \zeta$ in place of $ax, by, cz, f\xi, g\eta, h\zeta$, so that now

$$x + y + z + \xi + \eta + \zeta = 0,$$

the equation of the line 7 is expressed in the two equivalent forms $x + y + z = 0, \xi + \eta + \zeta = 0$. We have for Ω its original value

$$\Omega = x^2\xi^2 + y^2\eta^2 + z^2\zeta^2 - 2yz\eta\zeta - 2zx\zeta\xi - 2xy\xi\eta,$$

$= (x\xi - y\eta - z\zeta)^2$, a perfect square. The like proof applies to the equations 8, 9, 10; and then writing herein $x = -x - y, \xi = -\xi - y\eta$, we have $x\xi - y\eta - z\zeta = xy + y\xi$ and consequently $\Omega = (x\eta + y\xi)^2 = (\xi y - y\xi)^2$, a perfect square. The like proof applies to the equations 8, 9, 10; and then writing herein $z = -x - y, \zeta = -\xi - \eta$, we have to apply this result to the equations 11, 12, 13, 14; 15, 16, 17, 18; 19, 20, 21, 22.

We come now to the equation of the line 23, say this is in the first instance taken to be

$$\frac{x}{bc - b_1c_1} + \frac{y}{ca - c_1a_1} + \frac{z}{ab - a_1b_1} = 0;$$

if from this equation, by means of the two fundamental equations

$$\begin{aligned} ax + by + cz + f\xi + g\eta + h\zeta &= 0, \\ a_1x + b_1y + c_1z + f_1\xi + g_1\eta + h_1\zeta &= 0, \end{aligned}$$

we eliminate first (y, z) , secondly (z, x) , and thirdly (x, y) , it is found that ξ, η, ζ will also disappear in the three cases respectively, and that the resulting equations are

$$\begin{aligned}\frac{x}{bc - b_1c_1} + \frac{\eta}{bb_1(ca - c_1a_1)} + \frac{\zeta}{cc_1(ab - a_1b_1)} &= 0, \\ -\frac{y}{aa_1(bc - b_1c_1)} + \frac{\eta}{ca - c_1a_1} + \frac{\zeta}{cc_1(ab - a_1b_1)} &= 0, \\ \frac{\xi}{aa_1(bc - b_1c_1)} - \frac{\eta}{bb_1(ca - c_1a_1)} + \frac{z}{ab - a_1b_1} &= 0,\end{aligned}$$

so that each of these is, in fact, equivalent to the original equation in (x, y, z) : and observe that, adding together the three equations, we reproduce the original equation in (x, y, z) .

Write for shortness

$$\begin{aligned}P &= bc - b_1c_1, & \mathfrak{a} &= bc - b_1c_1, \\ Q &= ca - c_1a_1, & \mathfrak{B} &= ca - c_1a_1, \\ R &= ab - a_1b_1, & \gamma &= ab - a_1b_1,\end{aligned}$$

then the equation in (x, y, z) is

$$QRx + RPy + PQz = 0,$$

so that, eliminating y and z , we find

$$\begin{vmatrix} b, & c, & ax + f\xi + g\eta + h\zeta \\ b_1, & c_1, & a_1x + f_1\xi + g_1\eta + h_1\zeta \\ RP, & PQ, & QRx \end{vmatrix} = 0.$$

In this equation, the coefficient of x is

$$\begin{aligned}(bc_1 - b_1c)QR + (cm_1 - c_1a)RP + (ab_1 - a_1b)PQ, \\ = (bc_1 - b_1c)[aPb + a_1b_1c_1 - aa_1(bc_1 - b_1c)] \\ + (cm_1 - c_1a)[bP + a_1b_1c_1 - bb_1(ca_1 - c_1a)] \\ + (ab_1 - a_1b)[cP + a_1b_1c_1 - cc_1(ab_1 - a_1b)], \\ = -aa_1(bc_1 - b_1c)^2 - bb_1(ca_1 - c_1a)^2 - cc_1(ab_1 - a_1b)^2, \\ = -aa_1\gamma.\end{aligned}$$

The coefficient of ξ is

$$= RP(cf_1 - c_1f) - PQ(bf_1 - b_1f),$$

which (observing that $cf_1 - c_1f = \frac{c}{a_1} - \frac{c_1}{a} = \frac{1}{aa_1}Q$ and $bf_1 - b_1f = \frac{b}{a_1} - \frac{b_1}{a} = \frac{1}{aa_1}R$) is $= 0$.

The coefficient of η is

$$RP(cg_1 - c_1g) - PQ(bg_1 - b_1g),$$

which is

$$\begin{aligned}
&= RP \frac{1}{bb_1} (bc_1 - b_1c_1) - PQ \frac{1}{bb_1} (b^2 - b_1^2), \\
&= \frac{P}{bb_1} [(bc_1 - b_1c_1)(ab - a_1b_1) - (b^2 - b_1^2)(ca - c_1a_1)], \\
&= \frac{P}{bb_1} [b_1^2ca + b^2c_1a_1 - bb_1(ac_1 + a_1c)], \\
&= -\frac{P}{bb_1} \gamma \mathfrak{a};
\end{aligned}$$

and similarly the coefficient of ζ is

$$= -\frac{P}{cc_1} \mathfrak{a}\beta.$$

We have thus in x, η, ζ the equation

$$-aa_1\gamma x - \frac{P}{bb_1} \gamma \mathfrak{a} \eta + \frac{P}{cc_1} \mathfrak{a}\beta \zeta = 0,$$

or, what is the same thing,

$$\frac{x}{P} + \frac{\eta}{bb_1\beta} + \frac{\zeta}{cc_1\gamma} = 0;$$

and in like manner, by the elimination of (z, x) and (x, y) respectively,

$$-\frac{\xi}{aa_1\mathfrak{a}} + \frac{y}{Q} + \frac{\zeta}{cc_1\gamma} = 0,$$

$$\frac{\xi}{aa_1\mathfrak{a}} + \frac{\eta}{bb_1\beta} + \frac{z}{R} = 0,$$

which are the required three equations.

We have next to show that the line is a bitangent—write for a moment $aa_1\mathfrak{a}, bb_1\beta, cc_1\gamma = \lambda, \mu, \nu$, then the three equations give

$$x, y, z = P\left(-\frac{\eta}{\mu} + \frac{\zeta}{\nu}\right), \quad Q\left(-\frac{\zeta}{\nu} + \frac{\xi}{\lambda}\right), \quad R\left(-\frac{\xi}{\lambda} + \frac{\eta}{\mu}\right),$$

and we thence have

$$\begin{aligned}
\Omega &= P^2\left(-\frac{\xi\eta}{\mu} + \frac{\xi\zeta}{\nu}\right)^2 \\
&\quad + Q^2\left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda}\right)^2 \\
&\quad + R^2\left(-\frac{\xi\zeta}{\lambda} + \frac{\eta\zeta}{\mu}\right)^2 \\
&\quad - 2QR\left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda}\right)\left(-\frac{\xi\zeta}{\lambda} + \frac{\eta\zeta}{\mu}\right) \\
&\quad - 2RP\left(-\frac{\xi\zeta}{\lambda} + \frac{\eta\zeta}{\mu}\right)\left(-\frac{\xi\eta}{\mu} + \frac{\xi\zeta}{\nu}\right) \\
&\quad - 2PQ\left(-\frac{\xi\eta}{\mu} + \frac{\xi\zeta}{\nu}\right)\left(-\frac{\eta\zeta}{\nu} + \frac{\xi\eta}{\lambda}\right);
\end{aligned}$$

THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR
CAYLEY
VOL. XII

BOOK PAGES 101–125 (PAPERS 819 CONCL.–825 CHS. I–II)

Arthur Cayley (1821–1895)

Published 1889

*Transcribed from the original Cambridge University Press edition, 1889.
This file covers book pages 101–125; restored from scan to correct prior errors.*

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819. On Two Cases of the Quadric Transformation between Two Planes (concluded)

[p. 101] Writing for convenience (a, b, c) for the coordinates of the fixed point, and (x_1, y_1, z_1) , (x_2, y_2, z_2) for those of the other two points, the formulæ with A, B, C give thus the correspondence

$$x_2 : y_2 : z_2 = bcx_1^2 - a^2y_1z_1 : cay_1^2 - b^2z_1x_1 : abz_1^2 - c^2x_1y_1,$$

which is the first of the two cases in question. These equations give reciprocally

$$x_1 : y_1 : z_1 = bcx_2^2 - a^2y_2z_2 : cay_2^2 - b^2z_2x_2 : abz_2^2 - c^2x_2y_2,$$

or the correspondence is a $(1, 1)$ quadric correspondence.

The formulæ with P, Q, R give in like manner

$$x_2 : y_2 : z_2 = a(ax_1^2 + by_1^2 + cz_1^2) - a_1(a''y_1z_1 + b'z_1x_1 + c'x_1y_1), \text{ \&c.},$$

or if for shortness

$$\Omega_1 = ax_1^2 + by_1^2 + cz_1^2, \quad \Theta_1 = a''y_1z_1 + b'z_1x_1 + c'x_1y_1,$$

then

$$x_2 : y_2 : z_2 = a\Omega_1 + a_1\Theta_1 : b\Omega_1 + b_1\Theta_1 : c\Omega_1 + c_1\Theta_1,$$

which is the second of the two cases. We have reciprocally

$$x_1 : y_1 : z_1 = a\Omega_2 + a_1\Theta_2 : b\Omega_2 + b_1\Theta_2 : c\Omega_2 + c_1\Theta_2,$$

where

$$\Omega_2 = ax_2^2 + by_2^2 + cz_2^2, \quad \Theta_2 = a''y_2z_2 + b'z_2x_2 + c'x_2y_2,$$

and the correspondence is thus in this case also a $(1, 1)$ quadric correspondence.

820. On a Problem of Analytical Geometry

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 209.]

[p. 102] THE object of the present note is only to call attention to a problem of Analytical Geometry which presents itself in connexion with the reduction of an algebraical integral, and which is solved, pp. 21, 22 of Clebsch and Gordan's *Theorie der Abel'schen Functionen* (Leipzig, 1866); viz. the problem is, considering a curve $f = 0$ of the order n , to find the conic $\Omega = 0$ of the order $n - 2$ passing through the remaining $n - 2$ points of intersection of the line with the curve f , through the double points of f , and through as many other given points as are required for the determination of the curve. If, for instance, f is a quartic curve without double points, then Ω is the quadric curve which passes through the remaining two intersections of the line with Ω , and through three given points. Take (ξ, η, ζ) , (ξ_1, η_1, ζ_1) the coordinates of the two given points on the curve f (f is, say of the order $n = 4$); $\theta(x, y, z) = 0$ the equation of the required curve. By the equation of suppositions originally of the form, that the line $x = x' + \rho x''$, &c. joining (x', y', z') , (x'', y'', z'') meets the curve θ where $\theta = 0$ in two points, the equation in ρ originally of the fourth order in (λ, μ) but which divides by $\lambda\mu$, and which when this factor is thrown out becomes

$$\alpha\lambda^2 + 2\beta\lambda\mu + \gamma\mu^2 = 0;$$

where

$$\alpha = (\xi_1, \eta_1, \zeta_1)^2 \theta, \quad \beta = (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta) \theta, \quad \gamma = (\xi, \eta, \zeta)^2 \theta,$$

where for shortness f_1, f are written to denote $f(\xi_1, \eta_1, \zeta_1), f(\xi, \eta, \zeta)$ respectively.

[p. 103] The condition as to the two points obviously is that, making the same substitution $x, y, z = \lambda\xi_1 + \mu\xi, \lambda\eta_1 + \mu\eta, \lambda\zeta_1 + \mu\zeta$ in the equation $\Omega = (a, \dots)(x, y, z)^2 = 0$, we must obtain the same quadric equation in $\lambda : \mu$. We have thus two conditions, which, introducing an indeterminate multiplier θ , are expressed by the three equations

$$\begin{aligned} (a, \dots)(\xi_1, \eta_1, \zeta_1)^2 &= \theta\alpha, \\ (a, \dots)(\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta) &= \theta\beta, \\ (a, \dots)(\xi, \eta, \zeta)^2 &= \theta\gamma. \end{aligned}$$

The conditions as to the three points are obviously

$$\begin{aligned} (a, \dots)(x_1, y_1, z_1)^2 &= 0, \\ (a, \dots)(x_2, y_2, z_2)^2 &= 0, \\ (a, \dots)(x_3, y_3, z_3)^2 &= 0, \end{aligned}$$

and these equations determine the ratios of a, b, c, f, g, h . But to complete the solution the convenient course is to regard the function $\Omega = (a, \dots)(x, y, z)^2$ as a quantity to be determined, and consequently to join to the foregoing the equation

$$(a, \dots)(x, y, z)^2 = \Omega;$$

we have thus seven equations from which (a, b, c, f, g, h) may be eliminated, the result being expressed by means of a determinant of the seventh order

$$\begin{vmatrix} (x, y, z)^2 & \Omega \\ (\xi_1, \eta_1, \zeta_1)^2 & \theta\alpha \\ (\xi_1, \eta_1, \zeta_1)(\xi, \eta, \zeta) & \theta\beta \\ (\xi, \eta, \zeta)^2 & \theta\gamma \\ (x_1, y_1, z_1)^2 & 0 \\ (x_2, y_2, z_2)^2 & 0 \\ (x_3, y_3, z_3)^2 & 0 \end{vmatrix} = 0,$$

viz. this is an equation of the form $A\Omega = \theta V$, where A is a constant determinant of the sixth order (i.e. a determinant not involving x, y, z), V a determinant of the seventh order, a quadric function of (x, y, z) , obtained from the foregoing determinant by writing therein $\Omega = 0$ and $\theta = 1$; the multiplier θ is and remains arbitrary; but it is convenient to take it to be $= 1$, viz. we have then not only find the equation $\Omega = 0$ of the required conic, but we put a determinate value for Ω , say $\Omega = A^{-1}V$, which is, in fact $(a, \dots)(x, y, z)^2$, viz. this gives $a = (\xi_1\beta_1 + \eta_1\delta_1 + \zeta_1\theta_1)f_1$, and so on for $(x, y, z) = (\xi, \eta, \zeta)$, we have

$$\Omega_1 = \gamma_1 = (\xi_1\beta_1 + \eta_1\delta_1 + \zeta_1\theta_1)f_1.$$

821. On the Geometrical Representation of an Equation between Two Variables

[From the *Johns Hopkins University Circulars*, No. 15 (1882), p. 210.]

[p. 104] AN equation between two variables cannot be represented in a satisfactory manner by a curve, for this serves only to represent the corresponding real values of the variables; to represent the imaginary values the natural course is to represent each variable by a point in a plane, viz. the variable $z, = x + iy$, will be represented by a point the coordinates of which are the components x and y of the variable, and similarly the variable $z', = x' + iy'$, by a point the coordinates of which are the components x' and y' of the variable; the equation between the two variables will then establish a correspondence between the two variables points respectively, or say a correspondence between the planes which contain the two points respectively: and it is this correspondence of two planes which is the proper geometrical representation of the equation between the two variables; to exhibit the correspondence we may in either of the planes draw a network of curves at pleasure, and then draw in the other plane the network of corresponding curves. This well-known theory [can be] illustrated by the case $z' = \sqrt{z^2 - 1}$: taking in the infinite half-plane y positive about the origin as centre a system of semi-circles, to these correspond in the infinite plane of $x'y'$ a series of lemniscate-shaped curves: and by means of these it is easy to show in the second plane the path corresponding to a given path of the point $z, = x + iy$, in the first half-plane.

822. On Associative Imaginaries

[From the *Johns Hopkins University Circulars*, No. 15 (1882), pp. 211, 212.]

[p. 105] THE imaginaries (x, y) defined by the equations

$$\begin{aligned}x^2 &= ax + by, \\xy &= cx + dy, \\yx &= ex + fy, \\y^2 &= gx + hy,\end{aligned}$$

will not be in general associative; to make them so, we must have 8 double relations corresponding to the combinations $x^2.x$, $xy.x$, $yx.x$, $x^2.y$, $xy.y$, $yx.y$, $y^2.x$, $y^2.y$ respectively, viz. the first of these gives $x.x.x = x.x.x$, that is, $0 = x(ax + by) - (ax + by)x = b(xy - yx) = b\{(c - e)x + (d - f)y\}$; and similarly for each of the other terms. We thus obtain apparently 16, but really only 12, relations between the 8 coefficients a, b, c, d, e, f, g, h , viz. the relations so obtained are

$$\begin{aligned}b(c - e) &= 0 \text{ (twice)}, & b(d - f) &= 0, & g(c - e) &= 0, & g(d - f) &= 0 \text{ (twice)} \\bg - cd &= 0 \text{ (twice)}, & bg - ef &= 0 \text{ (twice)}, \\c^2 + dg - cg - eh &= 0, & d^2 + bc - cd - af + bg - ah &= 0, \\a(c - e) + de - cf &= 0, & h(f - d) - cf + de &= 0.\end{aligned}$$

From the first four equations it appears that either $b = 0$ and $g = 0$, or else $c = e$ and $d = f$: the latter solution may be set aside for the moment, giving the commutative system

$$\begin{aligned}x^2 &= ax + by, \\xy &= yx = cx + dy, \\y^2 &= gx + hy.\end{aligned}$$

[p. 106] In order that this may be associative, we must still have the relations

$$\begin{aligned}bg - cd &= 0, \\c^2 + dg - cg - ch &= 0, \\d^2 + bc - ad - bh &= 0,\end{aligned}$$

which are all three of them satisfied by $g = \frac{cd}{b}$, $h = \frac{d^2 + bc - ad}{b}$, viz. we thus have the associative and commutative system

$$\begin{aligned}x^2 &= ax + by, \\xy &= yx = cx + dy, \\y^2 &= \frac{cd}{b}x + \frac{d^2 + bc - ad}{b}y.\end{aligned}$$

I did not perceive how to identify this system with any of the double algebras of B. Peirce's *Linear Associative Algebra*, see pp. 120–122 of the Reprint, *American Journal of Mathematics*, t. iv. (1881); but it has been pointed out to me by Mr C. S. Peirce that my system, in the general case $ad - bc \neq 0$, is reducible to the form (a_1) , see p. 122 (l.c.). The object of the present Note is to exhibit in the simple case of two imaginaries the whole system of relations which must subsist between the coefficients in order that the imaginaries may be associative; that is, the system of equations which are solved implicitly by the establishment of the several multiplication tables of the memoir just referred to.

823. On the Geometrical Interpretation of Certain Formulæ in Elliptic Functions

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 238.]

[p. 107] I HAVE given in my *Elliptic Functions* expressions for the sn of $u + \frac{1}{2}K$, $u + \frac{1}{2}iK'$, $u + \frac{1}{2}K + \frac{1}{2}iK'$; but it is better to consider the dn, sn, cn of these combinations respectively, and to write the formulæ thus

$$\begin{aligned} \operatorname{dn}^2(u + \tfrac{1}{2}K) &= k' \frac{\operatorname{dn} u - (1 - k') \operatorname{sn} u \operatorname{cn} u}{\operatorname{dn} u + (1 - k') \operatorname{sn} u \operatorname{cn} u}, &= k' \frac{1 - k'x - (1 - k')y}{1 - k'x + (1 - k')y}; \\ \operatorname{sn}^2(u + \tfrac{1}{2}iK') &= \frac{1}{k} \frac{(1 + k) \operatorname{sn} u + i \operatorname{cn} u \operatorname{dn} u}{(1 + k) \operatorname{sn} u - i \operatorname{cn} u \operatorname{dn} u}, &= \frac{1}{k} \frac{(1 + k)x + iy}{(1 + k)x - iy}; \\ \operatorname{cn}^2(u + \tfrac{1}{2}K + \tfrac{1}{2}iK') &= -\frac{ik'}{k} \frac{\operatorname{cn} u - (k + ik') \operatorname{sn} u \operatorname{dn} u}{\operatorname{cn} u + (k + ik') \operatorname{sn} u \operatorname{dn} u}, &= -\frac{ik'}{k} \frac{1 - x - (k + ik')y}{1 - x + (k + ik')y}, \end{aligned}$$

where in the last set of ratios x, y are used to denote $\operatorname{sn} u \operatorname{cn} u$ and $\operatorname{sn} u \operatorname{dn} u$ respectively; and the three combinations above show that for u are written respectively the values just mentioned. The curve $y^2 = (1 - x^2)(1 - k^2x^2)$ has an inflexion at infinity on the line $x = 0$; and the three tangents from the inflexion are at the points $x, y = 0, 1, 0$ and $(\frac{1}{k}, 0)$ respectively; touching the curve at the points $x, y = (0, 0), (1, 0), (\frac{1}{k}, 0)$ respectively; hence these points are sextactic points.

[p. 108] We may from any one of these, for instance the point $(0, 0)$, draw four tangents to the curve, $(1 + k)x + iy = 0$, $(1 + k)x - iy = 0$; $(1 - k)x + iy = 0$, $(1 - k)x - iy = 0$; where the first and second of these lines form a pair, and the third and fourth of them form a pair, viz. the two tangents of a pair touch in points such that the line joining them passes through the point of inflexion; in particular, for the first-mentioned pair, the equation of the line joining the points of contact is $1 + kx = 0$. The tangents from belonging to a pair of tangents are precisely those which present themselves in the formulæ; thus if $T_1 = (1 + k)x + iy$, $T_2 = (1 + k)x - iy$, the second of the three formulæ is $\operatorname{sn}^2(u + \frac{1}{2}iK') = \frac{1}{k} \frac{T_1}{T_2}$; and the other two formulæ correspond in

like manner to pairs of tangents from the sextactic points $(\frac{1}{k}, 0)$, and $(1, 0)$ respectively. The formulæ are connected with the fundamental equations expressing the functions sn, cn, dn as quotients of theta-functions.

824. Note on the Formulæ of Trigonometry

[From the *Johns Hopkins University Circulars*, No. 17 (1882), p. 241.]

[p. 108] THE equations $a = c \cos B + b \cos C$, $b = c \cos A + a \cos C$, $c = b \cos A + a \cos B$, which connect together the sides a, b, c and the angles A, B, C of a plane triangle, may be presented in an algebraical rational form, by introducing in place of the angles A, B, C the functions $\cos A + i \sin A$, $\cos B + i \sin B$, $\cos C + i \sin C$, viz. calling these $\frac{x}{x'}$, $\frac{y}{y'}$, $\frac{z}{z'}$ respectively, or, what is the same thing, writing $2 \cos A = \frac{x}{x'} + \frac{x'}{x}$, $2 \cos B = \frac{y}{y'} + \frac{y'}{y}$, $2 \cos C = \frac{z}{z'} + \frac{z'}{z}$, then the foregoing equations may be written

$$\begin{aligned} (-2a)xx' + y(zy' + z'y)b + z(xy' + x'y)c &= 0, \\ x(yz' + y'z)a + (-2b)yy' + z(xy' + x'y)c &= 0, \\ x(yz' + y'z)a + y(zx' + z'x)b + (-2c)zz' &= 0, \end{aligned}$$

that is, as a system of bipartite equations linear in (a, b, c) and cubic in (x, y, z, x', y', z') respectively.

Similarly in Spherical Trigonometry, writing as above for the angles, and for the sides writing in like manner $2 \cos a = \frac{\alpha}{\delta} + \frac{\delta}{\alpha}$, $2 \cos b = \frac{\beta}{\delta} + \frac{\delta}{\beta}$, $2 \cos c = \frac{\gamma}{\delta} + \frac{\delta}{\gamma}$, we have a system of bipartite equations separately homogeneous in regard to (x, y, z, x', y', z') and $(\alpha, \beta, \gamma, \delta)$ respectively.

825. A Memoir on the Abelian and Theta Functions

[Chapters I to III, *American Journal of Mathematics*, t. v. (1882), pp. 137–179;
Chapters IV to VII, *ib.*, t. vii. (1885), pp. 101–167.]

[p. 109] THE present memoir is based upon Clebsch and Gordan's *Theorie der Abel'schen Functionen*, Leipzig, 1866 (here cited as C. and G.); the employment of differential rather than of integral equations is a novelty; but the chief addition to the theory consists in the determination which I have made for the cubic curve, and also (but not as yet in a perfect form) for the quartic curve, of the differential expression $d\Pi_{12}$ (or as I have it $d\Pi$) in the integral of the third kind $\int_a^b d\Pi_{12}$ in the final normal form (endliche Normalform) for which we have (p. 117) $\int_a^b d\Pi_{12} = \int_1^2 d\Pi_{ab}$, the limits and parametric points interchangeable. The want of this determination presented itself to me as a locus in the theory during the course of lectures on the subject which I had the pleasure of giving in the Johns Hopkins University, Baltimore, U.S.A., in the months January to June, 1882, and I succeeded in effecting it for the cubic curve; but it was not until shortly after my return to England that I was able partially to effect the like determination in the far more difficult case of the quartic curve. The memoir contains, with additional developments, a reproduction of the course of lectures just referred to. I have endeavoured to simplify as much as possible the notations and demonstrations of Clebsch and Gordan's admirable treatise; to bring some of the geometrical results into greater prominence; and to illustrate the theory by examples in regard to the cubic, the nodal quartic, and the general quartic curves respectively. The various chapters are: I, Abel's Theorem; II, Proof of Abel's Theorem; III, The Major Function (continued); IV, The Major Function (continued); V, Miscellaneous Investigations; VI, The Nodal Quartic; VII, The Functions T , U , V , Θ . The paragraphs of the whole memoir will be numbered continuously.

Chapter I. Abel's Theorem.

The Differential Pure and Affected Theorems. Art. Nos. 1 to 5.

[p. 110] 1. We have a fixed curve and a variable curve, and the differential pure theorem consists in a set of linear relations between the displacements of the intersections of the two curves; in the affected theorem, a linear function of the displacements is equated to another differential expression. I state the two theorems, giving afterwards the necessary explanations.

The pure theorem is

$$\Sigma (x, y, z)^{n-3} du = 0.$$

The affected theorem is

$$\Sigma \frac{(x, y, z)^{n-3} du}{012} = - \frac{\delta\phi_1}{\phi_1} \frac{\delta\phi_2}{\phi_2} * .$$

2. We have a fixed curve $f = 0$, or say the curve f , or simply the fixed curve, of the order n , with δ dps, and therefore of the deficiency $\frac{1}{2}(n-1)(n-2) - \delta = p$. The expression "the dps" means always the δ dps of f .

And we have a variable curve $\phi = 0$, or say the curve ϕ , or simply the variable curve, of the order m , passing through the dps and besides meeting the fixed curve in $mn - 2\delta$ variable points.

Moreover, du is the displacement of the current point 0, coordinates (x, y, z) , on the fixed curve, viz. the equation $f = 0$ being satisfied, we have

$$\frac{df}{dx} dx + \frac{df}{dy} dy + \frac{df}{dz} dz = 0,$$

and we thence have

$$\frac{df}{dx} : \frac{df}{dy} : \frac{df}{dz} = y dz - z dy : z dx - x dz : x dy - y dx,$$

so that we have three equal values each of which is put $= du$, viz. we write

$$\frac{y dz - z dy}{\frac{df}{dx}} = \frac{z dx - x dz}{\frac{df}{dy}} = \frac{x dy - y dx}{\frac{df}{dz}} = du,$$

and du as thus defined is the displacement.

[p. 111] 3. $(x, y, z)^{n-3} = 0$ is the minor curve, viz. the general curve of the order $n - 3$ which passes through the dps; and the function $(x, y, z)^{n-3}$ in the proper major function, viz. the implicit factor of the function is so determined that, taking $0 = 1$, the current point at 1, that is, writing (x_1, y_1, z_1) for (x, y, z) , the function $(x, y, z)^{n-3}$ reduces itself to the polar function $\left(x_1 \frac{d}{dx_2} + y_1 \frac{d}{dy_2} + z_1 \frac{d}{dz_2}\right) f_2$, afterwards written $n.1^{n-1} 2$, of f : this implies that taking $0 = 2$, the current point at 2, the function reduces itself to the polar function $n.1 2^{n-1}$.

ϕ_1 is what ϕ becomes on writing therein (x_1, y_1, z_1) for (x, y, z) ; and similarly ϕ_2 is what ϕ becomes on writing therein (x_2, y_2, z_2) for (x, y, z) .

δ denotes differentiation in regard only to the coefficients of ϕ ; viz. writing $\phi = (a, \dots)(x, y, z)^m$ we have $\delta\phi = (da, \dots)(x, y, z)^m$, and similarly $\delta\phi_1$ and $\delta\phi_2 = (da, \dots)(x_1, y_1, z_1)^m$ and $(da, \dots)(x_2, y_2, z_2)^m$ respectively.

The sum Σ extends to all the variable intersections of the two curves.

3. As to the meaning of the theorems, consider first the pure theorem. The variable intersections are not all of them arbitrary points on the fixed curve; a certain number of them taken at pleasure on the fixed curve will determine the remaining variable intersections; and there are thus a certain number of integral relations between the coordinates of the variable intersections; to each such integral relation there corresponds a linear relation between the displacements du of these points, or say a displacement-relation; and it is precisely these displacement-relations which are given by the theorem, viz. the equation

$$\Sigma (x, y, z)^{n-3} du = 0$$

breaks up into as many linear relations as there are arbitrary constants in the function $(x, y, z)^{n-3}$, which equated to zero gives a curve of the order $n - 3$ passing through the dps; for instance, if there be no singularities $\delta = 0$, then the relation gives the single relation $\Sigma du = 0$; but $n = 4$, $\delta = 0$, the equation gives the three relations $\Sigma x du = 0$, $\Sigma y du = 0$, $\Sigma z du = 0$.

[p. 112] 4. It is of course important to show, and it will be shown, that the number of independent displacement-relations given by the theorem is equal to the number of independent integral relations between the variable intersections.

5. Observe that the pure theorem gives all the displacement-relations between the variable intersections; the coefficients of ϕ are hereby led to see the nature of the affected theorem. The variable intersections, the coefficients of ϕ are thereby determined in terms of the coordinates of the assumed variable intersections*, and hence the value of $-\frac{\delta\phi_1}{\phi_1} \frac{\delta\phi_2}{\phi_2}$ is given as a linear function of the corresponding displacements du ; and, substituting this value, the affected theorem gives a linear relation between the displacements du of the several variable intersections. But any such linear relation must clearly be a mere linear combination of the displacement-relations $\Sigma (x, y, z)^{n-3} du = 0$ given by the pure theorem.

Examples of the Pure Theorem—The Fixed Curve a Cubic. Art. Nos. 6 to 12.

6. The pure theorem is not applicable to the case $n = 2$, the fixed curve a conic: it in fact gives no displacement-relation; and this is as it should be, for the variable intersections are all of them arbitrary.

The next case is $n = 3$, $\delta = 0$, the fixed curve a cubic. For greater simplicity the equation is taken in Cartesian coordinates. In general for such an equation, writing in the homogeneous formula $z = 1$, we have

$$du = \frac{dx}{-\frac{df}{dy}} = \frac{dy}{\frac{df}{dx}},$$

the two values being of course equal in virtue of $\frac{df}{dx} dx + \frac{df}{dy} dy = 0$; taking the former value and considering $\frac{df}{dy}$ as expressed in terms of x , let this be called P (of course, P is an irrational function of x); then we have $du = \frac{dx}{P}$; and similarly $du_1 = \frac{dx_1}{P_1}$, &c.

The fixed curve being then a cubic, let the variable curve be a line; this meets the cubic in three points, say 1, 2, 3; and any two of these determine the line, and therefore the third point; there should therefore be one integral relation, and consequently one displacement-relation; and this is in fact the case by the theorem, viz. we have $\Sigma du = 0$, or, $du_1 + du_2 + du_3 = 0$, that is in the same thing,

$$\frac{dx_1}{P_1} + \frac{dx_2}{P_2} + \frac{dx_3}{P_3} = 0.$$

[p. 113] The corresponding integral equation is the equation which expresses that the points 1, 2, 3 are in a line, viz. considering y_1, y_2, y_3 as given functions of x_1, x_2, x_3 respectively, this is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0,$$

or, in the notation already made use of for such a determinant, $123 = 0$.

7. This equation $du_1 + du_2 + du_3 = 0$, here $du = \frac{dx}{P}$, has a peculiar interpretation when we consider the coefficients of the cubic as arbitrary constants, and therefore the cubic curve, we in effect determine y_1 as a function of x_1 and the constants; the points 1 and 2 determine the line and the third point; we have thus x_3, y_3 determined as a function of x_1, x_2 and the constants.

Considering x_3 as thus expressed, we have $dx_3 = \frac{dx_3}{dx_1} dx_1 + \frac{dx_3}{dx_2} dx_2$, an equation which must agree with $du_1 + du_2 + du_3 = 0$, that is, with $dx_3 = -\frac{P_3}{P_1} dx_1 - \frac{P_3}{P_2} dx_2$. It follows that we have $\frac{dx_3}{dx_1} : \frac{dx_3}{dx_2} = P_1 : P_2$, and taking the logarithms and differentiating with regard to x_1 and x_2 , we find $\frac{d}{dx_1} \frac{d}{dx_2} \log \left(\frac{dx_3}{dx_1} : \frac{dx_3}{dx_2} \right) = 0$, a partial differential equation of the third order, independent of any particular cubic curve, and satisfied by x_3 considered as a function of x_1, x_2 and the nine constants. Observe that starting from the expression for x_3 , and proceeding to the differential coefficients of the third order, we have ten equations from which the nine constants can be eliminated, that is, we ought to have a partial differential equation of the third order; and conversely that the equation for x_3 , as containing nine arbitrary constants, is a complete solution of the partial differential equation: the complete solution of the partial differential equation in question is thus the equation which expresses that 3 is the remaining intersection of the line through 1 and 2 with a cubic.

8. As a particular illustration, we have $\frac{dx}{P} = du$, $\Sigma du = 0$, and the integral relation $123 = 0$. I give two new ways of obtaining this—first by differentiation.

We have the equation $123 = 0$. I give a direct verification.

To find $\frac{dx_3}{dx_1}, \frac{dx_3}{dx_2}$, where 1, 2, 3 are on the cubic, $f = 0$. Let ϕ be any conic, $\phi = 0$, taken so that $\frac{d\phi}{dx} \div f = du$, and as the equation of ϕ at 0 of the curve, and it may be regarded as the determination of ϕ ; with the dps and the ϕ other points $a, \dots, h$ on the curve.

This being so, considering as variable the original coordinates f , we have by differentiation

$$\Sigma \frac{1}{P} \delta \log \frac{f}{(ax^3 + by^3 + cz^3 + 6lxyz)} = 0,$$

or what is the same, taking the logarithms and differentiating with regard to one of the constants a , the equation $\Sigma du = 0$ is satisfied.

We have shown also that the cubic curve $f = 0$, and the cube derived line $\phi = 0$ are now equivalent to the two cases.

9. As a particular example, let the cubic be $x^3 + y^3 - 1 = 0$; then $y = \sqrt[3]{1 - x^3}$, and $du = \frac{dx}{y^2} = \frac{dx}{(1 - x^3)^{\frac{2}{3}}}$; and with these values as before the differential relation $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. I give a direct verification. To find x_3, y_3 the coordinates of the third intersection, we may in the equation of the cubic write $x_3, y_3 = \lambda_1 + \mu x_1, \lambda y_1 + \mu y_1$, where $\lambda + \mu = 1$, and we obtain for the determination of λ, μ the equation $\lambda_1 \cdot 1^2 2 + \mu \cdot 12^2 = 0$.

This being so, from the equation $123 = 0$ we obtain by differentiation

$$\Sigma (y_2 - y_3) dx_1 - dy_1 \cdot (x_2 - x_3) = 0,$$

the sum consisting of three terms, the second and third of them obtained from the one given by the cyclical interchange of the numbers 1, 2, 3. But we have $x_1 dx_1 + y_1 dy_1 = 0$, and the equation thus is

$$\Sigma \frac{dx_1}{y_1^2} \{y_1^2(y_2 - y_3) + x_1^2(x_2 - x_3)\} = 0;$$

this will reduce itself to $\Sigma \frac{dx_1}{y_1^2} = 0$, if only the three coefficients in $\{ \}$ are equal, that is, we ought to have

$$y_1^2(y_2 - y_3) + x_1^2(x_2 - x_3) = y_2^2(y_3 - y_1) + x_2^2(x_3 - x_1).$$

Comparing for instance the first and second terms, the equation is

$$-y_1(y_3 + y_1^2)(x_2 - x_3) + (x_1^2 + x_2^2)y_1(x_2 - x_3) + y_1(y_2 - y_3)(x_3 + x_2) = 0,$$

or, as this may be written,

$$-(\lambda_1 + \mu_1)(y_1^2 + y_2^2) + (\lambda_1 + \mu)x_1(x_1^2 + x_2^2) + (\lambda_1 - \mu)(x_1^2 + x_2^2)y_1 + (\lambda + \mu)(y_2 + y_1^2) = 0,$$

whence the whole coefficient of λ is $-x_2^2(y_1^2 + y_3^2) + x_3^2(x_1^2 + x_3^2) + (\lambda + \mu)y_3(y_1 + y_3^2) = 0$, which we see similarly the whole coefficient of μ is $-x_2^2(y_2^2 + y_3^2) + x_2^2(x_2^2 + x_3^2) + (\lambda + \mu)y_2(y_2 + y_3^2) = 0$, which is true; and similarly for the second and third terms; we have thus the required result,

$$\frac{dx_1}{y_1^2} + \frac{dx_2}{y_2^2} + \frac{dx_3}{y_3^2} = 0,$$

the integral of which is $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$, but it is in fact a particular integral, as it is in fact the only general integral.

[p. 115] $du_1 + du_2 + du_3 = 0$, and the integral relation $123 = 0$. This last equation is an integral of the differential equation $du_1 + du_2 + du_3 = 0$; but it is, in fact, a particular integral.

But regard one of the three points, say 3, as a fixed point, that is, the line passes through the fixed point on the cubic; then we may write $du_1 + du_2 = 0$, and the differential equation contains du_1 and du_2 only, the constants of u unaltered.

Let the variable curve be a conic; say the intersections with the cubic are 1, 2, 3, 4, 5, 6. Any five of these points determine the conic, and the displacement-relation is therefore the one displacement-relation

$$\Sigma du = 0,$$

which corresponds to the equation $123456 = 0$, which expresses that the six points are on a conic.

We have thus $123456 = 0$ as a particular integral of

$$du_1 + du_2 + du_3 + du_4 + du_5 + du_6 = 0.$$

If, however, we take 5 a fixed point on the cubic, then we have the same equation as the general integral of $du_1 + du_2 + du_3 + du_4 = 0$.

We have here obviously $\Sigma du = 0$ as a particular integral of the cubic; and the integral relation $123456 = 0$ as the general integral.

But taking also 6 a fixed point on the cubic, the equation $123456 = 0$, which expresses that the six points are on a conic, of any equation $123456 = 0$ for 4 fixed and 5 fixed point on the cubic, becomes a particular integral of $du_1 + du_2 + du_3 + du_4 = 0$.

[p. 116] determinate point, independent of the particular conic through 1, 2, 3, 4 and 5 used for the construction. We have thus a particular integral of the differential equation $du_1 + du_2 + du_3 + du_4 = 0$.

12. We might go on to the case where the variable curve is a cubic; there are here nine intersections; any eight of these do not determine the variable cubic, but they do determine the ninth intersection; and there is between the nine intersections one integral relation, and corresponding to it one displacement-relation $\Sigma du = 0$; that is, $du_1 + du_2 + \dots + du_9 = 0$, given by the pure theorem. But as in this case the number of arbitrary constants in the conic $\Omega = 0$, the corresponding equation $\Sigma du = 0$ is an identity; the number of independent displacement-relations is 0.

Examples of the Affected Theorem—Fixed Curve a Conic. Art. Nos. 13 and 14.

13. The fixed curve is taken to be the conic $x^2 + y^2 - 1 = 0$, and the parametric points 1 and 2 to be the points (1, 0) and (0, 1) on this circle. The variable curve is taken to be a line, say the line $ax + by - 1 = 0$, meeting the circle in the points 3 and 4, coordinates (x_3, y_3) and (x_4, y_4) respectively.

Starting from the formula

$$\Sigma \frac{(x, y, 1)^{n-3} du}{012} = - \frac{\delta\phi_3}{\phi_3} \frac{\delta\phi_4}{\phi_4},$$

where the summation extends to the points 3 and 4, $(x, y, 1)^{n-3}$ is here a constant, $= 2.12$, that is, $2.(x_1x_2 + y_1y_2 - 1)$, which for the points 1, 2 in question is $= -2$. We have 012 denoting the determinant

$$012 = \begin{vmatrix} x & y & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix},$$

which is $= -x - y + 1$; and $du = \frac{dx}{2y}$. Also $\frac{\delta\phi_3}{\phi_3} = \frac{x_3 \delta a + y_3 \delta b}{ax_3 + by_3 - 1}$, is $= \frac{da}{a - 1}$, and similarly $\frac{\delta\phi_4}{\phi_4}$

is $= \frac{db}{b-1}$. The formula thus is

$$\Sigma \frac{da}{y(x+y-1)} = -\frac{da}{a-1} + \frac{db}{b-1}.$$

The coefficients a and b are determined by means of the points 3 and 4, that is, they are functions of x_3, x_4 ; and considering them as thus expressed, then (inasmuch as there is no linear relation between the displacements $\frac{dx_3}{y_3}$ and $\frac{dx_4}{y_4}$ of the two arbitrary points 3 and 4 on the circle) the equation must become an identity in regard to the two lines dx_3, dx_4 respectively. [p. 117] 14. Writing $P, Q, R = y_3 + y_4, x_3 - x_4, x_3y_4 - x_4y_3$; also L_3 and $L_4 = x_3 + y_3 - 1$ and $x_4 + y_4 - 1$ respectively, we have $a = P + R, b = Q + R$; and the equation is found to be

$$\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4} = \frac{1}{(Q-R)(R-P)} \{(Q-R)dP + (R-P)dQ + (P-Q)dR\},$$

where, substituting for dy_3, dy_4 their values in terms of dx_3, dx_4 , we have

$$dP, dQ, dR = \frac{1}{y_3} x_3 dx_3 - \frac{1}{y_4} x_4 dx_4, \frac{1}{y_3} y_3 dx_3 - \frac{1}{y_4} y_4 dx_4, \frac{1}{y_3} (x_3y_4 + y_3y_4) dx_3 - \frac{1}{y_4} (x_3y_4 + y_3y_4) dx_4,$$

and with these values, and by aid of the relations

$$Q - R, R - P, P - Q = x_3y_3 - x_4y_4, y_3L_4 - y_4L_3, -L_3 + L_4,$$

the equation is found to be

$$\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4} = \frac{L_3L_4(x_3x_4 + y_3y_4 - 1)}{(x_3y_4 - x_4y_3)(y_3L_4 - y_4L_3)} \left(\frac{dx_3}{y_3L_3} + \frac{dx_4}{y_4L_4} \right);$$

viz. this will be truly identity if only

$$L_3L_4(x_3x_4 + y_3y_4 - 1) = (x_3y_4 - x_4y_3)(y_3L_4 - y_4L_3),$$

that is,

$$-x_3y_4L_3 + L_3L_4(x_3x_4 - 1) + (y_3y_4 + x_3y_3)(x_3 + y_3 - 1) = 0.$$

But from the values of L_3, L_4 we have $x_3 = \frac{1}{2}L_3^2 + L_3, x_3y_3 = L_3 + L_3$, and the coefficient of L_3L_4 is $= L_3L_4 + L_4 + L_3 + 1$, the equation is thus verified.

The example would perhaps have been more instructive if the points 1 and 2 had been left arbitrary points on the circle, but the working out would have been more difficult.

The Variable Intersections of the Two Curves—Number of Independent Integral Relations. Art. Nos. 15 to 19.

15. Suppose $n = 3, \delta = 0$ ($p = 1$), the fixed curve a cubic; and suppose successively $m = 1, 2, 3, \dots$, the variable curve a line, conic, cubic, &c.

If $m = 1$, then two points on the cubic determine the line, and consequently the remaining intersection with the cubic; hence the number α depends.

If $m = 2$, then 5 points on the cubic determine the conic, and the remaining intersection with the cubic, hence the number α, f^2 , will be a determinate point, &c.

If $m = 3$, then 8 points on the cubic determine the cubic, and consequently the remaining intersection with the cubic, hence the number α, f^3 , will be a determinate point.

[p. 118] If $m = n - 3$, the variable curve is a generic functions of p, q, r , but the dependence on the parameters considered all of 1, 2, 3, &c., f which is a determinant of the variable curve.

16. Suppose next $n = 4, \delta = 0$ ($p = 3$); the fixed curve a general quartic; and as before suppose successively $m = 1, 2, 3, \dots$, the variable curve a line, conic, cubic, &c.

If $m = 1$, then two points on the quartic determine the line, and therefore the remaining two intersections; the number of integral relations is $= 2$.

If $m = 2$, then five points on the quartic determine the conic, and therefore the remaining three intersections; the number of integral relations is $= 3$, and similarly if $m = 3$, the number of integral relations is $= 3$.

If $m = 4$, then thirteen points on the fixed quartic do not determine the variable quartic, but they do determine the remaining three intersections. For draw through the thirteen points a no-matter-what quartic $\chi = 0$; the general quartic through the thirteen points is $\chi + \alpha f = 0$, and this meets the fixed quartic in the points $\chi = 0, f = 0$, that is, in the thirteen points and in three other points independent of α and thus completely determinate; and the number of integral relations is $= 3$, but as α is in general for any higher value of m , the number is still $= 3$.

17. Suppose $n = 5, \delta = 0$ ($p = 6$); the fixed curve a general quintic; and as before suppose $m = 1, 2, 3, \dots$, successively.

If $m = 1$, then two points on the quintic determine the line, and therefore the remaining three intersections; the number of integral relations is $= 3$.

If $m = 2$, then five points on the quintic determine the conic, and therefore the remaining five intersections; the number of integral relations is $= 5$.

If $m = 3$, then 9 points on the quintic determine the cubic, and therefore the remaining six intersections; the number of integral relations is $= 6$, and so if $m = 4$, or if $m = 5$ or any greater number, in the first case directly, and in the other cases by consideration of the quintic $\chi + \alpha f = 0$, &c., we find that the number of integral relations is always $= 6$.

[p. 119] 18. The reasoning is scarcely altered in the case where the fixed curve has dps; thus considering the general case of a fixed curve n, δ, p :

If $m = 1$, then $2 - \delta$ points on the fixed curve (this implies $\delta \geq 2$) determine the line, and therefore the remaining $n - 2\delta - (2 - \delta), = n - 2 - \delta$ intersections; the number of integral relations is $= n - 2 - \delta$.

If $m = 2$, then $5 - \delta$ points on the fixed curve (this implies $\delta \geq 5$) determine the conic, and therefore the remaining $2n - 2\delta - (5 - \delta) = 2n - 5 - \delta$, and so for any value of $m \geq n - 3$, and indeed for the values $n - 2$ and $n - 1$; here $\frac{1}{2}m(m + 3) - \delta$ points on the fixed curve (this implies $\delta \geq \frac{1}{2}m(m + 3) - \delta$) determine the remaining $mn - 2\delta - \{\frac{1}{2}m(m + 3) - \delta\}, = mn - \frac{1}{2}m(m + 3) - \delta$ intersections. Hence the number of integral relations is $= mn - \frac{1}{2}m(m + 3) - \delta$, that is, $= p - \frac{1}{2}(n - m - 1)(n - m - 2)$. And thus in the cases $m = n - 2$ and $n - 1$ the number is $= p$.

If $m = n$, then $\frac{1}{2}n(n + 3) - 1 - \delta$ points on the fixed curve do not determine the variable curve, but they do determine the remaining p intersections, and the number of integral relations is thus $= p$; and so, for any higher value of m , the number is still $= p$.

19. The conclusion thus is

$$\begin{aligned} m \geq n - 3, \quad \text{the number of integral relations} &= p - \frac{1}{2}(n - m - 1)(n - m - 2), \\ m = n \text{ or } > n - 3, \quad „ &= p. \end{aligned}$$

The integral equations spoken of throughout are of course independent relations.

The Variable Intersections of the Two Curves. Number of Independent Displacement Relations given by the Pure Theorem. Art. No. 20.

20. The number of displacement-relations given by the pure theorem is $=$ number of constants in minor function $(x, y, z)^{n-3}$, which equated to zero represents a curve through the dps, viz. this is always

$$\frac{1}{2}(n - 1)(n - 2) - \delta, = p.$$

But for $m \geq n - 2$, these relations are not independent. For instance, for $n = 4, \delta = 0, m = 1$, the displacement-relations are

$$\Sigma (x, y, z)^2 du = 0, \text{ that is, } \Sigma x du = 0, \Sigma y du = 0, \Sigma z du = 0,$$

and conversely from these last equations we have $\Sigma(x, y, z)^2 du = 0$. But in this case the variable curve is a line meeting the quartic in four points; the equation $\Sigma x du = 0$ is satisfied for each of the intersections of the line with the quartic, and the corresponding equation $\Sigma(x, y, z)^2 du = 0$ is an identity. Hence the number of independent displacement-relations is $3 - 1, = 2$.

So for $n = 5, \delta = 0, m = 1$, the displacement-relations are

$$\Sigma(x, y, z)^2 du = 0, \text{ that is, } \Sigma(x^2, y^2, z^2, yz, zx, xy) du = 0,$$

and so on for $n = 6, \delta = 0, m = 1$, the displacement-relations are

$$\Sigma(x, y, z)^3 du = 0.$$

[p. 120] and six equations give conversely $\Sigma(x, y, z)^2 du = 0$, in particular they give $\Sigma x du = 0, \Sigma y du = 0, \Sigma z du = 0, \Sigma xy du = 0$. But if (x, y, z) denote any of the intersections of the line with the curve, then for each of the five intersections $(x, y, z) = 0$, that is, the fast-mentioned three equations are identities. Hence the number of independent displacement-relations is $6 - 3, = 3$.

So for $n = 5, \delta = 0, m = 2$. Here if the variable curve is $\phi = (a, \dots)(x, y, z)^2$, then taking $(x, y, z) = (x_1, y_1, z_1), (x_2, y_2, z_2), \dots$, the corresponding equation $\Sigma(x, y, z)^2 du = 0$ is an identity; the number of independent displacement-relations is $6 - 1, = 5$.

The reasoning is the same when δ is not $= 0$, and we see generally that for $m \geq n - 2$,

$$\begin{aligned} m \geq n - 3, \quad \text{the number of independent displacement-relations} \\ = p - \frac{1}{2}(n - m - 1)(n - m - 2); \end{aligned}$$

while for

$$m = n \text{ or } > n - 2, \text{ the number is } = p;$$

since in this case the relations given by the theorem are all of them independent. It thus appears *à posteriori*, that in every case the number of independent displacement-relations given by the pure theorem is equal to the number of independent integral relations.

As to the dps of the Fixed Curve. Art. No. 21.

21. I conclude with a general remark applicable to the whole of the three chapters. There is no necessity to attend to all or indeed to any of the dps of the fixed curve. Suppose that the fixed curve has $\delta + \delta'$ dps, where δ, δ' may be either of them or each $= 0$, and attend only to the δ dps, the δ' dps being wholly disregarded, and accordingly let the expression the dps mean as before the δ dps of the fixed curve. No attention at all is required: only, if δ' be not $= 0$, then $p = \frac{1}{2}(n - 1)(n - 2) - \delta - \delta'$ will no longer be the deficiency. To obtain the best theorem we use all the $\delta + \delta'$ dps; but disregarding the δ' dps, we obtain theorems, as for a curve with δ dps, which are true and may frequently be useful either in their original form or with simplifications introduced therein by afterwards taking account of the δ' dps.

Chapter II. Proof of Abel's Theorem.

Preparation. Art. Nos. 22 and 23.

22. Starting from the equation $\phi = (a, \dots)(x, y, z)^m = 0$ of the variable curve, we have

$$\begin{aligned} \frac{d\phi}{dx} dx + \frac{d\phi}{dy} dy + \frac{d\phi}{dz} dz + \delta\phi &= 0, \\ \frac{d\phi}{dx} x + \frac{d\phi}{dy} y + \frac{d\phi}{dz} z &= 0, \end{aligned}$$

where $\delta\phi = (da, \dots)(x, y, z)^m$. Let τ denote an arbitrary linear function, $= ax + by + cz$; multiply the two equations by $ax + by + cz$, $= \tau$, and $-(x dz + b dy + c dz)$, $= -d\tau$ respectively, and add: we obtain

$$(y dz - z dy) \left(a \frac{d\phi}{dz} - c \frac{d\phi}{dy} \right) + (z dx - x dz) \left(b \frac{d\phi}{dx} - a \frac{d\phi}{dz} \right) + (x dy - y dx) \left(a \frac{d\phi}{dy} - b \frac{d\phi}{dx} \right) + \tau \delta\phi = 0;$$

introducing du , this becomes

$$du \left[\frac{df}{dx} \left(b \frac{d\phi}{dz} - c \frac{d\phi}{dy} \right) + \frac{df}{dy} \left(c \frac{d\phi}{dx} - a \frac{d\phi}{dz} \right) + \frac{df}{dz} \left(a \frac{d\phi}{dy} - b \frac{d\phi}{dx} \right) \right] + \tau \delta\phi = 0,$$

or observing that a, b, c are the differential coefficients $\frac{d\tau}{dx}, \frac{d\tau}{dy}, \frac{d\tau}{dz}$, the term in $[]$ is $J(f, \tau, \phi)$, or say $J(\phi, f, \tau)$, and the equation is

$$du J(\phi, f, \tau) + \tau \delta\phi = 0,$$

where $J(\phi, f, \tau)$ is the Jacobian, or functional determinant

$$\begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} & \frac{d\phi}{dz} \\ \frac{df}{dx} & \frac{df}{dy} & \frac{df}{dz} \\ \frac{d\tau}{dx} & \frac{d\tau}{dy} & \frac{d\tau}{dz} \end{vmatrix}, = \frac{d(\phi, f, \tau)}{d(x, y, z)};$$

and we hence have

$$du = - \frac{\tau \delta\phi}{J(\phi, f, \tau)}.$$

23. The two theorems thus become

$$\Sigma(x, y, z)^{n-3} \frac{\tau \delta\phi}{J(\phi, f, \tau)} = 0,$$

$$\Sigma(x, y, z)^{n-3} \frac{-\tau \delta\phi}{012 J(\phi, f, \tau)} = - \frac{\delta\phi_1}{\phi_1} \frac{\delta\phi_2}{\phi_2}.$$

But further, if in the first equation we retain τ , while in the second equation we retain τ , using it to denote the linear function 012, the equations become

$$\Sigma(x, y, z)^{n-3} \frac{\delta\phi}{J(\phi, f)} = 0,$$

$$\Sigma(x, y, z)^{n-3} \frac{-\delta\phi}{J(\phi, f, \tau)} = - \frac{\delta\phi_1}{\phi_1} \frac{\delta\phi_2}{\phi_2};$$

where, in the first equation, $J(\phi, f)$ denotes the Jacobian

$$\begin{vmatrix} \frac{d\phi}{dx} & \frac{d\phi}{dy} \\ \frac{df}{dx} & \frac{df}{dy} \end{vmatrix}, = \frac{d(\phi, f)}{d(x, y)}.$$

[p. 122] In these equations the only differential symbol is the δ affecting the coefficients $a, b, \dots$ of ϕ, ϕ_1, ϕ_2 ; the equations are, in respect to the coordinates (x, y, z) of the several variable intersections, but they are mere algebraical equations, which are in fact given by Jacobi's

Fraction-theorem about to be explained. But for the further reduction of the affected theorem I interpose the next three articles.

The Coordinates (ρ, σ, τ). Art. Nos. 24 to 26.

24. The letter τ has just been used to denote the determinant 012; there is often occasion to consider three points 1, 2, 3 coordinates (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) respectively; and then writing ρ, σ, τ to denote the determinants 023, 031, 012 respectively, and Δ the determinant 123, we have identically

$$\begin{aligned}\Delta x &= x_1 \rho + x_2 \sigma + x_3 \tau, \\ \Delta y &= y_1 \rho + y_2 \sigma + y_3 \tau, \\ \Delta z &= z_1 \rho + z_2 \sigma + z_3 \tau,\end{aligned}$$

which equations, regarding therein the point 0, coordinates (x, y, z) , as a current point, are in fact equations for transformation from the coordinates (x, y, z) to the coordinates ρ, σ, τ , belonging to the triangle of reference 123. The points 1 and 2 have been already taken to be on the fixed curve, and it will be assumed that 3 is also a point on this curve.

25. If the function f , which equated to zero gives the equation of the fixed curve, be transformed to the new coordinates (ρ, σ, τ) , the coefficients of the transformed function are polar functions, each divided by ∇^n , viz. the coefficient of ρ^n is $\frac{1}{\nabla^n} 1^n$, which by reason that 1 is a point on the curve is $= 0$ (and similarly the coefficients of σ^n and of τ^n are each $= 0$); the coefficient of $\rho^{n-1}\sigma$ is $\frac{1}{\nabla^n} n \cdot 1^{n-1} 2$; that of $\rho^{n-1}\tau$ is $\frac{1}{\nabla^n} n \cdot 1^{n-1} 3$; that of $\rho^{n-2}\sigma^2$ is $\frac{1}{\nabla^n} \frac{1}{2} n(n-1) \cdot 1^{n-2} 2^2$; and so in other cases. I write this in the form

$$f = \frac{1}{\nabla^n} (1^n, \dots)(\rho, \sigma, \tau)^n;$$

or we might also use the symbolic form

$$f = \frac{1}{\nabla^n} (\rho 1 + \sigma 2 + \tau 3)^n.$$

[p. 123] The terms independent of τ contain, it is clear, the factor $\rho\sigma$, and separating these terms from the others, the equation may be written

$$f = \frac{1}{\nabla^n} \rho\sigma (n \cdot 1^{n-1} 2, \dots)(\rho, \sigma)^{n-2} + \text{terms involving } \tau.$$

26. The equations $\tau = 0, (\dots)(\rho, \sigma)^{n-2} = 0$, determine the residues of the points 1, 2, and hence the major function $(x, y, z)^{n-3}$, expressed in terms of ρ, σ, τ , and writing therein $\tau = 0$, must reduce itself to $(\dots)(\rho, \sigma)^{n-2}$ multiplied by a constant factor which can be found to be $= \frac{1}{\nabla^{n-1}}$: for taking the current point at 1 we have $(\rho, \sigma, \tau) = (\Delta, 0, 0)$, and the corresponding value of the major function is thus equal, and in this case $\frac{1}{\nabla^{n-1}} n \cdot 1^{n-1} 2, = n \cdot 1^{n-1} 2$, as it ought to be. We have thus

$$(x, y, z)^{n-3} = \frac{1}{\nabla^{n-1}} (n \cdot 1^{n-1} 2, \dots)(\rho, \sigma)^{n-2} + \text{terms involving } \tau;$$

and we hence see that, for $\tau = 0$,

$$(x, y, z)^{n-3} = \frac{\Delta f}{\rho\sigma}.$$

an equation which will be useful.

The Preparation for the Affected Theorem resumed. Art. No. 27.

27. In the affected theorem instead of (x, y, z) we introduce the new coordinates (ρ, σ, τ) . We have

$$J(\phi, f, \tau) = \frac{d(\phi, f, \tau)}{d(\rho, \sigma, \tau)} \frac{d(\rho, \sigma, \tau)}{d(x, y, z)},$$

where the first factor is $= \frac{d(\phi, f)}{d(\rho, \sigma)}$, say that this is $J(\phi, f)$, the Jacobian in regard to ρ, σ : and the second factor is at once found to be $= \Delta!$. We have consequently

$$\frac{1}{J(\phi, f, \tau)} = \frac{1}{\Delta! J(\phi, f)},$$

and the equation for the affected theorem becomes

$$\Sigma (x, y, z)^{n-3} \frac{\delta \phi}{J(\phi, f)} = -\Delta! \left(-\frac{\delta \phi_1}{\phi_1} \frac{\delta \phi_2}{\phi_2} \right),$$

where $(x, y, z)^{n-3}$ is to be regarded as standing for its value in terms of (ρ, σ, τ) .

Jacobi's Fraction-Theorem. Art. Nos. 28 to 31.

28. This is the extension of a well-known theorem, which, in a somewhat disguised form, may be thus written: viz. if U be any rational and integral function $(x, 1)^m$, then we have

$$\frac{1}{U} = \Sigma \frac{1}{x - x'} \frac{1}{J(U)},$$

[p. 124] or, introducing an arbitrary constant A by the equation $AU = X$, say this is

$$\frac{A}{X} \left(= \frac{1}{U} \right) = \Sigma \frac{1}{x - x'} \frac{1}{J(U)},$$

where U' is the same function (of x') that U is of x ; $J(U) = \frac{dU'}{dx'}$, is the Jacobian of U' , and the summation extends to all roots x' of the equation $U' = 0$: obviously this is nothing else than the formula for the decomposition of $\frac{1}{U}$ into simple fractions.

29. Take now $U = (x, y, 1)^m$, $V = (x, y, 1)^p$ functions of x, y of the degrees m and n respectively, and assume

$$\begin{aligned} AU + BV &= X, & \text{a function of } (x, 1)^{mn}, \\ CU + DV &= Y, & \text{„ of } (y, 1)^{mn}; \end{aligned}$$

viz. let $X = 0$ and $Y = 0$ be the equations obtained by elimination from $U = 0$ and $V = 0$ of the y and the x respectively. The forms are

$$\begin{aligned} A &= (x, y', 1)^{mn-m}, & B &= (x, y', 1)^{mn-n}, \\ C &= (x^{m-1}, y, 1)^{mn-m}, & D &= (x^{m-1}, y, 1)^{mn-n}; \end{aligned}$$

where these equations denote the kind of them; that is A is a rational and integral function of the degree $mn - m$ in x and y jointly, but only of the degree $mn - 1$ in y ; and so for the other equations. It follows that

$$AD - BC = (x^{mn-1}, y^{mn-1}, 1)^{mn-mn-m}.$$

The theorem now is

$$\frac{AD - BC}{XY} = \Sigma \frac{1}{x - x' \cdot y - y'} \frac{1}{J(U', V')},$$

where U', V' are the same functions of (x', y') that U, V are of (x, y) ; $J(U', V')$ is the Jacobian $\frac{d(U', V')}{d(x', y')}$, $= 0$; and the summation extends to all the simultaneous roots x', y' of the equations $U = 0, V = 0$.

30. For the proof, observe that $AD - BC$ is a sum of terms of the form $\alpha x^\alpha y^\beta$ where α and β are each of them at most $= mn - 1$; hence X being of the degree mn we have $\frac{x^\alpha}{X}$ a sum of fractions $\frac{L_y}{x - x'}$, where x' is any root of $X = 0$; and similarly $\frac{y^\beta}{Y}$ a sum of fractions $\frac{M_y}{y - y'}$, where y' is any root of $Y = 0$; multiplying the two expressions and taking into account the several terms $\lambda x^\alpha y^\beta$ of $AD - BC$ we obtain a formula

$$\frac{AD - BC}{XY} = \Sigma \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to all the combinations of the mn values of x' with the mn values of y . But such a formula existing, the coefficients K may be determined.

[p. 125] in the usual manner, viz. multiplying by XY and then writing $x = x', y = y'$, there is on the right-hand only one term which does not vanish, and we find

$$(AD - BC)_{xy'} = K \left(\frac{X}{x - x'} \right)_x \left(\frac{Y}{y - y'} \right)_{y'}, = K \left(\frac{dX}{dx} \frac{dY}{dy} \right)_{xy'};$$

where the factor which multiplies K does not vanish.

We distinguish the cases where (x', y') are corresponding or non-corresponding roots of $X = 0, Y = 0$, viz. corresponding roots are those for which $U = 0, V = 0$, but for non-corresponding roots these equations do not hold good; there are obviously mn pairs of corresponding roots.

In the latter case $(AD - BC)U = DX - BY$, $(AD - BC)V = -CX + AY$, and since for these values have $X = 0, Y = 0$, the foregoing equation for K gives then $K = 0$.

31. The formula thus is

$$\frac{AD - BC}{XY} = \Sigma \frac{K}{x - x' \cdot y - y'},$$

where the summation extends to corresponding roots only; or, for which we have $U = 0, V = 0$.

THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR
CAYLEY
VOL. XII

SCAN PDF PAGES 151–175 (BOOK PP. 131–155; PAPER 825, CHS. II CONCL.–IV)

Arthur Cayley (1821–1895)

Published 1889

*Transcribed from the original Cambridge University Press edition, 1889.
This file covers scan PDF pages 151–175 (book pages 131–155); restored from scan.*

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825. A Memoir on the Abelian and Theta Functions (continued)

Chapter II. Proof of Abel's Theorem (concluded).

[book p. 131] 41. But for $\tau = 0$, we have

$$A_x \phi_1 + B_x f_x = a \rho^{mn},$$

$$C_x \phi_1 + D_x f_x = \beta \sigma^{mn},$$

and hence

$$(AD - BC)_x f_x = A_x \beta \sigma^{mn} - C_x a \rho^{mn},$$

and the right-hand side thus becomes, say

$$-\Delta! \text{ const. of } \left(-\frac{A_x}{a \rho^{mn}} + \frac{C_x}{\beta \sigma^{mn}} \right) \delta \phi_x.$$

But in calculating the constant of $\frac{A_x}{a \rho^{mn}} \delta \phi_x$, we may suppose not only $\tau = 0$, but also $\sigma = 0$: we then have $\phi_x = (x, y, z)^m = \left(\frac{\rho}{\Delta} \right)^m (x_1, y_1, z_1)^m = \left(\frac{\rho}{\Delta} \right)^m \phi_1$, and hence also

$$\delta \phi_x = \left(\frac{\rho}{\Delta} \right)^m \delta \phi_1.$$

Similarly, in calculating the constant of $\frac{B_x}{\beta \sigma^{mn}} \delta \phi_x$, we may suppose not only $\tau = 0$, but also $\rho = 0$: we then have $\phi_x = (x, y, z)^m = \left(\frac{\sigma}{\Delta} \right)^m (x_2, y_2, z_2)^m = \left(\frac{\sigma}{\Delta} \right)^m \phi_2$, and hence

$$\delta \phi_x = \left(\frac{\sigma}{\Delta} \right)^m \delta \phi_2.$$

Moreover, in the equations

$$A_x \phi_1 + B_x f_x = a \rho^{mn},$$

$$C_x \phi_1 + D_x f_x = \beta \sigma^{mn},$$

writing in the first equation $\sigma = 0$, we find $A_x \left(\frac{\rho}{\Delta} \right)^m \phi_1 = a \rho^{mn}$; that is, $\frac{A_x}{a \rho^{mn}} = \left(\frac{\Delta}{\rho} \right)^m \frac{1}{\phi_1}$;

and similarly writing in the second equation $\rho = 0$, we find $C_x \left(\frac{\sigma}{\Delta} \right)^m \phi_2 = \beta \sigma^{mn}$, that is,

$\frac{C_x}{\beta \sigma^{mn}} = \left(\frac{\Delta}{\sigma} \right)^m \frac{1}{\phi_2}$; and the expression thus becomes

$$= -\Delta! \left(-\frac{\delta \phi_1}{\phi_1} + \frac{\delta \phi_2}{\phi_2} \right),$$

giving the equation which was to be proved. This completes the proof of the affected theorem

$$\Sigma \frac{(x, y, z)_2^{n-2} d\omega}{012} = -\frac{\delta \phi_1}{\phi_1} + \frac{\delta \phi_2}{\phi_2}.$$

Chapter III. The Major Function $(x, y, z)_2^{n-3}$.

Analytical Expression of the Function. Art. Nos. 42 to 49.

[book p. 132] **42.** The function has been defined by the conditions that the curve $(x, y, z)_2^{n-3} = 0$ shall pass through the dps, and also through the $n-2$ residues of the parametric points 1, 2; and moreover, that on writing therein (x_1, y_1, z_1) for (x, y, z) , the function shall become $= n \cdot 1^{n-2} 2$. Obviously the function is not completely determined: calling it Ω (or when required Ω_{12}), then if Ω' be any particular form of it, the general form is $\Omega = \Omega' + (x, y, z)^{n-3} 012$, where $(x, y, z)^{n-3} = 0$ is the general minor function, viz. $(x, y, z)^{n-3} = 0$ is a curve passing through the dps: the major function thus contains $\frac{1}{2}(n-1)(n-2) - \delta$, $= p$, arbitrary constants.

Agreeing with the definition, we have the before-mentioned equation

$$\Omega = \frac{1}{\Delta^{n-1}} (n \cdot 1^{n-1} 2, \dots) (\rho, \sigma)^{n-2} + \text{terms involving } \tau,$$

viz. from this expression for Ω it appears that the curve $\Omega = 0$ meets the line through 1, 2 in the $n-2$ residues of these points, and moreover, for $(x, y, z) = (x_1, y_1, z_1)$ and therefore $(\rho, \sigma, \tau) = (\Delta, 0, 0)$, the value of Ω is $= n \cdot 1^{n-2} 2$.

43. We can without difficulty write down an equation determining Ω' as a function $(x, y, z)^{n-3}$, which, on putting therein $\tau = 0$, becomes equal to the foregoing expression $\frac{1}{\Delta^{n-1}} (n \cdot 1^{n-1} 2, \dots) (\rho, \sigma)^{n-2}$, and which is moreover such that the curve $\Omega' = 0$ passes through the dps; which being so, we have as before, $\Omega = \Omega' + (x, y, z)^{n-3} 012$, for the general value of Ω .

To fix the ideas, consider the particular case $n = 4$, the fixed curve a quartic: Ω' , on putting therein $\tau = 0$, should become

$$= \frac{1}{\Delta} (4 \cdot 1^3 2, 6 \cdot 1^2 2^2, 4 \cdot 1 2^3) (\rho, \sigma)^2;$$

and it is to be shown that this will be the case if we determine Ω' , a quadric function of (x, y, z) , by the equation

$$\begin{vmatrix} (x, y, z)^2 & , & \Omega' \\ 1(x_1, y_1, z_1)^2 & , & 4 \cdot 1^3 2 \\ 2(x_1, y_1, z_1 \xi x_2, y_2, z_2), 6 \cdot 1^2 2^2 & , & \\ 1(x_2, y_2, z_2)^2 & , & 4 \cdot 1 2^3 \\ a, b, c, f, g, h, & , & 0 \end{vmatrix} = 0,$$

where the left-hand side is a determinant of seven lines and columns, the top line being $x^2, y^2, z^2, 2yz, 2zx, 2xy, \Omega'$, and similarly for the second line; the third line is

[book p. 133] $2x_1x_2, 2y_1y_2, 2z_1z_2, 2(y_1z_2 + y_2z_1), 2(z_1x_2 + z_2x_1), 2(x_1y_2 + x_2y_1), 6 \cdot 1^2 2^2$; and in each of the last three lines we have six arbitrary constants followed by a 0. The equation is of the form $\Omega + M\Omega' = 0$, where Ω is a quadric function $(x, y, z)^2$, and M is a constant factor.

44. If the quartic curve has a dp, suppose at the point a , coordinates (x_a, y_a, z_a) , then in order that the curve $\Omega' = 0$ may pass through the dp, we must for one of the last three lines substitute $(x_a, y_a, z_a)^2, 0$; and so for any other dp or dps of the quartic curve. And the conditions as to the dp or dps (if any) being satisfied in this manner, we may if we please, taking $(x_\beta, y_\beta, z_\beta)$ as the coordinates of an arbitrary point β (not of necessity on the fixed curve), write any line not already so expressed, of the last three lines, in the form $(x_\beta, y_\beta, z_\beta)^2, 0$; the effect being to make the curve $\Omega' = 0$ pass through the arbitrary point β .

45. To show that the equation on putting therein $\tau = 0$ does in fact give the required value, $\Omega' = \frac{1}{\Delta}(4.1^3 2, 6.1^2 2^2, 4.1 2^3) (\rho, \sigma)^2 = \Phi$ suppose, it is to be observed that, effecting a linear substitution upon the first six columns, the equation may be written

$$\begin{vmatrix} (\rho, \sigma, \tau)^2 & , & \Omega' \\ 1(\rho_1, \sigma_1, \tau_1)^2 & , & 4.1^3 2 \\ 2(\rho_1, \sigma_1, \tau_1 \xi \rho_2, \sigma_2, \tau_2), 6.1^2 2^2 & , & \\ 1(\rho_2, \sigma_2, \tau_2)^2 & , & 4.1 2^3 \\ a', b', c', f', g', h', & , & 0 \end{vmatrix} = 0,$$

where $(\rho_1, \sigma_1, \tau_1), (\rho_2, \sigma_2, \tau_2)$ are what (ρ, σ) become on writing therein for (x, y, z) the values (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively, viz. we have $(\rho_1, \sigma_1, \tau_1) = (\Delta, 0, 0)$; $(\rho_2, \sigma_2, \tau_2) = (0, \Delta, 0)$; the equation thus is

$$\begin{vmatrix} \rho^2, & \sigma^2, & \tau^2, & 2\sigma\tau, & 2\tau\rho, & 2\rho\sigma, & \Omega' \\ \Delta^2, & 0, & 0, & 0, & 0, & 0, & 4.1^3 2 \\ 0, & 0, & 0, & 0, & 0, & 2\Delta^2, & 6.1^2 2^2 \\ 0, & \Delta^2, & 0, & 0, & 0, & 0, & 4.1 2^3 \\ a', & b', & c', & f', & g', & h', & 0 \end{vmatrix} = 0,$$

and then by another linear substitution upon the columns, the last column can be changed into $\Omega' - \Phi, 0, 0, 0, 0, 0$, whence writing $\tau = 0$, the equation becomes

$$\begin{vmatrix} \rho^2, & \sigma^2, & 0, & 0, & 0, & 2\rho\sigma, & \Omega' - \Phi \\ \Delta^2, & 0, & 0, & 0, & 0, & 0, & 0 \\ 0, & 0, & 0, & 0, & 0, & 2\Delta^2, & 0 \\ 0, & \Delta^2, & 0, & 0, & 0, & 0, & 0 \\ a', & b', & c', & f', & g', & h', & 0 \end{vmatrix} = 0,$$

[book p. 134] or, omitting a constant factor, it is

$$\begin{vmatrix} \rho^2, & \sigma^2, & 2\rho\sigma, & \Omega' - \Phi \\ \Delta^2, & 0, & 0, & 0 \\ 0, & 0, & 2\Delta^2, & 0 \\ 0, & \Delta^2, & 0, & 0 \end{vmatrix} = 0,$$

that is, $\Omega' - \Phi = 0$, or $\Omega' = \Phi, = \frac{1}{\Delta}(4.1^3 2, 6.1^2 2^2, 4.1 2^3) (\rho, \sigma)^2$, the required value.

46. Considering the equation for Ω' as expressed in the before-mentioned form $\Omega + M\Omega' = 0$, the value of the constant factor M is

$$M = \begin{vmatrix} (x_1, y_1, z_1)^2 \\ (x_1, y_1, z_1 \xi x_2, y_2, z_2) \\ (x_2, y_2, z_2)^2 \\ a, b, c, f, g, h, \end{vmatrix},$$

or if, instead of each line such as a, b, c, f, g, h , we have a line $(x_a, y_a, z_a)^2$, then we have

$$M = \begin{vmatrix} (x_1, y_1, z_1)^2 \\ (x_1, y_1, z_1 \xi x_2, y_2, z_2) \\ (x_2, y_2, z_2)^2 \\ (x_3, y_3, z_3)^2 \\ (x_4, y_4, z_4)^2 \end{vmatrix},$$

a value which is

$$= \begin{vmatrix} x_1 & y_1 & z_1 \\ x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \end{vmatrix} \cdot \begin{vmatrix} x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \end{vmatrix} \cdot \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \cdot \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_4 & y_4 & z_4 \end{vmatrix},$$

or say this is $= 12s \cdot 12\beta \cdot 12\gamma \cdot a\beta\gamma$.

47. It is obvious that the foregoing process is applicable to the general case of the fixed curve of the order n with δ dps, and gives always Ω' , by an equation of the foregoing form $\Omega + M\Omega' = 0$, where Ω is a function $(x, y, z)^{n-3}$ of the coordinates, and M is a constant factor. Supposing that in the determinant for Ω' , each of the lower lines is written in the form $(x_a, y_a, z_a)^{n-3}$, 0 instead of a, b, c , &c., will indicate this function to be $= 0$; and for the points α , are on a curve of the order $n - 3$.

[book p. 135] **48.** Preceding the case $n = 4$, above considered, we have, of course, the case $n = 3$, $\delta = 0$, the fixed curve a cubic; the equation for Ω' is here

$$\begin{vmatrix} x & y & z & \Omega' \\ x_1 & y_1 & z_1 & 3 \cdot 1^2 2 \\ x_2 & y_2 & z_2 & 3 \cdot 1 2^2 \\ x_3 & y_3 & z_3 & \end{vmatrix} = 0,$$

giving

$$\frac{1}{3}\Omega' = \frac{1^2 \cdot 023 + 1^2 \cdot 031}{12s},$$

or if we write herein 3 for a , this is

$$\frac{1}{3}\Omega' = \frac{1^2 \cdot 023 + 1^2 \cdot 031}{123},$$

and we have hence the general form

$$\frac{1}{3}\Omega = \frac{1^2 \cdot 023 + 1^2 \cdot 031}{123} + K \cdot 012,$$

where K is an arbitrary constant.

49. There is, however, a more simple particular solution $\frac{1}{3}\Omega = \text{polar function } 012$ ($f = x^3 + y^3 + z^3$; then $012 = xx_1x_2 + yy_1y_2 + zz_1z_2$), to avoid a confusion of notation, we may write $= \overline{012}$. We at once verify this; for, expressing the coordinates (x, y, z) in terms of (ρ, σ, τ) , we have $\frac{1}{3}\Omega = \overline{012} = \frac{1}{\Delta}(1^2 \cdot \rho + 1^2 \cdot \sigma + \overline{125} \cdot \tau)$, which, for $\tau = 0$ becomes $= \frac{1}{\Delta}[1^2 \cdot \rho + 1^2 \cdot \sigma]$.

We must, of course, have an identity of the form

$$\overline{012} = \frac{1^2 \cdot 023 + 1^2 \cdot 031}{123} + K \cdot 012,$$

and to find K , writing here $0 = 3$, we have $K = \frac{\overline{123}}{123^2}$, or we have the identity

$$123 \overline{012} - \overline{123} 012 = 1^2 \cdot 023 + 1^2 \cdot 031.$$

Single Letter Notation for the Polar Functions of the Cubic. Art. Nos. 50 and 51.

50. The notation of single letters for the polar functions is not much required in the case of the cubic, but, in the next following case of the quartic it can hardly be dispensed with, and I therefore establish it in the case of the cubic: viz. I write

$$2\overline{3}, 3\overline{1}, 1\overline{2} = f, g, h; \quad 23\overline{1}, 31\overline{2}, 12\overline{3} = i, j, k; \quad \overline{123} = l,$$

or, what is the same thing, the expression for the cubic function f in terms of ρ, σ, τ is

$$\Delta^2 . f = 3l\rho^2\sigma + 3jp^2\tau + 6l\rho\sigma\tau + 6lp\sigma\tau + 3g\sigma^2\tau + 3i\sigma\tau^2 + 3i\sigma\tau^2;$$

[book p. 136] an equation which, writing 0^3 instead of f , may also be written

$$\Delta! 0^3 = (3l, 3j, 3i, 3k, 6l, 6l, 3i, 3i, 3j, 3i) \left(\rho^3, \sigma^3, \tau^3, \rho^2\sigma, \rho^2\tau, \rho\sigma^2, \rho\tau^2, \sigma^2\tau, \sigma\tau^2 \right),$$

and I join to it the series of equations

$$\begin{aligned} \Delta! . 0^2 1 &= (0, 2k, 2j, k, 2l, 2l, 2l, g) (\rho\sigma, \rho\tau, \sigma\tau, \dots), \\ , , 0^2 2 &= (k, 2l, 2i, 2l, 2k, l), \\ , , 0^2 3 &= (j, 2l, 2g, l, 2h, 2l), \\ \Delta . 0 1^2 &= (0, h, l\xi\rho, \sigma, \tau), \\ \Delta \overline{012} &= (h, l, l\xi \dots), \\ , , 0^2 &= (k, 0, f\xi \dots), \\ , , \overline{025} &= (l, f, l\xi \dots), \\ , , 0 3^2 &= (j, i, 0\xi \dots), \end{aligned}$$

51. In particular, we have $\Delta \overline{012} = h\rho + l\sigma + l\tau$, and the above-mentioned identity $123\overline{012} - \overline{123}012 = 1^2 . 023 + 1^2 . 031$ is simply $h\rho + l\sigma + l\tau - \bar{l}l = h\rho + l\sigma$.

Single Letter Notation for the Polar Functions of the Quartic. Art. No. 52.

52. I write here

$$\begin{aligned} 2\bar{3}, 3\bar{1}, 1\bar{2} &= f, g, h; \quad 23\bar{1}, 31\bar{2}, 12\bar{3} = i, j, k; \\ 1\bar{2}3, 1\bar{2}3, 1\bar{2}3 &= l, m, n; \quad 23, 31, 1\bar{2} = p, q, r; \end{aligned}$$

so that the expression for the quartic function f in terms of ρ, σ, τ is

$$\begin{aligned} \Delta! . f &= 4l\rho^3\sigma + 4jp^3\tau + 12lp^2\sigma^2 + 6gp^2\sigma^2 \\ &\quad + 4lp\sigma + 12mp\sigma\tau + 12n\sigma\tau + 6lp^2\sigma + 6n\rho\tau^2 + 4j\sigma\tau^3, \end{aligned}$$

which, putting 0^4 for f , may also be written

$$\begin{aligned} \Delta . 0^4 &= (4k, 4j, 12l, 6g, 4l, 12m, 12n, 4j, 6l, 6n, 4j) \\ &\quad (\rho^4, \rho^3\sigma, \rho^3\tau, \rho^2\sigma^2, \rho^2\tau^2, \rho\sigma^2\tau, \rho\sigma\tau, \rho\sigma\tau^2, \rho^2\tau, \sigma^4, \sigma^3\tau, \sigma^2\tau^2, \sigma\tau^3), \end{aligned}$$

and I join to it the series of equations

$$\begin{aligned} \Delta . 0^3 1 &= (0, 3k, 3j, 3m, 3l, 3lm, 3lm, 3m, 3lm, 3l, l) (\rho^3, \rho^2\sigma, \rho^2\tau, \rho\sigma^2, \rho\sigma\tau, \rho\tau^2, \sigma^3, \sigma^2\tau, \sigma\tau^2, \tau^3), \\ , , 0^3 2 &= (k, 3l, 3m, 6m, 3m, 3l, 3l), \\ , , 0^3 3 &= (j, 3l, 3g, 3p, 3p, 3l), \end{aligned}$$

[book p. 137]

$$\begin{aligned}
\Delta . 0^2 1^2 &= (0, 2k, 2k, 2l, q\xi p, p, p, p), \quad ,, ,, \\
,, 0^2 12 &= (k, 2k, 3l, 3lm, n\xi , , , , , ,), \\
,, 0^2 13 &= (j, 2l, 2k, l, k\xi p, p, p, p), \quad ,, ,, \\
,, 0^2 2^2 &= (k, 2m, 2n, l, p\xi , , , , , ,), \\
,, 0^2 23 &= (l, 2m, 2m, f, l\xi p, p, p, p), \quad ,, ,, \\
,, 0^2 3^2 &= (q, 2m, 2p, p, 2l, 0\xi , , , , , ,), \\
\Delta . 01^3 &= (0, h, l, l\xi \rho, \sigma, \tau), \\
,, 01^2 2 &= (h, l, l\xi \dots), \\
,, 012^2 &= (l, m, n\xi \dots), \\
,, 01^2 3 &= (j, l, m\xi \dots), \\
,, 0123 &= (l, m, n\xi \dots), \\
,, 013^2 &= (q, n, p\xi \dots), \\
,, 02^3 &= (k, l, l\xi \dots), \\
,, 02^2 3 &= (l, m, n\xi \dots), \\
,, 023^2 &= (m, f, l\xi \dots), \\
,, 03^3 &= (q, k, 0\xi \dots),
\end{aligned}$$

which will be convenient in the sequel.

Major Function—The Fixed Curve a Cubic. Art. No. 53.

53. It has been already seen that a simple particular form is $\frac{1}{3}\Omega = \overline{012}$: and that the general form is $\Omega = \Omega' + K . 012$.

Major Function—The Fixed Curve a Quartic. Art. No. 54.

54. It is to be shown that a particular form is

$$\frac{1}{2}\Omega' = \frac{-01^3 . 02^2 + 01^2 2 . 012 + 012 . 1^2 2}{1^2 2}.$$

In fact, by the foregoing values of $\Delta . 01^3$, &c., the numerator of this expression, multiplied by $\Delta!$, is =

$$\begin{aligned}
& - (ha + jr)(la + fr) \\
& + (kp + ra + lr)(c\rho + l\sigma + nr) \\
& + r(h\rho^2 + 2r\rho\sigma + 2lr\sigma + l\rho\sigma + lnr\sigma + n\sigma^3),
\end{aligned}$$

which is

$$= 2lr\rho^2 + 3r^2\rho\sigma + (lm - jk + 3lr)\rho\sigma + 2lr\sigma^2 + r(-fh + kl + 3n\sigma\tau + r\tau^2),$$

and this, for $\tau = 0$, becomes

$$= r(2k\rho^2 + 3r\rho\sigma + 2l\sigma^2).$$

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[book p. 138] Hence for $\tau = 0$, we have

$$\Omega' = \frac{1}{\Delta!} (4k\rho^2 + 6r\rho\sigma + 4l\sigma^2),$$

that is,

$$= \frac{1}{\Delta} [4 \cdot 1^3 2 \cdot \rho^2 + 6 \cdot 1^2 2^2 \cdot \rho\sigma + 4 \cdot 1 2^3 \cdot \sigma^2];$$

and Ω' is thus a form of the major function $(x, y, z)_2^2$. Of course the general form is $\Omega = \Omega' + (x, y, z) \cdot 012$.

Syzygy of the Major Function. Art. No. 55.

55. Writing now $(x, y, z)_k^{n-3} = \Omega_{ik}$; and taking on the fixed curve a new point 3, consider the like functions Ω_{23} and Ω_{31} : it is to be shown that we have identically

$$\Omega_{23} \cdot 011 + \Omega_{31} \cdot 012 \cdot 023 + \Omega_{12} \cdot 031 - (123)^n \cdot 011 \cdot 012 (x, y, z)^{n-4},$$

where $(x, y, z)^{n-4} = 0$ is a properly determined minor function; or, considering herein 0 as a point on the fixed curve and writing therefore $f = 0$, the equation is

$$\frac{\Omega_{23}}{023} \cdot 011 + \frac{\Omega_{31}}{031} \cdot 012 + \frac{\Omega_{12}}{012} \cdot (x, y, z)^{n-4}.$$

56. Write for a moment $X = \Omega_{23} \cdot 011 \cdot 012 + \Omega_{31} \cdot 023 \cdot 031 - (123)^n (x, y, z)^{n-4}$, then k being an arbitrary coefficient, we have $X - kf = 0$, a curve of the order n , passing through the points 1, 2, 3, and the residues of 1, 2; in fact, take for any other point in the curve $\Omega_{12} \cdot 012 = 0$, $033 = 0$, and therefore $X = 0$ a point on the curve. Again, at any residue of 2, 3, we have $\Omega_{23} = 0$, $023 = 0$, and therefore $X = kf = 0$ will have a dp at 1; and clearly by symmetry, it will also have a dp at 2, and at 3; the theorem is thus proved.

57. Taking an arbitrary point a coordinates (x_a, y_a, z_a) , and writing

$$D = x_a \frac{d}{dx} + y_a \frac{d}{dy} + z_a \frac{d}{dz},$$

* This is the differential theorem corresponding to C , and G .'s integral theorem, p. 30, viz. this is $\Omega_{23} + \Omega_{31} + \Omega_{12} = 1$, a sum of three integrals of the third kind—an integral of the first kind.

[book p. 139] we have to find k , so that the curve $D(X - kf) = 0$ shall pass through the point 1. Observing that $D023 = a23$, &c., we have

$$\begin{aligned} D(X - kf) &= D\Omega_{23} \cdot 031 \cdot 012 + \Omega_{23} (031 \cdot a12 + a31 \cdot 012) \\ &\quad + a23 (\Omega_{31} \cdot 012 + \Omega_{12} \cdot 031) \\ &\quad + 023 [\Omega_{31} \cdot a12 + \Omega_{31} \cdot a31 + D\Omega_{31} \cdot 012 + D\Omega_{12} \cdot 031] \\ &\quad - kDf, \end{aligned}$$

and, to make the curve pass through 1, writing herein 0 = 1, we have

$$0 = 123 (\Omega_{12}^2 \cdot a12 + \Omega_{1,a} \cdot a31) - k(Df)_1,$$

where the suffixe (1) denotes that we are in Ω_{23} , &c., and f respectively to write 0 = 1. We have here $\Omega_{1,a} = a \cdot 1^{n-3} 3$, $\Omega_{12,1} = n \cdot 1^{n-3} 2$, $(Df)_1 = a \cdot 1^{n-1}$, and the equation thus is

$$n \cdot 123 (1^{n-2} 3 \cdot a12 + 1^{n-2} 2 \cdot a31) = ka \cdot 1^{n-1}.$$

But we have identically $1^{n-2} 1 \cdot a23 + 1^{n-2} 2 \cdot a31 + 1^{n-2} 3 \cdot a12 = 1^{n-1} \cdot 123$, where $1^{n-1} = 1^n$ is, in fact, = 0; the factor $1^{n-2} a$ thus divides out, and the equation becomes $k = (123)^n$, viz.

k having this value, the curve $X - kf = 0$ will have a dp at 1; and clearly by symmetry, it will also have a dp at 2, and at 3; the theorem is thus proved.

The Syzygy, Fixed Curve a Cubic. Art. No. 58.

58. The syzygy may be verified independently in the case where the fixed curve is a cubic. Observe that the syzygy, if satisfied for any particular form of Ω , will be generally satisfied; we may therefore take $\frac{1}{3}\Omega_{12} = 012$. Writing thus

$$\frac{\frac{1}{3}\Omega_{23}}{012} \cdot \frac{\overline{012}}{012} = [012] \text{ suppose,}$$

and taking 0 to be a point on the cubic curve, we ought to have $[023] + [031] + [012] = a$ constant; the value of this constant comes out to be $= [123]$, and the syzygy in its complete form thus is

$$[023] + [031] + [012] = [123].$$

We have

$$\Delta \overline{023}, \Delta \overline{031}, \Delta \overline{012} = lp + f\sigma + i\tau, jp + h\sigma + g\tau, hp + l\sigma + l\tau,$$

and the equation thus is

$$\frac{lp + f\sigma + i\tau}{\rho} + \frac{jp + h\sigma + g\tau}{\sigma} + \frac{hp + l\sigma + l\tau}{\tau} - l = 0;$$

this, multiplied by $\rho\sigma\tau$, becomes

$$h\rho^2\sigma + jp^2\tau + 3l\rho\sigma\tau + g\sigma^2\tau + f\sigma\tau^2 + i\sigma\tau^2 = 0,$$

which is, in fact, $\frac{1}{3}f = 0$, the equation of the cubic curve.

Observe that the new symbol $[012]$ is, in virtue of its determinant denominator, an alternate function, $[012] = -[102]$, $[012] = [201] = [012]$. The syzygy is a relation between any four points 1, 2, 3, 0 of the curve, and it may be also expressed in the form

$$[123] - [230] + [301] - [012] = 0.$$

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[book p. 140] *The Syzygy, Fixed Curve a Quartic.* Art. No. 59.

59. Taking Ω_{12} as before, we have

$$\frac{1}{2}\Omega_{12} = \frac{-01^3 \cdot 02^2 + 01^2 \cdot 2 \cdot 012 + 01 \cdot 2 \cdot 1^2 \cdot 2}{012 \cdot 1^2 \cdot 2} = [012] \text{ suppose;}$$

and taking 0 to be a point on the quartic curve, we ought to have

$$[023] + [031] + [012] = (x, y, z)^0, \text{ a linear function of } (x, y, z),$$

or, what is the same thing, considering the left-hand side as expressed in terms of ρ, σ, τ , the sum should be

$$= (\rho, \sigma, \tau)^0, \text{ a linear function of } (\rho, \sigma, \tau).$$

By a preceding formula we have

$$[012] = \frac{1}{\Delta^2 \rho \sigma} [2lr\rho^2 + 3r^2\rho\sigma + km - jk + 3lr)\rho\sigma \\ + 2lr\sigma^2 + (-fh + kl + 3n\sigma\tau)\tau^2],$$

which is

$$= \frac{1}{\Delta!} \left\{ \left(2k + \frac{km - jk}{r} \right) \rho + \left(3m + \frac{-fh + kl}{r} \right) \sigma + \left(n + \frac{-fj + lm}{r} \right) \tau \right\} + \frac{1}{\Delta!} \frac{2k\rho^2 + 3r\rho\sigma + 2l\sigma^2}{\tau}.$$

And hence, forming the sum [023] + [031] + [012], we have first a fractional part which is found to be integral, viz. this is

$$\begin{aligned} & \frac{1}{\Delta^2} \left\{ \frac{2fr^2 + 3r\rho\sigma + 2ir^2}{p} + \frac{2gp^2 + 3r\rho\sigma + 2lp^2}{q} + \frac{2kp^2 + 3r\rho\sigma + 2l\sigma^2}{r} \right\}, \\ &= \frac{1}{\Delta^2 \rho\sigma\tau} \{ 2hp\sigma + 2lp\rho + 3l\rho^2\sigma + 3gp^2\sigma + 2lp\sigma\tau + 2k\rho\sigma\tau + 2k\rho\tau \}, \\ &= \frac{1}{\Delta! \rho\sigma\tau} \left[\frac{1}{2} \Delta! f - 6lp\sigma\tau - 6n\sigma\tau - 6n\sigma\tau \right], \end{aligned}$$

or since $f = 0$, this is

$$= \frac{1}{\Delta} (-6l\rho - 6n\sigma - 6n\tau).$$

We then have integral terms which are at once deduced from the above integral terms of $0^1 2$; collecting the several terms, we find

$$[023] + [031] + [012] =$$

$$\frac{1}{\Delta} \left\{ \rho \left(l + \frac{mn - gk}{p} + \frac{in - gh}{q} + \frac{kn - gh}{r} \right) + \sigma \left(m + \frac{fn - hi}{p} + \frac{lh - lk}{q} + \frac{kl - fh}{r} \right) + \tau \left(n + \frac{lm - fj}{p} + \frac{g}{p} \right) \right\}$$

which is the required result.

[book p. 141] *Preparation for the Conversion—The Symbol ∂ .* Art. Nos. 60 to 63.

60. I use ∂ as the symbol of a quasi-differentiation, viz. U being any function of (x, y, z) , ∂U denotes $\frac{1}{d\omega}$ multiplied by the differential $\frac{dU}{dx} dx + \frac{dU}{dy} dy + \frac{dU}{dz} dz$; in such a differential, the increments dx, dy, dz do not in general present themselves in the combinations $yz - zdy, zdx - xdz, xdy - ydx$; but they will do so if U is a function of the degree zero in the coordinates x, y, z , that is, if U be the quotient of two homogeneous functions of the same degree(s); and this being so, we say by the equations

$$\frac{yz - zdy}{\frac{df}{dx}} = \frac{zdx - xdz}{\frac{df}{dy}} = \frac{xdy - ydx}{\frac{df}{dz}} = d\omega$$

get rid of the increments, and ∂U will denote a function of (x, y, z) derived in a definite manner from the function U . The symbol ∂ will be used only in the case in question of a function of the degree zero. Of course ∂_1 will denote the like operation in regard to (x_1, y_1, z_1) , and so ∂_2 , &c.; and we may for greater clearness write ∂_0 in place of ∂ .

61. Consider then $\partial \frac{P}{Q}$, where P, Q are functions $(x, y, z)^m$ of the same degree; we have

$$\partial \frac{P}{Q} = \frac{1}{Q^2 d\omega} (QdP - PdQ),$$

and then

$$dP = \frac{dP}{dx} dx + \frac{dP}{dy} dy + \frac{dP}{dz} dz, \quad \frac{1}{m} P = \frac{dP}{dx} x + \frac{dP}{dy} y + \frac{dP}{dz} z,$$

with the like formula for Q . Substituting, we find

$$\partial \frac{P}{Q} = \frac{1}{mQ^2 d\omega} \left\{ \frac{d(Q, P)}{d(y, z)} (ydz - zdy) + \frac{d(Q, P)}{d(z, x)} (zdx - xdz) + \frac{d(Q, P)}{d(x, y)} (xdy - ydx) \right\},$$

that is,

$$\partial \frac{P}{Q} = \frac{1}{mQ^2} \left\{ \frac{df}{dx} \frac{d(Q, P)}{d(y, z)} + \&c. \right\} = \frac{1}{mQ^2} \frac{d(f, Q, P)}{d(x, y, z)}, = \frac{1}{mQ^2} J(f, Q, P),$$

or say

$$\partial \frac{P}{Q} = -\frac{1}{mQ^2} J(P, Q, f).$$

62. As an example, consider

$$\partial [012], = -\frac{1}{(012)^2} J(\overline{012}, 012, f).$$

The determinant is

$$\begin{vmatrix} \frac{d}{dx} \overline{012}, & y_1 z z - y_2 z_1, & \frac{df}{dx} \\ \frac{d}{dy} \overline{012}, & z_1 z_2 - z_2 x_1, & \frac{df}{dy} \\ \frac{d}{dz} \overline{012}, & x_1 y_2 - x_2 y_1, & \frac{df}{dz} \end{vmatrix}$$

[book p. 142] and the coefficient herein of $\frac{d}{dz} \overline{012}$ is $(x_1 z_2 - x_2 z_1) \frac{df}{dy} - (x_1 y_2 - x_2 y_1) \frac{df}{dx}$, which is

$$= x_1 \left(x_2 \frac{df}{dx} + y_2 \frac{df}{dy} + z_2 \frac{df}{dz} \right) - x_2 \left(x_1 \frac{df}{dx} + y_1 \frac{df}{dy} + z_1 \frac{df}{dz} \right), = 3(0^2 \cdot x_1 - 0^2 \cdot x_2),$$

and so for the other terms.

The determinant is thus

$$= 3 \left[0^2 1 \left(x_1 \frac{d}{dx} + y_1 \frac{d}{dy} + z_1 \frac{d}{dz} \right) \overline{012} - 0^2 2 \left(x_2 \frac{d}{dx} + y_2 \frac{d}{dy} + z_2 \frac{d}{dz} \right) \overline{012} \right]$$

say this is

$$= 3[0^2 1 \cdot \overline{D} - 0^2 2 \cdot \overline{D}] \overline{012}.$$

But we have $\overline{D} \overline{012} = 1^2 2$, $\overline{D} \overline{012} = 1 2^2$, and the determinant is then $= 3(0^2 \cdot 1^2 2 - 0^2 \cdot 1 2^2)$, whence finally, writing ∂_0 instead of ∂ ,

$$\partial_0 [012] = -3 \cdot \frac{0^2 \cdot 1^2 2 - 0^2 \cdot 1 2^2}{(012)^2}.$$

63. By cyclical interchange of the 0, 1, 2, we have

$$\partial_1 [012] = -3 \cdot \frac{1^2 \cdot 0^2 2 - 0^2 \cdot 0 2^2}{(012)^2},$$

$$\partial_2 [012] = -3 \cdot \frac{0 2^2 \cdot 0 1^2 - 1^2 \cdot 0^2 1}{(012)^2},$$

and thence adding, we find

$$(\partial_0 + \partial_1 + \partial_2) [012] = 0,$$

an important property, which, joined to the equation before obtained,

$$[023] + [031] + [012] = [123],$$

completes the theory of the function $[012]$.

Conversion of the Major Function (Interchange of Limits and Parametric Points). Art. No. 64.

64. Write in general

$$\frac{(x, y, z)_k^{n-3}}{012} = Q_{k,01}.$$

$Q_{k,01}$ is an alternate function in regard to the points 1, 2 ($Q_{k,01} = -Q_{k,10}$), and in regard to the coordinates of the points 0, 1, 2, it is rational, but not integral, of the degrees $n-3, 0, 0$ respectively; it can therefore be operated upon with ∂_0 or ∂_1 , but (except in the case $n=3$) not with ∂_2 .

[book p. 143] The conversion relates not to the general major function $(x, y, z)_k^{n-3}$, but to this function with the arbitrary constants properly determined, and consists in a relation between two functions $Q_{k,01}$ and $Q_{k,1k}$ (each of them a function of three out of four arbitrary points 1, 2, 4, 5 on the fixed curve), viz. the conversion is

$$\partial_2 Q_{k,01} = \partial_4 Q_{k,4k};$$

an equation which may be written in four different forms, viz. we may in the form written down interchange 1, 2 and also 4, 5.*

The determination of the constants is a very peculiar one, inasmuch as it is not algebraical; viz. in the case of the cubic curve, about to be considered, it appears that $Q_{k,01}$ contains the term $\int_2^1 d\omega \{0\overline{36}\}$, which is a transcendental function of the coordinates of the parametric points 1 and 2.

The Conversion, Fixed Curve a Cubic. Art. No. 65.

65. We may write $Q_{k,01} = [012] + K$, where K is a constant, that is, it is independent of the point 0, but depends on the parametric points 1 and 2. I assume K to be properly determined, and give an *a posteriori* verification of the equation

$$\partial_2 Q_{k,01} = \partial_4 Q_{k,4k}.$$

The value is $K = \int_2^1 d\omega \{0\overline{36}\} - [123]$, where 3, 4 are arbitrary points on the cubic curve, and where in the definite integral, regarded as an integral $\int U du$ with a current variable u , the meaning is that this variable has as the limits the values u_1, u_2 , which belong to the points 1 and 2 respectively: a fuller explanation might be proper, but the investigation will presently be given in a form not depending on any integral at all.

Substituting for K its value, we have

$$Q_{k,01} = [012] + \int_2^1 d\omega \{0\overline{36}\} - [123],$$

or, as this may also be written,

$$= -[023] - [031] + \int_2^1 d\omega \{0\overline{36}\}.$$

We have thence

$$\partial_2 Q_{k,01} = -\partial_2 [031] + \int_2^1 d\omega \{0\overline{36}\},$$

* The meaning of the property is better seen from the integral form: $Q_{k,01}$ is a function of the points 0, 1, 2 and $Q_{k,4k}$ the like function of the points 0, 1, 4, 5, and these in their meaning (alluded to in the heading) that there exists for the integral of the third kind Ω a canonical form Π , such that $\int_0^{(\alpha\beta)} d\omega \{036\}_{\alpha\beta} = \int_a^{(\alpha\beta)} d\omega Q_{k,a\beta}$ which equation operated upon with ∂_0 gives the formula of the text. And there is then the existing (alluded to in the heading) that there exists for the integral of the third kind a canonical form Π , in which the symmetry, expressed by the interchange of the limits and the parametric points. The expression for $Q_{k,a\beta}$ mentioned further on in the text for the case, must cover a cubic, shows that in this case the canonical form of the integral of the third kind is $\int_a^b d\omega \{[026] + \left(\int_a^b d\omega \{[036] - [126]\}\right)\}$.

[book p. 144] and consequently

$$\partial_2 Q_{k,01} = -\partial_2 [431] + \partial_2 [136],$$

$$\partial_4 Q_{k,4k} = -\partial_4 [134] + \partial_4 [436];$$

hence, observing that $[431] = -[134]$, &c., we have

$$\begin{aligned} \partial_2 Q_{k,01} - \partial_4 Q_{k,4k} &= -\partial_2 [134] + \partial_2 [136] - \partial_4 [436], \\ &= -\partial_2 [134] + \partial_2 [136] - \partial_4 [436], \end{aligned}$$

which, observing that we have $\partial_2 [641] = 0$, is

$$= \partial_2 ([136] - [364] + [641] - [413]), = 0,$$

the required theorem.

To avoid, in the proof, the use of the integral sign, we have only to consider the required function $Q_{k,01}$ as given by the foregoing differential formula

$$\partial_2 Q_{k,01} = -\partial_2 [031] + \partial_2 [136],$$

for we have then the values of $\partial_1 Q_{k,01}$ and $\partial_2 Q_{k,01}$: the rest of the proof the same as before.

The Conversion, Fixed Curve a Quartic. Art. Nos. 66 to 73.

66. We have

$$Q_{k,01} = [012] + (x, y, z),$$

where (x, y, z) is a linear function of (x, y, z) , but depending also on the parametric points 1 and 2, which has to be determined so as to satisfy the conversion equation

$$\partial_2 Q_{k,01} = \partial_4 Q_{k,4k}.$$

Observing that we have $[023] + [031] + [012] =$ a linear function of (x, y, z) , the linear function (x, y, z) of $Q_{k,01}$ may be taken to be $= \Theta_{k,12} = [023] - [031] - [012]$; that is, we may assume

$$\begin{aligned} Q_{k,01} &= [012] + \Theta_{k,12} = (-[023] + [091]) + [012], \\ &= -[023] - [031] + \Theta_{k,12}, \end{aligned}$$

where $\Theta_{k,12}$ is a linear function of (x, y, z) , but depending also on the points 1 and 2, which has to be determined. We have

$$Q_{k,01} = [012] + \Theta_{k,12} + \partial_1 \Theta_{k,12},$$

and thence

$$\partial_1 Q_{k,01} = -\partial_1 [431] + \partial_1 \Theta_{k,12},$$

$$\partial_2 Q_{k,01} = -\partial_2 [134] + \partial_2 \Theta_{k,12},$$

giving an equation for Θ ,

$$\partial_1 \Theta_{k,12} - \partial_2 \Theta_{k,12} = \partial_2 [431] - \partial_2 [134];$$

here 4 is an arbitrary point of the quartic, and we may instead of it write 0, the equation thus becoming

$$\partial_1 \Theta_{k,12} - \partial_2 \Theta_{k,12} = \partial_2 [031] - \partial_2 [130].$$

[book p. 145] **67.** Of the terms on the left-hand side, the first is a linear function of (x, y, z) , or say it is an integral function of 0, and the second is a linear function of (x_1, y_1, z_1) , or say it is an integral function 1'; the first depends on the parametric points 1, 3, and $\partial_1(0, 1, 3)$ is a known function, suppose, as regards the coordinates (x, y, z) of the point 0, replacing the corresponding point 3, 4 or 5 by the new arbitrary point 6, 7 or 8. Further 045, &c., denote determinants as above; so that in H each of the last three terms is, in fact, as regards the point 0, a mere linear function of the coordinates (x, y, z) of this point.

We have $\partial_1(1; 3, 2; 6, 4, 5) = 13'246$, and hence this function multiplied by $\frac{045}{345}$ is $= 01'2456$; and so for the third and fourth terms of H : thus each of the four terms of H is $= 01'2456789$, of the degree 1 in the coordinates of the points of the other points 2, 3, 4, 5, 6, 7, 8 respectively, but of the degree 0 in regard to the coordinates of the point 1. The before-mentioned function $\Omega(0; 1, 2; 3, 4, 5)$ is a function satisfying these conditions 1Ω as $= s \cdot 0121^22$ + arbitrary linear function (x, y, z) , of (x_1, y_1, z_1) , of the points of of (x, y, z) , which presents itself in the case of the cubic curve.

Nöther's conversion-theorem consists herein, that the function

$$H(0; 1, 2; 3, 4, 5; 6, 7, 8)$$

is unaltered by the interchange of the two points 0, 1; or putting for shortness

$$H(0; 1, 2; 3, 4, 5; 6, 7, 8) = H_1(0),$$

the theorem is

$$H_1(0) = H_0(1).$$

74. We have, No. 59,

$$\frac{1}{2}\Omega_{12} = \frac{-01^3 \cdot 02^2 + 01^2 2 \cdot 012 + 01 2 \cdot 1^2 2}{012 \cdot 1^2 2}, = [012];$$

or as for greater simplicity I write it

$$= 012,$$

viz. 012 is now written instead of $[012]$ to denote the function just given as the value of $\frac{\frac{1}{2}\Omega_{12}}{012}$. Ω_{12} is thus $= r \cdot 012 \cdot 012$, viz. this is a particular form of Ω_{12} satisfying the conditions that $\Omega_{12} = 0$ is a conic passing through the residues of the points 1, 2, and such that Ω_{12} on writing therein (x_1, y_1, z_1) for (x, y, z) becomes $= s \cdot 0121^22$ + arbitrary linear function (x, y, z) . The before-mentioned function $\Omega(0; 1, 2; 3, 4, 5)$ is a function satisfying these conditions 1Ω as $= s \cdot 0121^22$ + arbitrary linear function (x, y, z) , of (x_1, y_1, z_1) , of the points of of (x, y, z) , which presents itself in the case of the cubic curve.

[book p. 146] we find, by a process such as that of No. 62,

$$\begin{aligned}\partial_1 [013] &= \frac{1}{a} \{-01^3 \cdot 01^2 \cdot 02 + 09^2 \cdot j\} \\ &+ \frac{1}{q} \{2 \cdot 01^3 (03^2 \cdot 01^2 \cdot 2 + (013)^2)\}_\beta \\ &+ \frac{1}{q} \{(2 \cdot 01^3 \cdot 12) \cdot (012 \cdot 03^2 + 2(2 \cdot 013 \cdot 01^2 \cdot 01^2 \cdot 2 + 01^2 \cdot 03))\}_\beta \\ &+ \frac{1}{q^2} \{-2 \cdot 01^3 (-01^2 \cdot 01^2 + 01^2 \cdot 03^2)\}.\end{aligned}$$

Substituting herein the value $01^3 = \frac{1}{a^3}(ha + jr)$, &c., we have $\frac{1}{a^3}$ multiplied by a cubic function (ρ, σ, τ) ; writing down very simply the integral terms, and then the others, we have

$$\begin{aligned}\partial_1 [013] &= \frac{1}{a^3} \{(-6hn + 3jm) + \frac{1}{q}(8hkl - 3qjp + 3jp + 2jh)\rho\sigma \\ &+ \frac{1}{q^2}(\frac{15}{4}h^2 - 7hp + 8jm)\rho^2 + \frac{1}{q}(-2pjk + 8jh - 5jp + 4h^2)\rho\sigma + \frac{1}{q^2}(2j^2hl - 2pjn - 2pjm + 2j^2n)\} \\ &(\text{say this linear function of } \rho, \sigma, \tau \text{ is } = \square)\end{aligned}$$

$$+ \frac{1}{\Delta^2 p^2} [ar^3 \cdot 2j^2 + p^2\sigma\rho + p^2\sigma\rho + p\sigma\tau + 6jp + 6jp + 6jn\sigma] - *$$

70. The expression of $\partial_1 [103]$ is deduced from this by the interchange of 0, 1: and I write

$$\begin{aligned}\partial_1 [013] - \partial_1 [103] &= \square - * \\ &+ \frac{1}{a^3 \Delta! p} \{p^2 \rho^2 2j^2 + p^2 \sigma 3jh + p^2 \tau 3l + p^2 \sigma 3lk + p\rho\sigma + p\sigma\tau + p^2 \rho + p\sigma\tau + 2j\sigma\tau 3jn\} \\ &- \Delta^2 \Delta \cdot 2(09\bar{1})\sigma - \Delta! \sigma (-3 \cdot 01^2 \cdot 02^2 + 0\bar{2}^2 \cdot 025) - \Delta! 3 \cdot 01^3 \cdot 01^2 \\ &+ \Delta\sigma (-2 \cdot 01^3 \cdot 02 + 0\bar{2}^2 (02 + 01\bar{2})) - 2 \cdot 01^3 \cdot 01^2\end{aligned}$$

where, and in what follows, the $*$ denotes the function included immediately to the left of it, interchanging therein the 0, 1. It will be observed that the $\square$, and linear function of (ρ, σ, τ) , that is, of (x, y, z) , is a series of the required function $\partial_1 (0; 1, 3)$: the remaining portion has to be denoted by means of the expressions for $\Delta^2(013)$, &c., in terms of ρ, σ, τ .

71. We obtain

$$\begin{aligned}\partial_1 [013] - \partial_1 [103] &= \square - * \\ &+ \frac{1}{\Delta!} [\sigma(2fj - 3kp - 3hq + 15lm - 9nr) + (-3gr - 3jp + 9mq)] \\ &+ \frac{1}{\Delta! p} [\sigma^2(12fk - 12hn + 18m^2 - 9pr) + \sigma\tau(6fp - 6jr + 18nq) + \tau^2 \cdot 4n\sigma] \\ &+ \frac{1}{\Delta! p^2} [\rho^3(18gn - 9pkp) + \rho^2\sigma(12hp - 8tk + 9mqp) + \sigma\tau(4fp + 6lm)] \\ &+ \frac{1}{\Delta! p^2} [\rho^3 \cdot 4f^2 + \rho^2\sigma \cdot 6fp + \rho\sigma \cdot 4fj].\end{aligned}$$

[book p. 147] The terms of the second line may be transformed as follows:

$$\frac{\sigma}{\Delta!} (2fj - 3kp - 3hq + 18lm - 9nr)$$

$$\begin{aligned}
&= \frac{1}{2} \frac{\sigma}{\Delta!} (2fj - 3kp - 3hq + 18lm - 9nr) - * \\
&+ \frac{1}{\Delta!p} [\sigma^2 (-12fk + 12hn - 18m^2 + 9pr) + \sigma\tau \{ \frac{1}{2}fp + 3pk + \frac{3}{4}jr - \frac{3}{4}(p - 9nm) \}] \\
&+ \frac{1}{\Delta!p^2} [\sigma^3 (-20fm + 15hp) + \sigma^2\tau (-\frac{3}{2}fr + 12lk - 18np) + \sigma\tau^2 - 6\sigma p] \\
&+ \frac{1}{\Delta!p^2} [\sigma^2\tau (-10fp^2 + a^2\tau - 15fp + a^2\tau - 4mn^2) a\sigma\tau + a\sigma\tau a^2\tau + a\sigma\tau a\sigma\tau - 6p] \\
&+ \frac{1}{\Delta!p^3} [\sigma^2\tau (-10fn^2 + a^2\tau - 12fp + 8ik - 9im\rho) a\sigma\tau - 9ip],
\end{aligned}$$

and

$$\begin{aligned}
&\frac{1}{\Delta!} (-3gr - 3jp + 9mq) \\
&= \frac{1}{2} \frac{1}{\Delta!} (-3gr - 3jp + 9mq) - * \\
&+ \frac{1}{\Delta!p} [\sigma\tau (-\frac{3}{2}fq + \frac{3}{2}lr + \frac{3}{4}(p - 9nm) + \frac{1}{2}\sigma\tau (p - 9nm)) + \tau^2 - 6\sigma p] \\
&+ \frac{1}{\Delta!p^2} [\sigma^2\tau (-6fp + a^2\tau - 12fp + a^2\tau - 6fq + a^2\tau - 6fq + a^2\tau) a\sigma\tau - 6jp] \\
&+ \frac{1}{\Delta!p^3} [\sigma^2\tau (-6fp + a^2\tau - 12fp + a^2\tau - 6fq + a^2\tau) a\sigma\tau - 6jp],
\end{aligned}$$

substituting these values, the whole third line is destroyed, and we find

$$\begin{aligned}
&\partial_1 [013] - \partial_1 [103] = \square - * \\
&+ \frac{1}{2} \cdot \frac{1}{\Delta!} [\sigma (2fj - 3hq - 3kp + 18lm - 9nr) + \tau (-3gr - 3jp + 9mq)] - * \\
&+ \frac{1}{\Delta!p^2} [\sigma^2 (-12fm + 6fp) + \sigma\tau (-9fn + 7lk - 9mp - 9np) + \sigma\tau a^2\tau (-2fp - 9np) + \sigma^2\tau - 5\sigma p] \\
&+ \frac{1}{\Delta!p^3} [\sigma^2 (-6f^2 + \sigma^2\tau a^2\tau - 12fp + a^2\tau - 6fq + a^2\tau) a\sigma\tau - 6\sigma p].
\end{aligned}$$

Ultimately the last two lines of this expression are found to be

$$= \frac{1}{\Delta!} [\rho (-2hn + 4jm + 2l^2 - 2qr) + \sigma (-2fj + 2kp + 2hq - 10lm + 5nr) + \tau (2gr + hi - jp + 4hr - 6mq)] - *$$

so that the whole is now a sum of three linear functions of (ρ, σ, τ) . - *

19-2

[book p. 148] **72.** Collecting the terms, we have

$$\begin{aligned}
&\partial_1 [013] - \partial_1 [103] = \\
&\frac{1}{\Delta!} \left\{ \sigma \left[\rho (-8hn + 5jm + 2l^2 - 2qr) + \frac{1}{q} (4phl - 3qjr - 3lhj + 2jh) + \frac{1}{q^2} (-2jhk + 4qhjn - 2jnl) \right] \right. \\
&+ \sigma \left[\frac{1}{q} (-4fp + 4jq - 2lm + nr) + \frac{1}{q} (2fhi - 3jp - 9jp + 6jpm + 2lr) + \frac{1}{q^2} (2klm^2 - 8ik - 18np - 2lhr) \right] \\
&+ \sigma \left[\frac{1}{q} ((gr + 2hi + jp + n + 14hi) + \frac{1}{q} (-2pkn + 2lkjn - 2pnm) + \frac{1}{q^2} (2pkl - 2qhi - 2qjm + 2j^2n)) \right. \\
&\quad \left. \left. - * \right\}
\end{aligned}$$

73. The right-hand side depends on the points 0, 1, 3 and 2: viz. we have therein $\rho = 023$, $\Delta = 123$, &c., but the left-hand side depending on only the points 0, 1 and 2, the right-hand side cannot really contain 3, and it must then remain unaltered, if for 2 we substitute as another function of 0, 1, 3 and 5, viz. ρ , Δ , f , &c., will become 063, 163, 0/3, &c.; we have then $\phi(0, 1, 3) =$ the above linear function with 2 thus replaced by 6; say

$$\phi(0, 1, 3) = \frac{1}{\Delta!} [\rho() + \sigma() + \tau()],$$

a given function of the points 0, 1, 3 and the arbitrary point 6, on the quartic curve; we therefore write it $\phi(0; 1, 3, 6)$. There is no obvious value for $X(0, 1, 3)$ which will produce any simplification: I therefore take this function to be $= 0$; and the final result is

$$Q_{k,01} = [012] + \Theta_{k,12} = -([025] + [031] + [012]),$$

where $\Theta_{k,12}$ is a function integral and linear as regards the coordinates (x, y, z) of the point 0, but transcendental as regards the parametric points 1, 2; and containing besides the arbitrary points 3, 6, of the quartic curve, its value being determined by the differential formulae

$$\partial_1 \Theta_{k,12} = \phi(0', 1, 3, 6), \quad \partial_2 \Theta_{k,12} = -\phi(0', 2, 3, 6),$$

where $\phi(0', 1, 3, 6)$ is a given function as above. I do not see the meaning of the very complicated linear function of (ρ, σ, τ) , nor how to reduce it to any form such as the simple one $\partial_1 [036]$, which presents itself in the case of the cubic curve.

Cambridge, England, October 5, 1882.

Chapter IV. The Major Function $(x, y, z)_k^{n-3}$ continued.

The Conversion, Fixed Curve a Quartic, continued. Art. Nos. 74 to 82.

[book p. 149] **74.** I resume the question considered *ante* Nos. 66 to 73. The general problem, where the fixed curve is any given curve whatever, has recently been solved in a very complete and elegant form by Dr Nöther, in the two notes “Zur Reduction algebraischer Differentialausdrücke auf die Normalformen” and “Ueber die algebraischen Differentialausdrücke, 2^e Note,” *Sitzungsb. der phys.-med. Soc. zu Erlangen*, 10 Dec. 1883 and 14 Jan. 1884. I consider here the case of the quartic curve, $n = 4$, and connect his result with my former investigations.

We have the differential

$$\frac{(x, y, z)_k^{n-3} d\omega}{012} = \frac{\Omega_{12} d\omega}{012},$$

where Ω_{12} , or as I also write it $\Omega(0; 1, 2; 3, 4, 5)$, is a rational and integral function of the degree $(n-2) = 2$ in the current coordinates (x, y, z) : it depends also on the parametric points 1, 2, which are points on the quartic, coordinates (x_1, y_1, z_1) , (x_2, y_2, z_2) respectively; and on $(p =) 3$ other points 3, 4, 5 are arbitrary points which we take to be in fact the points (x_3, y_3, z_3) , (x_4, y_4, z_4) , (x_5, y_5, z_5) respectively. The curve $\Omega = 0$ is a conic, passing through the points 3, 4, 5 and through the $(n-2) = 2$ residues of the parametric points 1, 2, of the quartic curve, $n = 4$, and connect his result with my former investigations.

We have hence

$$\Omega(0; 1, 2; 3, 4, 5) = \frac{1}{\Delta} (a 1^{n-2} 2 1 1 1, \dots) (x, y, z)^{n-2}; \text{ see No. 2.}$$

the function determined by the equation

$$\begin{vmatrix} (x, y, z)^2 & , & \Omega \\ 1(x_1, y_1, z_1)^2 & , & 4.1^3 2 \\ 2(x_1, y_1, z_1 \xi x_2, y_2, z_2), 6.1^2 2^2 & , & 4.1 2^3 \\ 1(x_2, y_2, z_2)^2 & , & 0 \\ (x_3, y_3, z_3)^2 & , & 0 \\ (x_4, y_4, z_4)^2 & , & 0 \\ (x_5, y_5, z_5)^2 & , & 0 \end{vmatrix} = 0,$$

$$^* \left(x_a \frac{d}{dx_b} + y_a \frac{d}{dy_b} + z_a \frac{d}{dz_b} \right) (x_b, y_b, z_b)^{n-2} = \text{see No. 2.}$$

[book p. 150] this is of the form

$$M\Omega + \square = 0,$$

and as appears in No. 46, $M = 123.124.125.345$. Hence writing $\square(0; 1, 2; 3, 4, 5)$ for $\square$, we have

$$\Omega_n = \Omega(0; 1, 2; 3, 4, 5) = \frac{-\square(0; 1, 2; 3, 4, 5)}{123.124.125.345}.$$

Hence further writing

$$Q_n = Q(0; 1, 2; 3, 4, 5) = \frac{\Omega(0; 1, 2; 3, 4, 5)}{012},$$

so that the differential is $Qd\omega, = Q(0; 1, 2; 3, 4, 5) d\omega$, we have

$$Q = Q(0; 1, 2; 3, 4, 5) = \frac{-\square(0; 1, 2; 3, 4, 5)}{012.123.124.125.345},$$

which is of the form

$$Q = \frac{012\overline{2345}}{012\overline{2345}}, = 012\overline{2345},$$

viz. Q is a rational fraction where the numerator is of the degree 2, and the denominator of the degree 5 as regards the coordinates (x, y, z) of the current point; but the numerator and denominator are each of the degree 4 as regards the coordinates of the points 1, 2 separately; and the degree 2 as regards the coordinates of the points 3, 4, 5 separately.

76. The signification of the symbol of quasi-differentiation ∂ (applicable only to a function of the degree 0 in the coordinates to which the differentiations have reference) is explained *ante* No. 60. The function Q just mentioned is of the degree 0 in regard to the coordinates of each of the points 1, 2, 3, 4, 5; and it can thus be operated upon by the symbols $\partial_1, \partial_2, \partial_3, \partial_4, \partial_5$ respectively. Observe, in particular, that we have $\partial_3 Q(0; 1, 2; 3, 4, 5) = 012\overline{2345}$, viz. it is of the degree 1 in the coordinates of the points 0 and 1 respectively, but of the degree 0 in regard to the coordinates of the points 2, 3, 4, 5 respectively.

77. This being so, we may consider the function

$$\begin{aligned} H(0; 1, 2; 3, 4, 5; 6, 7, 8) &= \partial_3 Q(0; 1, 2; 3, 4, 5) \\ &\quad + \partial_3 Q(1; 3, 2; 6, 4, 5) \cdot \frac{045}{345} \\ &\quad + \partial_4 Q(1; 4, 2; 3, 7, 5) \cdot \frac{053}{453} \end{aligned}$$

$$+\partial_5 Q(1; 5, 2; 3, 4, 8) \cdot \frac{034}{534},$$

where 6, 7, 8 are arbitrary points on the quartic; the functions

$$\partial_3 Q(1; 3, 2; 6, 4, 5), \partial_4 Q(1; 4, 2; 3, 7, 5), \partial_5 Q(1; 5, 2; 3, 4, 8),$$

[book p. 151] are functions of the same form as $\partial_3 Q(0; 1, 2; 3, 4, 5)$, and derived from it by changing in each case the current point 0 into the parametric point 1, and the points 3, 4, 5 respectively, and using the corresponding point 3, 4 or 5 by the new arbitrary point 6, 7 or 8. Further 045, &c., denote determinants as above; so that in H each of the last three terms is, in fact, as regards the point 0, a mere linear function of the coordinates (x, y, z) of this point.

We have $\partial_1 Q(1; 3, 2; 6, 4, 5) = 13'\overline{2456}$, and hence this function multiplied by $\frac{045}{345}$ is $= 01'\overline{2456}$; and so for the third and fourth terms of H : thus each of the four terms of H is $= 01'\overline{2456789}$, of the degree 1 in the coordinates of the points 0 and 1 respectively, but of the degree 0 in regard to the coordinates of the other points 2, 3, 4, 5, 6, 7, 8.

Nöther's conversion-theorem consists herein, that the function

$$H(0; 1, 2; 3, 4, 5; 6, 7, 8)$$

is unaltered by the interchange of the two points 0, 1; or putting for shortness

$$H(0; 1, 2; 3, 4, 5; 6, 7, 8) = H_1(0),$$

the theorem is

$$H_1(0) = H_0(1).$$

78. We have, No. 59,

$$\frac{\frac{1}{2}\Omega_{12}}{012} = \frac{-01^3 \cdot 02^2 + 01^2 2 \cdot 012 + 01 2 \cdot 1^2 2}{012 \cdot 1^2 2}, = [012],$$

or as for greater simplicity I write it

$$= 012,$$

viz. 012 is now written instead of $[012]$ to denote the function just given as the value of $\frac{\frac{1}{2}\Omega_{12}}{012}$. Ω_{12} is thus $= r \cdot 012 \cdot 012$, viz. this is a particular form of Ω_{12} satisfying the conditions that $\Omega_{12} = 0$ is a conic passing through the residues of the points 1, 2 on writing therein (x_1, y_1, z_1) for (x, y, z) becomes $= s \cdot 012 1^2 2 +$ arbitrary linear function (x, y, z) . The before-mentioned function $\Omega(0; 1, 2; 3, 4, 5)$ is a function satisfying these conditions on writing therein for 4, 5, 6 in succession the values 3, 4, 5, becomes $= 0$; we have therefore

$$\frac{\frac{1}{2}\Omega(0; 1, 2; 3, 4, 5)}{012} = 012 - 012 \frac{034}{345} - 012 \frac{054}{354} - 012 \frac{045}{354},$$

viz. the value of Q given by this equation, on writing therein $0 = 3, 4$, or 5 , becomes $= 0$ as it should do.

[book p. 152] **79.** We thus have

$$Q(0; 1, 2; 3, 4, 5) = 012 - 012 \frac{045}{345} - 012 \frac{053}{453} - 012 \frac{034}{534},$$

and Nöther's conversion-equation becomes

$$\partial_1 \left\{ 012 - 012 \frac{045}{345} - 012 \frac{053}{453} - 012 \frac{034}{534} \right\}$$

$$\begin{aligned}
& +\partial_3 \left\{ 1'02 - 6'32 \frac{145}{645} - 4'32 \frac{156}{456} - 5'32 \frac{164}{564} \right\} \frac{045}{345} \\
& +\partial_4 \left\{ 1'02 - 3'42 \frac{175}{375} - 7'42 \frac{153}{753} - 5'42 \frac{137}{573} \right\} \frac{053}{453} \\
& +\partial_5 \left\{ 1'02 - 3'52 \frac{148}{348} - 4'52 \frac{183}{483} - 8'52 \frac{134}{834} \right\} \frac{034}{534} \\
& = \partial_0 \left\{ 1'02 - 6'32 \frac{145}{645} - 4'32 \frac{156}{456} - 5'32 \frac{164}{564} \right\} \\
& +\partial_3 \left\{ 0'62 - 6'32 \frac{045}{645} - 4'32 \frac{056}{456} - 5'32 \frac{064}{564} \right\} \frac{045}{345} \\
& +\partial_4 \left\{ 1'02 - 3'72 \frac{075}{375} - 7'42 \frac{053}{753} - 5'72 \frac{037}{573} \right\} \frac{053}{453} \\
& +\partial_5 \left\{ 0'82 - 3'82 \frac{048}{348} - 4'82 \frac{083}{483} - 8'82 \frac{034}{834} \right\} \frac{034}{534},
\end{aligned}$$

an equation where the functions operated on with the ∂ 's are only functions such as 012; for there is not any determinant operated upon containing the number which is the suffix of the ∂ operating upon it.

80. Taking all the terms over to the left-hand side, there are in all 32 terms; but of these 3 + 3 destroy each other, and 6 + 6 unite in pairs into 6 terms: there are thus in all 7 + 7 + 6, = 20 terms; viz. multiplying the whole equation by 345, it is found that the equation becomes:

as so this may be written where

$$\begin{aligned}
345 (\partial_1 012 - \partial_0 1'02) & \quad 345 \boxed{012} & = \partial_1 012 - \partial_0 1'02, \text{ \&c.} \\
-045 (-\partial_1 102 + \partial_0 012) & \quad -045 \boxed{312} \\
-053 (-\partial_1 142 + \partial_0 012) & \quad -305 \boxed{412} \\
-034 (-\partial_1 152 + \partial_0 012) & \quad -340 \boxed{512}
\end{aligned}$$

[book p. 153]

$$\begin{aligned}
-145 (\partial_1 032 - \partial_2 302) & \quad -145 \boxed{032} \\
-153 (\partial_1 042 - \partial_2 402) & \quad -315 \boxed{042} \\
-134 (\partial_1 052 - \partial_2 502) & \quad -341 \boxed{052} \\
+013 (-\partial_1 542 + \partial_2 452) & \quad +301 \boxed{452} \\
+014 (-\partial_1 352 + \partial_2 532) & \quad +140 \boxed{532} \\
+015 (-\partial_1 432 + \partial_2 342) & \quad +015 \boxed{342} \\
= 0, & \quad = 0,
\end{aligned}$$

viz. the equation is

$$\Sigma \pm 345 \boxed{012} = 0.$$

the nine terms which follow the first term $345 \boxed{012}$ of the sum being obtained by the interchanges of 0, 1 (one or each) with the 3, 4, 5, each interchange giving rise to a sign -.

81. In obtaining the foregoing result, we have, for instance, a pair of terms

$$\partial_1 542 \frac{-137 \cdot 035 + 037 \cdot 135}{537}, = \partial_1 542 \frac{-013 \cdot 537}{537}, = \partial_1 542 (-013),$$

viz. this depends on the equation

$$-137.035 + 037.135 - 537.013 = 0,$$

or say

$$-137.035 + 037.135 + 013.357 = 0,$$

an identity which, in a form which will be readily understood, may be written

$$\det \begin{vmatrix} 0137 \\ 013735 \end{vmatrix} = 0.$$

Similarly, the two terms which contain $\partial_3 452$ combine into the single term $\partial_3 452 (013)$; and the two new terms taken together are

$$013 (-\partial_3 542 + \partial_2 452) = 301 \boxed{452}.$$

C. XII.

20

[book p. 154] **82.** The proof of the identity,

$$\Sigma \pm 345 \boxed{012} = 0,$$

depends on the property of the function

$$\boxed{012}_1 = \partial_1 012 - \partial_0 1'02,$$

enunciated No. 67, and proved *à posteriori* by the tedious calculation Nos. 69 to 73, viz. in No. 67, writing 2 in place of 3, this is— $\partial_1 012 - \partial_0 1'02$ is equal to the difference of two functions, the first of them linear in the coordinates (x, y, z) of the point 0, but depending also on the coordinates of the points 1 and 2, the second of them linear in the coordinates (x_1, y_1, z_1) of the point 1, but depending also on the coordinates of the points 0 and 2. Or, what is the same thing, the property is

$$\boxed{012}_1 = A_{12}x + B_{12}y + C_{12}z - (A_{01}x_1 + B_{01}y_1 + C_{01}z_1),$$

where A_{12}, B_{12}, C_{12} are functions of $(x_1, y_1, z_1), (x_2, y_2, z_2)$, and A_{01}, B_{01}, C_{01} are the like functions of $(x, y, z), (x_2, y_2, z_2)$.

Substituting such values in the sum $\Sigma \pm 345 \boxed{012}$, but writing down only the terms which contain x , these are

$$\begin{aligned} & 345 (A_{12}x - A_{01}x_1) \\ & - 045 (A_{12}x_1 - A_{01}x_1) \quad - 145 (A_{02}x - A_{20}x_0) \quad + 301 (A_{42}x_4 - A_{42}x_5) \\ & - 305 (A_{42}x_1 - A_{01}x_4) \quad - 315 (A_{02}x - A_{40}x_0) \quad + 140 (A_{52}x_5 - A_{52}x_3) \\ & - 340 (A_{52}x_1 - A_{01}x_5) \quad - 341 (A_{02}x - A_{50}x_0) \quad + 015 (A_{32}x_3 - A_{32}x_4), \end{aligned}$$

This is

$$\begin{aligned} & = A_{12} (x.345 - x_1.015 + x_2.015) \\ & + A_{02} (x.345 - x_0.345 - x_2.340) \\ & + A_{42} (x_4.145 - x_1.305 + x_0.340) \\ & + A_{52} (x_5.134 + x_1.340 + x_0.140) \\ & + A_{01} (-x_1.345 + x_4.305 + x_5.340) \\ & + A_{42} (-x.341 + x_5.140 + x_3.015), \end{aligned}$$

where the coefficient of each of the A 's is identically $= 0$; and similarly, the terms in y and the terms in z are each $= 0$. We have thus the proof of the identity

$$\Sigma \pm 345 \begin{bmatrix} 012 \end{bmatrix} = 0,$$

that is, of the conversion-equation $H_1(0) = H_0(1)$.

[book p. 155] *The Syzygy—Fixed Curve a Quartic.* Art. No. 83.

83. I revert to the theory of the Syzygy, *ante* No. 59.

We have

$$Q(0; 1, 2; 3, 4, 5) = 012 - 012 \frac{045}{345} - 012 \frac{053}{453} - 012 \frac{034}{534},$$

or if for convenience we take instead of 1, 2, the parametric points to be a, β coordinates (x_a, y_a, z_a) and $(x_\beta, y_\beta, z_\beta)$ respectively, then this equation is

$$Q_{a\beta} = Q(0; a, \beta; 3, 4, 5) = 0a\beta - 0a\beta \frac{045}{345} - 0a\beta \frac{053}{453} - 0a\beta \frac{034}{534}.$$

Considering a new parametric point γ , and forming the like functions $Q_{\beta\gamma}$ and $Q_{\gamma a}$, it is to be shown that we have identically

$$Q_{a\beta} + Q_{\beta\gamma} + Q_{\gamma a} = 0.$$

To prove this, observe that, in the equation at the end of No. 59, $\Delta, \rho, \sigma, \tau$ denote 123, 023, 031, 012 respectively. Hence writing therein a, β, γ in place of 1, 2, 3 respectively, and putting A, B, C for the coefficients (including therein the factor $\frac{1}{\Delta!}$) of ρ, σ, τ respectively, the equation is

$$0\bar{\beta}\gamma + 0\gamma a + 0a\beta = A.0\beta\gamma + B.0\gamma a + C.0a\beta;$$

where A, B, C are absolute constants (functions, that is, of the coefficients of the quartic) each divided by $(a\beta\gamma)^6$. We hence obtain

$$\begin{aligned} (Q_{\beta\gamma} + Q_{\gamma a} + Q_{a\beta}).345 &= 345(A.0\beta\gamma + B.0\gamma a + C.0a\beta) \\ &\quad - 045(A.3\beta\gamma + B.3\gamma a + C.3a\beta) \\ &\quad - 053(A.4\beta\gamma + B.4\gamma a + C.4a\beta) \\ &\quad - 034(A.5\beta\gamma + B.5\gamma a + C.5a\beta). \end{aligned}$$

On the left-hand side the whole coefficient of A is $= 0$; viz. the coefficient has the value $\det \begin{vmatrix} 0345 \\ 0345\beta\gamma \end{vmatrix}$, which is $= 0$. Similarly, the whole coefficient of B is $= 0$, and the whole coefficient of C is $= 0$; and we have thus the required result

$$Q_{\beta\gamma} + Q_{\gamma a} + Q_{a\beta} = 0.$$

The syzygy is thus obtained in a more perfect form than in No. 59; viz. by considering (instead of $0\bar{a}\beta$) the new form $Q_{a\beta}$ then, instead of a sum which is a linear function of the coordinates (x, y, z) , we obtain a sum $= 0$. 20-2

THE COLLECTED MATHEMATICAL PAPERS OF ARTHUR
CAYLEY
VOL. XII

SCAN PDF PAGES 176–225 (BOOK PP. 156–205; PAPER 825, CHS. IV CONCL.–VII)

Arthur Cayley (1821–1895)

Published 1889

*Transcribed from the original Cambridge University Press edition, 1889.
This file covers scan PDF pages 176–225 (book pages 156–205); restored from scan.*

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825. A Memoir on the Abelian and Theta Functions (continued)

Chapter IV (concluded). The Fixed Curve a Cubic — Syzygy and Conversion.

The Fixed Curve a Cubic — Syzygy and Conversion. Art. Nos. 84 and 85.

[book p. 156] **84.** In the case when the fixed curve is a cubic (see Nos. 58 and 64), the analogous formulæ are

$$\frac{1}{2}\Omega(0; 1, 2; 3) = \frac{1 \cdot 2 \cdot 023 + 12 \cdot 031}{12s} = \frac{\{012\}}{12s} \{012\} \quad (\text{see No. 49}),$$

that is,

$$Q(0; 1, 2; 3) = \frac{\{012\}}{012} \cdot \frac{\{012\}}{12s} = \{012\} = \{1\overline{2}3\},$$

where 1, 2 are the parametric points: 3 any other point on the cubic: the brackets $\{ \}$ are of course here necessary in order to distinguish $\{012\}$ from the determinant 012. It will be remembered that $\{012\}$ is an alternate function

$$\{012\} = -\{102\} = \{120\}, \text{ \&c.}$$

If instead of 1, 2 we take the parametric points to be α, β , coordinates $(x_\alpha, y_\alpha, z_\alpha)$ and $(x_\beta, y_\beta, z_\beta)$ respectively, then the formula is

$$Q_{\alpha\beta} = \frac{1}{2} Q(0; \alpha, \beta; 3) = \{0\alpha\beta\}\{3\alpha\beta\}.$$

Hence taking on the cubic a new point γ , coordinates $(x_\gamma, y_\gamma, z_\gamma)$ and forming the functions $Q_{\beta\gamma}$ and $Q_{\gamma\alpha}$ we have

$$Q_{\alpha\beta} + Q_{\beta\gamma} + Q_{\gamma\alpha} = \{0\beta\gamma\}\{3\beta\gamma\} + \{0\gamma\alpha\}\{3\gamma\alpha\} + \{0\alpha\beta\}\{3\alpha\beta\}.$$

But by the formula No. 58,

$$\{0\beta\gamma\} + \{0\gamma\alpha\} + \{0\alpha\beta\} = 0;$$

hence also

$$\{3\beta\gamma\} + \{3\gamma\alpha\} + \{3\alpha\beta\} = 0;$$

and we have thus

$$Q_{\alpha\beta} + Q_{\beta\gamma} + Q_{\gamma\alpha} = 0,$$

the syzygy for the cubic.

85. For the conversion, the definition of H is

$$H(0; 1, 2; 3, 6) = \partial_3 Q(0; 1, 2; 3) - \partial_6 Q(0; 3, 2; 6),$$

viz. this is

$$\begin{aligned} H &= H(0; 1, 2; 3, 6) = \partial_3(\{012\} - \{123\}) + \partial_6(\{132\} - \{326\}), \\ &= \partial_3\{012\} - (\partial_3 + \partial_6)\{123\} - \partial_6\{326\}, \end{aligned}$$

which, in virtue of $(\partial_1 + \partial_2 + \partial_3 + \partial_6)\{123\} = 0$ (see No. 63), becomes

$$H = H(1) = (\partial_1 + \partial_2)\{012\} - \partial_2\{123\} - \partial_6\{326\}.$$

[book p. 157] Interchanging the 0 and 1, we thence have

$$H_1(0) = \partial_0\{102\} + \partial_2\{023\} - \partial_6\{326\}.$$

Hence the difference $H_1(1) - H_1(0)$ is

$$= \partial_1\{012\} - \partial_0\{102\} + \partial_2\{123\} - \partial_2\{023\},$$

viz. this is

$$= (\partial_1 - \partial_0)\{012\} + \partial_2(\{123\} - \{023\}),$$

where the first term is

$$= -\partial_2\{012\},$$

and the whole therefore is

$$\begin{aligned} &= \partial_2(\{123\} - \{230\} - \{012\}), \\ &= -\partial_2\{301\}, \end{aligned}$$

in virtue of

$$\{123\} - \{230\} + \{301\} - \{012\} = 0;$$

the whole is consequently $= 0$.

We have thus

$$H_1(1) - H_1(0) = 0,$$

the conversion-equation in the case of the cubic.

Chapter V. Miscellaneous Investigations.

The Differential Symbol $d\omega$. Art. Nos. 86 and 87.

86. The definition is

$$\frac{y \, dz - z \, dy}{\frac{df}{dx}} = \frac{z \, dx - x \, dz}{\frac{df}{dy}} = \frac{x \, dy - y \, dx}{\frac{df}{dz}} = d\omega,$$

and it hence follows that we have

$$d\omega = \frac{\begin{vmatrix} dx & dy & dz \\ x & y & z \\ \lambda & \mu & \nu \end{vmatrix}}{\lambda \frac{df}{dx} + \mu \frac{df}{dy} + \nu \frac{df}{dz}},$$

where (λ, μ, ν) are arbitrary constants or, if we please, arbitrary functions of (x, y, z) : viz. the expression just written down is altogether independent of the values of λ, μ, ν , and is consequently equal to the value obtained by writing any two of these symbols $= 0$, that is, the expression is equal to any one of the foregoing three equal values of $d\omega$. The expression was first given by Aronhold (1863), in the memoir presently referred to.

[book p. 158] It is to be remarked that, considering (λ, μ, ν) as the coordinates of a point, the denominator $\lambda \frac{df}{dx} + \mu \frac{df}{dy} + \nu \frac{df}{dz}$ equated to 0 is the polar $(n-1)$ thic of the point λ, μ, ν in regard to the fixed curve.

If instead of λ, μ, ν we write $bc' - b'c, ca' - c'a, ab' - a'b$, where (a, b, c) (a', b', c') are constants, then the numerator is

$$= (ax + by + cz)(a' \, dx + b' \, dy + c' \, dz) - (a'x + b'y + c'z)(a \, dx + b \, dy + c \, dz),$$

or introducing ρ, σ to denote the arbitrary linear functions $ax + by + cz$ and $a'x + b'y + c'z$ respectively, the numerator is $= \rho d\sigma - \sigma d\rho$: moreover, observing that a, b, c and a', b', c' are the differential coefficients of ρ, σ in regard to the coordinates (x, y, z) , the denominator is $= J(f, \rho, \sigma)$; and the value of $d\omega$ is

$$d\omega = \frac{\rho d\sigma - \sigma d\rho}{J(f, \rho, \sigma)},$$

where, in accordance with a previous remark, the denominator equated to 0 is the polar $(n - 1)$ th of the intersection of the lines $\rho = 0, \sigma = 0$ in regard to the fixed curve.

Obviously, by taking for ρ, σ any two of the three coordinates x, y, z , we reproduce the original three forms of $d\omega$.

87. The last-mentioned form of $d\omega$ suggests the expression for this symbol in the case where the fixed curve, instead of being a plane curve, is a curve of double curvature defined by two equations $f = 0, g = 0$ between the four coordinates (x, y, z, w) : viz. ρ, σ being now arbitrary linear functions

$$ax + by + cz + dw, \quad \text{and} \quad a'x + b'y + c'z + d'w$$

of the four coordinates, the expression is

$$d\omega = \frac{\rho d\sigma - \sigma d\rho}{J(f, g, \rho, \sigma)};$$

and by taking for ρ, σ any two of the four coordinates x, y, z, w , we have for $d\omega$ six values which must of course be equal to each other; it is easy to verify

[book p. 159] *a posteriori* that this is so.

In the case where the curve of double curvature is not the complete intersection of two surfaces, the denominator (regarded as the Jacobian of the curve and of the arbitrary planes ρ, σ) will have a definite meaning, but what this is I do not at present consider.

The last-mentioned expression for $d\omega$ will be applied further on to the case of the quadri-quadric curve.

Integral Formulæ. Art. Nos. 88 to 90.

88. In what precedes, $d\omega$ has been used as a single symbol to denote any one of the equal differential expressions

$$\frac{y dz - z dy}{\frac{df}{dx}} = \frac{z dx - x dz}{\frac{df}{dy}} = \frac{x dy - y dx}{\frac{df}{dz}};$$

there is no quantity ω . The expressions are of the order $-(n - 3)$ in the coordinates (x, y, z) , and since (x, y, z) are as to their absolute magnitudes altogether arbitrary (only their ratios being determinate), a symbol such as

$$\omega = \int d\omega = \int \frac{y dz - z dy}{\frac{df}{dx}},$$

would, except in the case $n = 3$, be altogether meaningless. In fact, the integral would be

$$\int z^{n-3} \frac{d\left(\frac{y}{z}\right)}{\rho\left(\frac{y}{z}, \frac{x}{z}\right)} = \int \frac{d\left(\frac{y}{z}\right)}{z^{-(n-3)} \rho\left(\frac{y}{z}, \frac{x}{z}\right)},$$

where $\frac{x}{z}$ is, by the equation of the fixed curve, given as a function of $\frac{y}{z}$; but the other factor z^{n-3} is an absolutely indeterminate variable value, and the expression is meaningless.

But we have integrals $\int Q d\omega$, where Q is a homogeneous function of the order $n-3$ in the coordinates (x, y, z) ; and, in particular, we have such integrals where (corresponding to the forms which present themselves in the differential pure and affected theorems respectively) Q is either a rational and integral function $(x, y, z)^{n-3}$, or a rational and integral function $(x, y, z)^{n-2}$ divided by a linear function (x, y, z) : for in every such case, the form of integral is

$$\int \frac{d\left(\frac{y}{z}\right)}{\Phi\left(\frac{y}{z}, \frac{x}{z}\right)},$$

where $\frac{x}{z}$ is a given function of $\frac{y}{z}$, and the factor of $d\left(\frac{y}{z}\right)$ is thus a mere function of $\frac{y}{z}$. More definitely, in the integrals $\int Q d\omega$ which are considered, Q is either a minor function $(x, y, z)^{n-3}$, or it is the quotient of a major function $(x, y, z)^{n-2}$ by the linear function 012.

[book p. 160] In the case $n=2$, there is no rational and integral function $(x, y, z)^{n-3}$, but the function may be of the form belonging to the affected theorem, viz. it is unity divided by a linear function $(x, y, z)^2$; or say the integral is $\int \frac{d\omega}{ax + \beta y + \gamma z}$, where the (x, y, z) are connected by a quadric equation $(a, \dots)(x, y, z)^2 = 0$: it will be shown presently that this integral is obtainable as a logarithmic function.

In the case $n=3$, we have the rational and integral function $(x, y, z)^{n-3}$, a constant, or say $=1$, so that there is here an integral $\int d\omega$: we do not call this ω , but introducing a new letter, say u , and fixing at pleasure the inferior limit of the integral, we write $u = \int d\omega$.

89. In the foregoing form $\int Q d\omega$, so long as we retain the symbol $d\omega$, there is nothing to show what is the variable in regard to which the integration is to be performed; we may, for instance, writing

$$d\omega = \frac{y dz - z dy}{\frac{df}{dx}},$$

make it to be z , or in like manner to be any other of the six quotients. We thus cannot attribute a value to the inferior or superior limit of such an integral, but we may take the limits to be each of them a point on the fixed curve: for instance, if 0, 1 be points on the fixed curve, then the integral $\int_0^1 Q d\omega$ means the integral taken from the value at the point 0 to the value at the point 1 of the variable in regard to which the integration is performed; or when there is no expressed superior limit, then the integral is to be taken from the value for the expressed or known inferior limit to the value at the current point (x, y, z) of the variable in regard to which the integration is performed. The actual value of the integral will of course depend upon the path of the variable; but this is a question which is not here entered upon.

If using Cartesian Coordinates x, y , we write for instance

$$d\omega = \frac{dx}{\frac{df}{dy}}, \quad \text{then} \quad \int \frac{Q dx}{\frac{df}{dy}}$$

will denote an integral $\int \Phi x dx$ in regard to the variable x , and the inferior and superior limits will be as usual values of x ; or if there is no expressed superior limit, then the integral will be $\int_0^x \Phi x dx$ the integral taken from the inferior limit x_0 to the current value x .

[book p. 161] We may, if we please, consider the coordinates (x, y, z) as depending upon a parameter θ , viz. the ratios $x : y : z$ may be regarded as given functions of θ , and the integral $\int Q d\omega$ is then an integral $\int \Theta d\theta$, which taken from a constant inferior limit up to the value θ_1 belonging to a given point 1 of the curve, is a given function of θ_1 , or say of the point 1. But except in the case of the cubic (or generally $p = 1$), we do not have the coordinates actually given as known functions of a parameter θ (say they are potentially known functions of θ), and it is further to be noticed that the functions which present themselves are functions not of a single point, but of p or more points: thus in the case of the quartic, $n = 4$, $p = 3$; we have $\int_0^1 x d\omega$, $\int_0^1 y d\omega$, $\int_0^1 z d\omega$, each standing for a given function of the parameter θ_1 , but these integrals do not present themselves singly, but in combinations such as

$$\int_0^1 + \int_0^2 + \int_0^3 (x d\omega, y d\omega, z d\omega),$$

say these sums of integrals are u, v, w : each of the functions u, v, w is a potentially known function of the parameters $\theta_1, \theta_2, \theta_3$, which belong to the points 1, 2, 3, respectively, and is consequently regarded as a given function of these four points.

90. Consider as before, in the case of a cubic curve, the integral $u = \int d\omega$: it will presently be seen that for the general curve as given by a cubic equation $f = 0$ of any form whatever, we arrive at a form of elliptic function: but the ordinary elliptic functions $\text{sn}, \text{cn}, \text{dn}$ connect themselves most readily with the cubic curve $y^2 = x \cdot 1 - x \cdot 1 - k^2 x$. We have here

$$d\omega = \frac{dx}{2y} = \frac{dx}{2\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

or, in the equation $u = \int d\omega$, taking the inferior limit to be 0, say

$$u = \int_0^x \frac{dx}{2\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

an equation which determines u as a function of x , or conversely, x as a function of u . We might thence, by means of Abel's theorem as applied to the curve in question, investigate the properties of the function $x = \lambda(u)$ thus arising, and so establish the theory of elliptic functions: but it is more convenient, treating the elliptic functions as known functions, to write for λu its value; viz. to take for x as given by this equation, the value $x = \text{sn}^2 u$: we thence have $y = \text{sn} u \text{ cn} u \text{ dn} u$; viz. these values $x = \text{sn}^2 u$, $y = \text{sn} u \text{ cn} u \text{ dn} u$, satisfy the equation $y^2 = x \cdot 1 - x \cdot 1 - k^2 x$ of the curve, and give, moreover, $d\omega = du = \frac{dx}{2y}$: and we can with these values, and the formulæ for elliptic functions, verify any results given by Abel's theorem. This will be done in considerable detail: but at present I wish only to remark that the formulæ give the coordinates x, y of a point on the cubic curve expressed as one-valued functions of

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[book p. 162] a parameter or argument u : but that this argument u is not a one-valued function of the coordinate x , or even of the coordinates x, y of the given point on the curve: say the argument u has not a unique value for a given point (x, y) of the curve. There are, in fact, an infinity of values $u = u + 2mK + 2m'iK'$, where m, m' are any positive or negative integers: that this is so, depends on the multiplicity of values, according to the different paths of the variable, of the integral

$$\int \frac{dx}{2\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

or, regarding the elliptic functions as known functions, it depends upon the double periodicity of these functions.

Aronhold's Quadric Integral. Art. Nos. 91 to 93.

91. I reproduce the investigation contained in Aronhold's paper "*Ueber eine neue algebraische Behandlungsweise u.s.w.*," Crelle, t. LXI. (1863), pp. 95–145. We take the general quadric function $(a, b, c, f, g, h)(x, y, z)^2$: $\alpha x + \beta y + \gamma z$ an arbitrary linear function of x, y, z : the theorem is $\frac{d\omega}{\alpha x + \beta y + \gamma z} =$ differential of logarithm of an algebraic function of (x, y, z) ; viz. taking (ξ, η, ζ) for the coordinates of either of the points of intersection of the line $\alpha x + \beta y + \gamma z = 0$ with the quadric $(a, \dots)(x, y, z)^2 = 0$, and writing also

$$H = 2(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch)(\alpha, \beta, \gamma)^2,$$

then the theorem is

$$\frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\sqrt{H}} d \log \frac{(a, \dots)(x, y, z)(\xi, \eta, \zeta)}{\alpha x + \beta y + \gamma z},$$

or, what is the same thing,

$$\int \frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\sqrt{H}} \log \frac{(a, \dots)(x, y, z)(\xi, \eta, \zeta)}{\alpha x + \beta y + \gamma z} + \text{const.}$$

It is to be observed, in reference to this equation, that the two sides respectively are in regard to (α, β, γ) homogeneous functions of the degree -1 , and in regard to (ξ, η, ζ) homogeneous of the degree 0 ; viz. on the right-hand side the effect of a change in the absolute magnitudes of ξ, η, ζ , say the change into $k\xi, k\eta, k\zeta$, is merely to change by $\log k$ the constant of integration.

It is to be remarked also that the equation $(a, \dots)(\xi, \eta, \zeta)(x, y, z) = 0$ represents the tangent to the conic at the point (ξ, η, ζ) of intersection with the line $\alpha x + \beta y + \gamma z = 0$; calling the linear function in question T , the value of the integral is $\frac{1}{\sqrt{H}} \log \frac{T}{\alpha x + \beta y + \gamma z}$; the two (ξ_1, η_1, ζ_1) (ξ_2, η_2, ζ_2) are the coordinates of the points of intersection respectively, and the two values thus are $\frac{1}{\sqrt{H}} \log \frac{T_1}{\alpha x + \beta y + \gamma z}$, $\frac{1}{\sqrt{H}} \log \frac{T_2}{\alpha x + \beta y + \gamma z}$, then in passing from one of these to the other we change the sign of the radical $\sqrt{H}$, $\frac{1}{\sqrt{H}} \log \frac{T_1}{\alpha x + \beta y + \gamma z}$ and $-\frac{1}{\sqrt{H}} \log \frac{T_2}{\alpha x + \beta y + \gamma z}$. These must

[book p. 163] differ by a constant only: viz. we should have $\log \frac{T_1 T_2}{(\alpha x + \beta y + \gamma z)^2} = \text{a const.}$ And, in fact, T_1 and T_2 being the tangents to the conic f at its intersections with the line $\alpha x + \beta y + \gamma z = 0$, we have it is clear $f = \lambda T_1 T_2 + \mu(\alpha x + \beta y + \gamma z)^2$, that is, (x, y, z) referring to a point of the conic $f = 0$, we have $\frac{T_1 T_2}{(\alpha x + \beta y + \gamma z)^2} = \text{a constant}$, which is right.

92. We require the coordinates (ξ, η, ζ) of an intersection: these are determined by the equations $\alpha\xi + \beta\eta + \gamma\zeta = 0$, $(a, \dots)(\xi, \eta, \zeta)^2 = 0$, or as these may be written

$$\alpha\xi + \beta\eta + \gamma\zeta = 0,$$

$$\xi(a\xi + h\eta + g\zeta) + \eta(h\xi + b\eta + f\zeta) + \zeta(g\xi + f\eta + c\zeta) = 0;$$

we have thence ξ, η, ζ proportional to the determinants

$$\begin{array}{ccc} \alpha\xi + h\eta + g\zeta, & h\xi + b\eta + f\zeta, & g\xi + f\eta + c\zeta \\ \alpha, & \beta, & \gamma \end{array}$$

say these determinants are $\Omega\xi$, $\Omega\eta$, $\Omega\zeta$, where Ω is a value as yet undetermined. The equations are $\gamma(h\xi + b\eta + f\zeta) - \beta(g\xi + f\eta + c\zeta) = \Omega\xi$, &c., viz. these are

$$(\gamma h - \beta g)\xi + (\gamma b - \beta f - \Omega)\eta + (\gamma f - \beta c)\zeta = 0$$

$$(\alpha g - \gamma a - \Omega)\xi + (\alpha f - \gamma h)\eta + (\alpha c - \gamma g)\zeta = 0$$

$$(\beta a - \alpha h)\xi + (\beta h - \alpha b)\eta + (\beta g - \alpha f - \Omega)\zeta = 0;$$

eliminating (ξ, η, ζ) , we have an equation which may be written

$$\begin{vmatrix} A - \Omega, & B, & C \\ A', & B' - \Omega, & C' \\ A'', & B'', & C'' - \Omega \end{vmatrix} = 0,$$

that is,

$$\begin{vmatrix} A, & B, & C \\ A', & B', & C' \\ A'', & B'', & C'' \end{vmatrix} - \Omega(BC' - B'C + C'A'' - C''A' + A'B'' - A''B') + \Omega^2(A + B' + C'') - \Omega^3 = 0.$$

We find very easily that the determinant and $A + B' + C''$ are each $= 0$; and the equation thus reduces itself to

$$\Omega^2 = BC' - B'C + C'A'' - C''A' + A'B'' - A''B',$$

[book p. 164] **93.** We may now verify the theorem; in the general expression for $d\omega$ by λ, μ, ν the values ξ, η, ζ , the equation to be verified becomes

$$\frac{d\omega}{\alpha x + \beta y + \gamma z} = \frac{1}{\Omega} \frac{(\alpha, \dots)(\xi, \eta, \zeta)(dx, dy, dz)}{(\alpha x + \beta y + \gamma z)(\alpha, \dots)(\xi, \eta, \zeta)(x, y, z)},$$

and, in fact, each of the three lines is separately $= 0$, so the equation to be verified becomes

$$\frac{1}{(ax + hy + gz, \dots)(\xi, \eta, \zeta)}(dx, dy, dz) = \frac{1}{\Omega} \frac{N}{(\alpha x + \beta y + \gamma z)(\dots)};$$

viz. this is

$$\Omega\{dx, dy, dz, 2dydz, 2dzdx, 2dxdy\}(\alpha x + \beta y + \gamma z)($$

Here, on the right-hand side, the coefficient of dx is

$$\begin{aligned} &= \alpha(a\xi + h\eta + g\zeta)(\alpha x + \beta y + \gamma z) - (ax + hy + gz)\{\alpha(\alpha\xi + \beta\eta + \gamma\zeta) - \Omega\xi\} \\ &= \alpha\Omega\xi(\alpha x + \beta y + \gamma z), \end{aligned}$$

which is right; and similarly, the coefficients of dy and dz have the same values on the two sides of the equation respectively.

Aronhold's Quadric Integral deduced from the Affected Theorem. Art. Nos. 94 to 99.

94. Let the fixed curve be a conic, say $f = (a, b, c, f, g, h)(x, y, z)^2 = 0$; and let the variable curve be a line meeting the conic in the points 2 and 4. The affected theorem is

$$\Sigma \frac{\delta\phi_2 d\omega}{012} = \frac{\delta 24}{124} + \frac{\delta 24}{234},$$

where (x_1, y_1, z_1) and (x_2, y_2, z_2) are the coordinates of the points 1 and 2 respectively; 12 denotes the constant $(x_1, \xi x_2, y_2, z_2)$, and 012, &c., denote determinants as usual.

The left-hand side is here

$$\frac{1}{12} \left(\frac{\delta \phi_2 d\omega_2}{\delta_4} + \frac{\delta \phi_4 d\omega_4}{\delta_4} \right),$$

on the right-hand side, δ refers to the variation of the constants of ϕ , that is, of the variations of the points 1 and 2; or we may write $\delta = d_1 + d_2$, where the suffixed d 's are independent, and the equation, being satisfied as all, must be satisfied separately

[book p. 165] in regard to the variations of δ , and in regard to the variations of 4: we must therefore have

$$\frac{d\omega_2}{12} = \frac{d_2 24}{124} + \frac{d_2 34}{234},$$

and the like equation obtained therefrom by the interchange of the numbers 2 and 4.

95. The equation just written down relates to any four points 1, 2, 3, 4 of the conic; and if 3 is to vanish, 0, 2 respectively, the equation is reduced to

$$\frac{1}{12} \frac{d\omega_2}{d_2 24} = \frac{d_2 24}{124} - \frac{d_2 01}{012},$$

which may be verified as follows: the equation is obvious; as to the second, observe that writing therein 0 for 4, it becomes $\frac{d_2 02}{012} = -\frac{d_2 02}{012}$, that is, $0 = 2 \frac{d_2 02}{012}$, an identity. And the formula above obtained are

$$\frac{d\omega_2}{12} = \frac{d_2 02}{012} = \frac{d_2 12}{12}, \quad \text{that is,} \quad \frac{d\omega_2}{12} = \frac{12}{12} \left(\frac{d_2 02}{012} - \frac{d_2 12}{12} \right);$$

the equation in question.

96. We have, as a property of any four points 0, 1, 2, 3 of a conic,

$$\frac{23 \ 01}{12 \ 12} = 0, \quad \text{or say} \quad \frac{23 \ 01 \ 23}{012 \ 12 \ 12} = \text{const.}, \quad \text{that is,} \quad \frac{23 \overline{01}}{12 \ 30} = \frac{01}{12};$$

hence considering as a variable point 1, and differentiating the logarithms,

$$d \log \frac{23 \overline{01}}{12} = \frac{d 23}{23} + \frac{d \overline{01}}{01} - \frac{d 12}{12},$$

and the foregoing equation $12 \frac{d\omega_2}{d_2 24} = d \log \frac{23}{12}$ thus becomes $12 \frac{d\omega_2}{d_2 24} = -d \log \frac{01}{12}$, or restoring u in value of 012,

$$12 \frac{d\omega_2}{d_2 24} = d \log \frac{12}{01}.$$

Taking now $\alpha x + \beta y + \gamma z = 0$ for the equation of the line 012; this meets the conic in the points 1 and 2, the coordinates being to be taken: hence 0, 1 are

$$\alpha, \beta, \gamma = y_1 z_2 - y_2 z_1, \ z_1 x_2 - z_2 x_1, \ x_1 y_2 - x_2 y_1,$$

$$12 = ax_1 x_2 + by_1 y_2 + cz_1 z_2 + f(y_1 z_2 + y_2 z_1) + g(z_1 x_2 + z_2 x_1) + h(x_1 y_2 + x_2 y_1),$$

(the second term being of course = 0), viz.

$$\begin{aligned} \overline{12} &= ax_1 x_2 + by_1 y_2 + cz_1 z_2 + f(y_1 z_2 + y_2 z_1) + g(\dots) + h(\dots), \\ &= -(a, b, c, f, g, h) (y_1 z_2 - y_2 z_1, z_1 x_2 - z_2 x_1, x_1 y_2 - x_2 y_1)^2 + (\dots). \end{aligned}$$

[book p. 166] or say $12 = \Omega$, if $\Omega = -(bc - f^2, \dots)(\alpha, \beta, \gamma)^2$ as before; and the equation thus is

$$\frac{\Omega d\omega}{\alpha x + \beta y + \gamma z} = d \log \frac{(a, \dots)(x_1, y_1, z_1)(x, y, z)}{\alpha x + \beta y + \gamma z};$$

or finally, writing (ξ, η, ζ) instead of (x_1, y_1, z_1) , $12 = \Omega$ becomes the coordinates of one of the intersections of the line $\alpha x + \beta y + \gamma z = 0$ with the conic, the equation becomes

$$\frac{\Omega d\omega}{\alpha x + \beta y + \gamma z} = d \log \frac{(a, \dots)(\xi, \eta, \zeta)(x, y, z)}{\alpha x + \beta y + \gamma z},$$

which is Aronhold's quadric integral.

97. (The foregoing property, which may be also written

$$\frac{23\overline{01}}{012\overline{123}} = \frac{01}{23},$$

is verified very easily in the case of four points 0, 1, 2, 3 of a circle: in fact, calling the differential $\Omega\alpha, \beta, \gamma, \delta$, and the proper distances between the points

$$\overline{20} = \alpha\beta \cdot 2 \sin \frac{1}{2}\alpha, \quad \overline{a01} = 2 \sin \frac{1}{2}\beta \cdot 2 \sin \frac{1}{2}\gamma, \quad \overline{2012} = 4 \sin \frac{1}{2}\delta \cdot \dots$$

and so for the other like expressions; each side of the equation thus reduced to $1 \cdot \sin \frac{1}{2}\delta \cdot 1 \cdot 4 \sin \frac{1}{2}\delta \cdot 4 \sin \frac{1}{2}\beta \cdot 4 \sin \frac{1}{2}\gamma$,

$$01\overline{2} = (1 - x + y \sin \frac{1}{2}\delta + 2) 4 \sin \frac{1}{2}\delta \cdot 1 \cdot 4 \sin \frac{1}{2}\beta,$$

$\sin \overline{20} = (1 + \cos \overline{20}) 2 \sin \frac{1}{2}\gamma \cdot \frac{1}{2}\delta$, $LD = 2D\delta$, $LE = 2E\delta$, and so to the squarer in like manner to require $T = 1 - t$, $\alpha = 1$. We can see at once when $\overline{12} = 0$, viz. when the second is the right one.)

98. The foregoing investigation gives for these points $\alpha, \beta, \gamma, \delta = 1, -\delta$. Hence

$$\frac{d\omega}{\sqrt{1-x^2}} = -\frac{1}{2} d \log(x - iy),$$

viz. this is

$$\sin^{-1} x = -\frac{1}{2i} \log(x - iy) = \frac{1}{2i} \log(x + iy).$$

Fixed Curve a Cubic: the Parametric Points 1, 2 consecutive points on the Curve. Art. Nos. 99 to 106.

99. The major function $(x, y, z)^{n-3}$ is taken to be $= \frac{1^2 \cdot 023 - 12^2 \cdot 013}{12s}$, so that, calling the differential $Q d\omega$, we have

$$Q = \frac{1^2 \cdot 023 - 12^2 \cdot 013}{12s \cdot 012}.$$

I take for convenience the cubic to be $f, = (x^2 + y^2 + z^2)w, = 0$; the coordinates of 1 are (x_1, y_1, z_1) , those of 2 are $(x_1 + dx_1, y_1 + dy_1, z_1 + dz_1)$, so that for $\overline{2}$ we have

$$\frac{x_1 dx_1 + y_1 dy_1 + z_1 dz_1}{w} = \overline{2} \quad \text{or} \quad x_1 dx_1 + y_1 dy_1 + z_1 dz_1 = 0;$$

write 1, 1, 1 for x_1, y_1, z_1 , so that the formulæ become for shortness

$$\overline{012} = x dx_1 + y dy_1 + z dz_1, \quad \overline{12s} = x_1 dx_1 + y_1 dy_1 + z_1 dz_1, \quad \dots$$

100. We have

$$\overline{023} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_3, & y_3, & z_3 \end{vmatrix}, \quad \overline{012} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ dx_1, & dy_1, & dz_1 \end{vmatrix} = O(1, 3),$$

say. Moreover

$$\overline{023} = \begin{vmatrix} x, & y, & z \\ x_1, & y_1, & z_1 \\ x_2, & y_2, & z_2 \end{vmatrix} = -(yz_1 - y_1z) + x_1(yz_2 - y_2z) - x(y_1z_2 - y_2z_1) (\&c.),$$

which by definition is $\overline{O23}$.

[book p. 168] as the second term may be written; moreover

$$12s \cdot 023 - 12 \cdot 013 + 12^2 \cdot 012 + 2(2ws + s^2) 012 = wz_1 012 \cdot 012 \cdot 012 \cdot 012,$$

$$12s \cdot 013 = \overline{O2s} \cdot 01 - \overline{O1s} \cdot 02 + 2(2ws + s^2 + s^3) 012 + 2(2ws + s) 012 \cdot 012,$$

and hence

$$1^2 \cdot 023 - 12^2 \cdot 013 = -(yz_1 - y_1z) \overline{O2s} + (yz_2 - y_2z_1) \overline{O1s} - 2(2ws + s) \overline{O1s} \overline{012},$$

or reducing by

$$2(s_1z_2 - s_2z_1 + y_1y_2(z_2 - z_1) + z_1z_2(y_2 - y_1)) = (\partial_y y_1 + \partial_z z_1) 012,$$

this is

$$= -2(s_1\overline{s_2} - s_1z_2) 013 - (\partial_y y_2 + \partial_z z_2) 012,$$

which is of the third order.

101. We may show that each of the terms contains the factor $(ds_1)^2$: we have, in fact,

$$y_1 (ds_1)^2 = \frac{l}{y_1} \cdot 2y_1 dz_1 - 2lz_1 \cdot 2z_1 dy_1 + \frac{l}{y_1} (\dots),$$

$$z_1 (ds_1)^2 = \frac{l}{z_1} \cdot 2z_1 dx_1 - 2lx_1 \cdot 2x_1 dz_1 + \frac{l}{z_1} (\dots),$$

$$x_1 (ds_1)^2 = \frac{l}{x_1} \cdot 2x_1 dy_1 - 2ly_1 \cdot 2y_1 dx_1 + \frac{l}{x_1} (\dots),$$

hence, first multiplying by x, y, z and adding, we have

$$(xy_1z_1 + yz_1x_1 + zx_1y_1) (ds_1)^2 = \frac{l}{x_1y_1z_1} \{ (xy_1 - 2lx_1) dy_1 - 2lx_1 (yz_1) dx_1 \} - \frac{2l}{x_1y_1z_1} (\dots).$$

But in virtue of $ax_1^2 + by_1^2 + cz_1^2 = 0$; hence the equation is

$$(xy_1z_1 + yz_1x_1 + zx_1y_1) 012 = (a' + b' + c') 012,$$

the required expression for the first term.

102. Again, multiplying by x_1, y_1, z_1 and adding, we have

$$\Sigma x_1y_1z_1 (ds_1)^2 = \frac{l}{x_1y_1z_1} \{ 4y_1z_1 dx_1 - 2l(x_1 + y_1 + z_1) dx_1 + 4z_1x_1 dy_1 + 4x_1y_1 dz_1 \}.$$

[book p. 169] when, in virtue of $x_1^2 + y_1^2 + z_1^2 = 0$, the first line is

$$= -\frac{2}{x_1 y_1 z_1} \{ (3y_1 z_1 + y_1^2 + z_1^2) dx_1 + \dots \},$$

and this again is $= -2(x_1 dx_1 + y_1 dy_1 + z_1 dz_1)$: in fact, we have identically

$$\begin{aligned} & x_1 y_1 z_1 (2y_1 z_1 + 3\partial_y y_1 + \partial_z z_1) + (x_1^2 + y_1^2 + z_1^2) dx_1 + \dots \\ &= (x_1 y_1 z_1 + 2lx_1 + 2ly_1) dx_1 + (\partial_y y_1 + \partial_z z_1) dx_1 + \dots \end{aligned}$$

which, in virtue of $ax_1^2 + by_1^2 + cz_1^2 = 0$ and $a^2 + b^2 + c^2 = 0$, becomes

$$= a(\partial_y y_1 + \partial_z z_1) (dx_1 + dy_1 + dz_1).$$

Hence the equation is

$$\Sigma x_1 y_1 z_1 (ds_1)^2 = 2(ax_1 + by_1 + cz_1) 012,$$

or finally

$$x_1 y_1 z_1 (ds_1)^2 = (ax_1 + by_1 + cz_1) 012,$$

the required expression for the second term.

103. Writing for shortness

$$1^2 \cdot 023 - 12^2 \cdot 013 = \frac{1}{4}(a, b, c, f, g, h)(x, y, z)^2,$$

we have

$$1^2 \cdot 023 - 12^2 \cdot 013 = -\frac{1}{4}(a, b, c, f, g, h)(x_1, y_1, z_1)(012) \cdot (ds_1)^2,$$

and hence dividing by

$$012 \cdot 12s, = 012 \cdot \frac{1}{2l}(dx_1)^2,$$

we have

$$Q = \frac{1^2 \cdot 023 - 12^2 \cdot 013}{012 \cdot 12s} = -\frac{l}{2}(a, b, c, f, g, h)(x_1, y_1, z_1)(0613, \dots).$$

But this can be further reduced: its numerator, multiplied by 3, is

$$= -(xy_1 z_1 + 2lx_1) \{x_1 \dots\} + \frac{x_1^2}{x_1 y_1 z_1} \{2yz_1 (2ly_1 z_1) + \dots\},$$

which is

$$= y \{x_1 (y_1 - z_1) - 2(x_1^2 - 2ly_1 z_1)\} + z \{(\dots)\} \frac{1}{x_1 y_1 z_1} (\dots),$$

where $a_1(y^2 - z^2)$, $y_1(z^2 - x^2)$, $z_1(x^2 - y^2)$ are the coefficients of the tangential of the point 1 in regard to the cubic, etc. in usual notation; and we have thus

$$Q = \frac{12 \cdot 023 - 12^2 \cdot 013}{012 \cdot 123} = \frac{12 \cdot 613}{012 \cdot 123}$$

where $0\bar{1}.0$ is the equation of the tangent of the line 012; this meets the conic in the points 1, $\bar{1}$ regard to the cubic, where $\bar{1}$ is what may be called the harmonic of the original.

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[book p. 170] **104.** The identity just referred to is proved very easily. Computing on each side the coefficient of yz , $-yx$, the factor a , divides out and we ought to have

$$-(xy_1 z_1 + 2lx_1 + yz_1 y_1) + (x_1 z_1 + 2lz_1)(y_1^3 - z_1^2) x z_1 = (y_1^2 - z_1^2) a x_1,$$

that is,

$$(yz_1^3 - 2lz_1^3y_1)(y_1^2 - z_1^2) = ax_1.$$

and, in fact, from $x_1^2 = ax_1$, we have for a an integral $\int \frac{dx}{a}$; and the equation is thus verified.

105. The proof has been given in the present case, $f = ax_1^2 + by_1^2 + cz_1^2$, where the cubic is general, the equation is similarly the coefficients of ax_1 , ay_1 , az_1 , equal as the cubic, and the equation is thus verified.

106. We have

$$\overline{028} + \overline{001} + \overline{012} = \overline{12s},$$

or writing this in the form

$$\overline{012} - \overline{[12]} + \overline{021} + \overline{012} = 0;$$

suppose 2 is here the consecutive point $1 + d1$; then

$$\overline{012} - \overline{[12]} + \overline{021} + \overline{012} = 0$$

becomes $= 02(1d1)$; we have also

$$\overline{021} + \overline{012} = -01d12,$$

and the result is

$$-\overline{012} \cdot d\overline{12} = 0,$$

that is, $\overline{012}[\overline{12} + \overline{12} + d61] + d61y012 = 0$, is

$$= \frac{1}{0!} \frac{1}{12!} (d^2d^2 - d^2d^2) \cdots (d^2 - d^2);$$

in the differential of $\overline{012}$ in regard to the parametric point 1 is $\overline{012}_2$, the symbol for the case where the parametric line is the tangent at 1.

Formulæ Relating to the Tangents from the Vertices.

121. I assume some formulæ relating to the tangents to the curve from the vertices A, B, C respectively. We have from each vertex four tangents: say $\rho = 0, \sigma = 0$, symmetrically situate in regard to the axis, and $\sigma = 0, \sigma = 0$, symmetrically situate in regard to the axis: the four tangents from the vertex A are accordingly equations $\rho = 0, \sigma = 0$, denoted by $\rho = 0, \sigma = 0, \rho = 0, \sigma = 0$, the tangents at $\sigma = 0, \rho = 0$, the tangents at $\sigma = 0, \sigma = 0$.

[book p. 171] *Fixed Curve a Cubic: the Parametric Points 1, 2 corresponding points.* Art. Nos. 107 to 110.

107. The parametric points 1, 2 are taken to be corresponding points, that is, such that the tangents at these points meet at a point of the cubic. We may take, in the very particular case considered, the cubic $y^2 = 1 - x \cdot 1 - k^2x$. The four points 1, 1', 2', 2 are then such that the lines 11' and 22', viz. the lines $\tau = 0$ and $\tau' = 0$, meet at a point 7 on the cubic, T being a properly determined linear function of (x, y, z) ; it is a properly determined linear function of (x, y, z) such that the tangents from T to the cubic, that is the line $T = 0$ being any line through the two points 1, 2 meet in a point 9 as above; viz. if we write

$$f = yz^2 + 2lxyz + (x^2 + 2ay + bz)xz = 0$$

and writing

$$d\omega = \frac{x dy - y dx}{\frac{df}{dz}},$$

which, from the equation of the curve written in the form

$$z(yx + 2ay + bz)x + (1 + 2ay + bz)xy + \xi z(xz + by) + lxyz = 0,$$

we have

$$\frac{df}{dz} = \Sigma(xz + by + cz),$$

or say

$$x(xy + by + cz) + xyz = -2lyz\left(\frac{l}{x}\right),$$

becomes

$$= \frac{-2lyz}{\xi},$$

and we have here

$$d\omega = \frac{\xi(dy - ydx)}{-2lyz}, \quad = -\frac{1}{2l}\xi \frac{dx}{(dy)},$$

where $\xi' dx + 6y + lz$. To find the meaning of ξ , observe that the line $x = My = 0$ tangents at the points 1 and 2, which is the case in which the case where, $a = My - 6$ are equal $T = 9$; whence in the case where the tangents at the point 1 and 2, the lines $\overline{1y} = \overline{12}$ and $\overline{6y} = \overline{12}$ meet at a point T on the cubic, T being a properly determined function $\overline{T}$ of (x, y, z) , this line will be a tangent if

$$(yM + \xi)(M^2 + h) = 0,$$

this line will be a tangent at

$$(yM + \xi)(M^2 + \lambda) - 0;$$

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[book p. 172] and we then have the point of contact $(\lambda M + \xi y + \lambda = 0)$, and writing this in the form

$$\xi xy + 2ay + cz = 0;$$

we see that the equation $x = My + \xi z = 0$, the curve $\xi = \xi z + cy + Aps = 0$. Moreover, from the equation of the curve written in the foregoing form

$$\xi(\xi x + 2ay + cy)x + 2yz = 0;$$

it appears that the lines $x = My + \xi z = 0$, $x = My + cz = 0$, viz. the curve, that the line through the two parametric points 1, 2 meets the curve at a point $\overline{T}$ on this it is essential it is essential it is essential.

108. The equation $\xi = 1$ becomes $= ds$, in fact,

$$\begin{aligned} 2. \overline{12} &= (a, b, c)(x_1, y_1, z_1)(x_2, y_2, z_2), \quad (x_1, y_1, z_1) = (1, 0, 0), \\ &= \frac{1}{l} c \xi_2, \\ &= \frac{1}{l} c (xy_2 + \xi_2 x_1 + 2cy_2 + \dots), \quad (x_2, y_2, z_2) = (1, 0, 0), \\ &= b. \end{aligned}$$

We have thus $\xi =$ satisfying the required condition for the major function: and the differential $Q d\omega$ may therefore be taken to be $= \frac{a}{ay}$, that is, we have

$$Q d\omega = \frac{x}{ay} \left(\frac{dx}{ay} \right).$$

The affected theorem then becomes

$$2 \frac{Q \cdot d\omega}{012} = \frac{d\phi_1}{\phi_1} + \frac{d\phi_2}{\phi_2},$$

Integrating each side, and assuming that the superior limits are given by a line ϕ which cuts the cubic in the points 4, 5, 6 and the inferior limits by a line ϕ' cutting it at the points a, β, γ , we have

$$\log \frac{x_1 y_1 z_1 z_1}{x_2 y_2 z_2 y_2} = 2 \log \frac{\phi_1}{\phi'_1},$$

that is,

$$\frac{x_1 y_2 z_3 z_4}{x_4 y_5 z_6 y_2} = \frac{(\phi_1)^2}{(\phi'_1)^2},$$

where ϕ_1, ϕ_2, ϕ_3 denote the values of the linear function ϕ at the points 1 and 2 respectively. We have a cubic cut by the lines ϕ, ψ, x, y in the points 4, 5, 6, 1, $a, \beta, \gamma, 2$, and we have first the points where in course we have ϕ, y replaced by their values, and the formula for the integral of the points (α, β) is

$$\left(\frac{xv}{a}\right) \left(\frac{\phi_1 t}{\phi'_1}\right) \left(\frac{\phi_2 t}{\phi'_2}\right) = \left(\frac{\phi_1 t}{\phi'_1}\right).$$

[book p. 173] that is,

$$\frac{x_1 \bar{y}_2 z_4 z_5 z_6}{x_2 \bar{y}_2 z_a z_\beta z_\gamma} = \frac{\phi_1 \phi_2 \phi_3 \phi_4 \phi_5 \phi_6}{\phi'_1 \phi'_2 \phi'_3 \phi_a \phi_\beta \phi_\gamma},$$

which, dividing out the $\phi_1 \phi_2$, and writing 1, 2 in place of $1', 2'$, becomes

$$= \frac{(\phi_1 \phi_2)^2}{(\phi'_1 \phi'_2)^2},$$

agreeing with the result just obtained.

Aronhold's Cubic Transformation. Art. Nos. 111 to 119.

111. This was obtained in the paper "*Algebraische Reduktion des Integrals $\int F(x, y) dx$, u.s.w.*," Berl. Monatsb., April, 1861, pp. 462–468. I give in the first place the analytical results, independently of the general theory, with the values for the canonical form $f, = \frac{1}{4}(x^2 + y^2 + z^2 + 6lxyz), = 0$, of the cubic. We have

$$T = \text{auxic invariant,} \quad = 1 - 20l^3 - 8l^6,$$

$$S = \text{quartic} \quad (\text{Aronhold's} = 4(l - l^4),$$

$$R = \text{discriminant,} \quad = (1 + 8l^3)^4,$$

$$P = 9b^2 l^2 = [-30l^2 + (1 + 20l^3 + 8l^6)] a + \&c.,$$

$$Q = f y l^2 = (a^3 + 2lc) y x + \&c.,$$

$$B = f y i^2 = (a^3 + 22lac) y x + \&c.,$$

$$C = f x^2 y = (a^3 + 23) y x + \&c.,$$

$$D = f x^2 z = a^2 + yx + x^2 + 6lxyx.$$

where $a, b, c \equiv (P - x)^3, \bar{S}(x^3 - x^3), \bar{x}(z^2 - x^3)$.

Then we have

$$27Q^2 + 54P^2 = (1 - lP)^3 QP \quad = R^2(8C^2 - 8RD),$$

viz. this equation, where each side is a quartic function $(x, y, z)^2$, is an identity when (a, B, c) are connected by the equation, $f a^3 = a^3 + b^3 + 6lab c = 0$; and further,

$$Q4P^2 - PdQ - Rfz_2 = R^2(R^2 - 7R)fz_2 = R^2(R^2 - 7R)fz_2.$$

Hence writing

$$\Lambda = \frac{6loRQ}{(BC)^2} - \frac{2P}{Q} - \frac{Q^3}{Q^3} = \frac{1 - 30W^2 + (1 + 27)Wlc + \&c.}{(b^3 + 22l)a + \&c.},$$

we have

$$Q^*(\Lambda - 3\Lambda + 27) = R^2(C^2 - 8RD).$$

and

$$Q^* d\Lambda_a = 2(QQ^2 - PdQ) = -2R^2(c\sqrt[3]{cd_a - dd_a} + \&c.).$$

[book p. 174] **112.** Supposing now that (x, y, z) are the coordinates of a point on the cubic, that $D = 0$; and taking the square root of each side of the first equation, we may write

$$Q^4 = 3RR - 3RS A^2 = -2(P^2 + (yI + dy + \&c.)),$$

We have

$$d\omega = \frac{ax dy + dy dcx}{x^2 + 2lcy} = \frac{a dx - x dz}{a + 2lcy} = \frac{a dy - y dx}{x^3 + 2lcy},$$

whence

$$d\omega = \frac{x(y dx - x dy)}{x^3 + 2lcy},$$

and we consequently have

$$\frac{d\omega}{\sqrt{x^3 - 3x^3 - LR^2\sqrt{x}}} = \frac{1}{\sqrt{R}} \frac{dy}{y^2 - 3y^2 - LR},$$

which, if 128, bP are put $= p, \bar{p}$, respectively, takes the Weierstrassian form

$$\frac{d\omega}{\sqrt{x^3 - px - \bar{p}}} = \frac{1}{\sqrt{R}} \frac{dy}{\sqrt{y^3 - py - \bar{p}}}.$$

The conclusion is that for the cubic curve, taking Λ a quotient of two linear functions of (x, y, z) , the differential $d\omega$ is transformed into $d\Lambda$ a square root of a cubic function of Λ : viz. we have from a form of differential $d\omega$, which belongs to the ordinary theory of elliptic functions, and which has been adopted by Weierstrass as a canonical form.

113. The transformation depends on the arbitrary point (a, β, γ) of the cubic: the point (a, β, γ) is the tangential of this point, viz. the point of intersection of the tangent at (a, β, γ) with the cubic: we can from (a, b, c) draw three tangents to the cubic, viz. the tangent at (a, b, γ) and three other tangents: the equations of the four tangents being $\frac{4P}{Q} - \frac{8P}{f}y = a, \lambda, \lambda, \lambda$, respectively; where $\lambda, \lambda_1, \lambda_2$ are the roots of the equation $\lambda^3 - 3R\lambda - 12 = 0$.

Suppose for a moment that $(\bar{a}, \bar{\beta}, \bar{\gamma})$ is a point on the cubic curve, and write $A = a' + B y + \bar{6}^2$ Such. We have

$$A'D^2 + 4A'C^3 + 6PD^2 - 3DPC^2 - 6A'LCD = 0,$$

for the equation of the six tangents which we know drawn from the point (a, β, γ) to the cubic: when $(\bar{a}, \bar{\beta}, \bar{\gamma})$ is a point on the cubic, $A = 0$, and the equation reduces itself to a quartic

function, $f\bar{a} = a' + \beta + \gamma + 6lab\gamma, = 0$; and the equation $\frac{4P}{Q} - \frac{8P}{f} = 0$, throwing out the factor B , is an identity.

The last-mentioned expression for the $d\Lambda$ depends on the points A, B, C, D as the four tangents to the cubic from the points $-3K, 1, 2K, 1, 6, 27K y \rightarrow b$, those points are the four tangents from (a, β, γ) to the cubic.

Fixed Curve the Quartic $y^2 = (1 - x^2)(1 - k^2x^2)$. Art. Nos. 145 to 145.

145. This is a curve having a uniquely twin to inflexion: there are at this curve being. The points a, β, β' are double points. The equation of the curve are equal $A = a^2, K = a^2$ resolved into seven points 1, 2, 3, 4, 5, 6, 7, with the order of points be $y^2 = (1 - x^2)(1 - k^2x^2), = 0$; and the cubic forms a circle and is involves more terms in the more concrete fashion of 1 and 2. We have $a, b'a = (1 + a)c$; and $b'a = -a'(1 + a^2)c$, those of the four tangents from $(0, 0, 0)$ as expressed by $\frac{6}{4 - x_0}$ are equal $\frac{6}{4 - x_0}$.

[book p. 175] The foregoing expression for $Q dP = P dQ$, viz. the equations of the four tangents at the points of contact thus are obtained, where the dynamical conditions that Aronhold obtained his results.

114. The foregoing expression for $Q dP - P dQ$, viz.

$$Q dP - P dQ = (Aa + By + Cz)(Adx + B dy + C dz) + Aa da + Bb db + Cc dc - (BC' - CB' da) \cdots,$$

may be verified without difficulty. Writing for a moment

$$Q dP - P dQ = (Aa + By + Cz)(\Lambda dx + 3\Lambda dy + Adx) - (Add a + Bdy db + S dc) - (BC' - CB')(L a + L db + L da) \cdots,$$

we have

$$\begin{aligned} RX &= -CR + a[-2yQP - (1 + 20l^3)X] - (1 - 5l)(1 + 20X^3 + 5l^2)aY - \&c., \\ &= -(1 + 30l^3)(a^3 + 2lac)a\dot{x} + (1 - 5l)(1 + 5l^2(1 + 8l^3))aYY - \&c., \\ &= -8l^3(1 + 20l^3)(a^3 + 2lc)a\dot{x}, \\ &= -16l(a^3 + 2lac)a\dot{x}, \end{aligned}$$

which prove the theorem.

115. I content myself with a partial verification of the identity

$$2TQ^4 + 6T^2Q^2 - 9T^4P^2 - 4T^4 - (1 + 8L^3)^4(8C^2 - 8RD).$$

Writing herein $a, b, c = 1, 0, 0$, we have $D = 0$, and the equation becomes

$$2TQ^4 + 6T^2Q^2 - 9T^4P^2 - 4T^4 = (1 + 8L^3)^4(8C^2 - 8RD);$$

where now

$$Q = a - 2l(a + \beta - 8l\gamma), \quad P = (a - \beta) - 5l^2(a + \beta - 5l\gamma) + (1 + 25\gamma^2);$$

which, putting therein $a^2 + a^2 + \beta^3 = 6la\beta\gamma$, becomes

$$= -(a - \beta)(a + B - 5l\gamma),$$

Hence writing

$$X = a + \beta - 5l\gamma, \quad Y = -5l(a + \beta) - (1 + 25\gamma^2)\gamma,$$

we have

$$Q = P, \quad C = (a - \beta) X, \quad (a - \beta) Y, \quad (a - \beta) X;$$

substituting these values, the factor $(a - \beta)^4$ divides out, and the equation becomes

$$2TX^2 + 6T^2X - 9T^4 + 4T^4 = (1 + 8T^4)^4(8Y - 8X^4);$$

To complete the verification, observe that we have $Y = 8T^4X^2 - (1 + 8T^4)X$, whence

$$-Y = (1 + 8T^4)(YX + 8(PY - X^2)YT X^4)YT^4X^2.$$

[book p. 176] and herein $-y' = a^2 + 8laa\beta_1$, whence

$$-y' + a^2X = -y + (1 - 5T^2)(8a + 8BX) + 8T^2 + 16yYX + 8T^2yX,$$

that is,

$$-(1 + 8T^2)a + X^2 = a^2 + 5T^2y(X) + 8(15)y(XY) - (X^2X)X^2 + 16T^2X^2.$$

Hence the equation to be verified becomes

$$\begin{aligned} 2TX^2 + 6XT^2X &= (1 + 8X^2)(20X^4 - 4X) + 12LX^2X. \\ &= (1 - 5T^2)(2y - 4y_2)(2X^4 + 16X) + (12L^2)Y \\ &\quad - (X + X^3)Z X^3, \\ &= 0X^2 + 8T^2(y'X)X. \end{aligned}$$

which is again the factor X , and which can be thrown out: the equation thus becomes

$$2TX + 6T^2 + 6TX = (1 + 8T^2)(20X^4 - 16) + (12L^2)(X + 4y) = 0.$$

This may be written

$$2TX + 6T^2 + 6TX = (1 + 8T^2)(20X^4 - 16) + (12L^2)(X + 4y) = 0;$$

or, finally, it is

$$2TX + 6T^2 + 6TX - 30T^4 - 8T^4 + 4(1 + 8T^2)(X^2 + 8T^2)(X + 4y) = 0;$$

$+ - a^2 - X - 4y(1 + 8T^2) + 2T(2 + 16T^2)(X^2 + 8T^2)X = 0$, substituting for T, X their values $1 - 200 - 8l$ and $-a + 4l - 5y$ respectively, the coefficients of X^2 and $(1 + 8l^3)$ are separately $= 0$, and the equation is then proved.

116. The foregoing equation $\Lambda = \frac{6loyQ}{(b^3)^2} - \frac{8P}{fy}$, regarding therein Λ as an arbitrary parameter and (x, y, z) as current coordinates, is the equation of an arbitrary line through the point (a, b, c) of the cubic: it meets the cubic in two other points (depending, of course, on the value of Λ); and the equation considered as relating to these points is

[book p. 177] $\frac{Efa}{fy} = \frac{C\Lambda}{\sqrt{P^3 - 3\Lambda S^2 + T}}$. The expressions answer a peculiarly simpler form when (a, β, γ) , instead of being an arbitrary point of the cubic, is a point of inflexion on the cubic; and it is easy to see *a priori* why this is so: in fact, if we assume

$$x : y : z = a + \lambda M, \quad \beta + \lambda N, \quad \gamma + \lambda R;$$

where a, β, γ are linear functions of Λ and the modulus a , the inflexion point of the cubic from which we proceed; observe also, that depending on a, β, γ , take an integral, by elimination, such a proper inflexion at the point of inflexion, we have

$$x : y : z = a + \Lambda M, \beta + \Lambda N, \gamma + \Lambda R.$$

where, λ, ν, ρ are constants! And then with these values of Λ, M , and for elimination, Λ, ν, ρ are also linear function of the equation of Λ , viz. $(\lambda)(\beta - \lambda P X) c) \rho = X, y(\lambda - \beta W)$.

117. To work this out, we start from the equation

$$\Lambda = \frac{6loyQ}{(BC)^2} - \frac{2P}{fy}, \quad fa = (1 + 8l^3)a, \quad B = 2aa^2, \quad C = 12lc^2a,$$

hereon $Aa + By + Cz = 0$ is the equation of the line through the point $a, b, c = 0, 0, a$; or as before, $a = 0$. We can the same equation written, $Ab + By + 0, Cya$.

We have $a, b, c = 0, 0, a$; $B = a, y, z = 4a, b, c, ax^3, C = 12lc^2 = 12y, LD = 6lz, \xi = D$; in ξ at $(0, 0, a)$, the cubic function reduces to a in the form

$$Ax + By + Cz = -4ay^2 + 12lyz + 6lyz = (-4a + 12ly + 6lz)y;$$

from which we as yet unknown coefficient LR is to be obtained. This is most easily effected by assuming $a, y, a, 0$, values which give Λ obtained $-4a - 1$; whence

$$C' = a^4 - 6la(yxy + y + (1 + 8l^3)Ly'(y'a^4) - 6lxy sc) = 2L(y'),$$

$\delta C = 2L(y(yx + yy) + 6ly a(yx + yx) - 24l(ya)c)$. from which on yet unknown coefficient L is to be obtained. This most easily effected by assuming $a, y, a, 0$, values which give $\Lambda = a$ obtained. We need easily obtain by writing $y = a^2, 0, 0$, which from the previous results so above

$$C' = (1 + 4a^2(1 + 8l^3)aY \cdots ()).$$

the whole equation becomes a quartic in y , satisfied by all, must be satisfied separately

$$C = U + (3l^2(C + 8l^3)L((PA^2 + 3(LA^2)L(CPB + A^2)(CL(A^2)A^2 + 121\rho A^2).$$

[book p. 178] where for C is to be substituted the value $LX + 2RAS$. We may also reduce by reducing also the right-hand side, the whole equation becomes

$$C' = L'(x' + By + 6lG\bar{a})(C + LX);$$

which after reduction is found to contain only the factor a , and omitting this factor and reducing also the right-hand side, the whole equation becomes

$$\begin{aligned} &L(x^3 + Dy + 6ly) + L^2A^2 = -3yC, \\ &+LM(By + 34la)B + LM(C + B,)B^2(LX)(LX) \\ &+(LX)(Q + 26la^2)(C + 20) \\ &+LB(LX + 6lB + 9(LL)B + 4l(LL)(CL)L^2 + 0)4 \\ &+(BL^2Q + LX^2B + 2L)M(P) + 2LLM + 0)P \\ &+4(LB - 4lQ)(B + 3l^2)Q + LM(P)P + 12X^2 + 0)P \\ &+1(2 + LA^2)P; \end{aligned}$$

omitting from these the factor C , on putting therein the values of C , D , the equation contains a^2 as a factor; and dividing also throughout by aS , the equation becomes a quadric function in Λ and the equation thus contains Λ , LX , LX as the coefficients of the resulting quadric.

We find very easily that the differentiating, always by $a + 6y + 6lyz = 0$;

$$\frac{1}{2}(2P - 9R) = ly^2 + 6lyzX' + 2L(X + BX);$$

But we have

$$(L = (LL) + 6lLX + RY(C LX) + 2lL(RL)X)$$

the equation in (x, y) is

$$By = -3(2x + 9y)Y;$$

or

$$y^2 + 9y = d^2 \sqrt{X}.$$

We had evident difficulty, that taking always $y = 4 + 6y(LXY)X = 0$, the factor which divides out being in each of these cases

$$\sqrt{2} = \sqrt{x}(1 + a),$$

[book p. 179] which is

$$= (LL - M)(LA^2)LLM + B(L LX)XL - LLXL - LLXL;$$

But we have

$$(L - M)(z + LLX)LX + (L - M)YXY(XYXL)XL - LL,$$

showing that the solution involves the factor $\sqrt{LX^3 + 5LX^2 - X^2}$.

$$\frac{1}{2}(QL - LXL)XL = (1 + 8L^3)XL - 2L(L + l + 2L^2)L(L^2YL^2).$$

showing that the equation in (z, y) is

$$A = \frac{-6L^2 - L(L^2 + 2T^2 + 8X^2)y}{x + y - 5Lz};$$

the equation in (x, y) is, in fact,

$$(B + 8L^3)y^2 + (1L + Lc)y + (-6L^2 + L(L^2 + 2L^2))y = 0,$$

or, as this may be written,

$$(L + 8L^3)x_a + 2LYy_a + 8L^2X_a + 8X_a^2X_a + 4l(LL(L, M))y = 0;$$

viz. we thus have

$$\sqrt{1 - 25X^2} = (1 + 25L)a + \sqrt{2L + 8L^2 - 6T};$$

or, retaining M in the relation

$$y = \sqrt{2}(a + 5L)a + 5L\sqrt{1 - 25X^2 - 8L^4 - 8X^2};$$

$x = \sqrt{2}(a + 5L)a + 5L\sqrt{1 - 25X^2 - 8L^4 - 8X^2}$; viz. we thus have

$$x = \sqrt{2}(1 + 8L^4 - 8L^3 - 8L^4 - 8L^4);$$

where also

$$x = \sqrt{2}(2 + a),$$

these values satisfying obviously

$$(a + 6y)(1 - L^2)^2 = (a + 6y)yL^4 - 27T.$$

119. Starting from these values we, in fact, easily obtain

$$\frac{a dy + y dx}{a^4 + y^4 + a^2} = \frac{-\sqrt{x} dx}{\sqrt{x - 8X^4}};$$

$$2y d\dot{x} + (1 - 6L^3)(d\dot{x}) = -(12 + 12L^3 + 8x^3) - 4(-72 - 8L)a^2,$$

$a^2 - 25y_4 = 2(11x_a)$, and hence

$$d\omega = \frac{x dy + y dx}{a^4 + y^4 + a^2} = \frac{1\sqrt{x} dx}{\sqrt{x - 8X^4 - 8LX}}.$$

The same result might of course have been obtained from the values of a , x , or y , a : the factor which divides out being in each of these cases respectively.

23-2

[book p. 180] *The Cubic* $y^2 = x \cdot (1 - x) \cdot (1 - k^2x)$. Art. Nos. 120 to 130 (*several sub-headings*).

120. The curve is a semi-cubical parabola, symmetrical in regard to the axis of x ; and if, as usual, k^2 is to be taken to be real, positive and less than 1, the curve consists of an oval, and an infinite portion which may be called the sagitta. (See Figure.)

The equation is satisfied by

$$x = \operatorname{sn}^2 u,$$

$$y = \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u.$$

Observe that the periods for these combinations of the elliptic functions are $2K$, $2iK'$; in fact, we have

$$\operatorname{sn}(u + 2K) = -\operatorname{sn} u, \operatorname{sn}(u + 2iK') = \operatorname{sn} u,$$

$$\operatorname{cn} = -\operatorname{cn} u, \operatorname{cn} = -\operatorname{cn} u,$$

$$\operatorname{dn} = \operatorname{dn} u, \operatorname{dn} = -\operatorname{dn} u.$$

whence the sn^2 and the $\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$ both unaltered by the change of u into $u + 2K$ or $u + 2iK'$. Hence to a given u as regards sn may be equally well taken to be a determinate value (for sn^2 is a determinate value $\operatorname{sn} u$), we may, for instance, attribute to u any one of the four points belonging to those values of u respectively; thus, $\{u, u + K, u + 2iK', u + K + iK'\}$.

121. If, to fix the ideas, we select for each point of the curve one of the geometrical arguments, viz., that arising from a positive value of the radical, then taking 1, $1'$, 2, $2'$ as in the figure: for these points the arguments are $u, u + 2K, -u, 2K + u$ respectively.

[book p. 181] infinity $v = 0$, and $u' = -K$; below the axis $v = -v + iK'$, $v = u + iK'$, where $u + 2iK'$ are conjugate; thus $u + 2 = u + 2iK'$. Hence these two arguments are negative integers: more generally, $u' = u + 2(mK + m'iK')$; for the present we may consider only A, A', A, a, a, C ; but the two values AC, a ; but the next have, $aC = u + AK'$, etc.

122. The poor theorem gives for these points u_1, u_2, u_3 a line

$$du_1 + du_2 + du_3 = 0.$$

and thence $u_1 + u_2 + u_3 = C$. The constant C cannot have a determinate value (for if it had assuming the values of u_1, u_2, u_3 are points where, by some interpretation of u , the arguments are points on the cubic; the constant of $u_1 + u_2 + u_3 = AK'$; for the constant $u_1 + u_2 + u_3 = u_1 + u_2$, and substituting in the form $u_1 = u + 2iK'$, $u_1 = -u_2 + iK'$; for points "on the sagitta" ($AC + a = -2iK' - iK'$), and substituting in

$$u_1 + u_2 + u_3 = C \quad \text{viz.} \quad u + u_3 + 2iK' = C,$$

the values are simplifying obviously

$$(K + 1)C = (1 - K_a)(1 - K_a)(1 - K_2)(1 - K_3),$$

and thence

$$a_1 = au_1 - K;$$

the values of P, G are easily found to be

$$F = y_1^2 + 2(y_2 + y_3) - z_1(1 + z_1z_2 + z_2z_3 + z_3z_1) + 2k^2(1 - z_1)(1 - z_2)(1 - z_3),$$

$$G = 2y_1y_2(y_2 + y_3) + 2y_2y_3(y_3 + y_1) + 2y_3y_1(y_1 + y_2) k k k k (\dots).$$

[book p. 182] or, as these may be written,

$$F = a_1 + y_2 + y_3 + a_1 + (a_1^2 - a_1) + (b_1^2 - b_1) + (\dots),$$

$$G = a_1y_1 - \frac{1}{4k}a_1; \quad y_1 = \frac{1}{4} + 4K; \quad a_1 = a_1(1 - K)(1 - K)(1 - K),$$

where of course x, y, a are to be replaced by their values $a, a_1, y_1y_2y_3$, etc. This is, in fact, the ordinary process of finding the third point of intersection of a cubic by a line which meets it in two given points.

Writing $aK - a_1$, and a, b, c, d to denote a_1, b_1, c_1, d respectively, we have

$$\frac{a_1}{a_2a_3a_4} \frac{a_2}{a_2a_3a_4} = \frac{1}{abcd};$$

whence

$$a_1^2 = -\frac{1}{a^2a^2}; \quad a_2a_3a_4a_1 - \frac{a^2a^2a^2}{a^2a^2a^2} - \frac{a^2a^2}{a^2a^2};$$

and the relative equation lookings:

$$(1 - k^2a^2a^2)a^2 + (1 - k^2a^2a^2)k^2(a^2)[(k)] = 0,$$

corresponding to the four tangents which can be drawn from the point u to the cubic.

The Four Tangents from a Point of the Cubic.

123. Suppose that the line is a tangent to the cubic, and let us regard the cubic at the point u , and again meets it at the point u' : then instead of u_1, u_2, u_3 , we have u', u, u' and the relation become $2u + u'C$.

Here a being given, u' is uniquely determined; this gives u' , the unique tangent to the cubic at the point u , viz., to the value $u' = a_1, a_1, u' = a_1$, corresponding to the four tangents which can be drawn from the point u to the cubic.

[book p. 183] The tangents are real for a point of the sagitta, and for such a point we may write $u = iK + v$, where v is real and included between the values $\pm K$; the corresponding values of u' are

$$u' = -2u = -2iK' - 2v = -2v + 2K; \quad \text{for inconsistent to coval,}$$

$$\begin{aligned}
u' &= -\frac{1}{2}u + K = -\frac{1}{2}v - \frac{1}{2}iK' + K, & \text{extramediate to oval,} \\
u' &= -\frac{1}{2}u + iK' = -\frac{1}{2}v + iK' = -\frac{1}{2}v + \frac{1}{2}iK', & \text{intermediate to coval;} \\
u' &= -\frac{1}{2}u + K + iK' = -\frac{1}{2}v + \frac{1}{2}iK' + K, & \text{intermediate to oval.}
\end{aligned}$$

The first and second of these belonging to tangents to the oval, the third and fourth to tangents to the sagitta. We may further distinguish a tangent according as it passes between an oval and not passes between: the writing in the tangents from $K = -2v + 2K$, that is, the tangents from the points of the oval $K = K(K + 1)$; but the tangents from the point $iK' + v$ are conjugate. We may, for distinction, call them 1822 and 1823, etc.

124. We may make a corresponding division of the real lines which meet the curve in three real points: any such line meets the oval twice (and twice or three times) and taking the singular three times; and taking the arguments to be a, u_2, u_3 , for the tangents from the point $iK' + v$, we have

$$\begin{aligned}
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK', & \text{for intermediate line meeting oval twice,} \\
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK', & \text{extramediate line meeting oval twice,} \\
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK', & \text{extramediate line meeting sagitta three times,} \\
\frac{1}{2}(u_1 + u_2 + u_3) &= \frac{1}{2}iK' + K, & \text{intermediate line meeting sagitta three times,}
\end{aligned}$$

above and below the axis respectively.

125. Returning to the tangents, the point $iK' + v$ may be an inflexion: we have then the point of contact of the intermediate tangent to the sagitta coinciding with the point itself, viz. writing $iK' + v$ for u and $-\frac{1}{2}v + \frac{1}{2}iK'$ for u' , this requires $v = \frac{1}{2}iK' + \frac{1}{2}K$ and $-\frac{1}{2}v + \frac{1}{2}iK' = -K$; and for these values of the argument v for u , we have, $iK' + \frac{1}{2}K$, the inflexion above and below the axis respectively.

In the case of an inflexion, $u = u_1$, and we have

$$u' = C - 2u_1 = u_1 + 2u_2(u'_a - u_1),$$

to satisfying the four tangents from the point of inflexion $u = u_1, u = u_1, u = u_1, u = u_1, u = u_1$.

If the line be a tangent, this will be satisfied by $u = u_1$, the condition for this is $2v(P + 2)(P + 1)Q = 0$, and in any of the two other points determined by the equation

$$(2y^2 + 1)GH + g(P + 1)(LH + bL) = 0.$$

this line will be a tangent if

$$(yM + 1\sigma M^2 + 1\sigma) - 0;$$

[book p. 184] giving for the equation

$$3Mw + bC Du + 6RDa - B = 0,$$

or for B, C, D writing their values $1, -(1 + k^2)P, kP$ this is

$$3kw - 4(1 + k^2)a + B^2 = 0;$$

for the a -coordinates of the inflexion. The form $u = -iK + K + iK'$, and $-iK + K + iK'$, give us $-iK = K + iK'$ in $iK + iK' = K(1 + iK')$, and others corresponding terms by the interaction of the numbers 3 and 4.

It is clear that, in addition to a roots; positive or negative, of these roots are real, all positive or all negative; whilst it is also a $-1, -1, -1$, which is the third value. In an inflexion above and the inflexion below the axis being symmetrically situate in regard to the axis.

The Sextactic Points.

128. The vertices A, B, C are each of them a sextactic point: in fact, writing the equation in the form $y^2 = (1 - x^2)(1 - k^2x^2)$, we see at once that the conic $y^2 = (1 - x)(1 - k^2x)$, $= 0$ in regard to coincide and C are real inflexions; and besides these none on the cubic in the points A, A, A, A, C, C .

129. The conics having a vertex at one of these points have each a contact of the second order with the cubic; but inquire how the others may.

Formulæ Relating to the Tangents from the Vertices.

129. I assume some formulæ relating to the tangents to the curve from the vertices A, B, C respectively. We have from each vertex four tangents: say at C , $\rho = 0$, $\rho' = 0$, symmetrically situate in regard to the axis, $\sigma = 0$, $\sigma' = 0$, symmetrically situate in regard to the axis: the four tangents from C are accordingly equations $\rho = 0$, $\sigma = 0$, $\rho' = 0$, $\sigma' = 0$, denoted by $\rho = 0$, $\rho' = 0$.

[book p. 185] eight angles to the axis, and that joining the points of contact of ρ' , σ' is the line $x = w'$ also perpendicular to the axis.

Values ρ , Tangents, Imaginary.

130.

$$\begin{aligned}\rho, \rho_1, \rho_2 &= x - y(1 + i)x, 1 + ix, 0, 1 + k; \\ \sigma, \sigma_1, \sigma_2 &= x + y(1 + i)x, 1 + ix, 0, 1 - k; \\ \rho', \rho'_1, \rho'_2 &= x - y(1 - i)x, 1 - ix, 0, 1 + k; \\ \sigma', \sigma'_1, \sigma'_2 &= x + y(1 - i)x, 1 - ix, 0, 1 - k;\end{aligned}$$

the equation of the curve is

$$f = yp - r(x + y)\rho\rho'; \quad \text{Coefficients of Powers of Coords.}$$

Values ρ , Tangents, Imaginary. Coordinates.

$$\rho, \sigma = x - y - (1 + i)kx, kx - x - (1 - i), \quad x = \frac{1}{i+k}, y = \frac{(1+i)(1-k)}{k};$$

$$\rho', \sigma' = x - y - (1 - i)kx, kx - x - (1 + i), \quad x = \frac{1}{i+k}, y = \frac{(1+i)(1+k)}{k};$$

the equation of the curve is

$$f = y\rho(x + i)\rho\sigma' = y(-x - i)\sigma\sigma'; \quad \text{Coords.}$$

Values σ , Tangents, Real. Contacts.

$$\rho, \sigma = x - y - (1 + k)(1 + i)kx, 1 - x - (1 + i), \quad x = \frac{1}{1+i+k}, y = \frac{(1+k)(1-k)}{1+i};$$

$$\rho', \sigma' = x - y - (1 - k)(1 - i)kx, 1 - x - (1 - i), \quad x = \frac{1}{1-i+k}, y = -\frac{(1+k)(1-k)}{1-i};$$

the equation of the curve is

$$f = y(-1 + k)\rho\rho' = y(-1 - k)\sigma\sigma'; \quad \text{Coords.}$$

130. These functions ρ , σ , &c., considering therein x , y as depending therein on the elliptic function $\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u$, are functions of u : I propose to demonstrate from the values of the constants Nos. 163 of my *Elliptic Functions Memoir*, p. 78; viz. we have

$$\operatorname{sn}(v + \tfrac{1}{2}K) = \frac{1}{\sqrt{k}} \frac{1+k-k(1-k) \operatorname{sn}^2 v}{1+k-k(1+k) \operatorname{sn}^2 v},$$

$$\operatorname{cn}(v + \tfrac{1}{2}K) = \frac{1}{\sqrt{k}} \frac{\operatorname{cn} v + i \operatorname{dn} v}{(1+k')(1-\operatorname{cn} v)}, \quad (\text{vertex C});$$

which is

$$= -\frac{1-y-i(1-x)}{1+y-i(1+k)};$$

and

$$\operatorname{sn}(u + \tfrac{1}{2}K + \tfrac{1}{2}iK') = -i \frac{\operatorname{cn} v - i \operatorname{dn} v}{(1+k')(1-\operatorname{cn} v)}; \quad (\text{vertex A});$$

which is

$$= -\frac{1-y+i(1-k)}{1+y-i(1+k)}.$$

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[book p. 186] which is

$$= \frac{1-y+i(1-kx)}{1+y-i(1+kx)} = \frac{1}{\sigma};$$

Observe that this is the inverse formula, the inflexion which is not a sextactic point, viz. that is, $z = u(1-x) + i(y+kx) - 1 \cdot u$, $\rho\pi$, etc. But neither of these is constant; and so for any other vertex. This is, in fact, no inflexion, and consequently we have

$$\rho\rho'\sigma\sigma' = \rho^2\sigma'^2,$$

and the identity is shown by some calculation of $\operatorname{sn}(u + \tfrac{1}{2}K) - i\rho$, which can be in like manner shown.

Fixed Curve a Quartic in Space; the Quadri-quadric Curve $y^2 = 1 - x^2$, $z^2 = 1 - k^2x^2$. Art. Nos. 131 to 144.

131. It is assumed that k is real, positive, and less than unity: the curve may be regarded as the intersection of the two cylinders $z^2 = 1 - x^2$, $y^2 = 1 - k^2x^2$, but the conception of the curve as belonging to an elliptic cylinder $\frac{2}{1+k^2} - \frac{y^2}{1+k^2} = 1$, and a circular cylinder, $\frac{x^2}{1+k^2+2k^2z}$, are also adopted by Weierstrass as a canonical form.

132. The foregoing expression for $Q dP - P dQ$ is satisfied by the equations $y = \operatorname{cn} u$, $z = \operatorname{dn} u$, $w = \operatorname{sn} u$, $y = \operatorname{cn} u$, $z = \operatorname{dn} u$, $y = \operatorname{sn} u$.

[book p. 187] five points, and then the generating lines come each of them to coincide with the tangent at A , we have the osculating planes meeting the curve in the points A counting four times. We will, therefore, taking A and A' as opposite vertices of a parallelogram, but A' as one of two parallelograms; but A' , A'' , B' , B'' , B'' being the vertices of a quadrilateral.

132. The equations are satisfied by writing therein

$$x, y, z = \operatorname{sn} u, \operatorname{cn} u, \operatorname{dn} u;$$

the periods are here $4K$, $4iK'$; whence it is easy to see that the points A , A' , B , C have for arguments 0 , $2K$, iK' , K , $K + iK'$, $K - iK'$, K , $K - iK'$, $K + iK'$ which are $4 - 0 \cdot iK + 0$.

Similarly the Reformatted form depends on $u + 2K + iK'$, and so reduce the number is like manner to $K + 1 - 1 + 0$, with that the the form which might has been to taken to be

$$x = u, y = u(K - iK') \&(\text{some equation});$$

$0 = K - iK'$, intermediate. Hence &c. in fact, from the values u for which $a = 0$, viz. $u = iK'$, in this case the values u, z, y are like the case of the conic.

133. Returning to the tangents, the point $iK' + v$ may be an inflexion: we have then the point of contact of the intermediate tangent to the sagitta coinciding with the point itself, viz. writing $iK' + v$ for u and $-\frac{1}{2}v + \frac{1}{2}iK'$ for u' , this requires $v = \frac{1}{2}iK' + \frac{1}{2}K$ and $-\frac{1}{2}v + \frac{1}{2}iK' = -K$. The same reduction will be shown directly. Moreover we have

$$BX^2 - 4lac(1 - 2lac)x = a^2 + 4lac(1 - 2lac)x = 0,$$

$$+B - C(1 + a + 1 + Y)B + L - 5(C - LK + 6Y + 4y + 4;$$

$$+B - LK a + ((B + 4l)a + (a + 1)(b - K) + 0)x.$$

$$+2(c - LK)B = -4y(b - K) + a(b + L)(1 - K)b - L(LK) + 0, + -16Z(b - K)x + LK - 1(b + L)L(LK) + 0(LL + 0X),$$

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[book p. 188] which are the required expressions. If we wish to introduce $u_4 = -u_1 - u_2 - u_3$, the resulting equations give the values $\text{sn } u_4, \text{cn } u_4, \text{dn } u_4$ of the fourth point, and in this way identifiable with the ordinary results.

132. The determinant equation may be written

$$(x_1 - x_2, \frac{1}{y_2} \frac{x_1}{y_1} - \frac{x_2}{y_2}) + (x_2 - x_3, \frac{1}{y_3} \frac{x_2}{y_2} - \frac{x_3}{y_3}) + (x_1 - x_3, \frac{1}{y_2} \frac{x_1}{y_1} - \frac{x_3}{y_3}) + (\dots) = 0,$$

and, in fact, each of the three lines is separately $= 0$. This appears from the following three formulæ:—

$$\begin{aligned} \frac{\text{cn}(u_1 + u_2)}{\text{cn}(u_1 - u_2)} &= \frac{x_2 - x_1}{x_2 + x_1}, & \frac{x_2 - x_1}{x_2 + x_1} &= \frac{y_2 - y_1}{y_2 + y_1}, \\ \frac{\text{sn}(u_1 + u_2)}{\text{sn}(u_1 - u_2)} &= \frac{y_2 - y_1}{y_2 + y_1}, & \frac{x_1}{y_1} - \frac{x_2}{y_2} &= -\frac{2(x_1 - x_2)}{y_1 + y_2}, \\ \frac{\text{dn}(u_1 + u_2)}{\text{dn}(u_1 - u_2)} &= \frac{z_2 - z_1}{z_2 + z_1}, & \frac{1}{y_2} \frac{1}{y_1} - \frac{1}{y_2} &= \frac{1}{2}(z_1 - z_2), \end{aligned}$$

which are themselves at once deducible from the formulæ

$$\text{sn}(u + v) + \text{sn}(u - v) = \frac{2 \text{sn } u \text{ cn } v \text{ dn } v}{1 - k^2 \text{sn}^2 u \text{sn}^2 v}, \quad \text{etc.}$$

In fact, the numerators of $\text{cn}(u_1 + u_2) + \text{cn}(u_1 - u_2)$, $\text{sn}(u_1 + u_2) - \text{sn}(u_1 - u_2)$ thus become $= 2(x_1 - x_2)(1 - k^2 x_1 x_2)$, $(2(x_1 - x_2)(1 - k^2 x_1 x_2))y_1 y_2 z_1 z_2$, respectively; so that, taking the numerator of $\text{cn}(u_1 + u_2) + \text{cn}(u_1 - u_2)$ as unity, we obtain by division the formulæ in question.

134. Then above formulæ, putting therein $u_1 = u_3, = 2u$, become, on writing the functions of $2u$, hence

$$\frac{1}{2} \frac{dx}{dx} = -\frac{x_1 - x_2}{y_3 - y_1}, \quad \text{giving} \quad -Dx_1 = \frac{2(x_2 - x_1)y_1}{1 - x_1^2 + (x_1 - x_2)(x_2 + 1)} = \frac{2(x_2 - x_1)y_1}{x_2 - x_1};$$

$$\frac{Dy_2}{2} = \frac{y_2 - y_1}{x_2 - x_1}, \quad = \frac{y_1 y_2 + y_2 y_1 - (1 + k^2)}{x_2 - x_1};$$

$$\frac{Dy_2}{2dy} = \frac{x_2 - x_1}{x_2 - x_1} = \frac{(1 + k^2)(x_1 - x_2)(1 + x_1)}{(x_1 + x_2)(x_2 - x_1)};$$

equations which must of course give each of them the same value for u' : the equations belong to the relation $u = u_1 + u_2$, $u_3 - u_2$, since $(x_2 - x_2)$ are conjugates of (x_1, y_1, x_1) and (x_2, y_2, z_2) respectively.

134. Then above formulæ, putting therein $u_3 = u_4 = -2u$, and the corresponding functions of $2u$, become.

[book p. 189] Write

$$a, \beta, \gamma, \delta, p, q = u_1 - u_2, u_2 - u_3, u_3 - u_1, u_1 + u_2 + u_3, u_1 - u_3, u_2 - u_3;$$

$a', \beta', \gamma', \delta' = u + 1$, writing $\xi = u_1$, it becomes, in place of a, β, γ, δ , in any of $a, \beta, \gamma, \delta, \beta, \beta, \gamma, \delta, \gamma, \gamma, \delta, \delta, \delta$. The determinant equation is satisfied by these formulæ:—

$$a^2 + \beta^2 + \gamma^2 + \delta^2 = -\frac{1}{12k}(L^2 + 2y) + L^2P^2 = 0,$$

$$ax + by + cz + d = a + \beta + \gamma + \delta(d - L) + 8(L^2 + 2 + 4y), \quad au + ax + by + bx + dx = 0;$$

By what precedes, it appears that not only is this so but that we have separately

$$au^2 = a + \beta - 5l\gamma, \quad bu^2 - 5l(a + B) - (1 + 25\gamma^2)\gamma + (1 + 2L^2)P^2 = 0, \quad cu^2 = 0,$$

then not only the original equation, but each of the three equations, holds good.

The Cubic Curve $xy = 2y + (4 + 1)^2z$; $xy - 1 - \sqrt{1 - x^2}\sqrt{1 - k^2}$. Art. Nos. 136 and 137.

135. Writing the equation in the form

$$(xy - 1)^4 = x^4(1 - k^2x^2), \quad \text{or} \quad xy - 1 = \sqrt{1 - x^2}\sqrt{1 - k^2x^2},$$

we see that the general form is as shown in the figure: the end portions of the curve

[book p. 190] curve lie between the values $x = -\infty, -\frac{1}{k}$, and $\frac{1}{k}, \infty$. The curve may be made to depend on an elliptic function by means of the equation; we may here write

$$x = \text{sn } u = \frac{\text{sn}(u + \frac{1}{2}K) \text{ dn}(u + \frac{1}{2}K)}{\text{cn}(u + \frac{1}{2}K)}, \quad \text{say} \quad x = \frac{\text{sn } v \text{ dn } v}{\text{cn } v},$$

where the function sn, cn, dn belong to the second, that is, to the second equation; and herein for u writing $v + \frac{1}{2}K$, and for k writing $k' = \sqrt{1 - k^2}$, we have, of course, the formulæ given.

But from the formulæ, p. 63, of my *Elliptic Functions*,

$$\text{sn}(2v, k) \text{ cn}(2v, k) = \frac{\text{sn}(v, k') \text{ dn}(v, k')}{\text{cn}(v, k')}, \quad \text{dn}(2v, k) = \frac{\text{dn}(v, k')}{\text{cn}(v, k')},$$

and herein for v writing $\frac{1}{2}v$ and for k writing $k' = \sqrt{1 - k^2}$, we have, of course, the formulæ given.

Fixed Curve the Cubic $y^2 = x(1 - x)(1 - k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by

$$\frac{012}{012} = \left[\int_a^x (d\omega)_3 (d\omega)_1 (d\omega)_2 \dots \right],$$

the last term being the properly determined constant K , attached to the variable term $\overline{012}$, in order to obtain a standard form of integral. The object of the present articles is to show what these formulæ become for the before-mentioned form of cubic curve $y^2 = x(1-x)(1-k^2x)$, which is most directly connected with elliptic functions: and to exhibit the connexion of the formulæ with the elliptic integrals of the second and the third kinds respectively.

139. We have, in general,

$$\overline{012}, \overline{0\overline{12}} = \left| \begin{array}{c} (1+k')x_1y_1, -(x_1+x_2+y_2)y_1 \\ \xi(x_1+1) + (\xi+1)x_2, (c_0, c_1, x_1) \\ +y(x_1)x_2, \end{array} \right|, \quad \left| \begin{array}{ccc} x, & y, & 1 \\ x_1, & y_1, & 1 \\ x_2, & y_2, & 1 \end{array} \right|$$

[book p. 191] where the sn, cn, dn refer to the modulus k' , a, z, d denote $\text{sn } v, \text{cn } v, \text{dn } v$, k modulus k .

But from the formulæ, p. 63, of my *Elliptic Functions*,

$$\text{sn}(2v, k) \text{cn}(2v, k) = \frac{\text{sn}(v, k') \text{dn}(v, k')}{\text{cn}(v, k')}, \quad \text{dn}(2v, k) = \frac{1}{\text{cn}(v, k')},$$

and herein for v writing $\frac{1}{2}v$ and for k writing $k' = \sqrt{1-k^2}$, we have, of course, the formulæ given.

Fixed Curve the Cubic $y^2 = x(1-x)(1-k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by $\overline{012}/\overline{0\overline{12}}$.

139. We have

$$\text{sn}(v + \frac{1}{2}K) = \frac{\text{cn } v}{\text{cn}(v + \frac{1}{2}K)}, \quad \text{dn}(v + \frac{1}{2}K) = \frac{1+k}{1+k} \frac{\text{cn } v}{\text{cn}(v + \frac{1}{2}K)},$$

$$\text{cn}(v + \frac{1}{2}K) = \frac{1}{1+k} [1+k + (1-k) \text{cn } v], \quad \text{dn}(v + \frac{1}{2}K) = \frac{1-k^2}{1+k} [1+k - (1-k) \text{cn } v];$$

and the formula above clear obtained are

$$\text{cn}(1+k, k) = (1+k)(1+ka, k') \quad \text{giving} \quad \text{cn}(1+k, k) = \frac{(1+k)(1+ka, k')}{1+ka(1+k)};$$

$$\frac{1}{\text{cn}(2+k, k)} = \frac{1}{1+k} + a(1-k) \frac{1-x}{1+x}, \quad \text{cn}(2+k, k) = \frac{1+k}{1+k} \frac{1+ka}{1+ka(1+k)};$$

$$\frac{\text{dn}(1+k, k)}{\text{cn}(1+k, k)} = \frac{1}{1+k(1+ka)} \frac{(1-x)a}{a+1}, \quad \text{dn}(2+k, k) = \frac{1-ka}{1+ka};$$

where, as before, a, c, d denote $\text{sn } v, \text{cn } v, \text{dn } v$, k modulus k , in fact, the formulæ of the second line of the table, *Elliptic Functions*, p. 63.

Fixed Curve the Cubic $y^2 = x(1-x)(1-k^2x)$. Art. Nos. 138 to 142.

138. It was shown, No. 63, how for the affected theorem, when the fixed curve is a cubic, the form of differential is the multiplied by

$$\overline{012} = \int_a^x [\chi d\omega]_3, (d\omega)_1, (d\omega)_2, \dots],$$

the last term being the properly determined constant K , attached to the variable term $\overline{012}$, in order to obtain a standard form of integral. The object of the present articles is to show what these formulæ become for elliptic integrals of the second and the third kinds respectively.

139. We have, in general,

$$\overline{012}, \overline{012} = \begin{vmatrix} (1+k)x_1y_1, & -(y_1+y_2)y_1 \\ \xi(y_1) + (\xi+1)x_2, & (c_0, c_1, x_1) \\ y_1(y_1)x_2, & \end{vmatrix}, \quad \begin{vmatrix} x, & y, & 1 \\ x_1, & y_1, & 1 \\ x_2, & y_2, & 1 \end{vmatrix}.$$

[book p. 192] Taking here $2, = \emptyset$, for the point, coordinates $x_1, y_1 = 0, 0$, we have

$$\overline{012} = -\frac{x_1y_2}{y_2(x_2-x_1)(y_2-y_1)} = -\overline{012},$$

and, retaining 1 to denote the point coordinates (x_1, y_1) belonging to the argument u_1 , and $1'$ for the point belonging to the argument $u_1 + \frac{1}{2}iK'$; the coordinates of $1'$ are $\frac{1}{k^2x_1}, \frac{1}{k^2x_1^3}y_1$; and the formulæ become

$$\overline{1'0} = -k^2x_1(x - \frac{1}{k^2}x_1(y-0)y_1), \quad = 1+k^2,$$

the numerator herein multiplied by ν , is

$$= -k^2x_1(x - \frac{1}{k^2}x_1(y-0)y_1) = (-\Omega 1)y,$$

which, substituting for y, x, z as their values in terms of u, v, w , becomes

$$= \frac{0 + \nu^2(\nu - z_1z_2 - z_2z_1)}{(\nu\nu_1 + \nu_2\nu_3)(\nu_2 - \nu_1)} = -k^2x_1;$$

Operating on each side with $\frac{a}{dx_2}, = 0$, we obtain

$$\partial_2\overline{012} = -\partial_2,$$

$\partial_3 \overline{012} = w^2 - 12y + L^2(2y - z) - z_1$. The differentiation being, in fact, that which covers in establishing the fundamental property of the elliptic integral of the second kind. Hence we have

$$Z(u+a, k) = (1 - \frac{R}{8})u + L)(n du,$$

viz. we have

$$ZU = ZU_1 - L^2a - L^2z = L(u - u_1) - LK(u - u_1) = \Omega(\nu - x_1);$$

and thence

$$\partial_x ZU = \partial_x ZU_1 - L = LK^2aU(u - u_1)k(u - u_1) = \Omega'.$$

Observe that, L referring to u , all the variable's might be written U' .

140. Continuing the discussion, the expression for the right-hand side is $= \omega + (a-2)\omega'$, and this divided by u^2 , the equation $\frac{1}{2} = 0$. We have here

$$\partial_x \omega - \omega + \omega' = \overline{0(u-u_1)} = (u_1 - u_2)(u_2 - u_3)(u_3 - u_1);$$

viz. this is, $\overline{u_1} = -\overline{u_2}$, in $\overline{u_1}$, in $\overline{u_2}$. From this, and adding, the expression in question becomes

$$\partial_x w(z_1 - z_2) \frac{x_1}{a_1 a_2 a_3} = -\partial_x w(z_1 - z_2) \frac{x_1}{a_1 a_2 a_3},$$

the differentiation being, in fact, that which covers in establishing the fundamental property of the elliptic integral of the second kind.

[book p. 193] **140.** To introduce into the formula 1 instead of $1'$, we have only to write $u_2, -K'$ instead of u_1 , putting also for shortness c, d, f, g, h for the functions of u and a , respectively, we then obtain

$$[01'] = -\frac{x}{y} a_1 (c_1 + x a_1),$$

where observe that, interchanging u and a_1 , we have

$$[110] = \frac{a_1 d_1}{c_1 (c + x_1 c_1)},$$

that is, $[10] = -[01]$, as it should be. The formula may be written

$$[01'] = -[10'] = -\frac{a a_1 d}{x c_1 (c + a a_1)},$$

and

$$\partial_y [01'] = -\frac{1}{(c + a a_1) a a_1},$$

whence

$$\partial_x [10'] = -\frac{1}{a^2 (x - x_1) a c (c + a a_1)},$$

we have, moreover,

$$[012] = [12'] - \frac{x}{y_1} [\overline{036}] - \frac{x_1}{y_1} [306],$$

and

$$\partial_x [012] = \frac{x_1 y_1}{a_1 a_1 (x - x_1)} - \frac{1}{x_1} \frac{1}{x c_1 (c + a a_1)},$$

which last equation gives $(\partial_x + \partial_x + \partial_x) [012] = 0$, as it should be.

141. Supposing that the differential $d\Pi_a$ is defined by the equation

$$d\Pi_a = du_1 [012] + du_2 [01'] + du_3 [026] - [126],$$

we have

$$\int d\Pi_a = du_1 [012] + du_2 [01'2'] + du_3 [026] - [126];$$

and thence

$$\begin{aligned} \partial_x \int d\Pi_a &= \partial_x [012] + du_2 [126] - \partial_y [126]; \\ &= [126] - [126], \\ &= \frac{1}{(c + a a_1)} + \frac{(1 + k^2 a_1 - 2 a_1) a_1}{(1 + k a_1)(1 + k a_2)}; \\ &= \frac{1}{(c + a a_1)} - \frac{1}{(1 + k^2 a_1)} (1 + k a_2)(1 + k a_3). \end{aligned}$$

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[book p. 194] or establishing between the constants u, a , the relation $\text{sn}^2(u_a - u_1) = \frac{K}{K - R}$, this becomes

$$= 1 + \frac{R}{R - k} k^2 \text{sn}^2(u_a - u_1) a k^2 K^2,$$

which is

$$= \partial_x \log \frac{\Theta(u_1 + a + iK') \Theta(u - u_1 - iK')}{\Theta(u_1 + iK') \Theta(u - iK')},$$

where Θ is Jacobi's theta-function, see my *Elliptic Functions*, p. 144. The expression is, in fact, $= \frac{1}{2} k \log \Pi(u - u_1, a - u_1)$, where the integral $\partial = \int_0^a \log \Theta$.

But we have

$$\log \Theta(u) = \log \frac{\Theta(0 + iK') \Theta(u - iK')}{\Theta(0 + iK') \Theta(0 - iK')} (1 + k(\dots)) + \int_0^u n \, du;$$

that is, $\partial = \int_0^a \log \Theta$, and consequently we have $\partial_x [\int_0^a \log \Theta] = \Omega$,

$$= \frac{1}{\sqrt{(\nu_1 - \nu_3)(1 - k^2)}} \log \frac{(1 - k\nu_1)\sqrt{2K}}{(1 - k\nu_1)\sqrt{R}} \frac{LK}{(LK)} - 2k(1 - k^2)(1 - k^2)a(1 - k^2a(1 - k^2)),$$

that is, $\partial = \frac{K}{B - K} \sqrt{1 - k} a$, and consequently we have $\partial_x \sqrt{(1 - k\nu_1^2)(1 - k\nu_2^2)} a$,

$$\begin{aligned} &= -\frac{R}{R - k} \partial_x \sqrt{1 - k^2} a^2 - 2k(1 + k^2)(1 - k^2)a^2 + 2ka^2, \\ &= -\frac{R}{R - k} \partial_x \sqrt{(1 - k^2)(1 - k^2)} a \partial_x \sqrt{1 - k^2} a b a, \end{aligned}$$

The difference of the two functions on the right-hand side is $= -\partial_x (u - a)(1 - k \operatorname{sn}^2(u - a))$, which is $= -\operatorname{sn}^2 u a + \operatorname{sn}^2 a$; and the integral being $= 0$ on the lower limit, we readily see that the expression on the right-hand side is, in fact, the function which is required. The transformation also is given in summary by the form which I had obtained. I consider with this remark.

Fixed Curve the Quartic $y^2 = x(1 - x^2)(1 - k^2)$. Art. Nos. 145 to 145.

143. This is a curve having a tacnode at infinity on the line $w = 0$; as may be seen by writing the equation in the homogeneous form $y^2 w^2 = x(1 - x^2)(1 - k^2 x^2)$, and putting for shortness $X = w(1 - x^2)(1 - k^2 x^2)$, $X = w(1 - x^2)(1 - k^2 x^2)$, when X, w small as compared to a , the equation is reduced to $y^2 w^2 = 1$, viz. this is a $\left(\frac{yw}{1}\right)$ to which the sign $\pm$ is to be understood accordingly. I consider with

[book p. 195] reference to this curve only the affected theorem, in the particular form in which it must readily connects itself with the ordinary theory of the integral of the third kind.

144. I consider the differential $\frac{(x, y, z)_p d\omega}{012}$ in the particular case where the line 012 is a line parallel to the axis of y : taking its equation to be $x - x_1 = 0$, and putting for shortness $X = \sqrt{1 - x^2} \sqrt{1 - k^2 x^2}$, $X_1 = \sqrt{1 - x_1^2} \sqrt{1 - k^2 x_1^2}$, the parametric points are taken to be (x_1, X_1) , $(x_1, -X_1)$, and the residues are the intersections with the curve $f - X_1 = 0$ of the line $f = 0$, viz. this is the line passing through the two points of contact of the parallel tangents: the points (x, y) and the points (x_1, X_1) , $(x_1, -X_1)$ are on a conic Ω , containing the affected theorem, and we have for the form $a(x - x_1)$, containing the arbitrary constant a . We have moreover, in usual notation, for the form a containing the arbitrary constant a .

If the limits are taken to be points on the curve $a^2 + y^2 + ax + by + c = 0$, what we want to do is to express the summation of the integrals in the form

$$\int \frac{\sqrt{X}}{x - x_1} \frac{x - x_1}{X} dx + a \int \frac{\sqrt{X}}{x - x_1} \frac{dx}{X} - a \int \frac{\sqrt{X}}{x - x_1} \frac{dx - dx_1}{x_1 X};$$

where $X_1 = \sqrt{1 - x_1^2} \sqrt{1 - k^2 x_1^2}$, etc. Reference to the right curve as above $a^2 + y^2 = ax + b$, when $a^2 + y^2 - 2ax - a^2 = 0$; the right-hand side, by adding the like quantities for the other points a, β, γ .

We have

$$-\frac{1}{a^2} (a^2 - X^2 - a)^2 = \frac{a^2 - aB}{a^2 + a^2 X^2},$$

and the integral thus becomes

$$= -2 \int \frac{\sqrt{X} a d\omega}{a^2 - aB} \frac{a^2 - aB}{a^2 - aB^2 X^2} = a \int \frac{X^2}{a^2 - aB^2 X^2} \frac{dx}{\sqrt{X}}.$$

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[book p. 196] Taking here $a = 0, -c$, so $a = (X^2)$, $a = \frac{a}{\sin \theta}$, we have for such a $\Pi(a, k)$, and the result is

$$\int \frac{\sqrt{X}}{x - x_1} \frac{dx - dx_1}{X} = \int \frac{X dx}{x - x_1 + b a a X^2} - \frac{X a a X}{x - x_1} \sin \theta du,$$

where on the right-hand side the first term is $-\int(x, a, c)$ in Jacobi's form of the integral of the third kind, viz. *Elliptic Functions*, p. 146.

And thence

$$\partial_x \Pi(u, a) = \frac{2(\nu - 1)(z a) a + X(z a + 4u^2 - X a^2)(\nu + 2)(\nu + 2X)}{=} \left(1 - \frac{R}{R}\right),$$

or

$$\partial_z \Pi(u, a) = \frac{R^2(s^2 + r^2) a^2 - 2k^2(1 + k^2) k^2}{(1 - k^2 \nu^2 a) a^2} = \left(1 - \frac{R}{R}\right) + 1 x a^2;$$

or, this being represented in regard to u , (a , we have

$$\partial_z \Pi(u, a) = \omega a + L(0 + a - 2 a du);$$

and thence by integration, and a proper determination of the constants, $\Pi(z, a) = \Pi(z, a)$.

Chapter VI. The Nodal Quartic.

Nodal Quartic; the General and Polynomial Forms. Art. Nos. 146 to 148.

146. For a cubic, or when an x is, the nodal quartic is the next case as regards points on the curve, and corresponding theories with functions of a single argument (elliptic function): but for a node it is the next case as regards the number of points on the curve $\delta = 4, p = 8$; but it is, as in the case of the cubic, the next case as regards the number of points or, with a marked change in the form of the result.

[book p. 197] The most simple curve of deficiency 2 is the nodal quartic, $n = 4, p = 2$. Using homogeneous coordinates, the general form is $Ax^4 + 8Bx^3 + (\dots) = 0$. Using

$$A = (b, j, b)(y, z)^2,$$

$$B = (b, j, b)(y, z)^3,$$

$$C = (p, q, r, s)(y, z)^3,$$

and where we write also

$$D = AC^2 - 2(p, q, r, s)(y, z)^4 + 4(c, j, k,)(y, z)^4 - 4(c, j, k,)(y, z)^4.$$

Clearly the equation of the two tangents at the node is $A = 0$; and the coordinates of the six tangents from the node are $a - kx + by + cz = 0$; $a - kx + by + cz = 0$, the equation of the curve corresponds to the form: that is the case is unaltered if for β, γ, δ we substitute $\alpha + \delta, \beta + b, b + b, b + b$, and we have thus, for the conic from the tangents to the curve. $a' + Y' + Z' = 0$, where $\delta_1, \delta_1, \delta_2$, and δ_2 for the conic, when the cubic, the six tangents are entered by; or, in like manner, we have, $a' + Y' + Z' = 0, a + b + c = 0, b + e + d = 0$, the lines $\delta_1, \delta_2, \delta_3, \delta_2$, are corresponding points.

147. An important special case occurs when $B = 0$: we here have

$$\begin{aligned} A &= (a, h, i, h)(y, z)^2, \\ B &= 0, \\ C &= (p, q, r, s)(y, z)^3 = (a, b, c, d)(y, z)^3, \end{aligned}$$

or, omitting the factors i and j ,

$$(a + j)(b + j)(c + j) = (a + j)(b + j)(c + j)(a + j) = 0.$$

The origin is here a Sencocode; the tangents are $a - z + y = 0$, and there are six tangents to the cubic on the curve.

148. The general nodal depends on 11 constants, but by writing for $uby, uby + cy + dz, ukx + by + cy + ay$, we can reduce the number is like manner to $1 - 1 = 0$. Similarly the Reformatted form depends on 7 constants, but we can reduce the number is like manner to $1 + 1 - 1 = 0$, so that the final form might be taken to be

$$xy = (a - b)(a + b)(a - b)(b - a),$$

this line will be a tangent of

$$(yM + 1)(1\sigma M^2 + 1) = 0;$$

[book p. 198] but it is more convenient to retain the original form

$$x^2(ax + by)^2 = (a - x)(b - y)(c - x)(d - y),$$

bearing in mind that this is reducible to the form just referred to, and thus depends visually upon only 5 constants.

It is a general property that a curve of deficiency p greater than 1 can be by a rational transformation reduced to a curve of the corresponding deficiency $4p - 3$ (this number = 1 for $p = 2$), so far as it can be reduced; and the parametric in particular, $p = 2$, the nodal quartic is to be reduced to a curve $a = 0, \beta = 0$ corresponding properly, viz. the sextic. This will be done in the present chapter, but the form here adopted is not the standard form generally adopted, but the standard form first adopted by Weierstrass as a canonical form.

Reduction to the Plefonoidal Form. Art. Nos. 149 to 152.

149. Consider the general nodal quartic $Ax^4 + 2Bx^3 + (\dots) = 0$. The equation of the tangents from the node $(x, y, z = 1, 0, 0)$ to the curve is the discriminant in regard to ξ of $Ax^4 + 2Bx^3 + (\dots) = 0$. We may write

$$A_x = (a, h, j, h)(y, z)^2, C_x = (p, q, r, s)(y, z)^3$$

and the transformed equation is

$$Ax^4 + 2Bx^3 + (\dots) = A_x x^2 + 2B_x x + C_x = 0, \text{ \&c.}$$

say this equation is

$$Ax^2 + 2Bx + C = 0,$$

where A, B , &c. have the values of A_x, B_x , &c., written in x, y, z , in place of x', y', z' ; and the transformed equation $\Lambda x^2 - 2BxC = 0$ is the equation of the six tangents.

150. From the values of A, B, C we obtain after a little reduction the following formulæ:—

$$A' = (a, h, j, h)(y, z)^2,$$

$$B = (l, m, n, l, j, k)(y, z)^3,$$

$$C = (a, b, c, d, e, f, g)(y, z)^4.$$

and the transformed equation is

$$Ax^2 + 2Bx + C = 0,$$

where

$$\mathbf{A} = A',$$

$$\mathbf{B} = A^2C - (B^2),$$

$$\mathbf{C} = AC^2 - 2(B)^2 + (B^2)C + 2R,$$

[book p. 199] and thence

$$\mathbf{B}^2 - 4\mathbf{A}\mathbf{C} = A^2[A(AC^2 - 2(B)C^2 + (B^2)Cb)] - AC(A(Cbcebd));$$

We have here $B, = A(p, q, r, s)(y, z)^3$, a cubic function $(y, z)^3$ containing the factor $A_x, = (a, h, j, h)(y, z)^2$, and similarly for a, a, a, b . Calling the other factors B, V , we have $B = AB', V = AV'$, respectively; where b, b, c are merely the roots of the equation $AV = 0$.

We have what precedes, by the transformation $x = y + \beta_1, a = \beta, y = \beta$, passed from the form $Ax^4 + 2Bx^3 + Cx = 0$ to the form

$$\mathbf{A} = A, \mathbf{B} = V, \mathbf{C} = -V'(B, \mathbf{\Pi} - C),$$

where

$$\mathbf{\Pi} = (\xi, \eta, k)(y, z),$$

and thus the equation is $Ax^2 - 2Bx + C = 0$.

We have by what precedes, by the transformation, the canonical form $a = 0, y = 0$, viz. $\mathbf{\Pi}$ if we $\mathbf{\Pi}$ have the common factor $A(\mathbf{\Pi} - e^\eta \mathbf{\Pi} - 0)$.

152. Assume now

$$\Sigma B = AB', V = AV'; \mathbf{\Pi} = AB',$$

and therefore also

$$A = (t, u, b\xi yb)(y, z);$$

$$\mathbf{\Pi} = (\xi, \eta, k)(y, z)(xya);$$

$$\mathbf{\Pi} = AC^2 - 2(B)C^2 + (B^2)C;$$

viz. this equation is, the transformation, that of a curve of the order 5 ($p = 2, w, C$); and the equation thus becomes

$$Ax^2 - 2Bx + C = (xy + \mathbf{\Pi}\mathbf{\Pi}); \quad \text{the equation must become a tangent at}$$

and there is to the same equation $a = (\xi, \eta)(y, z)$,

$$xy = (\xi + \eta)(y, z)(xya);$$

where

$$\Xi = (c, f, g)(y, z);$$

$$\Pi = (\xi, \eta, k)(y, z); \quad \&c.;$$

151. Assume that the curve transforms by the substitution $x = y + Z\eta$. We have the change of Π, Ec, z, f, y , in A, C, b, y, c ; and thus

$$\Pi = Aa^2, \quad \Pi = Aa^2, \quad Z = -Cy^2 + AZ^2 + A\eta\xi(LVa);$$

which putting therein $(\xi, \eta, k)(y, z)$, the equation must become

$$Y = (a, b, c, d, e, f, g)(y, z);$$

and therein removed by

$$\Xi, Y = Z + Mp; \quad Y = Z + K - Mp - q;$$

and therefore would be in the in $Y Y$.

[book p. 200] then in the new coordinates (X, Y, Z) we have the equation

$$(i, j, k)(Y, Z)YZ + [(X - aY - bZ)^2 - X(p, q, r, s)(Y, Z)] - kZ$$

$$+ (1, b, c, d, k)(X, Y)(Y - 2bZ - kY) + (X X(Y - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)$$

$$+ 4(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - bZ - kY) = 0,$$

that is,

$$(t, j, k)(Y, Z)YZ + [(X - bY - bZ)^2 - X(p, q, r, s)(Y, Z)]$$

$$+ 4(X - aY - bZ)(X - aY - bZ)X(X - bY - bZ) = 0;$$

$$+ 1(X - bY - bZ)(X - aY - bZ)(X - bY - bZ)(X - bY - bZ)(X - bZ - kY) = 0;$$

when this equation $(X, Y, Z) = 0, Y(X - bY - bZ) = 0$, the equation transformations must become

$$(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)$$

$$+ 4(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ)(X - aY - bZ) = 0,$$

that is,

$$(p, q, r, s)(Y, Z)^4 - (Y - 2bZ - kY)^2(X - bY - bZ - bZ - kY)^2$$

$$- (X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ)^2(X - aY - bZ - kY)^2 = 0,$$

or rearrangely

$$X, Y, Z = X - aY - bZ, \quad Lz = Mb(bY - bZ) - kY,$$

transformed into the differential form as above.

It is similarly the case that the differential first appears, although there is a great many more steps to be made in establishing the truth of the previous, and the case where the case the parametric points are a curve of deficiency p , p being p .

Application of Abel's Theorem.

152. Taking the fixed curve to be $f, = (Ax^2 + 2Bx + C) = 0$, we have

$$\frac{df}{dx} = Ax + B;$$

if for shortness we write

$$(x, y)^2 = P, = P, Bx + Cy^2, y = (Ay - Cz)(z - y)P;$$

[book p. 201] *Application of Abel's Theorem.* Art. Nos. 153 to 157.

152. Taking the fixed curve to be $f, = (Ax^2 + 2Bx + C) = 0$, we have

$$\frac{df}{dx} = Ax + B,$$

if for shortness we write

$$(x, y)^2 = P, = Ax^2 + 2Bxy + Cy^2, P_x = (Ay - Cz)(y - z)(z - y) = (Ax + B)(\dots),$$

and we thence have

$$d\omega = \frac{x dy - y dx}{\sqrt{P}}.$$

The minor curve $(x, y, z)^{n-3} = 0$ is an arbitrary line passing through the node, that is, the point $x = 0, y = 0$; and there passes through this line one of two distinct relations $2x dz - dx = M(x dz - dz), = 0$, viz. this is an ordinary line through the points 1 and 2 of the curve.

The variable curve is taken to be an arbitrary curve passing through the intersections of the conic Ξ , with the conic $\mathbf{X}$ passing through the four points belonging to these values $2c, b, a, c$. Specifically the line is arbitrary, and the line being given, the equation determines the intersections of the line with $\mathbf{X}$ passing through these intersections of the line with $\mathbf{X}$ which are the points 1, 2, 3, 4, viz. we have

$$\frac{df}{dy} = Ay + B, \quad Lz = Mp, \quad LZ = K - Mp;$$

to complete the verification, observe that we have $Y = 8TX^2 - (1 + 8T^4)X$, whence

$$-T = (1 + 8T^4)(-YX + 8(PY - X^4) - 16YT^4 - 8X^4 = 16T^4X^4.$$

153. Observe that the intersections of the cubic with the fixed curve are given by the equation $\Omega + \mathbb{P}\mathbf{X}\mathbf{\Pi} = 0$, viz. $\Omega + \mathbb{P}\mathbf{X}\mathbf{\Pi} a, b, c, d, e, f, g)(x, y, z)^4$, an equation which determines a ; and the ratio $x : y$ is then determined rationally by the equation $Ax - By = 0$. Or what is the same thing, we may write

$$A = a^2 + 8aBy^2 = 6aBy^2, \quad B = -a^2y^2 + (1 + 16y^2 + 8y^2)8(8a + 8BX) + 8(15)yXX,$$

in like manner, the four equations are given when the points 1 and 2 are determined. Conversely if (y, y) and (z, z) are corresponding points the conic $\Omega + 0X^2$ pass through the points 1, 2, 3, 4.

154. Observe that the intersections of the cubic with the fixed curve are given by the equation $\Omega + \mathbb{P}\mathbf{X}\mathbf{\Pi} = 0$; the line $\mathbf{\Pi} = 0$, hence the conic passes through the points 5, 6, 7, 8, where 5, 6, 7, 8 are the intersections of the lines $\xi = 0$ with the conic. The variable line $\mathbf{T}P + \mathbf{X}P = 0$

corresponds to the cubic transformed integral; the deficiency is here equal to 2, integral relations, which agree with the conditions of a curve of deficiency p , p being 2.

[book p. 202] **155.** If we now assume

$$\begin{aligned} d\omega &= x d\omega_1 d\omega_2, & d\omega &= x d\omega_2, & d\omega_2, & d\omega &= x d\omega_2, \\ d\omega &= x d\omega_2 + y d\omega_1, & d\omega &= x d\omega_2 + y d\omega_2, & d\omega &= x d\omega_2 + y d\omega_2, \end{aligned}$$

or say

$$\begin{aligned} u, v &= \int_0^1 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} + \int_0^2 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} \right), \\ u', v' &= \int_0^1 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} + \int_0^2 (x, y) \left(\frac{x dy - y dx}{\sqrt{P}} \right), \end{aligned}$$

where u, v, u', v' are each of them, of course, regarded as functions of the four points 1, 2, 1', 2', then u, v, u', v' are functions which are connected together by the proper relations, and which present themselves either as singly, or simultaneously by means of the formula for σ functions.

In a basically completely determined way, the points 1, 2, 1, 2' are not perfectly determined, but are determined by elliptic integrals: but we have these functions, and the systems of points 1, 2, 1', 2' taken in connection with the various functions (u, v, u', v') in a system as 1, 2, 1', 2', etc. are points which it is the difference between the formula on the integrals. We have, in fact, that

$$\begin{aligned} \sqrt{u(u_1 - u_2) - 2(u_1 + u_2)u'} &= u(u_1 - u_3 + u_2)u'(\dots), \\ B(u_1 - u_2), &= u(u_3 - u_2)A(u_1 - u_3)(\dots), \\ A(u_1 - u_2), &= u(u_3 - u_2)A(u_1 - u_3)(\dots), \end{aligned}$$

and so on for the four other systems of equations $C(u, v)$, $D(u, v)$ respectively, formed in the case of an integral function of u, v of orders 1, 2, respectively (and not in the case of an arbitrary function ρ, ρ' of these integrals).

156. The differential relations thus become

$$du du' + dv dv' = 0, \quad x' du - y' dv = 0, \quad x du - y dv;$$

where T, J are constants which are determinable as definite integrals by the consideration that du, dv form a complete system. The equations $u_1 + u_2 = u, v_1 + v_2 = v$, then the points u, v are points where we have the consideration that the conic passes through the points 1, 2, 1', 2' on the cubic; whence considering the conic as a function of the conic f may be defined as a continuous function of these arguments: there is then a marked change in the integral to the number of constants in a curve of deficiency p .

156. The differential relations thus become

$$du du' + dv dv' = 0, \quad x du + y dv = 0, \quad x' du + y' dv = 0,$$

where J, J are constants which are determinable as definite integrals by the consideration that the conic f may be defined as a continuous function of these arguments: there is then a marked change in the integral to the number of constants in a curve of deficiency p .

[book p. 203] **157.** We may, in the first instance, disregarding altogether the consideration of the arguments u, v , attend only to the algebraic functions such as $\sqrt{X}, \sqrt{Y}, \sqrt{Z}$, of the coordinates of the pairs of points 1, 2, 1', 2', 3, 4, and so on as regards $\mathbf{X}$ as a function of the coordinates of the points only.

Origin of the Double-Letter Functions. Art. Nos. 158 and 159.

158. The cubic curve $Ax + B = 0$ may be taken to be a curve through two of the points of contact, say the points α, β : we have a Ω a function $(x, y)^2$; or throwing out the factor $(z - \beta y)(z - \beta y)y$, by the identity

$$(\alpha - x)(\beta - y) = -y(\alpha - y)(\beta - x) + (a, b, c, d, e, f, g, h)(x, y, z)^2,$$

and hence, introducing $\sqrt{a}, \sqrt{b}$ as quasi-coordinates of the curve, the original a becomes equal to the identity

$$(\sqrt{a} - x)(\sqrt{a} - y) + (a, b, c, d, e, f, g, h)(x, y, z)^2 = 0,$$

and hence, introducing a notation which it is convenient to use, we set

$$\begin{aligned} & \sqrt{a} \sqrt{b} - x \sqrt{a} y \sqrt{b} a (b - x \sqrt{a})(\sqrt{b} \sqrt{c})(c - b \sqrt{c})(b - x \sqrt{a}) a (b - x \sqrt{a})(\sqrt{c} b) \\ & - Y (b - x \sqrt{a})(b - \sqrt{a})(a - x \sqrt{a}) a y (a - x \sqrt{a})(b - x \sqrt{a}) a b (a - x \sqrt{a}) a b a b; \end{aligned}$$

159. Putting each of the y 's equal 1, we have the identity

$$(\alpha - x_1)(\alpha - x_2) - 2(1 + x_1)(1 + x_2)(1 - x_1)(1 - x_2)(1 - x_3)(1 - x_4) = 0,$$

and hence, putting in for shortness

$$a = \sqrt{a(1 + x_1)(1 + x_2)}, \quad x = \sqrt{(1 + x_1)(1 + x_2)x_1x_2}, \quad x' = \sqrt{(1 - x_1)(1 - x_2)x_1x_2};$$

and in like manner, $\beta = \sqrt{b(1 + x_1)(1 + x_2)b'b'a(bb' - c)(cb' - c)b}$, where the symbols $ab a, abb', a a', bcab', bbbca a'$ are determinate.

[book p. 204] We have thus

$$\frac{(a - x_1)(a - x_2)}{(a - x_1)(a - x_2)} = \frac{a^2}{a\sqrt{a}a\sqrt{a}}, \quad = \frac{ax_1ax_2}{ax_1\sqrt{a}ax_2\sqrt{a}ax_1ax_2} aabacd,$$

which last equation, taking the square root, becomes

$$\frac{a - x_1}{\sqrt{a}ax_1} = - \frac{ax_2ax_3}{\sqrt{a}ax_3ax_4},$$

or, substituting in the first equation, we have

$$\frac{x - x_1}{x - x_1} = \frac{ax_1ay_1ax_2ay_2a}{ax_1ax_2aa^2\sqrt{ay_1ay_2}ax_1\sqrt{ax_2}},$$

which clearly divides out being in each of these cases respectively.

159. Considering the dual DE as an abbreviation for the double letter ABC, DEF , the expressed dual being always accompanied by the letter E where we have left in the consideration of the double-letter functions

$$\Lambda_{ab} = \frac{1}{ax_1\sqrt{a}ax_2\sqrt{a}ax_1\sqrt{ax_2}} a(b\sqrt{ax_1}\sqrt{ax_2}\sqrt{a}a) \cdots,$$

in connexion with the already mentioned single-letter functions $\Lambda_a = \sqrt{ax_1ax_2}$, etc. viz. in this notation the equation just proved is

$$\Lambda_{ab} = \frac{\sqrt{(a-1)(1+a-c)}\sqrt{a-d}}{\sqrt{(a-1)(1+a-c)}\sqrt{a-d}} \frac{\Lambda_a\Lambda_a}{\Lambda_a};$$

and it thus appears that, by reason of the constancy of the symmetry of the first three letters and of the symmetry of the last three letters in the function Λ_{ab} , Λ_{ab} , the function Λ_{ab} , a , b is interchangeable by the suffices a and b , c , equal to that of the single-letter functions.

160. Considering the dual DE as an abbreviation for the double letter ABC , DEF , the expressed dual being always accompanied by the letter E where we have left in the consideration of the double-letter functions $\Lambda_{ab} = \sqrt{ax_1}\sqrt{ax_2}$, &c.

In connexion with the already mentioned single-letter functions $\Lambda_a = \sqrt{ax_1 ax_2}$, etc. we have a system of formulæ as $\sqrt{ax_1 ax_2}$, etc. and it is from these formulæ were considered by Aronhold in his (Göttingen) Memoir of 1865 (whence the term “double-letter functions”). It is in like manner be shown that the function $\Lambda_a = \sqrt{ax_1 ax_2}$, etc. is the quotient of a major function $(x, y, z)^{n-2}$ by the linear function 012.

Chapter VII. The Addition Theory.

The Addition Theory. Art. Nos. 160 to 163.

160. We have the six single-letter symbols A, B, C, D, E, F , viz. $\Lambda_a = \sqrt{(a-x_1)(a-x_2)}$, etc. and the ten double-letter symbols $AB, AC, AD, AE, AF, BC, BD, BE, CD, CE, DE$, viz. $\Lambda_{ab} = \frac{1}{a-b} [\sqrt{(a-x_1)(b-x_2)} - \sqrt{(a-x_2)(b-x_1)}]$, etc. from 10 functions being connected by algebraical relations which admit immediately deducible from the considerations of the various forms of $\Lambda_{ab} \Lambda_{ab}$, etc. The relations between the variables 1 and 2 being so connected that the systems form, in regard to one of these variables, say x_1 , an integral relation, which agrees with the symbol a , of the function above considered.

The relation between the variables x_1, x_2 is that they correspond to a line meeting the curve in two points which we call 1, 2, viz. taking the equation in the form

$$(ax^2 + 2bx + c)y^2 - (a-x)(b-x)(c-x)(d-x) = 0,$$

where it is a character expression of the function $(x, y, z)^{n-2} = 0$, of the line is to be taken to be

$$ay^2 + bx^2 + 2c'y + d = 0,$$

and we have thus the two relations

$$(ax_1^2 + 2bx_1 + c)y_1^2 = (a-x_1)(b-x_1)(c-x_1)(d-x_1),$$

$$(ax_2^2 + 2bx_2 + c)y_2^2 = (a-x_2)(b-x_2)(c-x_2)(d-x_2),$$

$$ay_1y_2 + bx_1x_2 + c'(x_1 + x_2) + d = 0;$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2 , and we have then the two relations

$$ax_1y_2 + bx_2y_1 + 2c'y_2 + d(\dots),$$

$$ay_1 + bx_1 + 2c'y_2 + d(\dots).$$

which determines the symmetric functions of x_1, x_2 .

[book p. 205] *The Addition Theory.* Art. Nos. 160 to 163.

160. We have the six single-letter functions A, B, C, D, E, F , viz. $\Lambda_a = \sqrt{(a-x_1)(a-x_2)}$, &c.; and the ten double-letter symbols $AB, AC, AD, AE, AF, BC, BD, BE, CD, CE, DE$, viz. $\Lambda_{ab} = \frac{1}{a-b} [\sqrt{(a-x_1)(b-x_2)} - \sqrt{(a-x_2)(b-x_1)}]$, &c.

These 10 functions being connected by algebraical relations which admit immediately deducible from the consideration of the various forms of $\Lambda_{ab} \Lambda_{ab}$ etc. The relations between the variables 1 and 2 being so connected that the systems form, in regard to one of these variables, say x_1 , an integral relation, which agrees with the symbol a , of the function above considered.

161. The relation is, in fact, as is given in any of the papers 1, 2 &c. of the double- $\mathbf{T}$ -function, Crelle, t. LXXVIII. (1865), pp. 74–81, [785]. Writing successively $z = a, \beta, b, b, b, a, \beta, b, b', a, \beta, b$, we have

$$\begin{aligned} ax_1^2 + 2bx_1 + c &= y_1 \sqrt{(a-x_1)(b-x_1)(c-x_1)(d-x_1)}, \\ ax_2^2 + 2bx_2 + c &= y_2 \sqrt{(a-x_2)(b-x_2)(c-x_2)(d-x_2)}; \end{aligned}$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2 , and we have then the two relations

$$\begin{aligned} ax_1y_2 + bx_2y_1 + 2c'y_2 + d(\dots), \\ ay_1 + bx_1 + 2c'y_2 + d(\dots), \\ ay_1 + bx_2 + 2c'y_2 + d(\dots), \\ ay_1 + bx_1 + 2c'y_2 + d(\dots). \end{aligned}$$

which equations serve to determine the ratios $a : \beta : k$ in terms of x_1, x_2, y_1, y_2 , and we have then the two relations

$$\begin{aligned} ay_1 + bx_1 + 2c'y_2 + d(\dots), \\ ay_1 + bx_2 + 2c'y_2 + d(\dots), \end{aligned}$$

which determines the symmetric functions of x_1, x_2 .